70篇插图英语阅读小册

新顺序版 · 第01—10篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
01	以为柠檬汁能隐身的劫匪	8
02	会烤化汽车的大楼	14
03	越狱的章鱼	20
04	喝下细菌的医生	26
05	新可乐的 79 天	33
06	被偷走的微笑	39
07	会算术的马	45
08	OK 原本是个玩笑	51
09	血里的指纹	57
10	第37手	63

70篇插图英语阅读小册

新顺序版 · 第11—20篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
11	千年虫：被成功抹去的危机	69
12	被街道重塑的大脑	76
13	两次卖掉埃菲尔铁塔的骗子	82
14	罗马混凝土的千年之谜	89
15	一夜之间换边开车	96
16	微塑料：无处不在，危害几何?	102
17	冰里的五千年凶案	108
18	一颗棉花糖的真相	114
19	两度摘得诺奖的女人	121
20	需要戴眼镜的望远镜	128

70篇插图英语阅读小册

新顺序版 · 第21—30篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
21	让蚂蚁变僵尸的真菌	135
22	禁区里的生机	141
23	十一天不睡觉的少年	148
24	集装箱：重塑世界的铁盒子	155
25	永不沉没的神话	162
26	法老的诅咒	169
27	孩子们发明的语言	175
28	发明了数码相机，却被它打败	182
29	青蒿一握，济世千万	188
30	洞穴里的十三个人	195

70篇插图英语阅读小册

新顺序版 · 第31—40篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
31	几个少年和一只狗发现的洞穴	202
32	黄石公园狼影再现	209
33	走出苏联的俄罗斯方块	215
34	多莉：一只羊的复制	222
35	不合拍的风：塔科马大桥的陨落	228
36	改变世界的科学巨匠	235
37	末日种子库的第一次取种	241
38	郁金香狂热：泡沫与传说	248
39	一座火山，造出了《弗兰肯斯坦》	255
40	从车库到世界之巅	262

70篇插图英语阅读小册

新顺序版 · 第41—50篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
41	夺命毒雾：伦敦1952	269
42	稻济天下	275
43	炸药大王的救心药	282
44	被拒绝了十二次的哈利·波特	289
45	从空气中造面包	296
46	失败的胶水，成功的便利贴	303
47	震动世界的一声巨响	310
48	光之殇	317
49	冰海求生：坚忍号传奇	323
50	一枚邮票改变通信	330

70篇插图英语阅读小册

新顺序版 · 第51—60篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
51	林下之网	336
52	向鹌鹑宣战	343
53	兴登堡号的 34 秒	350
54	比信先到家的鸟	357
55	大恶臭逼出了下水道	364
56	原子弹之父的审判	371
57	一颗刺果钩出的发明	378
58	破解霍乱之谜的医生	385
59	拥挤的苍穹	392
60	被降级的冥王星	399

70篇插图英语阅读小册

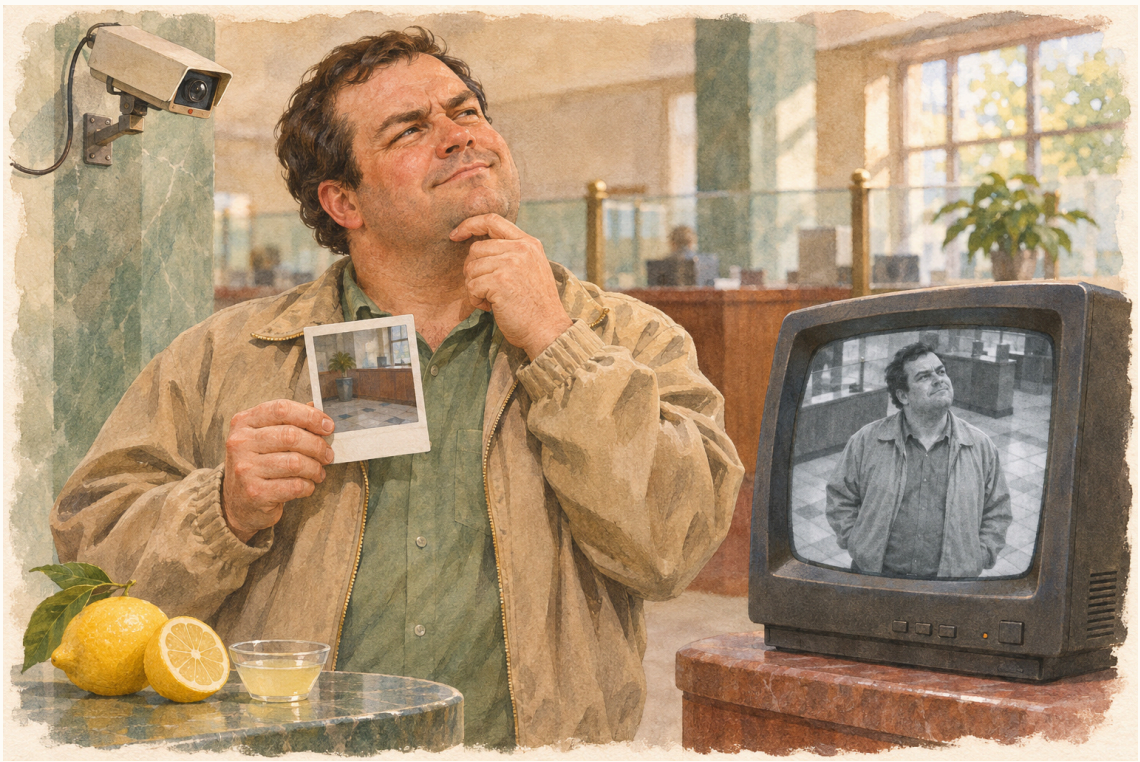
新顺序版 · 第61—70篇

篇号已按新顺序统一编号为01—70。点击篇目即可跳转；每篇依次包含插图、英文、中文、词汇与回想。

新篇号		全本页码
61	鲸歌易帜	405
62	为机器点燃思想的人	412
63	没有到来的零日	419
64	改变人类命运的霉菌	426
65	一封信惊醒了达尔文	433
66	入侵澳洲的毒蟾	440
67	从一道划痕到天花绝迹	447
68	阿波罗13号：成功的失败	454
69	拯救天空	461
70	写给外星人的一张唱片	468

以为柠檬汁能隐身的劫匪

The Robber Who Thought Lemon Juice Made Him Invisible



我们为什么有时看不见自己的判断失误?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 6 January 1995, McArthur Wheeler and an accomplice raided two Pittsburgh-area banks at gunpoint in broad daylight, without disguise. Their only camouflage was lemon juice, rubbed on in the earnest conviction that it would render them invisible to cameras.

The rationale had a kernel of truth. Lemon juice is a time-honoured invisible ink, and Wheeler made an inventive leap from paper to skin. His Polaroid self-portrait came out devoid of its subject, a result he took as conclusive.

The security cameras begged to differ. On 19 April the eleven o'clock news showed viewers the damning images, and Wheeler was arrested shortly after midnight. Confronted with the tapes, he protested in sheer disbelief that he had worn the juice.

Reading about the case, the Cornell psychologist David Dunning pondered a disquieting proposition. Perhaps incompetence can cloak itself, robbing people of the discernment to notice their deficiencies.

For a 1999 paper, Dunning and his student Justin Kruger quizzed Cornell undergraduates on humour, grammar, and logical reasoning. The bottom quarter of scorers averaged the 12th percentile but generously placed themselves near the 62nd, whereas high achievers showed undue modesty. The unskilled, the authors argued, bear a dual burden, since appraisal calls for the same skills as execution.

Detractors traced part of the Dunning-Kruger disparity to regression to the mean, a statistical mirage the paper partially acknowledged. Studies designed to avoid it still find a smaller but tangible effect. In 2023 the pair garnered the Grawemeyer psychology award.

In fairness, Wheeler had tested his method empirically before staking his liberty on it. The camera that missed him was the one he turned on himself.

中文译文

1995年1月6日，麦克阿瑟·惠勒和一名同伙在大白天持枪抢劫了匹兹堡地区的两家银行，连乔装都没有。他们唯一的伪装是抹在身上的柠檬汁：两人真心相信，这能让自己在摄像头前隐形。

这套想法多少有些事实依据。柠檬汁用作隐形墨水由来已久，惠勒别出心裁，把纸上的道理套到了皮肤上。他用宝丽来相机自拍，照片里却没有自己的身影，他便把这当成了确凿证据。

监控摄像头可不认同这套说法。4月19日晚11点的新闻向观众播出了那些铁证般的画面；午夜刚过不久，惠勒就被捕了。面对录像，他仍难以置信，辩解说自己明明抹了柠檬汁。

读到这个案子后，康奈尔大学心理学家戴维·邓宁思考起一种令人不安的可能：一个人能力不足，也许还会因此缺少判断力，察觉不到自己有哪些不足。

为撰写1999年的一篇论文，邓宁和他的学生贾斯汀·克鲁格测试了康奈尔大学本科生在幽默判断、语法和逻辑推理方面的能力。成绩排在末尾四分之一的学生，平均实际排名只超过约12%的参试者，却以为自己排在约62%的人前面；成绩优异者反而低估了自己的表现。两位作者认为，缺乏相应技能的人面临双重困难，因为评价表现与实际做事，需要的是同样的技能。

批评者认为，邓宁—克鲁格研究中的部分差距来自均值回归，是一种统计假象，原论文也在一定程度上承认了这一点。不过，为规避这个问题而设计的研究，仍发现了较小但确实存在的效应。2023年，两人获得了格文美尔心理学奖。

平心而论，在拿自己的自由做赌注之前，惠勒还真做过实际测试。那台没拍到他的相机，偏偏就是他对准自己的那一台。

读懂 percentile
百分位表示一个人在群体中的相对位置。第12百分位不等于答对了12%的题。

核心词汇

先记这22项。连着下方的原文短语一起回想。

accomplice n.
同伙；共犯
and an accomplice

raid v.
突袭；文中指持枪抢劫
raided two Pittsburgh-area banks

disguise n.
乔装；伪装
without disguise

conviction n.
坚定的信念
in the earnest conviction

render v.
使……处于某种状态
render them invisible to cameras

rationale n.
推理依据；理由
The rationale had a kernel of truth.

conclusive adj.
确凿的；足以定论的
a result he took as conclusive.

be confronted with 动词短语
面对（被出示的证据）
Confronted with the tapes

ponder v.
仔细思考
pondered a disquieting proposition

proposition n.
观点；待论证的命题
a disquieting proposition

incompetence n.
能力不足
incompetence can cloak itself

discernment n.
辨识力；判断力
the discernment to notice

deficiency n.
不足；能力缺陷
notice their deficiencies

paper n.
论文
For a 1999 paper

logical adj.
逻辑的；合乎逻辑的
logical reasoning

appraisal n.
评估（自身表现）
appraisal calls for the same skills

call for 动词短语
需要；要求
calls for the same skills

execution n.
执行；实际完成任务
the same skills as execution

disparity n.
差异；差距
the Dunning-Kruger disparity

tangible adj.
切实存在的
a smaller but tangible effect

empirically adv.
以观察或实验为依据
tested his method empirically

stake v.
拿……作赌注
staking his liberty on it

文中词汇与搭配

补齐另外34项。这里保留每个表达在本文中的意思。

in broad daylight 介词短语
光天化日之下

camouflage n.
伪装；掩饰手段

rub … on … 动词短语
涂抹在……上

earnest adj.
真诚认真的

kernel n.
一点真实成分

time-honoured adj.
沿用已久的

ink n.
墨水

inventive adj.
富有创意的（此处含调侃）

leap n.
思维上的跳跃

self-portrait n.
自拍照

devoid of 形容词短语
完全没有

beg to differ 动词短语
表示不赞同（拟人反讽）

viewer n.
电视观众

damning adj.
有力证明罪行的

sheer adj.
十足的；完全的

disbelief n.
不相信；难以置信

disquieting adj.
令人不安的

cloak v.
掩盖；遮蔽

rob v.
剥夺（某人拥有的东西）

quiz v.
测试；测验

undergraduate n.
本科生

scorer n.
测验中的答题者

generously adv.
“大方地”高估自己（反讽）

achiever n.
此处指高分者

undue adj.
过分的；不应有的

modesty n.
谦虚；自谦

unskilled adj.
缺乏技能的；文中指能力不足者

dual adj.
双重的

detractor n.
批评者

mirage n.
假象

partially adv.
部分地

garner v.
赢得（奖项）

in fairness 介词短语
平心而论

liberty n.
人身自由

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ had a kernel of truth.

02 His Polaroid self-portrait came out devoid of its subject, a result he took as _____.

03 The unskilled, the authors argued, bear a dual burden, since _____ calls for the same skills as execution.

04 Studies designed to avoid it still find a smaller but _____ effect.

文章主线

- 误信 劫匪惠勒把柠檬汁的隐形墨水原理误套到皮肤上；自拍没拍到自己，他便当作躲过镜头的验证。
- 落网 他涂着柠檬汁抢银行，仍被监控拍下；新闻播出后被捕，面对录像也不信办法失效。
- 研究 案件启发邓宁与克鲁格研究自评；幽默、语法和逻辑测试中，低分者常高估自己，高分者反而低估。
- 解释 两人认为，完成任务与判断水平需要相同技能，能力不足者可能也难察觉自己的不足。
- 限度 统计因素能解释部分差异；避开这些因素后仍有较小效应，原结论须谨慎看待。

参考答案　1 rationale 2 conclusive 3 appraisal 4 tangible

会烤化汽车的大楼

The Building That Melted a Car

弯曲的玻璃墙，怎么会烤坏汽车？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 2 September 2013 the news broke that a London skyscraper under construction had damaged a Jaguar. Martin Lindsay had left the saloon by the pavement on Eastcheap, opposite the tower's frontage. He returned to find its wing mirror warped, its panels deformed and its badge distorted. The glazed facade above was to blame.

Rafael Viñoly curved the south face of 20 Fenchurch Street inward, enlarging the upper storeys, where tenants pay a premium. That geometry acted like a magnifying glass, focusing sunlight into a beam up to six times brighter than direct sun. It was measured at about 110 degrees Celsius, and a reporter fried an egg in a pan using the reflection.

The developers issued an apology and reimbursed almost 1,000 pounds for repairs. Parking bays were suspended and a provisional barrier was erected. The beam struck for about two hours a day, two to three weeks annually, because of the sun's angle of elevation. Newspapers dubbed the Walkie-Talkie the Walkie-Scorchie, a taunt.

The architect was candid, admitting many mistakes. Viñoly had anticipated the focusing, he said, but not its intensity. Sun louvres in the original design were reportedly removed in cost-cutting. Nor was this his first architectural miscalculation. In 2010 the infamous glare from his curved, reflective Vdara hotel in Las Vegas burned a guest's scalp by the pool.

On 12 February 2014 a permanent fixture was announced: horizontal aluminium fins to diffuse the brightness, reportedly costing several million pounds. In September 2015 Building Design's critics gave the notorious tower the Carbuncle Cup, a trophy for the UK's worst new building. The ballot was unanimous.

中文译文

2013年9月2日，一则消息传开：伦敦一座在建摩天大楼损坏了一辆捷豹轿车。马丁·林赛将车停在伊斯特奇普街的人行道旁，正对大楼临街立面。回来时，他发现后视镜翘曲、车身板件变形、车标也已扭曲。问题出在上方那面玻璃幕墙。

建筑师拉斐尔·维诺利将芬丘奇街20号的南立面设计成内凹曲面，扩大了租金较高的上层楼面。这种几何形状像放大镜一样，将阳光聚成一道亮度最高可达直射阳光六倍的光束。受照处测得约110摄氏度，一名记者甚至用反射光在平底锅里煎熟了鸡蛋。

开发商道歉，并赔付近1000英镑修理费。路边停车位被暂停使用，现场搭起临时屏障。受太阳仰角影响，强光每年出现两至三周，在这些日子每天照射约两小时。报纸把“对讲机大楼”戏称为“烤焦机大楼”，以此挖苦它。

建筑师坦率承认犯了许多错误。维诺利说，他预料到会聚光，却没有想到强度如此之大。据报道，原设计的遮阳百叶在削减成本时被删掉了。这也不是他的第一次建筑失算：2010年，他设计的拉斯维加斯维达拉酒店曲面反光墙产生刺眼强光，灼伤了泳池旁一位客人的头皮。

2014年2月12日，开发商公布一项永久装置方案：加装水平铝制鳍片，分散反光，据报费用达数百万英镑。2015年9月，《建筑设计》杂志的评论家将“烂疮杯”颁给这座声名狼藉的大楼，这是授予英国最差新建筑的奖杯。这次表决全体一致。

核心词汇

先记这22项。连着下方的原文短语一起回想。

under construction 介词短语
正在建设中
a London skyscraper under construction

warped adj.
翘曲的；变形的
its wing mirror warped

deformed adj.
变形的
its panels deformed

distorted adj.
扭曲的；变形的
its badge distorted

facade n.
建筑立面；正面外墙
The glazed facade

premium n.
溢价；额外支付的费用
tenants pay a premium

geometry n.
几何形状；几何结构
That geometry acted

beam n.
光束
focusing sunlight into a beam

reflection n.
反射光；反射
using the reflection

reimburse v.
报销；赔付费用
reimbursed almost 1,000 pounds

provisional adj.
临时的；暂定的
a provisional barrier

erect v.
搭建；竖起
a provisional barrier was erected

candid adj.
坦率的
The architect was candid

anticipate v.
预料；预期
had anticipated the focusing

intensity n.
强度
not its intensity

cost-cutting n.
削减成本
removed in cost-cutting

miscalculation n.
误算；失算
architectural miscalculation

glare n.
刺眼强光
the infamous glare

fixture n.
固定装置
a permanent fixture

diffuse v.
使分散；散射
to diffuse the brightness

ballot n.
投票；表决
The ballot was unanimous

unanimous adj.
全体一致的
The ballot was unanimous

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

skyscraper n.
摩天大楼

saloon n.
轿车（英式用法）

pavement n.
人行道（英式用法）

frontage n.
建筑正面；临街立面

badge n.
车标；徽章

glazed adj.
镶玻璃的

enlarge v.
扩大

storey n.
楼层

tenant n.
租户

magnify v.
放大

sunlight n.
阳光

reporter n.
记者

pan n.
平底锅

developer n.
开发商

apology n.
道歉

suspend v.
暂停使用；中止

barrier n.
屏障；隔挡

annually adv.
每年

angle of elevation 名词短语
仰角

taunt n.
嘲讽；挖苦

architect n.
建筑师

sun louvre 名词短语
遮阳百叶

architectural adj.
建筑方面的

infamous adj.
臭名昭著的

reflective adj.
反光的

scalp n.
头皮

permanent adj.
永久的

horizontal adj.
水平的

aluminium adj.
铝制的

fin n.
鳍片；薄片构件

brightness n.
亮度

critic n.
评论家

notorious adj.
声名狼藉的

trophy n.
奖杯

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That _____ acted like a magnifying glass, focusing sunlight into a beam up to six times brighter than direct sun.

02 The developers issued an apology and _____ almost 1,000 pounds for repairs.

03 Viñoly had anticipated the focusing, he said, but not its _____.

04 The ballot was _____.

文章主线

- 事故 2013年，伦敦对讲机大楼反射的阳光，使路边捷豹车的后视镜、车身和车标变形。
- 原理 内凹的玻璃南立面扩大了高租金的上层面积，也聚集阳光；光束亮度最高为直射光六倍，受照处约110°C。
- 应急 开发商赔偿近千英镑并停用车位、设置屏障；因太阳角度，强光每年出现两三周，每天约两小时。
- 失算 建筑师承认低估强度；遮阳百叶据报因省钱被删；他设计的另一酒店，2010年就曾反光伤人。
- 后续 2014年公布加装横向铝片、分散反光的方案，据报需数百万英镑；2015年获英国最差新建筑奖。

参考答案　1 geometry 2 reimbursed 3 intensity 4 unanimous

越狱的章鱼

The Octopus That Escaped

章鱼的聪明，等于拥有意识吗？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

One night in early 2016, after keepers at New Zealand's National Aquarium in Napier had left, an octopus named Inky crept through an aperture in the mesh atop its tank. The fugitive presumably dropped to the floor, crossed the room, and compressed its elastic body into a drain, whose pipe stretched some 50 metres to the Pacific.

Staff reconstructed that route from the slime residue it deposited. Inky had resided in captivity since 2014, when a fisherman retrieved it from a crayfish pot off Napier. The breakout went unreported for about three months. Made public in April 2016, the story travelled the globe.

Such agility stems from an octopus's pliable body, which has no skeleton or shell. Its only hard part is its beak, so any crevice that accommodates the beak will do. Roughly two-thirds of its 500 million neurons lie in the limbs, which handle sensory input and react with some autonomy.

Its wits are as versatile as its anatomy. A giant Pacific octopus in Seattle learned to open screw-top jars, needing 15 minutes at the outset, later about two. In a 2010 study, octopuses came to discriminate between two strangers, a feeder and one who touched them with a bristly stick, adopting a different hue and posture for each.

In July 2012 the Cambridge Declaration on Consciousness listed octopuses among animals possessing the neurological substrates for conscious awareness. A November 2021 UK government review recommended treating them as sentient, and the Animal Welfare (Sentience) Act encompassed them in April 2022. None of this proves an inner life. Inky, somewhere beyond the pipe, remained wholly oblivious to the debate.

中文译文

2016年初的一个夜晚，新西兰内皮尔国家水族馆的饲养员离开后，一只名叫Inky的章鱼从水箱顶部网罩的开口悄悄爬了出来。这名“逃脱者”大概先落到地板上，穿过房间，再把柔软而有弹性的身体挤进排水口；那里的管道长约50米，一直通向太平洋。

工作人员根据它留下的黏液痕迹，还原了这条可能的路线。Inky自2014年起便生活在圈养环境中；当时，一名渔民从内皮尔附近海域的捕虾笼中取出了它。这次出逃约三个月都没有见诸报道，直到2016年4月公开后，故事才传遍全球。

章鱼如此敏捷，得益于柔韧的身体：它没有骨骼，也没有外壳，唯一坚硬的部位是喙。所以，只要缝隙能容下喙，身体便能通过。章鱼约5亿个神经元中，大约三分之二分布在腕足里；腕足会处理感觉信息，并能在一定程度上自主作出反应。

它的智力与身体构造一样，表现出多样的能力。西雅图的一只巨型太平洋章鱼学会了打开螺旋盖罐子，最初需要15分钟，后来只需约两分钟。在2010年的一项研究中，章鱼逐渐学会区分两个原本陌生的人：一个给它们喂食，另一个拿硬毛棒碰它们。面对不同的人，它们会呈现不同的颜色和姿态。

2012年7月，《剑桥意识宣言》把章鱼列为具有支撑意识的神经基础的动物之一。2021年11月，英国政府的一份评估报告建议把它们视为有感知能力的动物；2022年4月，《动物福利（感知）法》也将它们纳入适用范围。然而，这些都不能证明章鱼拥有主观的内在体验。在管道另一头某处的Inky，依然对人们的这场争论一无所知。

核心词汇

先记这22项。连着下方的原文短语一起回想。

aperture n.
开口；孔隙
an aperture in the mesh

presumably adv.
据推测；大概
fugitive presumably dropped

compress v.
压缩；挤压
and compressed its elastic body

reconstruct v.
根据线索还原
reconstructed that route

residue n.
残留物；痕迹
slime residue it deposited.

captivity n.
圈养；囚禁状态
in captivity since 2014,

agility n.
敏捷；灵活
Such agility

pliable adj.
柔韧的；易弯曲的
octopus's pliable body,

crevice n.
窄缝；裂隙
crevice that accommodates the beak

neuron n.
神经元
neurons lie in the limbs

sensory input n.
感觉信息；感官输入
handle sensory input

autonomy n.
自主性
react with some autonomy.

versatile adj.
多才多能的；功能多样的
as versatile as its anatomy.

anatomy n.
身体构造
as versatile as its anatomy

discriminate v.
区分；辨别
to discriminate between two strangers,

hue n.
颜色；色调
different hue and posture

posture n.
姿态
a different hue and posture

neurological adj.
神经系统的
the neurological substrates

substrate n.
活动所依托的物质基础
neurological substrates

awareness n.
意识；觉察
conscious awareness.

sentient adj.
有感知能力的
as sentient,

oblivious adj.
对……毫不知情的
wholly oblivious to the debate.

文中词汇与搭配

补充词汇共30项。保留每个表达在本文中的意思。

keeper n.
饲养员

creep v.
悄悄爬行

mesh n.
网；网罩

atop prep.
在……顶部

fugitive n.
逃脱者；逃亡者

elastic adj.
有弹性的

stretch v.
延伸

slime n.
黏液

deposit v.
留下；沉积

reside v.
居住；栖居

retrieve v.
取出；取回

crayfish pot n.
捕虾笼

breakout n.
逃脱；越狱

unreported adj.
未被报道的

globe n.
全球；世界

stem from phr. v.
源于

skeleton n.
骨骼

beak n.
喙

accommodate v.
容纳

limb n.
肢体；此处指腕足

wits n.
智力；机智

jar n.
罐子

at the outset phr.
起初

feeder n.
喂食者

adopt v.
采取；呈现

possess v.
具有；拥有

conscious adj.
有意识的

encompass v.
涵盖；纳入

inner adj.
内在的；内心的

wholly adv.
完全地

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Staff _____ that route from the slime residue it deposited.

02 Such _____ stems from an octopus's pliable body, which has no skeleton or shell.

03 Roughly two-thirds of its 500 million neurons lie in the limbs, which handle sensory input and react with some _____.

04 Inky, somewhere beyond the pipe, remained wholly _____ to the debate.

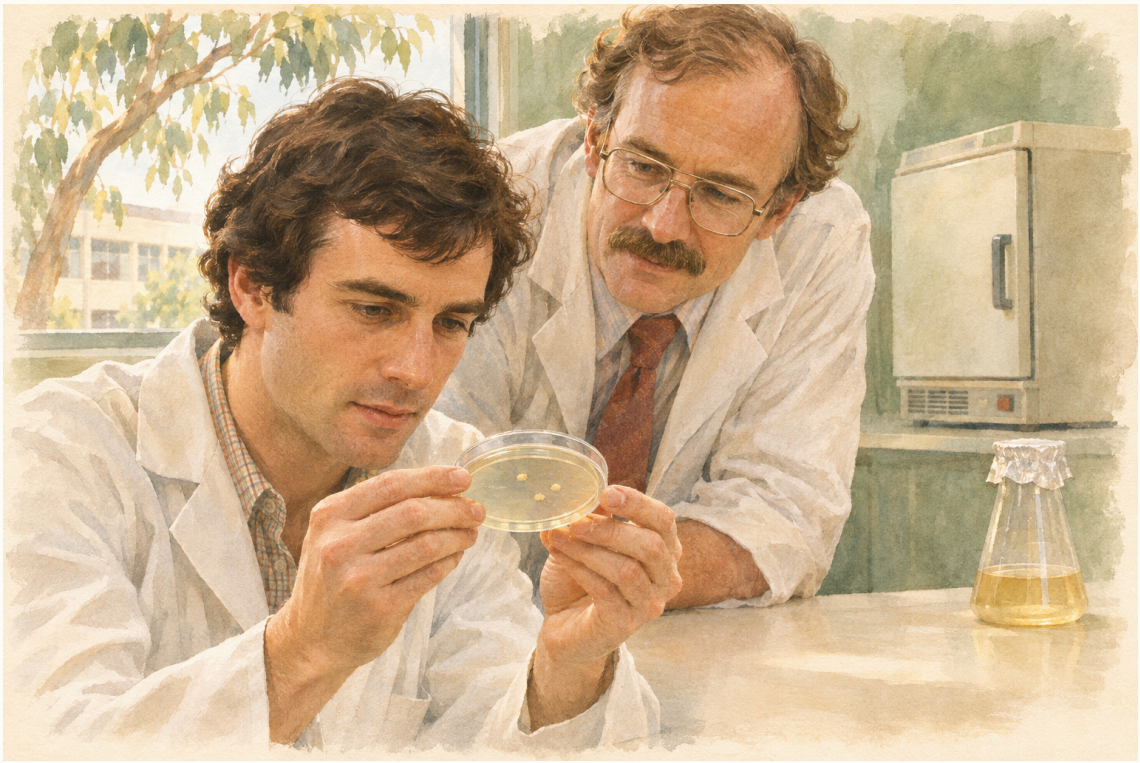
文章主线

- 出逃 章鱼Inky逃出水箱；工作人员据黏液痕迹推测，它钻入通向海洋的约50米排水管。
- 身体 章鱼没有骨骼和外壳；只要坚硬的喙能通过，柔软的身体就能挤过缝隙。
- 能力 章鱼约5亿个神经元中，约三分之二位于腕足，腕足可部分自主反应；章鱼还能学会开罐、区分喂食者与拿刺毛棒碰它的人。
- 认定 2012年《剑桥意识宣言》认为章鱼具有意识的神经基础；英国2022年法律也将章鱼视为有感知动物。
- 限度 这些仍不能直接证明章鱼确有内心体验。

参考答案　1 reconstructed 2 agility 3 autonomy 4 oblivious

喝下细菌的医生

The Doctor Who Drank Bacteria to Win an Argument



一杯菌液，如何改变对胃病的认识？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1984 Barry Marshall, a 32-year-old doctor at Fremantle Hospital, drank a brew of beef broth into which the microbial growth of two Petri dishes had been scraped.

The prevalent doctrine blamed ulcers on stress, spicy food, and excess acid secretion. No bacterium, the profession firmly held, could withstand so caustic and hostile a habitat.

Yet Robin Warren, a pathologist at Royal Perth Hospital, had seen curved, coiled bacteria in gastric biopsies from patients with gastritis. Marshall began to collaborate with him. Attempts to culture the germ were invariably unsuccessful until Easter 1982, when plates sat undisturbed in the incubator for about five days. They were ordinarily binned after two. Colonies blossomed. The organism was later named Helicobacter pylori.

In 1983 unimpressed assessors at the Gastroenterological Society of Australia relegated the pair's paper to the bottom tenth of that year's submissions. Efforts to infect swine were likewise in vain. Marshall made himself the guinea pig.

Within days Marshall was vomiting, his breath had a disagreeable odour and weariness set in. About ten days later an endoscopy revealed gastritis, an inflammation of the stomach's mucous membrane and a precursor to ulcers, albeit not an ulcer itself. The germ was cultured directly from his stomach.

Spurred on by his insistent spouse, Marshall took antibiotics, which cleared the infection. Ulcers came to be treated with such medication.

In 2005 Marshall and Warren shared the Nobel Prize in Physiology or Medicine for discovering Helicobacter pylori and its role in gastritis and peptic ulcer disease. Gastritis, the illness Marshall had willingly given himself, is named in the prize citation.

中文译文

1984年，弗里曼特尔医院32岁的医生巴里·马歇尔喝下了一份牛肉清汤混合液，里面加入了从两个培养皿中刮取的微生物培养物。

当时占主导地位的学说把溃疡归因于压力、辛辣食物和胃酸分泌过多。医学界坚信，没有细菌能够在这样具有腐蚀性、又不利于生存的环境中存活。

然而，皇家珀斯医院的病理学家罗宾·沃伦，已经在胃炎患者的胃部活检样本中看到了弯曲、盘绕的细菌。马歇尔开始与他合作。培养这种病菌的尝试一直未获成功，直到1982年复活节期间，培养皿在培养箱里原封不动地放了约五天。通常，这些培养皿两天后就会被丢弃。菌落这一次长了出来。这种微生物后来被命名为幽门螺杆菌。

1983年，澳大利亚胃肠病学会的评审者对这项研究并不看好，把两人的论文列入了当年投稿中排名最低的十分之一。让猪感染这种细菌的努力也同样没有成功。于是，马歇尔把自己当成了实验对象。

几天之内，马歇尔开始呕吐，呼出的气带有难闻的气味，身体也逐渐感到疲倦。大约十天后，胃镜检查发现他患了胃炎，也就是胃黏膜的炎症，属于溃疡的前期病变，但它本身并不是溃疡。研究者还直接从他的胃中培养出了这种病菌。

在妻子不断催促下，马歇尔服用了抗生素，感染随之被清除。后来，溃疡也逐渐开始采用这类药物治疗。

2005年，马歇尔与沃伦因发现幽门螺杆菌及其在胃炎和消化性溃疡中的作用，共同获得诺贝尔生理学或医学奖。马歇尔当年主动让自己患上的胃炎，也被写进了获奖评语。

本文的自我试验导致胃炎，并非溃疡；两者在叙述中需要严格区分。

核心词汇

先记这22项。连着下方的原文短语一起回想。

prevalent adj.
盛行的；普遍接受的
The prevalent doctrine blamed ulcers

doctrine n.
学说；信条
prevalent doctrine blamed ulcers

withstand v.
经受住；抵御
could withstand

hostile adj.
不利于生存的
so caustic and hostile a habitat.

pathologist n.
病理学家
a pathologist at Royal Perth

gastric adj.
胃部的
in gastric biopsies

biopsy n.
活体组织检查；活检样本
gastric biopsies

collaborate v.
合作
to collaborate with him.

culture v.
培养（细菌）
Attempts to culture the germ

invariably adv.
无一例外地；总是
were invariably unsuccessful

incubator n.
培养箱
in the incubator for about five days.

relegate v.
把……降至较低地位
relegated the pair's paper

in vain phr.
徒劳；没有结果
were likewise in vain.

guinea pig n.
实验对象；试验品
Marshall made himself the guinea pig.

inflammation n.
炎症
an inflammation

mucous membrane n.
黏膜
the stomach's mucous membrane

precursor n.
前兆；先发生的事物
and a precursor to ulcers,

albeit conj.
尽管；虽然
albeit not an ulcer itself.

spur on phr. v.
催促；激励
Spurred on by his insistent spouse,

antibiotic n.
抗生素
Marshall took antibiotics,

role n.
作用
its role in gastritis

prize citation n.
获奖评语；颁奖词
is named in the prize citation.

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

brew n.
调制的混合液

broth n.
肉汤；清汤

microbial adj.
微生物的

scrape v.
刮取；刮下

ulcer n.
溃疡

spicy adj.
辛辣的

acid n.
酸；此处指胃酸

secretion n.
分泌

bacterium n.
细菌（单数）

profession n.
此处指医学界

firmly adv.
坚定地

caustic adj.
有腐蚀性的

habitat n.
栖息环境

coiled adj.
盘绕的；卷曲的

attempt n.
尝试

germ n.
病菌

unsuccessful adj.
未成功的

undisturbed adj.
未被移动或打扰的

ordinarily adv.
通常

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

bin v.
丢进垃圾箱

blossom v.
繁盛生长

unimpressed adj.
不以为然的

assessor n.
评审人员

submission n.
提交的论文；投稿

effort n.
努力

swine n.
猪

likewise adv.
同样地

vomit v.
呕吐

disagreeable adj.
令人不快的

odour n.
气味

weariness n.
疲倦；乏力

set in phr. v.
开始出现并持续

reveal v.
显示；揭示

directly adv.
直接地

insistent adj.
坚持要求的

spouse n.
配偶

medication n.
药物

willingly adv.
自愿地

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 No bacterium, the profession firmly held, could _____ so caustic and hostile a habitat.

02 Marshall began to _____ with him.

03 Efforts to infect swine were likewise _____.

04 Spurred on by his insistent spouse, Marshall took _____, which cleared the infection.

文章主线

- 旧说 医学界曾把溃疡归于压力、辛辣食物和胃酸过多，认为细菌无法在胃中存活。
- 发现 沃伦在胃炎患者胃中看见细菌，与马歇尔合作培养成功；该菌后来被命名为幽门螺杆菌。
- 自试 马歇尔喝下菌液后患的是胃炎，并非溃疡；胃中也培养出了同种细菌。
- 治疗 抗生素清除了他的感染，溃疡治疗后来也逐步采用这类药物。
- 认可 两人因发现这种细菌及其在胃炎、消化性溃疡中的作用，共获2005年诺贝尔生理学或医学奖。

参考答案　1 withstand 2 collaborate 3 in vain 4 antibiotics

新可乐的 79 天

The 79 Days of New Coke

盲测胜出的新可乐，为什么遭到反对？

英文原文

Pepsi had set up booths in shopping centres across the US, inviting shoppers to sip two unmarked beverages and state a preference. Figures varied by region, but tasters generally picked Pepsi more often. Each stall, letting consumers judge by palate alone, doubled as publicity.

Coca-Cola's testing was thorough and extensive. Nearly 200,000 blind taste tests showed a sweeter formula beating the original and Pepsi. On 23 April 1985 the company announced it would replace its 99-year-old syrup, a year short of its centenary. The public called the soda New Coke. Its sweetness had given it a winning flavour, but launching the concoction was a wager.

The backlash was a grassroots outcry. Consumer hotline calls rose from about 400 a day to about 1,500, almost all furious complaints. Protest groups were organised, and some faithful drinkers stocked up on the original. In June 1985 one protester, Gay Mullins, filed a lawsuit to keep New Coke out of packaging like the original's. Its dismissal came a week later.

Pepsi's chief executive gave his employees a day off, treating the change as Coke's concession. After 79 days, on 11 July 1985, the original formula was relaunched as Coca-Cola Classic. The U-turn was a vindication for loyalists. ABC interrupted a soap opera to broadcast the comeback, which led two network news bulletins.

Coca-Cola had misjudged what the tests measured. They captured a mouthful, not the emotional attachment that nearly a century of familiarity had built. To many, the old taste carried nostalgia and heritage. The tests had hidden the brand from every taster, so they could not quantify what it meant to fiercely loyal drinkers.

中文译文

百事可乐曾在全美各地的购物中心设置摊位，请购物者小口品尝两种未标明品牌的饮料，并说出自己更喜欢哪一种。结果因地区而异，但品尝者总体上选择百事的次数更多。这些让消费者只凭味觉判断的摊位，同时也起着宣传作用。

可口可乐的测试既细致又广泛。近二十万次盲测显示，一种更甜的配方比原配方和百事更受欢迎。1985年4月23日，公司宣布，要更换使用了99年的糖浆配方，此时距它的百年诞辰只差一年。公众把这种汽水称为“新可乐”。甜味让它在口味比较中胜出，但推出这种调制饮料，却是一场冒险。

随之而来的是民众自发的强烈反对。消费者热线每天接到的电话，从约400通增至约1500通，几乎全是愤怒的投诉。抗议团体组织起来，一些忠实消费者开始囤积原版可乐。1985年6月，抗议者盖伊·马林斯提起诉讼，要求禁止新可乐使用与原版相似的包装。一周后，这项诉讼被驳回。

百事首席执行官把对手改换配方视为可口可乐的让步，给员工放了一天假。79天后，也就是1985年7月11日，原配方以“可口可乐经典”的名称重新上市。这次转向，让原配方的忠实拥护者觉得自己的坚持得到了证明。ABC电视台中断一部肥皂剧，播报原配方回归的消息；这条消息还成了两家电视网新闻节目的头条。

可口可乐误判了测试究竟衡量到了什么。测试捕捉到的是一小口饮料的味道，却没有测出近一个世纪的熟悉感所累积的情感依恋。对许多人而言，旧口味承载着怀旧情绪与传统。测试向每一位品尝者隐藏了品牌，因此无法量化：这个品牌对那些极其忠诚的消费者，究竟意味着什么。

核心词汇

先记这22项。连着下方的原文短语一起回想。

unmarked adj.
未作标记的；未标品牌的
two unmarked beverages

beverage n.
饮料
unmarked beverages

preference n.
偏好；更喜欢的事物
state a preference.

palate n.
味觉；品味能力
judge by palate alone

publicity n.
宣传；公众关注
doubled as publicity.

thorough adj.
全面细致的；彻底的
thorough and extensive.

extensive adj.
广泛的；覆盖面大的
thorough and extensive.

formula n.
配方
a sweeter formula

wager n.
赌注；冒险之举
launching the concoction was a wager.

backlash n.
强烈反对；反弹
The backlash was a grassroots outcry.

grassroots adj.
基层的；普通民众自发的
a grassroots outcry.

outcry n.
强烈抗议；公愤
a grassroots outcry.

faithful adj.
忠实的
some faithful drinkers

lawsuit n.
诉讼
filed a lawsuit

dismissal n.
驳回（诉讼）
Its dismissal came a week later.

concession n.
让步
Coke's concession.

relaunch v.
重新推出；重新上市
relaunched as Coca-Cola Classic.

vindication n.
证明正确；获得认可
a vindication for loyalists.

misjudge v.
误判
misjudged what the tests measured.

attachment n.
依恋；情感联系
the emotional attachment

nostalgia n.
怀旧；对往昔的眷恋
nostalgia and heritage.

quantify v.
量化
could not quantify what it meant

文中词汇与搭配

补充词汇共30项。保留每个表达在本文中的意思。

booth n.
摊位；小亭

shopper n.
购物者

sip v.
小口喝；啜饮

taster n.
品尝者

consumer n.
消费者

syrup n.
糖浆

centenary n.
一百周年；百年纪念

soda n.
汽水

sweetness n.
甜味

flavour n.
味道；口味

concoction n.
调制品；混合饮料

hotline n.
热线电话

furious adj.
极其愤怒的

complaint n.
投诉；抱怨

protest group n.
抗议团体

organise v.
组织；筹组

chief executive n.
首席执行官

employee n.
员工

U-turn n.
彻底转向；政策大反转

loyalist n.
忠实拥护者

interrupt v.
打断；中断

broadcast v.
播报；广播

comeback n.
回归；复出

news bulletin n.
新闻简报；新闻节目

mouthful n.
一口的量

emotional adj.
情感的

familiarity n.
熟悉；熟悉感

heritage n.
传统；传承下来的事物

fiercely adv.
强烈地；极其

loyal adj.
忠诚的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Coca-Cola's testing was thorough and _____.

02 Its sweetness had given it a winning flavour, but launching the concoction was a _____.

03 Its _____ came a week later.

04 Coca-Cola had _____ what the tests measured.

文章主线

- 竞争 百事用不标品牌的试饮吸引消费者，通常更受青睐；可口可乐随后做了近20万次盲测。
- 换方 更甜的新配方在盲测中胜过旧版和百事，公司于1985年用它替换已有99年历史的老配方。
- 反弹 热线日来电从约400增至1500，几乎全是投诉；顾客囤积旧款、结社抗议，还有人起诉但被驳回。
- 回归 仅79天后，老配方以经典可口可乐之名回归，成为电视新闻头条。
- 漏测 盲测只比较一口饮料的味道，遗漏了品牌与熟悉口味承载的感情、怀旧和传统。

参考答案　1 extensive 2 wager 3 dismissal 4 misjudged

被偷走的微笑

The Stolen Smile

一幅失踪的画，如何引来更多目光？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On Monday 21 August 1911, when the Louvre was customarily closed, the Mona Lisa was discreetly plucked from its wall. The disappearance went unnoticed until the next day, when a painter arriving to sketch found a conspicuous gap of bare plaster. Custodians shuttered the museum for a week.

When the museum reopened, curious onlookers queued to stare at the vacant space where the masterpiece had hung. On 7 September 1911 the poet Guillaume Apollinaire was held in custody for about a week, and Pablo Picasso was interrogated. Both were cleared of involvement in the burglary.

The intruder was Vincenzo Peruggia, an Italian who had temporarily worked at the Louvre as a glazier. Having lurked in a storeroom on Sunday, he strode out unchallenged in a workman's smock, the painting tucked beneath his clothes. For more than two years it lay stashed in a false-bottomed trunk in his Paris lodgings.

In December 1913 Peruggia contacted Alfredo Geri, an art dealer in Florence. The Uffizi's director authenticated the painting at Peruggia's hotel; Peruggia was apprehended on 11 December 1913. Citing patriotism, he said he wanted it repatriated, mistakenly believing Napoleon had looted it. In fact Leonardo da Vinci had accompanied it to France at the invitation of King Francis I.

After showings in Italy, the painting was reinstated in the Louvre on 4 January 1914. In June 1914 a tribunal imposed a penalty of a year and 15 days, but he was imprisoned for about seven months. Many historians argue the heist elevated one esteemed gem among many to the world's most renowned painting. If so, its fame began with a bare wall.

中文译文

1911年8月21日星期一，卢浮宫照例闭馆，《蒙娜丽莎》就在这天被人悄悄从墙上取走。直到第二天，一位前来画素描的画家发现，裸露的灰泥墙上有一块显眼的空缺，画作失踪才被察觉。管理人员随后将博物馆关闭了一周。

博物馆重新开放后，好奇的参观者排起队来，盯着原本悬挂那幅杰作的空位看。1911年9月7日，诗人纪尧姆·阿波利奈尔被拘押，羁押时间约为一周；巴勃罗·毕加索也受到讯问。两人最终都洗清了参与这起盗窃案的嫌疑。

闯入者是意大利人文森佐·佩鲁贾，他曾在卢浮宫短期担任玻璃工。他星期天便藏进储藏室，随后身穿工作罩衫，把画塞在衣服下面，大步走了出去，没有受到任何阻拦。此后两年多，画作一直藏在他巴黎住处一只带有夹层底的箱子里。

1913年12月，佩鲁贾联系了佛罗伦萨艺术品商人阿尔弗雷多·杰里。乌菲齐美术馆馆长在佩鲁贾下榻的旅馆鉴定了画作的真伪；佩鲁贾于1913年12月11日被捕。他以爱国为由，声称想把画送回祖国，误以为它曾被拿破仑掠走。事实上，列奥纳多·达·芬奇是应国王弗朗索瓦一世的邀请，带着这幅画来到法国的。

画作在意大利展出后，于1914年1月4日重新回到卢浮宫。1914年6月，法庭判处佩鲁贾一年零15天监禁，但他实际被关押了约七个月。许多历史学家认为，这起盗窃案让一幅原本只是众多备受推崇的珍品之一的画，跃升为世界上最著名的画作。若真如此，它的名声便始于一面空墙。

核心词汇

先记这22项。连着下方的原文短语一起回想。

discreetly adv.
不引人注意地；悄悄地
discreetly plucked from its wall.

disappearance n.
失踪；消失
The disappearance went unnoticed

unnoticed adj.
未被察觉的
went unnoticed until the next day

conspicuous adj.
显眼的；引人注意的
a conspicuous gap of bare plaster.

vacant adj.
空着的
to stare at the vacant space

masterpiece n.
杰作
where the masterpiece had hung.

in custody phr.
被拘押；在羁押中
held in custody for about a week

interrogate v.
讯问；审问
and Pablo Picasso was interrogated.

involvement n.
参与；牵涉
cleared of involvement

burglary n.
入室盗窃；盗窃案
involvement in the burglary.

unchallenged adj.
未受盘问或阻拦的
strode out unchallenged

stash v.
藏匿
stashed in a false-bottomed trunk

authenticate v.
鉴定真伪；证实为真
authenticated the painting

apprehend v.
逮捕
Peruggia was apprehended

patriotism n.
爱国主义；爱国之情
Citing patriotism,

repatriate v.
送回祖国；归还原属国
wanted it repatriated

loot v.
掠夺；劫掠
believing Napoleon had looted it.

reinstate v.
使恢复原位；放回原处
was reinstated in the Louvre

tribunal n.
法庭
a tribunal imposed a penalty

impose v.
施加；判处（刑罚）
imposed a penalty

heist n.
盗窃大案
the heist elevated one esteemed gem

renowned adj.
著名的；享有盛誉的
the world's most renowned painting.

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

customarily adv.
照例；按照惯例

pluck v.
摘下；取走

sketch v.
画素描；速写

gap n.
空缺；缺口

plaster n.
灰泥；墙面抹灰

custodian n.
管理人员；看管者

shutter v.
关闭

reopen v.
重新开放

curious adj.
好奇的

onlooker n.
围观者

stare at phr. v.
盯着看；凝视

intruder n.
闯入者

temporarily adv.
暂时地；短期地

lurk v.
潜伏；躲藏

stride v.
大步走

tuck v.
塞入；藏入

beneath prep.
在……下面

trunk n.
大箱子；旅行箱

lodgings n.
租住处

contact v.
联系；联络

dealer n.
商人；交易商

mistakenly adv.
错误地

accompany v.
陪同；此处指带着同行

at the invitation of phr.
应……的邀请

showing n.
展出；展示

penalty n.
刑罚；处罚

imprison v.
监禁；关押

elevate v.
提高；提升地位

esteemed adj.
受尊重的；备受推崇的

gem n.
珍品；瑰宝

to prep.
与elevate连用，表示提升到

fame n.
名声；声誉

bare adj.
空的；裸露的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The disappearance went unnoticed until the next day, when a painter arriving to sketch found a _____ gap of bare plaster.

02 Both were cleared of _____ in the burglary.

03 Citing _____, he said he wanted it repatriated, mistakenly believing Napoleon had looted it.

04 Many historians argue the heist elevated one esteemed gem among many to the world's most _____ painting.

文章主线

- 失窃 1911年《蒙娜丽莎》在卢浮宫闭馆日失窃，次日才被发现；重开后，人们排队看空墙。
- 手法 曾在卢浮宫做玻璃工的佩鲁贾藏入库房，穿工服把画夹在衣内带走，在巴黎藏了两年多。
- 寻回 诗人阿波利奈尔、画家毕加索曾被查，均洗清嫌疑；1913年佩鲁贾联系意大利画商，画经鉴定后寻回，他也被捕。
- 误信 佩鲁贾声称要让画回到意大利，却误以为是拿破仑所掠；画本由达芬奇带往法国。
- 影响 画于1914年回到卢浮宫；许多史学家认为，盗案让它从名画之一变成世界最著名的画。

参考答案　1 conspicuous 2 involvement 3 patriotism 4 renowned

会算术的马

The Horse That Could Count

汉斯是在算数，还是在读懂人的动作?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Wilhelm von Osten, a mathematics teacher and trainer in Berlin, owned a stallion, Hans. Asked to calculate a sum, attempt subtraction, or name a date, Hans tapped out the answer with a hoof. Such aptitude suggested brilliance, and demonstrations in his owner's courtyard made him seem intelligent. The New York Times reported the stunning act in 1904.

A commission of thirteen was formed to investigate. It included a veterinarian, a circus manager, a cavalry officer, schoolteachers, and the director of the Berlin zoological garden. Its members looked for trickery and collusion. In September 1904 they found no deception, in effect certifying the act authentic.

In 1907 the psychologist Oskar Pfungst tested the act systematically, altering one variable after another. He kept onlookers away, rotated questioners, fitted blinders, and sometimes used people ignorant of the answer. When Hans could see the questioner, he answered correctly 50 times in 56. With the questioner invisible, his accuracy fell to two in 35.

The cues came from the questioners. Each would lean forward as tapping began, and tension rose as the taps neared the answer. At the right tap it eased, and the questioner would straighten or jerk the head upward. Every gesture and motion was subconscious, made unwittingly by people with no wish to deceive.

Double-blind designs in psychology and animal cognition guard against the Clever Hans effect, a phenomenon of bias in which expectancy shapes behavioural results. In the laboratory, Pfungst took the animal's place and proved sensitive to the cues of attentive volunteers. Hans went, on his owner's death in June 1909, to a jeweller still convinced that such animals reasoned.

中文译文

柏林的数学教师兼驯马师威廉·冯·奥斯滕，养着一匹名叫汉斯的公马。让它计算加法、做减法，或者说出某个日期时，汉斯就会用一只蹄子敲出答案。这种本领看起来显示了非凡的才智；它在主人庭院里的表演，也让人觉得它很聪明。1904年，《纽约时报》报道了这场令人惊叹的表演。

一个由13人组成的委员会随后成立，负责调查。成员包括兽医、马戏团经理、骑兵军官、中小学教师，以及柏林动物园园长。委员们查找是否存在欺骗手法或暗中串通。1904年9月，他们没有发现有意欺骗的证据，实际上等于认可了表演的真实性。

1907年，心理学家奥斯卡·普丰斯特系统地检验了这项表演，逐一改变实验变量。他让围观者远离现场，轮换提问者，给马戴上遮眼罩，有时还请不知道答案的人提问。汉斯能够看见提问者时，56次作答中有50次正确；看不见提问者时，35次里却只对了2次。

提示来自提问的人。马蹄开始敲击时，人会向前倾身；次数越接近正确答案，人就越紧张。敲到正确的那一下时，紧张感便缓和下来，提问者会挺直身子，或者猛地抬一下头。这些姿态和动作全都是下意识的，是无意中做出来的；提问者并没有欺骗别人的打算。

在心理学和动物认知研究中，双盲实验设计可用来防范“聪明汉斯效应”：研究者的预期通过无意暗示，影响了所观察到的行为结果。在实验室里，普丰斯特还亲自代替那匹马，证明自己也能敏锐地察觉专注的志愿者发出的细微信号。汉斯的主人于1909年6月去世后，它归了一位珠宝商所有；那位珠宝商依然相信，这类动物能够进行推理。

没有发现有意作弊，并不等于已经证明马能够做算术。

核心词汇

先记这22项。连着下方的原文短语一起回想。

aptitude n.
天资；才能
Such aptitude suggested brilliance,

commission n.
调查委员会
A commission of thirteen

investigate v.
调查
to investigate.

collusion n.
暗中串通
trickery and collusion

deception n.
欺骗
they found no deception

certify v.
确认；证明
certifying the act authentic

authentic adj.
真实可信的
certifying the act authentic

systematically adv.
系统地；有步骤地
tested the act systematically

variable n.
变量
altering one variable after another.

accuracy n.
准确率；准确性
his accuracy fell to two in 35.

cue n.
提示；信号
The cues came from the questioners.

tension n.
紧张
and tension rose

gesture n.
手势；姿态
Every gesture and motion

subconscious adj.
下意识的
motion was subconscious

unwittingly adv.
无意中；不知不觉地
made unwittingly by people

deceive v.
欺骗
no wish to deceive

double-blind adj.
双盲的
Double-blind designs

cognition n.
认知
animal cognition

phenomenon n.
现象
a phenomenon of bias

bias n.
偏差；偏倚
a phenomenon of bias

expectancy n.
预期；期待心理
expectancy shapes behavioural results.

sensitive adj.
敏感的；能察觉细微变化的
proved sensitive to the cues

文中词汇与搭配

补充词汇共28项。保留每个表达在本文中的意思。

trainer n.
驯马师；训练者

stallion n.
公马

calculate v.
计算

subtraction n.
减法

brilliance n.
非凡才智

demonstration n.
演示；表演

courtyard n.
庭院

intelligent adj.
聪明的

stunning adj.
令人惊叹的

veterinarian n.
兽医

cavalry n.
骑兵

zoological adj.
动物学的

trickery n.
欺骗手法

psychologist n.
心理学家

rotate v.
轮换

questioner n.
提问者

blinder n.
马用遮眼罩

ignorant adj.
不知道的

correctly adv.
正确地

invisible adj.
不可见的；看不见的

lean v.
倾身；倾斜

straighten v.
挺直身子

jerk v.
猛地一抬；急动

motion n.
动作

psychology n.
心理学

behavioural adj.
行为的

attentive adj.
专注的

jeweller n.
珠宝商

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In 1907 the psychologist Oskar Pfungst tested the act systematically, altering one _____ after another.

02 With the questioner invisible, his _____ fell to two in 35.

03 The _____ came from the questioners.

04 Every gesture and motion was _____, made unwittingly by people with no wish to deceive.

文章主线

- 奇观 柏林一匹名叫汉斯的马用蹄子敲出算式答案；早期调查没发现作弊，表演因而被视为可信。
- 检验 心理学家普丰斯特改变视线等条件：能看见提问者时56次对50次，看不见时35次仅对2次。
- 机制 提问者会无意中在答案处抬头或放松；汉斯识别这些身体信号来停止敲蹄，被误以为会算术。
- 复现 普丰斯特亲自代替汉斯，也能根据志愿者的细微动作作答，说明这种影响不限于马。
- 防范 研究者的期待可经无意暗示影响结果，这叫聪明汉斯效应；双盲等实验设计用来减少这种偏差。

参考答案　1 variable 2 accuracy 3 cues 4 subconscious

OK 原本是个玩笑

The Joke That Became OK

一个拼写笑话，怎样变成了OK?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

OK, often said to be the planet's most frequently spoken word, began as a newspaper gag. From 1838 abbreviations were all the vogue in Boston's papers, and the craze turned comic through deliberate misspellings: O.W. for 'oll wright', K.Y. for 'know yuse'. The humour lay in the persona of an uneducated bumpkin.

On 23 March 1839 a Boston Morning Post satire took a swipe at a contemporary, the Providence Journal, over a mock society parodying an ordinance against ringing dinner bells. It carried the first known 'o.k.', glossed as 'all correct'. The letters stood for a knowingly illiterate 'oll korrect'. The presumed writer was the editor, Charles Gordon Greene.

Partisan politics propagated the letters nationwide. In 1840 Martin Van Buren, the incumbent president, sought re-election. His New York backers, punning on his nickname 'Old Kinderhook', after his birthplace, formed the Democratic O.K. Club, making the letters a double-meaning slogan. That November William Henry Harrison unseated him, 234 electoral votes to 60.

On 30 March 1840 the New York Morning Herald printed a fabricated smear: the former president Andrew Jackson, a supposedly inept speller, had marked paperwork O.K., for 'Ole Kurrek'. Jackson spelled reasonably well and never marked documents O.K. Within about twenty years, people did so in earnest. The bogus notation was merely premature.

Rival stories offered exotic pedigrees: Choctaw 'okeh' and Greek 'ola kala'. Authoritative dictionaries favoured Choctaw as late as 1961. In 1963–64 the linguist Allen Walker Read excavated the neglected Boston provenance from archival newspapers in six influential American Speech articles. Most dictionaries now embrace his derivation, a pedigree of bad spelling and dinner-bell satire.

中文译文

OK常被称为世界上最常说的词，起初却只是报纸上的一个笑话。从1838年起，波士顿报纸流行起缩写，又故意把词拼错来逗乐，例如用O.W.代表“oll wright”，用K.Y.代表“know yuse”。笑点在于装出一个没受过教育的乡下人的口吻。

1839年3月23日，《波士顿晨邮报》刊出一篇讽刺文，借一个戏仿禁敲开饭铃法令的虚构社团，挖苦同期的《普罗维登斯日报》。文中出现了已知最早的“o.k.”，并解释为“all correct”，即完全正确。这两个字母实际取自故意写错的“oll korrect”。文章的推定作者是编辑查尔斯·戈登·格林。

党派政治把这两个字母传播到了全国。1840年，在任总统马丁·范布伦谋求连任。他在纽约的支持者拿他根据出生地得来的绰号“Old Kinderhook”玩双关，成立民主党O.K.俱乐部，让OK成了一句兼具两重意思的竞选口号。当年11月，威廉·亨利·哈里森以234张选举人票对60张击败了他。

1840年3月30日，《纽约先驱晨报》刊出一则捏造的抹黑故事，说前总统安德鲁·杰克逊不擅拼写，曾在公文上批注O.K.，代表“Ole Kurrek”。实际上，杰克逊的拼写并不差，也从未这样批过文件。不过，在大约二十年内，人们果真开始认真使用这种批注。那种虚构的批注用法，不过是比实际流行早了一些。

其他词源说法另寻异域来历，例如乔克托语的“okeh”和希腊语的“ola kala”。直到1961年，权威词典仍倾向于乔克托语说。1963至1964年，语言学家艾伦·沃克·里德从报纸档案中发掘出被忽略的波士顿起源，并在《美国语言》发表了六篇影响深远的文章。如今，大多数词典都接受他的词源解释：这个词的身世，原来连着故意错拼和开饭铃讽刺文。

核心词汇

先记这22项。连着下方的原文短语一起回想。

abbreviation n.
缩写
From 1838 abbreviations were all the

all the vogue idiom
极为流行的
1838 abbreviations were all the vogue

deliberate adj.
故意的
comic through deliberate misspellings:

misspelling n.
拼写错误
comic through deliberate misspellings:

lie in phr.v.
在于
The humour lay in the persona of an

persona n.
扮演的角色；人物形象
The humour lay in the persona of an

satire n.
讽刺文章；讽刺作品
1839 a Boston Morning Post satire

take a swipe at idiom
挖苦；抨击
took a swipe at a contemporary, the

ordinance n.
法令；条例
an ordinance against ringing dinner

gloss as v.phr.
被解释为
first known 'o.k.', glossed as 'all

partisan adj.
党派的
Partisan politics propagated the

propagate v.
传播；推广
Partisan politics propagated the

incumbent adj.
在任的
Buren, the incumbent president, sought

pun on phr.v.
拿……玩双关
His New York backers, punning on his

slogan n.
口号
the letters a double-meaning slogan.

fabricated adj.
捏造的
Herald printed a fabricated smear:

inept adj.
不擅长的；笨拙的
a supposedly inept speller, had marked

bogus adj.
虚假的
The bogus notation was merely

premature adj.
过早的
bogus notation was merely premature.

authoritative adj.
权威的
Authoritative dictionaries favoured

provenance n.
起源；出处
the neglected Boston provenance from

derivation n.
词源；由来
now embrace his derivation, a pedigree

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

frequently adv.phr.
最频繁地；最常

gag n.
笑话；逗乐的噱头

craze n.
风潮；一时的狂热

comic adj.
滑稽的；逗乐的

humour n.
幽默；笑点

uneducated adj.
未受教育的

bumpkin n.
乡下土包子

contemporary n.
同期的报刊；同代者

mock adj.
虚构来戏仿的

parody v.
戏仿；滑稽模仿

knowingly adv.
明知地；有意地

illiterate adj.
不识字的；像文盲一样的

presumed adj.
推定的

editor n.
编辑

re-election n.
再次当选；连任

backer n.
支持者

birthplace n.
出生地

double-meaning adj.
有双重含义的

unseat v.
使落选；使失去职位

electoral vote n.phr.
选举人票

smear n.
抹黑；诽谤故事

supposedly adv.
据说；据称

paperwork n.
公文；文书

reasonably adv.
相当地；还算

in earnest idiom
认真地；当真地

notation n.
批注；记号

exotic adj.
异域的

pedigree n.
身世；来历

excavate v.
发掘；查出

neglected adj.
被忽略的

archival adj.
档案中的

influential adj.
有重要影响的

embrace v.
接受；采纳

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 From 1838 _____ were all the vogue in Boston's papers,

02 Partisan politics _____ the letters nationwide.

03 The bogus notation was merely _____.

04 Most dictionaries now embrace his _____,

文章主线

- 玩笑 1838年起，波士顿报纸流行故意拼错词再缩写；1839年出现已知最早的“o.k.”。
- 原义 o.k.取自all correct的故意错拼oll korrect，意为“完全正确”，笑点在模仿文盲口吻。
- 助推 1840年，范布伦阵营以绰号Old Kinderhook的缩写OK，兼指他本人和“完全正确”，用作竞选口号。
- 流传 范布伦败选，OK仍流传；杰克逊用它批文件的传闻是捏造，但约二十年内人们已认真如此批注。
- 考证 语言学家里德于1963至1964年查旧报，重建波士顿词源；多数词典采纳，取代早先乔克托语等说法。

参考答案　1 abbreviations 2 propagated 3 premature 4 derivation

血里的指纹

The Fingerprint in the Blood

DNA为何先替无辜者洗清嫌疑?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In November 1983 a 15-year-old girl was murdered in the Leicestershire village of Narborough, and the homicide went unsolved. In the summer of 1986 a second 15-year-old was slain near neighbouring Enderby. A 17-year-old local youth confessed to the second killing, and to detectives the perplexing investigation was seemingly resolved.

The revelation had come two years previously. On 10 September 1984 Alec Jeffreys, a geneticist at the University of Leicester, discerned on an X-ray film molecular markers whose layout varied considerably from person to person. He called them DNA fingerprints, since, like fingerprints, they differentiated individuals. The journal Nature promptly carried his paper in March 1985.

Police enlisted Jeffreys, and his genetic tests overturned the erroneous confession. The youth was not the perpetrator, and the same attacker had killed both girls. The youth became the first person known to be cleared by DNA evidence. The forensic innovation had freed an innocent before it implicated anyone.

In January 1987 police commenced the world's first large-scale DNA screening, asking more than 5,000 men in three villages for blood or saliva. None matched. The elusive killer, a local baker, had induced a co-worker to give a sample in his name, evading suspicion. In August 1987 someone overheard the co-worker discussing the substitution and notified police.

Colin Pitchfork was detained on 19 September 1987, and his DNA profile matched. In January 1988 he pleaded guilty and was jailed for life, the first person convicted of murder on the basis of DNA evidence. A tip-off, not the screening, had unmasked him. The test's first beneficiary had been a blameless juvenile who had falsely admitted the offence.

中文译文

1983年11月，一名15岁女孩在莱斯特郡的纳伯勒村遇害，这起凶杀案一直没有侦破。1986年夏天，又一名15岁女孩在邻近的恩德比附近被杀。一名17岁的当地少年供认了第二起命案；在警探们看来，这场令人困惑的调查似乎终于有了结果。

那项关键发现，其实早在两年前就已经出现。1984年9月10日，莱斯特大学遗传学家亚历克·杰弗里斯在一张X光胶片上辨认出一些分子标记，它们的排列在人与人之间差别很大。他把这些标记称为DNA指纹，因为它们和指纹一样，可以区分不同的人。《自然》期刊很快就在1985年3月刊登了他的论文。

警方请来杰弗里斯协助调查，他的基因检测推翻了那份错误供词。那名少年并非行凶者，而两名女孩确实是被同一个人杀害的。少年成为已知第一位凭DNA证据洗清嫌疑的人。在凭这项技术指认任何人涉案之前，这项法医学创新先让一名无辜者重获自由。

1987年1月，警方开始了世界上第一次大规模DNA筛查，请三个村庄的5000多名男子提供血液或唾液样本，结果没有一份吻合。那名难以追查的凶手是一位当地面包师；他事先诱使同事冒用自己的姓名提供样本，从而躲过怀疑。1987年8月，有人无意中听见这名同事谈论冒名顶替的事，便通知了警方。

1987年9月19日，科林·皮奇福克被拘留，他的DNA图谱与案件中的样本吻合。1988年1月，他认罪并被判终身监禁，成为第一位依据DNA证据被判谋杀罪的人。揭开他真面目的，是一条举报线索，而不是那次筛查本身。这项检测最先帮助的，却是一名清白的少年——他曾错误地承认自己犯下了那起罪行。

核心词汇

先记这22项。连着下方的原文短语一起回想。

homicide n.
凶杀；杀人案件
and the homicide went unsolved.

confess v.
供认；坦白
confessed to the second killing

investigation n.
调查；侦查
the perplexing investigation

geneticist n.
遗传学家
a geneticist

discern v.
辨认出；察觉
discerned on an X-ray film

molecular adj.
分子的
molecular markers

vary v.
存在差异；变化
layout varied considerably

differentiate v.
区分
they differentiated individuals.

overturn v.
推翻（原有判断或供词）
overturned the erroneous confession.

erroneous adj.
错误的
the erroneous confession.

perpetrator n.
作案者；行凶者
The youth was not the perpetrator,

forensic adj.
法医学的；刑事鉴证的
The forensic innovation

innovation n.
创新；新技术
The forensic innovation

innocent n.
无辜者
had freed an innocent

implicate v.
牵连；表明……涉案
before it implicated anyone.

commence v.
开始
In January 1987 police commenced

elusive adj.
难以找到或抓住的
The elusive killer, a local baker,

induce v.
诱使；促使
had induced a co-worker

substitution n.
替代；此处指冒名顶替
discussing the substitution

profile n.
此处指DNA图谱
his DNA profile matched.

tip-off n.
举报线索
A tip-off, not the screening,

beneficiary n.
受益者
The test's first beneficiary

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

unsolved adj.
未侦破的

slay v.
杀死（过去分词）

neighbouring adj.
邻近的

detective n.
警探；侦查人员

perplexing adj.
令人困惑的

seemingly adv.
看上去；似乎

resolve v.
解决

revelation n.
重要的新发现

previously adv.
先前；以前

marker n.
标记

layout n.
排列方式；布局

considerably adv.
相当地；很大程度上

promptly adv.
迅速地；及时地

enlist v.
请……协助

genetic adj.
基因的；遗传的

confession n.
供词；认罪陈述

attacker n.
袭击者

large-scale adj.
大规模的

saliva n.
唾液

co-worker n.
同事

evade v.
避开；躲过

suspicion n.
怀疑

overhear v.
无意中听见

notify v.
通知

detain v.
拘留

plead v.
作答辩；此处指认罪

jail v.
监禁

unmask v.
揭穿；揭露真面目

blameless adj.
无辜的；无可指责的

juvenile n.
未成年人；少年

falsely adv.
错误地；不实地

offence n.
罪行

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Police enlisted Jeffreys, and his genetic tests _____ the erroneous confession.

02 The youth was not the _____, and the same attacker had killed both girls.

03 In August 1987 someone overheard the co-worker discussing the _____ and notified police.

04 The test's first _____ had been a blameless juvenile who had falsely admitted the offence.

文章主线

- 发现 杰弗里斯发现DNA标记的排列因人而异，可像指纹一样辨认身份，由此提出DNA指纹。
- 洗冤 两名少女遇害后，一名少年供认第二起命案；DNA检验却证明他无辜，并确认两案出自同一凶手。
- 筛查 警方对五千多名男子开展世界首次大规模DNA筛查，却无一吻合；真凶让同事冒名采样，因而漏网。
- 破案 有人听见同事谈及顶替而举报；警方循线找到皮奇福克，DNA检验确认他是真凶。
- 结局 少年成为已知首位凭DNA证据获洗冤者；皮奇福克随后认罪、被判终身监禁，成为首位凭DNA证据被判谋杀罪者。

参考答案　1 overturned 2 perpetrator 3 substitution 4 beneficiary

第37手

Move 37

两步万分之一的棋，为什么令人惊叹？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Go is said to allow more configurations than the universe has atoms, a daunting computational challenge. In October 2015 DeepMind's AlphaGo reached a milestone, becoming the first program to beat a professional on a full board without a handicap. In March 2016 its opponent in a televised Seoul showdown was Lee Sedol, a South Korean strategist with 18 international titles.

In game two, on 10 March, AlphaGo's 37th move was a shoulder hit, a tactic placing a stone adjacent to an opponent's on the diagonal. On the grid's fifth line, it was hardly orthodox. Commentators found the unexpected placement bizarre; one took it for a mistake. Lee left the room, his pause for deliberation lasting over ten minutes.

DeepMind's diagnostic tools showed AlphaGo had put the odds of a human playing that stone at one in 10,000, a staggering rarity. AlphaGo had first learned to imitate strong amateurs online, then sharpened its play over tens of millions of simulated games against itself. Lee later acknowledged the stone's originality and beauty.

The fourth game, on 13 March, reversed the roles in neat symmetry. Lee's 78th move turned the tide, and many professionals heralded it as divine. DeepMind's team computed a professional's likelihood of finding it: about one in 10,000. Lee's only win was a consolation amid AlphaGo's overwhelming 4–1 victory.

In November 2019 Lee withdrew from the professional circuit, a farewell he tied to AI's unrivalled dominance. Calling AI an invincible entity, he said the summit was beyond him. Yet his one win came from a stone whose odds, by DeepMind's figures, were as slim as those of AlphaGo's audacious shoulder hit.

中文译文

据说，围棋可能出现的局面，比宇宙中的原子还多，这给计算带来了艰巨挑战。2015年10月，DeepMind的AlphaGo达成了一项里程碑：成为首个在全尺寸棋盘上、不让子的条件下击败职业棋手的程序。2016年3月，在首尔一场电视转播的对决中，它的对手是拥有18项国际冠军头衔的韩国围棋高手李世石。

3月10日的第二局，AlphaGo第37手下出一记肩冲，也就是在对方棋子的斜对角相邻处落子。这一手落在从棋盘边缘数起的第五线上，很不合常规。解说员觉得这个出人意料的落点十分怪异，其中一人还以为它下错了。李世石离开对局室，暂停下来思考了十多分钟。

DeepMind的诊断工具显示，AlphaGo曾估计，人类下出这一手的概率只有万分之一，罕见得令人惊叹。AlphaGo起初学习模仿网上实力强劲的业余棋手，随后通过数千万盘模拟自我对弈磨砺棋力。李世石后来也认可了这一手的独创性与美感。

3月13日的第四局，双方的角色巧妙地对调了。李世石第78手扭转局势，许多职业棋手将其誉为“神之一手”。DeepMind团队估算，职业棋手找到这一手的可能性约为万分之一。AlphaGo以4比1取得压倒性胜利，李世石唯一的一胜，则给他带来了一丝安慰。

2019年11月，李世石退出职业棋坛，并将这次告别与人工智能无可匹敌的优势联系起来。他称人工智能是不可战胜的存在，表示自己已无法企及那个巅峰。然而，带来他唯一胜局的那一手，按照DeepMind的数据，人类找到它的概率，和下出AlphaGo那记大胆肩冲的概率一样渺茫。

文中两次“万分之一”都是对人类选出某一步棋的概率估计，不是落下这步棋后的获胜概率。

核心词汇

先记这22项。连着下方的原文短语一起回想。

configuration n.
布局；局面
allow more configurations

daunting adj.
艰巨的；令人望而生畏的
a daunting computational challenge.

computational adj.
计算的
a daunting computational challenge.

milestone n.
里程碑
reached a milestone

handicap n.
让子；比赛中给对手的优势
without a handicap.

tactic n.
战术；招数
a tactic placing a stone

orthodox adj.
正统的；合乎常规的
hardly orthodox.

bizarre adj.
怪异的；异乎寻常的
found the unexpected placement bizarre

deliberation n.
审慎思考；斟酌
pause for deliberation

diagnostic adj.
诊断的；用于分析问题的
DeepMind's diagnostic tools

odds n.
概率；可能性
the odds of a human playing that stone

imitate v.
模仿
imitate strong amateurs online

simulated adj.
模拟的
simulated games against itself.

originality n.
独创性
the stone's originality and beauty.

symmetry n.
对称；相互呼应
neat symmetry.

turn the tide phr.
扭转局势
Lee's 78th move turned the tide

likelihood n.
可能性；概率
a professional's likelihood

consolation n.
安慰
Lee's only win was a consolation

overwhelming adj.
压倒性的
AlphaGo's overwhelming 4–1 victory.

unrivalled adj.
无可匹敌的
AI's unrivalled dominance.

invincible adj.
不可战胜的
an invincible entity

audacious adj.
大胆的；敢于冒险的
AlphaGo's audacious shoulder hit.

文中词汇与搭配

补充词汇共29项。保留每个表达在本文中的意思。

atom n.
原子

opponent n.
对手

televised adj.
电视转播的

showdown n.
对决；决战

strategist n.
战略家；善于布局的高手

adjacent to phr.
与……相邻的

diagonal n.
斜线；对角线方向

grid n.
网格；此处指棋盘格线

unexpected adj.
出人意料的

placement n.
放置；此处指落子位置

pause n.
暂停

staggering adj.
惊人的；令人震惊的

rarity n.
罕见；罕见之事

amateur n.
业余爱好者

sharpen v.
磨砺；提高（技艺）

acknowledge v.
承认；认可

reverse v.
颠倒；互换

neat adj.
巧妙的；恰到好处的

herald v.
称赞；誉为

as prep.
与herald连用，表示誉为

divine adj.
神妙的；此处指神之一手

compute v.
计算；估算

withdraw from phr. v.
退出；撤离

circuit n.
职业巡回赛事；职业赛场

farewell n.
告别

dominance n.
主导地位；优势

entity n.
实体；独立存在

summit n.
顶峰；巅峰

slim adj.
微小的；渺茫的（概率）

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In October 2015 DeepMind's AlphaGo reached a _____, becoming the first program to beat a professional on a full board without a handicap.

02 Commentators found the unexpected placement _____; one took it for a mistake.

03 Lee later acknowledged the stone's _____ and beauty.

04 Lee's only win was a _____ amid AlphaGo's overwhelming 4–1 victory.

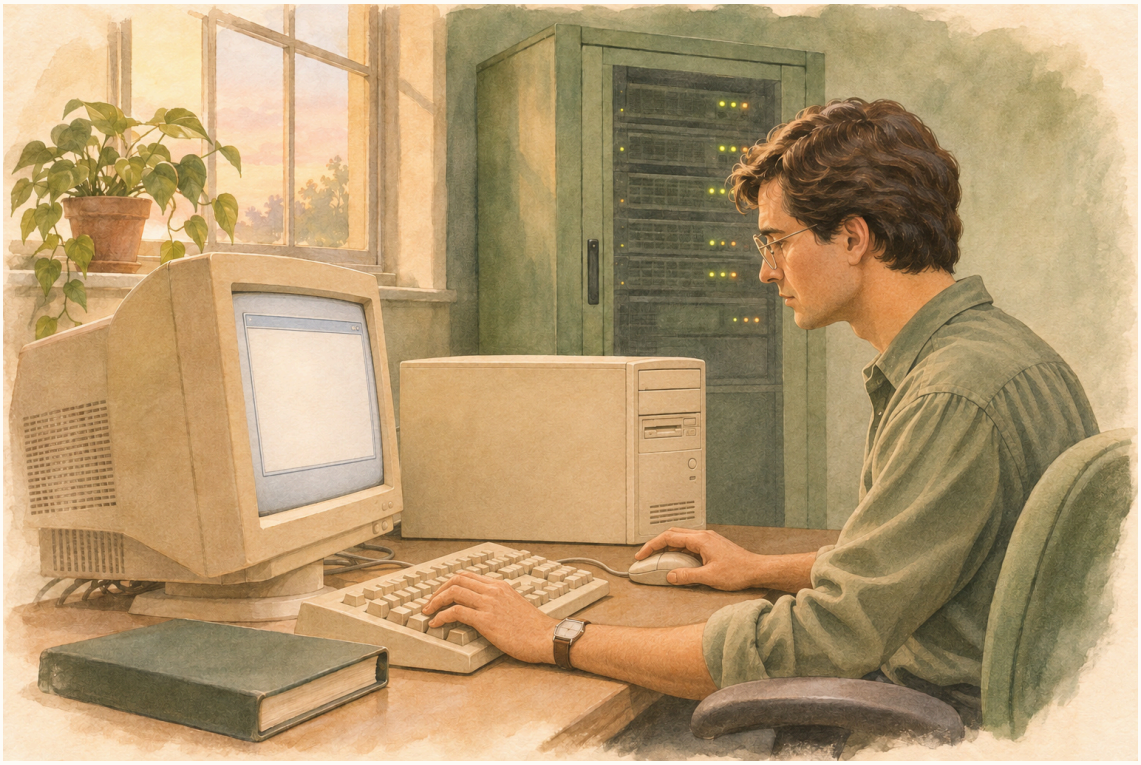
文章主线

- 训练 围棋程序AlphaGo先模仿高水平业余棋手，再经数千万盘自我对弈提升，2016年迎战李世石。
- 奇手 第二局第37手落在从棋盘边缘数的第五条线上，违背常规，起初被疑为失误，后来获得李世石赞赏。
- 回应 第四局，李世石第78手扭转局势，被誉为神之一手，带来他在五局中唯一的胜利。
- 赛果 AlphaGo最终4比1获胜；李世石2019年退役时称AI不可战胜，自己已难登顶。
- 对照 AlphaGo估计人类下出第37手的概率约万分之一；团队对职业棋手下出第78手的估计相同，双方都有罕见创意。

参考答案　1 milestone 2 bizarre 3 originality 4 consolation

千年虫：被成功抹去的危机

Y2K: The Crisis That Success Erased



危机没有发生，能说明预防多余吗？

英文原文

The Y2K bug, or millennium bug, began as a shortcut. When memory and hardware were costly, programmers stored years as two digits, writing "99" for 1999. That compact code hid an ambiguity: a machine might read "00" as 1900, corrupting calculations of interest, ages, or deadlines.

Governments and firms braced for chaos. In February 1998 President Clinton created a council to coordinate America's effort. Consultants audited legacy systems and patched faulty date logic. In November 1999 the Commerce Department estimated American governments and enterprises would spend about $100bn; projections of the global bill ran to hundreds of billions.

Then midnight struck on 31 December 1999, and little went wrong. Glitches that did crop up were sporadic and mostly minor. Radiation-monitoring equipment faltered at Japan's Shika nuclear plant without any threat to safety, and a Pentagon ground system that processes spy-satellite data failed. In February 2000 a Senate committee concluded that the expenditure had been justified.

Many saw it differently: the threat had been exaggerated, even a hoax. Yet calm prevailed largely because so much had been fixed. This is the preparedness paradox: when precautions work, the danger they ward off never appears, so it looks imaginary. Tedious repairs earned mockery rather than gratitude.

Critics noted that Italy and South Korea spent far less yet fared no worse. The comparison misleads: those low figures often omitted what firms spent, and much of the remedial work replaced obsolete systems overdue for upgrading. Nobody can say which dollars were essential and which excessive. Prevention that works may leave nothing but an invoice, to be weighed against a catastrophe that might have been.

中文译文

千年虫漏洞最初源于一种省事的做法。当内存和硬件还很昂贵时，程序员只用两位数字储存年份，例如以“99”代表1999年。这种精简代码暗藏歧义：计算机可能把“00”理解成1900年，导致利息、年龄或截止日期计算出错。

各国政府和企业严阵以待，准备应对可能出现的混乱。1998年2月，克林顿总统成立委员会，协调美国的应对工作。顾问们审查旧系统，修补有问题的日期处理逻辑。1999年11月，美国商务部估计，美国各级政府和企业将为此花费约1000亿美元；对全球总支出的预测则高达数千亿美元。

1999年12月31日午夜来临，结果几乎没有出大问题。偶尔出现的小故障也大多轻微。日本志贺核电站的辐射监测设备失灵，但没有威胁安全；美国国防部一套处理间谍卫星数据的地面系统也出了故障。2000年2月，美国参议院的一个委员会认定，这些支出是合理的。

许多人却持不同看法，认为威胁被夸大了，甚至是个骗局。但局面之所以平稳，很大程度上正是因为大量问题已经修好。这就是“防备悖论”：预防措施发挥作用，使所抵御的危险没有出现，危险反而看起来像是凭空捏造的。繁琐的修复工作换来的不是感谢，而是嘲笑。

批评者指出，意大利和韩国花的钱少得多，结果却并不更差。但这种比较具有误导性：较低的支出数字往往没有算上企业的花费，而且很多修补工作也替换了早该升级的过时系统。没人能逐笔说清，哪些开支必不可少，哪些花得过多。奏效的预防可能只留下账单，人们只能拿它与那场本可能发生的灾难作比较。

文中的约1000亿美元是当时的估计；平稳过渡不代表每笔支出都已被证明必要。

核心词汇

先记这22项。连着下方的原文短语一起回想。

bug n.
程序漏洞
The Y2K bug

shortcut n.
捷径；省事的办法
began as a shortcut

ambiguity n.
歧义
That compact code hid an ambiguity

corrupt v.
使出错；破坏
corrupting calculations

brace for 动词短语
准备应对（困难）
firms braced for chaos

coordinate v.
协调
to coordinate America's effort

audit v.
审查；核查
Consultants audited legacy systems

legacy adj.
沿用的；旧有的
legacy systems

patch v.
修补（程序）
patched faulty date logic

projection n.
预测
projections of the global bill

sporadic adj.
零星的；偶发的
sporadic and mostly minor

expenditure n.
支出
the expenditure had been justified

justified adj.
合理的；有正当理由的
the expenditure had been justified

exaggerate v.
夸大
the threat had been exaggerated

hoax n.
骗局
even a hoax

preparedness paradox n.
防备悖论
the preparedness paradox

precaution n.
预防措施
when precautions work

ward off 动词短语
抵御
the danger they ward off

omit v.
遗漏
omitted what firms spent

obsolete adj.
过时的
replaced obsolete systems

essential adj.
必不可少的
which dollars were essential

catastrophe n.
灾难
a catastrophe that might have been

文中词汇与搭配

补充词汇共41项。保留每个表达在本文中的意思。

millennium n.
千年；千禧年

memory n.
计算机内存

hardware n.
硬件

costly adj.
昂贵的

programmer n.
程序员

digit n.
数字

compact adj.
精简紧凑的

code n.
程序代码

deadline n.
截止日期

chaos n.
混乱

president n.
总统

council n.
委员会

consultant n.
顾问

faulty adj.
有缺陷的

Commerce Department n.
美国商务部

enterprise n.
企业

global adj.
全球的

midnight n.
午夜

strike v.
（钟）敲响

glitch n.
小故障

文中词汇与搭配 续

补充词汇共41项。保留每个表达在本文中的意思。

crop up 动词短语
意外出现

minor adj.
轻微的

falter v.
失灵；运转不稳

spy n.
间谍

conclude v.
得出结论

prevail v.
占上风；普遍存在

largely adv.
很大程度上

imaginary adj.
想象出来的

tedious adj.
繁琐乏味的

mockery n.
嘲笑

gratitude n.
感激

South Korea n.
韩国

fare no worse 动词短语
情况并不更差

comparison n.
比较

mislead v.
误导

remedial adj.
补救的

overdue for 形容词短语
早该进行……的

excessive adj.
过多的；过度的

prevention n.
预防

invoice n.
账单

be weighed against 动词短语
与……权衡比较

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That compact code hid an _____: a machine might read "00" as 1900, corrupting calculations of interest, ages, or deadlines.

02 Consultants audited legacy systems and _____ faulty date logic.

03 Glitches that did crop up were _____ and mostly minor.

04 This is the preparedness paradox: when _____ work, the danger they ward off never appears, so it looks imaginary.

文章主线

- 隐患 旧程序为省内存只用两位数字记年，可能将2000年的“00”误读为1900年，算错利息、年龄和期限。
- 应对 政府与企业排查旧系统、修补日期逻辑；美国1999年预计花约1,000亿美元，全球预测达数千亿美元。
- 跨年 2000年跨年大体平稳，故障零星轻微；美国参议院委员会认定开支合理，却有人把威胁当成骗局。
- 悖论 平稳很大程度上来自预先修复：预防奏效、灾难未现，危险反倒像凭空捏造，这就是“防备悖论”。
- 限度 意大利、韩国的低支出统计常漏算企业花费，修补也更新了过时系统；究竟哪些钱必要、哪些过量，无法逐笔断定。

参考答案　1 ambiguity 2 patched 3 sporadic 4 precautions

被街道重塑的大脑

The Brain That Learned London

熟记伦敦街道，会怎样改变大脑？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Name two addresses and a London cabbie will rattle off the quickest route. That prowess is hard-won. To earn a licence, aspiring drivers must conquer "the Knowledge", an arduous examination on about 25,000 streets within six miles of Charing Cross. Hopefuls typically spend three to four years committing this labyrinth to memory, and only about half qualify.

A study published in PNAS in April 2000 scanned 16 male cabbies and 50 other men. The hippocampus is a seahorse-shaped structure vital to memory and navigation. In the cabbies, its rear, or posterior, portion was larger than in the other men, and its front, or anterior, portion smaller. Longer service went with a larger rear portion.

Yet the link left the cause uncertain. Perhaps an innate flair for spatial tasks lured people to the occupation, or perhaps stressful driving was the culprit. A December 2006 study dispelled the second notion: cabbies had more rear grey matter, the tissue dense with nerve cells, than bus drivers, confined to predetermined routes, of equivalent experience and stress.

A December 2011 longitudinal study tackled the first, scanning 79 trainees and 31 non-trainees at enrolment and three to four years later. Their brains began alike. Only the 39 who qualified exhibited an expansion of rear grey matter, so the sequence suggests learning preceded the enlargement.

The gain appeared to entail a trade-off: qualified drivers trailed the non-trainees on one test of recalling an intricate figure. The cohorts were small and exclusively male, yet the studies imply the healthy adult brain can remodel itself through intensive learning. Every street learned, it seems, occupied room once claimed by something else.

中文译文

只要说出两个地址，伦敦出租车司机就能迅速报出最快的路线。这份本领来之不易。为了取得执照，准司机必须攻克一项叫作“知识大考”的艰难考试，熟记距查令十字六英里以内约2.5万条街道。备考者通常花三到四年把这座街道迷宫牢记于心，最终只有约一半能够合格。

2000年4月发表于《美国国家科学院院刊》的一项研究，扫描了16名男性出租车司机和50名其他男性的大脑。海马体形似海马，对记忆和认路至关重要。与其他男性相比，出租车司机海马体的后部更大，前部则更小。从业时间越长，后部也越大。

然而，这种关联并不能确定原因。也许是天生擅长空间任务的人更容易被这个职业吸引，也许是紧张的驾驶工作造成了差异。2006年12月的一项研究排除了后一种解释：与驾驶经验、压力水平相当的公交司机相比，出租车司机海马体后部的灰质仍然更多。灰质是一种密布神经细胞的组织，而公交司机的行驶路线是预先固定的。

2011年12月的一项纵向研究开始检验前一种解释。研究者在79名学员报名时，以及三到四年后，分别扫描了他们和31名未参训者的大脑。各组起初的大脑状况相似，只有最终通过考试的39名学员，出现了海马体后部灰质增加。这一先后顺序提示，学习发生在灰质增大之前。

这种收获似乎也伴随着某种取舍：在一项回忆复杂图形的测试中，通过考试的司机表现不如未参训者。研究的样本群体较小，而且全部为男性，但结果仍提示，健康的成人大脑可以通过高强度学习重塑自身。看来，每记住一条街道，都像是占用了原先属于其他事物的一点空间。

核心词汇

先记这22项。连着下方的原文短语一起回想。

prowess n.
高超本领
That prowess is hard-won.

arduous adj.
艰难费力的
an arduous examination

commit v.
此处指把……牢记于心
committing this labyrinth to memory

qualify v.
取得资格；合格
and only about half qualify.

hippocampus n.
海马体
The hippocampus

navigation n.
辨路；导航
vital to memory and navigation.

posterior adj.
后部的
its rear, or posterior, portion

anterior adj.
前部的
its front, or anterior, portion

uncertain adj.
无法确定的
Yet the link left the cause uncertain.

innate adj.
天生的
an innate flair for spatial tasks

spatial adj.
空间的
spatial tasks

dispel v.
消除；打消
study dispelled the second notion:

predetermined adj.
预先确定的
confined to predetermined routes

equivalent adj.
相当的；等同的
of equivalent experience and stress.

longitudinal adj.
长期追踪同一群体的
longitudinal study

expansion n.
扩大；增加
an expansion of rear grey matter

precede v.
发生在……之前
learning preceded the enlargement.

entail v.
意味着；必然涉及
to entail a trade-off:

trade-off n.
取舍；有得有失
appeared to entail a trade-off

cohort n.
接受研究的一组人
The cohorts were small

imply v.
提示；意味着
yet the studies imply

remodel v.
重塑；重新构建
can remodel itself

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

rattle off phr. v.
快速流利地说出

hard-won adj.
来之不易的

licence n.
执照；许可证

aspiring adj.
有志于成为……的

conquer v.
攻克；征服

examination n.
考试

hopeful n.
有望成功者；此处指备考者

typically adv.
通常

labyrinth n.
迷宫

scan v.
扫描检查

vital adj.
至关重要的

portion n.
部分

flair n.
天赋；才能

lure v.
吸引；诱使

occupation n.
职业

stressful adj.
造成精神压力的

culprit n.
问题的起因；罪魁祸首

notion n.
看法；想法

dense adj.
密集的

nerve n.
神经

confine v.
限制；使局限于

tackle v.
着手处理；检验

trainee n.
受训学员

enrolment n.
报名；入学

exhibit v.
表现出；显示

sequence n.
先后顺序

enlargement n.
增大

trail v.
落后于

intricate adj.
复杂精细的

exclusively adv.
全部；仅仅

intensive adj.
高强度的；集中的

occupy v.
占据

claim v.
此处指占用

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 To earn a licence, aspiring drivers must conquer "the Knowledge", an _____ examination on about 25,000 streets within six miles of Charing Cross.

02 The _____ is a seahorse-shaped structure vital to memory and navigation.

03 Yet the link left the cause _____.

04 The gain appeared to entail a _____: qualified drivers trailed the non-trainees on one test of recalling an intricate figure.

文章主线

- 训练 伦敦出租车司机通常花三四年记住约2.5万条街道，只有约一半学员通过考试。
- 差异 海马体参与记忆与认路；司机的海马体后部比对照男性大、前部小，最初无法判断原因。
- 排查 与经验、压力相近但走固定路线的公交司机相比，出租车司机的海马体后部灰质仍更多。
- 追踪 学员与对照者起初脑部相似；三四年后，仅考过者海马体后部灰质增加，支持成人大脑可随学习改变。
- 边界 样本小且全为男性；考过者在一项图形记忆测试中比未参训者差，提示认路能力增强可能伴随记忆取舍。

参考答案　1 arduous 2 hippocampus 3 uncertain 4 trade-off

两次卖掉埃菲尔铁塔的骗子

The Man Who Sold the Eiffel Tower Twice

铁塔怎么成了骗子的“货物”？

英文原文

In 1925 six scrap-metal traders met secretly at the Hôtel de Crillon in Paris. A deputy director-general of the Ministry of Posts and Telegraphs shared the momentous news, confidentially, that the Eiffel Tower was to be sold for scrap. He invited tenders and made each dealer solemnly vow secrecy, ostensibly to spare the public indignation.

The pretext was not absurd. Built for the 1889 World's Fair, the monument had originally been destined to come down after twenty years. The high-ranking deputy was Victor Lustig, a crook whose ministerial stationery was a forgery. He had read a newspaper article in which the authorities lamented the hefty cost of the tower's upkeep.

André Poisson, a businessman impatient to gain a foothold in Paris, won. Lustig intimated that his modest civil-service remuneration needed supplementing, and Poisson paid an additional bribe. Lustig left for Vienna with the takings.

Fleeced and too ashamed to go to the police, Poisson swallowed the loss quietly. Not a whisper of the deceit appeared in print.

About a month later Lustig had the gall to reappear with an encore of his script for new dealers. This time the winner grew distrustful and went to the police. The affair made the papers, and Lustig made his getaway. The much-repeated second sale is a fairy tale.

Later arrested in the United States for making fake money, Lustig broke out of detention, was recaptured 27 days later and served time on the penal island of Alcatraz. He died of pneumonia in 1947.

Of Lustig's two marks, the one who paid kept mum and the one who kept his money went to the police.

中文译文

1925年，六名废金属商人在巴黎克里雍酒店秘密会面。邮电部的一名副总负责人私下告诉他们一则重大消息：埃菲尔铁塔将当作废铁出售。他邀请商人投标，并要求每个人郑重发誓保密，表面上的理由是避免引起公愤。

这个借口并不荒唐。铁塔为1889年世博会而建，原本计划二十年后拆除。那位位高权重的“副总负责人”其实是骗子维克多·卢斯蒂格，带有政府部门抬头的信笺也是伪造的。他曾在报上看到，当局抱怨铁塔维护费用高昂。

急于在巴黎站稳脚跟的商人安德烈·普瓦松中标。卢斯蒂格暗示，自己微薄的公务员薪酬需要补贴，普瓦松便又付了一笔贿赂。随后，卢斯蒂格带着到手的钱前往维也纳。

普瓦松被骗了钱，却因羞于报警而默默咽下损失。报纸上连关于这场骗局的一点风声都没有。

大约一个月后，卢斯蒂格竟又厚着脸皮回来，向新一批商人重演旧把戏。这一次，中标者起了疑心，前去报警。事情登上报纸，卢斯蒂格逃之夭夭。广为流传的“第二次卖成铁塔”，其实只是虚构故事。

后来，卢斯蒂格在美国因制造假币被捕。他从拘留处逃出，27天后再次落网，并在恶魔岛监狱服刑。1947年，他死于肺炎。

卢斯蒂格的两个行骗对象中，付了钱的那人守口如瓶，保住钱的那人反而报了警。

标题里的“两次卖掉”是传说；原文明确，第二次骗局没有得手。

核心词汇

先记这22项。连着下方的原文短语一起回想。

scrap-metal adj.
废金属的
six scrap-metal traders

confidentially adv.
私下地；保密地
confidentially

tender n.
投标
He invited tenders

vow v.
发誓；郑重保证
solemnly vow secrecy

ostensibly adv.
表面上；声称地
ostensibly to spare

public indignation 名词短语
公愤；公众愤怒
the public indignation

pretext n.
借口；托词
The pretext was not absurd

destined adj.
原定的；注定的
destined to come down

crook n.
骗子；不法之徒
a crook

stationery n.
信笺；办公文具
ministerial stationery

forgery n.
伪造品；伪造文件
was a forgery

upkeep n.
维护；保养
the tower's upkeep

gain a foothold 动词短语
站稳脚跟；取得一席之地
gain a foothold in Paris

intimate v.
暗示
Lustig intimated

remuneration n.
报酬；薪酬
civil-service remuneration

bribe n.
贿赂；贿款
an additional bribe

fleece v.
骗取钱财；敲竹杠
Fleeced and too ashamed

deceit n.
欺骗；骗局
a whisper of the deceit

have the gall to 动词短语
竟敢厚着脸皮做……
had the gall to reappear

grow distrustful 动词短语
起疑；变得不信任
the winner grew distrustful

detention n.
拘留；羁押
broke out of detention

keep mum 动词短语
守口如瓶；保持沉默
the one who paid kept mum

文中词汇与搭配

补充词汇共35项。保留每个表达在本文中的意思。

trader n.
商人；交易者

secretly adv.
秘密地

deputy adj.
副职的

director-general n.
总负责人；总局长

momentous adj.
重大的；影响深远的

solemnly adv.
郑重地；严肃地

secrecy n.
保密；秘密状态

absurd adj.
荒唐的

monument n.
纪念建筑；纪念物

originally adv.
原本；最初

come down 动词短语
被拆除；倒下

high-ranking adj.
高级别的；位高权重的

ministerial adj.
政府部门的；部级机关的

lament v.
哀叹；抱怨

hefty cost 名词短语
高昂费用

businessman n.
商人

impatient adj.
急切的；迫不及待的

civil-service adj.
公务员的；文官的

文中词汇与搭配 续

补充词汇共35项。保留每个表达在本文中的意思。

supplement v.
补充；补贴

additional adj.
额外的

takings n.
到手的钱；收入

ashamed adj.
羞愧的；感到丢人的

quietly adv.
默默地；不声张地

whisper n.
一点风声；耳语

reappear v.
再次出现

encore n.
重演；再来一遍

script n.
台词；既定说辞

affair n.
事件；事情

getaway n.
逃脱；逃走

much-repeated adj.
被反复讲述的

fairy tale 名词短语
童话；虚构故事

fake adj.
假的；伪造的

recapture v.
再次抓获

serve time 动词短语
服刑

penal adj.
刑罚的；监禁的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ was not absurd.

02 The high-ranking deputy was Victor Lustig, a crook whose ministerial stationery was a _____.

03 Lustig intimated that his modest civil-service _____ needed supplementing, and Poisson paid an additional bribe.

04 Of Lustig's two marks, the one who paid _____ and the one who kept his money went to the police.

文章主线

- 冒充 1925年，卢斯蒂格伪造公文、冒充邮电部官员，邀六名废金属商秘密投标，谎称埃菲尔铁塔要卖废铁。
- 借口 铁塔原拟在世博会20年后拆除，维护费又高；卢斯蒂格利用这些真背景，并以免引公愤为由要求保密。
- 得手 急于在巴黎立足的普瓦松付了购塔款和贿赂，受骗后却羞于报警，让卢斯蒂格带钱逃走。
- 再骗 约一月后，卢斯蒂格重演骗局，新买家起疑报警且未付款；所谓第二次卖成铁塔的故事并不成立。
- 下场 卢斯蒂格后来在美国因造假币被捕，越狱27天后再落网，在恶魔岛监狱服刑，1947年死于肺炎。

参考答案　1 pretext 2 forgery 3 remuneration 4 kept mum

罗马混凝土的千年之谜

The Secret of Rome's Undying Concrete

古罗马混凝土为何如此耐久？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Nearly 1,900 years after its completion, in about AD 125–128, the Pantheon's dome still crowns the vast chamber below. About 43 metres in diameter, it remains the world's largest unreinforced concrete dome. It is an astonishing feat of architecture, and a stubborn puzzle.

Modern concrete is usually reinforced with embedded steel bars, which are prone to corrosion. As the steel rusts, it swells and fractures the concrete around it, and many structures are designed for a lifespan of around 50 years. Roman concrete is not stronger than its modern counterpart; its merit lies in durability.

Yet the ancient material seemed to have a shortcoming. It is dotted with white chunks of lime, known as lime clasts. Experts long ascribed these lumps to sloppy mixing or inferior raw ingredients, a defect that, they assumed, a more conscientious builder would have avoided.

A study published in Science Advances in January 2023 offered a different interpretation. The clasts, it suggested, were a by-product of "hot mixing" with quicklime, a reactive lime that heats up in water. When water seeps into a crack, the clasts dissolve and their calcium recrystallises as calcium carbonate, plugging the gap. In tests, water ceased to trickle through cracked hot-mixed samples within weeks, but kept flowing through samples without quicklime.

The implications extend beyond archaeology. Cement manufacture accounts for about 8 per cent of global carbon dioxide emissions, the think-tank Chatham House reported in June 2018. Concrete that mends its cracks could last longer, making demolition and rebuilding rarer and trimming emissions. The white lumps once scorned as signs of clumsy workmanship have prompted a rethink of construction.

中文译文

万神殿约在公元125至128年竣工。近1900年过去，它的穹顶依然高踞在宽阔的殿堂之上。穹顶直径约43米，至今仍是世界上最大的无钢筋混凝土穹顶。这是建筑史上的惊人壮举，也是一道始终难解的谜题。

现代混凝土通常用埋入其中的钢筋加固，而钢筋容易锈蚀。钢筋生锈后会膨胀，撑裂周围的混凝土；许多建筑的设计寿命约为50年。古罗马混凝土的强度并不比现代混凝土更高，它的优势在于耐久。

不过，古代材料看起来也有一个缺点：内部散布着白色石灰小块，称为石灰碎屑。专家长期把这些团块归咎于搅拌马虎或原料劣质，认为这是一种更认真负责的工匠本应避免的缺陷。

2023年1月，《科学进展》的一项研究提出了另一种解释。研究认为，这些碎屑是使用生石灰进行“热混合”留下的副产物；生石灰活性强，遇水会放热。水渗入裂缝时，碎屑会溶解，其中的钙重新结晶为碳酸钙，堵住缝隙。试验中，开裂的热混合样本在几周内就不再渗水，而不含生石灰的样本仍持续漏水。

这一发现的意义不止于考古。查塔姆研究所2018年6月报告称，水泥生产约占全球二氧化碳排放的8%。能自行修补裂缝的混凝土，可能延长建筑寿命，减少拆除与重建，进而降低排放。那些曾被轻视为手艺拙劣证据的白色团块，如今促使人们重新思考建造方式。

核心词汇

先记这22项。连着下方的原文短语一起回想。

feat n.
壮举；非凡成就
It is an astonishing feat of

prone to adj.phr.
容易发生；易于
which are prone to corrosion.

corrosion n.
锈蚀；腐蚀
which are prone to corrosion.

lifespan n.
使用寿命
for a lifespan of around 50 years.

counterpart n.
对应的事物
stronger than its modern counterpart;

durability n.
耐久性
its merit lies in durability.

shortcoming n.
缺点；不足
material seemed to have a shortcoming.

ascribe to phr.v.
与to连用：归因于
Experts long ascribed these lumps to

inferior adj.
劣质的
mixing or inferior raw ingredients,

conscientious adj.
认真负责的
more conscientious builder would have

interpretation n.
解释
offered a different interpretation.

by-product n.
副产物
were a by-product of "hot mixing" with

quicklime n.
生石灰
of "hot mixing" with quicklime,

reactive adj.
化学活性强的
reactive lime that heats up in water.

seep into phr.v.
缓慢渗入
When water seeps into a crack,

dissolve v.
溶解
the clasts dissolve and their calcium

implication n.
意义；可能影响
The implications extend beyond

extend beyond v.phr.
超出……范围
The implications extend beyond

account for phr.v.
占……比例
Cement manufacture accounts for about

emission n.
排放量
of global carbon dioxide emissions,

demolition n.
拆除
making demolition and rebuilding rarer

prompt v.
促使
have prompted a rethink of

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

completion n.
竣工；完成

dome n.
穹顶

crown v.
位于……顶部

vast adj.
宽广的；巨大的

chamber n.
大厅；殿堂

in diameter prep.phr.
直径为

unreinforced concrete n.phr.
无钢筋混凝土

astonishing adj.
令人惊叹的

architecture n.
建筑；建筑艺术

stubborn adj.
难以解决的

puzzle n.
谜题

reinforce with v.phr.
用……加固

embedded adj.
埋入内部的

steel bar n.phr.
钢筋

rust v.
生锈

swell v.
膨胀

fracture v.
使断裂；使开裂

structure n.
建筑物；结构

merit n.
优点；长处

dotted with adj.phr.
散布着

chunk n.
小块；厚块

clast n.
碎屑；碎屑颗粒

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

lump n.
团块

to prep.
标示所归因的对象

sloppy adj.
马虎的

raw ingredient n.phr.
原材料

defect n.
缺陷

assume v.
认为；未经证实地假定

heat up phr.v.
变热；升温

crack n.
裂缝

calcium n.
钙

plug v.
堵住

gap n.
缝隙

cease to v.phr.
不再；停止

trickle v.
细细流淌

within weeks prep.phr.
在几周之内

archaeology n.
考古学

manufacture n.
生产；制造

think-tank n.
智囊团；研究机构

mend v.
修补

trim v.
削减

scorn v.
轻视

clumsy adj.
拙劣的；笨拙的

rethink n.
重新考虑

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 which are prone to _____.

02 its merit lies in _____.

03 When water _____ a crack,

04 Cement manufacture _____ about 8 per cent of global carbon dioxide emissions,

文章主线

- 耐久 万神殿直径约43米的无钢筋穹顶已存近1900年；罗马混凝土长于耐久，并非强度高于现代混凝土。
- 疑点 现代钢筋锈胀会撑裂混凝土；罗马材料中的白色石灰块，过去却被当成搅拌差或原料劣的缺陷。
- 新解 2023年研究认为，生石灰热混合留下石灰块；水渗入裂缝后，石灰块溶解、钙重新结晶为碳酸钙，堵住裂缝。
- 试验 开裂的热混合样本几周内不再渗水，未加生石灰的对照仍漏水，支持这种自愈解释。
- 前景 水泥生产约占全球二氧化碳排放8%；若自愈材料能延长建筑寿命，就有望减少拆建及排放。

参考答案　1 corrosion 2 durability 3 seeps into 4 accounts for

一夜之间换边开车

The Night Sweden Switched Sides

全国换边行驶，怎样在一个清晨完成?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In a 1955 referendum, about 83 per cent of voters chose to keep driving on the left, an overwhelmingly clear rejection of change. On 10 May 1963, however, legislators in parliament overruled the electorate, approving right-hand traffic from 1967 in defiance of a populace among whom the plan was unpopular.

The case rested on conformity. Norway and Finland drove on the right, with about five million border crossings a year, so a left-hand orientation meant isolation and inconvenient journeys. Safety favoured alignment too. About 90 per cent of cars were left-hand drive, so motorists overtaking on narrow two-lane highways risked head-on collisions.

A state body ran the mammoth preparation under a logo, a large H with an arrow. Its propaganda included a song contest on television, and the winning jingle, whose lyrics urged people to keep right, reached fifth in the chart. Some 360,000 items of signage were changed, and a massive fleet of over 1,000 new buses with right-hand doors was procured.

The transition came on Sunday 3 September 1967. A curfew kept non-essential vehicles off the asphalt from one to six, with essential traffic exempt. At ten to five, all traffic stopped and crawled across, restarting at five, before dawn, with its direction flipped. Stockholm and Malmö had a longer closure, from Saturday morning to Sunday afternoon, while crews reconfigured junctions.

On the following Monday, 125 accidents were reported, against 130 to 198 on earlier Mondays, a statistical improvement put down to the heightened care of hesitant drivers. The nervousness apparently subsided and prudent habits faded, for by 1969 accident and fatality rates had reverted to their baseline.

中文译文

1955年瑞典举行公投，约83%的投票者选择保留靠左行驶，以压倒性多数拒绝改变。然而，1963年5月10日，议会推翻选民的选择，不顾这项计划在民众中缺乏支持，批准自1967年起改为靠右行驶。

改制的理由是与邻国统一。挪威和芬兰都靠右行驶，每年约有500万次跨境通行，瑞典靠左便带来隔阂和出行不便。安全也支持统一：约90%的汽车方向盘位于左侧，司机在狭窄的双车道公路上超车时，容易面临迎面相撞的危险。

一个国家机构负责这项庞大的准备工程，以带箭头的大写H为标识。宣传还包括电视歌曲比赛，获胜的广告歌用歌词劝人靠右行驶，甚至登上歌曲排行榜第五名。约36万件道路标识被更换，1000多辆右侧开门的新公交车也采购到位。

切换在1967年9月3日星期日进行。凌晨1点至6点，非必要车辆不得上路，必要车辆获得豁免。4点50分，仍在路上的车辆全部停下，再缓缓移到另一侧；5点、天亮之前，车辆统一按相反的行驶侧重新出发。斯德哥尔摩和马尔默的封路时间更长，从周六上午到周日下午，施工人员在此期间改造交叉路口。

随后的星期一报告了125起事故，此前的星期一则有130至198起。这一统计上的改善，被归因于司机对新制度尚不熟悉，开车格外小心。紧张感似乎后来消退了，谨慎习惯也随之淡去：到1969年，事故率和死亡率已回到原来的基准水平。

左侧方向盘与靠左行驶不是一回事；首个周一事故少，并不意味着长期事故率持续下降。

核心词汇

先记这22项。连着下方的原文短语一起回想。

referendum n.
全民公投
a 1955 referendum

legislator n.
立法者
legislators in parliament

overrule v.
推翻（决定或意见）
overruled the electorate

electorate n.
全体选民
overruled the electorate

in defiance of 介词短语
不顾；违抗
in defiance of a populace

conformity n.
一致；统一
The case rested on conformity

isolation n.
孤立
meant isolation

alignment n.
统一；保持一致
Safety favoured alignment too

overtake v.
超车
motorists overtaking

head-on collision n.
迎头相撞
risked head-on collisions

preparation n.
准备工作
the mammoth preparation

signage n.
各类标识
360,000 items of signage

procure v.
采购；取得
was procured

transition n.
转变；过渡
The transition came on Sunday

exempt adj.
获豁免的
with essential traffic exempt

reconfigure v.
重新设置；改造
crews reconfigured junctions

statistical adj.
统计上的
a statistical improvement

put … down to … 动词短语
把……归因于……
put down to the heightened care

subside v.
消退
The nervousness apparently subsided

prudent adj.
谨慎的
prudent habits faded

revert to 动词短语
回到；恢复到
had reverted to their baseline

baseline n.
基准水平
reverted to their baseline

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

overwhelmingly adv.
压倒性地

rejection n.
拒绝

parliament n.
议会

populace n.
民众

unpopular adj.
不得人心的

orientation n.
行驶侧；通行取向

inconvenient adj.
不便的

motorist n.
驾车者

lane n.
车道

highway n.
公路

mammoth adj.
庞大的

logo n.
标识

arrow n.
箭头

propaganda n.
宣传

contest n.
比赛

jingle n.
广告短歌

lyric n.
歌词

chart n.
歌曲排行榜

massive adj.
巨大的

fleet n.
车队

curfew n.
车辆限行令

non-essential adj.
非必要的

asphalt n.
沥青路面

crawl v.
缓慢移动

dawn n.
黎明

flip v.
翻转；改变方向

closure n.
封闭

junction n.
交叉路口

improvement n.
改善

heightened adj.
增强的

hesitant adj.
犹豫不决的

nervousness n.
紧张

fade v.
逐渐消失

fatality rate n.
死亡率

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In a 1955 _____, about 83 per cent of voters chose to keep driving on the left, an overwhelmingly clear rejection of change.

02 Some 360,000 items of _____ were changed, and a massive fleet of over 1,000 new buses with right-hand doors was procured.

03 A curfew kept non-essential vehicles off the asphalt from one to six, with essential traffic _____.

04 The nervousness apparently subsided and prudent habits faded, for by 1969 accident and fatality rates had reverted to their _____.

文章主线

- 缘由 瑞典靠左，挪威和芬兰靠右，过境车流频繁；瑞典九成汽车的方向盘又在左边，窄路超车有迎面相撞风险。
- 决定 1955年公投约83%反对改靠右，议会仍于1963年批准改制，定在1967年实施。
- 准备 政府广泛宣传，更换约36万件道路标识，购置1,000多辆右侧开门的公交车，并在切换期间限制非必要车辆上路。
- 切换 1967年9月3日4点50分，车辆停下后缓缓移到另一侧，5点统一靠右出发；部分城市另封路改造路口。
- 后续 切换后首个周一事故减少，被归因于司机格外谨慎；但到1969年，事故率和死亡率都回到原有水平。

参考答案　1 referendum 2 signage 3 exempt 4 baseline

微塑料：
无处不在，危害几何？

Microplastics: Everywhere, but How Harmful?

检出了微塑料，就能证明它有害吗？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In May 2004 Richard Thompson, a marine biologist at the University of Plymouth, and colleagues described an overlooked pollutant in the journal Science. Sifting through plankton samples kept since the 1960s, they found that microscopic plastic fragments and synthetic fibres had grown more abundant. They labelled the particles microplastics.

Since then the durable debris has proved ubiquitous, turning up on shorelines, in deep-sea sediment, in rainfall, and in Arctic ice. Regulation came before proof of harm to people: in December 2015 America passed a law banning plastic microbeads from scrubs and other cosmetics rinsed down drains.

The search turned inward. In March 2022 a small Dutch study found plastic in the blood of about three-quarters of 22 healthy donors. In March 2024 a study in the New England Journal of Medicine linked plastic in the fatty plaque of neck arteries to a higher risk of heart attack, stroke, or death. Its authors cautioned that correlation is not causation.

The most sensational figures are the shakiest. In June 2019 a report for WWF, a conservation group, reckoned people might swallow a credit card's worth of plastic a week; critics call that an overestimate. Two months later the World Health Organization was reassuring but tentative: on limited evidence, microplastics in drinking water posed no apparent risk at current levels. In February 2025 a study in Nature Medicine reported plastic in brain tissue in an amount popularised as a spoonful; others queried its methodology.

Detection has outrun understanding; human studies remain small or observational. Harm is plausible but unproven, and until the verdict is in, better measurement may guide policy more safely than panic.

中文译文

2004年5月，普利茅斯大学海洋生物学家理查德·汤普森和同事在《科学》杂志上描述了一种长期被忽略的污染物。他们细查自20世纪60年代起保存的浮游生物样本，发现其中微小的塑料碎片和合成纤维越来越多。他们把这些颗粒称为微塑料。

此后，人们发现，这些难以消失的碎屑几乎无处不在：海岸、深海沉积物、雨水和北极冰层中都有它们的踪迹。监管先于人体危害的确证出现：2015年12月，美国通过法律，禁止在磨砂膏等用后冲入下水道的化妆品中添加塑料微珠。

寻找微塑料的目光随后转向人体内部。2022年3月，一项小规模荷兰研究在22名健康献血者中，发现约四分之三的人血液里有塑料。2024年3月，《新英格兰医学杂志》发表的一项研究发现，颈部动脉脂肪性斑块中的塑料与较高的心脏病发作、中风或死亡风险相关。作者提醒：相关不等于因果。

最耸人听闻的数字，往往也最站不住脚。2019年6月，一份为自然保护组织WWF撰写的报告估计，人每周可能吞下相当于一张信用卡重量的塑料；批评者认为这个数字过高。两个月后，世界卫生组织给出了令人稍感安心、却仍属初步的判断：依据有限证据，饮用水中的微塑料在当时的含量下并无明显风险。2025年2月，《自然·医学》上的一项研究报告了脑组织中的塑料，其数量被大众转述成“一勺”；另一些人则质疑这项研究的方法。

检测走在了理解前面；人体研究仍然规模较小，或只属于观察性研究。微塑料可能有害，但尚未得到证实。在结论明确之前，用更准确的测量来指导政策，或许比任由恐慌主导更稳妥。

核心词汇

先记这22项。连着下方的原文短语一起回想。

pollutant n.
污染物
described an overlooked pollutant in

sift through phr.v.
细查；逐一筛查
Sifting through plankton samples kept

microscopic adj.
极微小的；显微镜下的
that microscopic plastic fragments and

synthetic adj.
合成的；人造的
plastic fragments and synthetic fibres

abundant adj.
大量存在的
fibres had grown more abundant.

durable adj.
耐久的；不易分解的
Since then the durable debris has

debris n.
碎屑；残片
Since then the durable debris has

ubiquitous adj.
无处不在的
durable debris has proved ubiquitous,

sediment n.
沉积物
on shorelines, in deep-sea sediment,

regulation n.
监管；规管
Regulation came before proof of harm

donor n.
献血者
three-quarters of 22 healthy donors.

artery n.
动脉
in the fatty plaque of neck arteries

correlation n.
相关性
Its authors cautioned that correlation

causation n.
因果关系
that correlation is not causation.

sensational adj.
耸人听闻的
The most sensational figures are the

overestimate n.
过高的估计
critics call that an overestimate.

tentative adj.
初步的；未定的
was reassuring but tentative:

methodology n.
研究方法；方法体系
others queried its methodology.

observational adj.
观察性的
studies remain small or observational.

plausible adj.
有可能的；合乎情理的
Harm is plausible but unproven,

verdict n.
定论；判断
and until the verdict is in, better

measurement n.
测量
is in, better measurement may guide

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

marine adj.
海洋的

biologist n.
生物学家

plankton n.
浮游生物

fragment n.
碎片

fibre n.
纤维

label v.
把……称为；命名

particle n.
颗粒；微粒

turn up phr.v.
出现；被发现

shoreline n.
海岸线

rainfall n.
降雨；雨水

pass a law v.phr.
通过一项法律

scrub n.
磨砂清洁产品

cosmetic n.
化妆品

rinse down phr.v.
随冲洗流入

turn inward v.phr.
转向内部；此处指人体内

fatty adj.
含脂肪的

plaque n.
血管内的斑块

stroke n.
中风

caution v.
告诫；提醒谨慎

shaky adj.
最不可靠的

conservation n.
自然保护

reckon v.
估计；估算

swallow v.
吞下

reassuring adj.
令人安心的

apparent adj.
明显的

tissue n.
生物组织

spoonful n.
一勺的量

query v.
质疑

detection n.
检测；发现

unproven adj.
尚未证实的

panic n.
恐慌

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 _____ plankton samples kept since the 1960s,

02 Its authors cautioned that _____ is not causation.

03 Two months later the World Health Organization was reassuring but _____:

04 Harm is _____ but unproven,

文章主线

- 发现 2004年，汤普森团队在存档海洋样本中发现微小塑料碎片和合成纤维增多，将其称为微塑料。
- 扩散 微塑料从深海到北极无处不在；美国2015年立法禁用冲洗类化妆品塑料微珠，当时人体危害尚未证实。
- 证据 22名健康献血者中，约四分之三的人血中检出塑料；另一研究发现颈部动脉斑块中的塑料与更高的心梗、中风或死亡风险相关，不能据此证明因果。
- 争议 每周吞下信用卡分量塑料的估算被批高估；脑内塑料量被转述成“一勺”，研究方法受质疑。
- 判断 世卫2019年据有限证据，初判饮水微塑料在当时含量下无明显风险；人体危害仍未定论。

参考答案　1 Sifting through 2 correlation 3 tentative 4 plausible

冰里的五千年凶案

The 5,000-Year-Old Murder in the Ice

冰人的箭伤，揭开了怎样的古老谜案?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 19 September 1991 two German hikers on a trek in the Ötztal Alps found a corpse by a glacier. It lay on a ridge at about 3,210 metres, on the border between Austria and Italy. The assumption was that a modern-day climber had died on an alpine slope, since glaciers sometimes yield frozen remains.

Taking it for a recent death, rescuers used a jackhammer that ripped flesh from his left hip. His possessions told another story: leather and grass clothing, a copper axe, a bow and arrows, all prehistoric artefacts. Known as Ötzi, the natural mummy was a man who lived about 5,300 years ago, between 3350 and 3105 BC.

In 2001, X-rays revealed a flint arrowhead lodged in his left shoulder, unseen by examiners a decade earlier. A scanner showed that the puncture had ruptured an artery, a mortal injury causing heavy bleeding, and left a large clot. Researchers believe he died within minutes. He was about 46.

Another person, it is thought, killed him. From the wound's depth researchers inferred that an archer shot him from the rear, at about 30 metres. The shooter probably caught him unawares. A 2018 study found he had dined on dried ibex meat and fat, red deer, and einkorn wheat, a hearty, nourishing meal reportedly consumed within two hours of his death.

Curators in Bolzano now keep Ötzi, whose 61 tattoos on joints and spine were perhaps an early form of acupuncture. The archer's identity is an enigma, and any motive is conjecture. Ötzi has a name, a last meal, and a cause of death, while his anonymous killer has none.

中文译文

1991年9月19日，两名德国徒步者在奥茨塔尔阿尔卑斯山远足时，于冰川旁发现一具遗体。它位于奥地利与意大利交界处、海拔约3210米的山脊上。人们最初以为，一名现代登山者死在高山坡地上，毕竟冰川有时会吐露冰冻的遗骸。

救援人员以为这是近期的死亡事件，使用风钻凿冰，结果撕损了死者左侧髋部的皮肉。但他的随身物品讲述了另一种故事：皮革和草编衣物、铜斧、弓箭，全是史前器物。这具被称为奥茨的天然木乃伊是一名男子，生活于约5300年前，即公元前3350至前3105年之间。

2001年，X光检查发现一枚燧石箭头嵌在他的左肩里，十年前的检查者并未发现它。扫描显示，穿刺伤撕裂了一条动脉，造成大量出血的致命损伤，并留下一个大血块。研究者认为，他在几分钟内便死亡，当时约46岁。

人们认为，他是遭人杀害的。研究者根据伤口深度推断，弓箭手从他背后约30米处射箭，很可能是在他毫无防备时出手。2018年的研究发现，他最后吃过干野山羊肉和脂肪、马鹿肉与单粒小麦。这是一顿丰盛而营养丰富的饭，据报道在死前两小时内吃下。

如今，博尔扎诺的馆藏保管人员保存着奥茨的遗体。他关节和脊柱上的61处文身，也许是一种早期针灸形式。弓箭手的身份仍是谜团，任何作案动机都只是猜测。奥茨有名字、最后一餐和死因；对于那名身份不明的凶手，我们却连这些信息都不知道。

核心词汇

先记这22项。连着下方的原文短语一起回想。

glacier n.
冰川
a corpse by a glacier

ridge n.
山脊
on a ridge at about 3,210 metres

assumption n.
假定；最初的判断
The assumption was

possession n.
随身物品；所有物
His possessions told another story

prehistoric adj.
史前的
all prehistoric artefacts

artefact n.
人工制品；考古器物
all prehistoric artefacts

mummy n.
木乃伊；保存下来的干尸
the natural mummy

flint n.
燧石
a flint arrowhead

lodge v.
嵌入；卡住
lodged in his left shoulder

puncture n.
穿刺伤；刺孔
the puncture had ruptured an artery

rupture v.
使破裂；撕裂
had ruptured an artery

mortal adj.
致命的
a mortal injury

infer v.
推断；推知
researchers inferred

unawares adv.
在没有防备时
caught him unawares

nourishing adj.
有营养的
a hearty, nourishing meal

curator n.
馆藏保管人员；策展人
Curators in Bolzano

acupuncture n.
针灸
an early form of acupuncture

identity n.
身份
The archer's identity

enigma n.
谜团
an enigma

motive n.
动机
any motive is conjecture

conjecture n.
猜测；推测
any motive is conjecture

anonymous adj.
身份不明的；不知姓名的
his anonymous killer

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

hiker n.
徒步者

trek n.
长途徒步；远足

corpse n.
尸体；遗体

border n.
边界；交界处

modern-day adj.
现代的；当代的

climber n.
登山者

alpine adj.
高山的；阿尔卑斯山的

slope n.
山坡；斜坡

jackhammer n.
风钻；气动破碎工具

rip v.
撕裂；扯下

flesh n.
皮肉；肌肉组织

hip n.
髋部

leather n.
皮革

copper adj.
铜制的

axe n.
斧子

unseen adj.
未被看见或发现的

by prep.
被；由

examiner n.
检查人员

scanner n.
扫描仪

bleeding n.
出血

clot n.
血块

it is thought 插入语
据认为；人们推测

archer n.
弓箭手

rear n.
后方；背后

shooter n.
射手

catch v.
使措手不及；趁其不备

dine on 动词短语
吃；进食

red deer 名词短语
马鹿

hearty adj.
丰盛的；充足的

consume v.
食用；吃下

tattoo n.
文身

spine n.
脊柱

killer n.
凶手

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ was that a modern-day climber had died on an alpine slope, since glaciers sometimes yield frozen remains.

02 From the wound's depth researchers _____ that an archer shot him from the rear, at about 30 metres.

03 Curators in Bolzano now keep Ötzi, whose 61 tattoos on joints and spine were perhaps an early form of _____.

04 The archer's identity is an enigma, and any motive is _____.

文章主线

- 误判 1991年阿尔卑斯山发现冰尸，最初被当作现代登山者，救援风钻还损伤了他的髋部。
- 身份 皮革和草编衣物、铜斧与弓箭显示他来自史前；冰人奥茨生活于约5300年前，死时约46岁。
- 死因 2001年发现肩内箭头，扫描显示动脉被撕裂；研究者认为他因此大量失血，数分钟内死亡。
- 重建 据伤口推测，射手从背后约30米放箭；胃中有野山羊肉和脂肪、鹿肉与小麦，据报道在死前两小时内吃下。
- 未解 关节和脊柱上的61处文身，或与早期针灸有关；但射手是谁、为何杀他，仍没有定论。

参考答案　1 assumption 2 inferred 3 acupuncture 4 conjecture

一颗棉花糖的真相

The Truth About One Marshmallow

一颗棉花糖能预测孩子的未来吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Picture a nursery school on Stanford's campus in the late 1960s and early 1970s. A child of about four or five sits in solitude before one treat, facing a blunt bargain: eat it now, or resist temptation for about 15 minutes and earn two. Impulsive children often surrender instantly; resolute ones hold out.

Follow-ups made the test legendary. A 1990 study in Developmental Psychology found that, as adolescents, longer waiters tended to attain higher scores on the SAT, a US college-entrance exam. A trivial snack became an emblem of virtue, and restraint at four was glorified as a predictor of destiny.

Yet the foundation was precarious. Only a few dozen children entered the SAT analysis, predominantly the sons and daughters of Stanford faculty and postgraduate students. Such a homogeneous, privileged group could scarcely represent ordinary families, and prosperous homes may foster patience and academic success alike.

A study published in Psychological Science in May 2018 ran a conceptual replication: the same idea, a new sample, modified methods. More than 900 youngsters from diverse backgrounds were assessed at four and a half and again at 15. The raw link was roughly half as strong as the original finding. Adjusting for family background and early cognitive ability shrank it by about two thirds, leaving most estimates statistically insignificant.

The test is not worthless; it may still reflect something substantive. Research published in Cognition in January 2013 adds context: children whose trust an adult had just betrayed waited markedly less than peers dealing with a reliable adult. How long a child perseveres depends partly on how trustworthy the adults around them seem.

中文译文

想象一下20世纪60年代末至70年代初，斯坦福大学校园里的一所幼儿园。一个四五岁的孩子独自坐在一份零食前，面对一个直截了当的选择：现在吃掉它，或者抵住诱惑，等上约15分钟，就能得到两份。冲动的孩子往往马上放弃等待，意志坚定的孩子则坚持下去。

后续追踪研究让这项测验广为人知。1990年发表于《发展心理学》的一项研究发现，进入青少年时期后，当初等得更久的孩子，往往在美国大学入学考试SAT中取得更高分数。一份不起眼的零食成了美德的象征，四岁时表现出的克制，也被过分推崇为预测人生走向的指标。

然而，这一结论的基础并不牢靠。进入SAT成绩分析的孩子只有几十名，而且主要是斯坦福大学教师和研究生的子女。这样一个背景相似、条件优越的群体，很难代表普通家庭。富裕的家庭环境，也可能同时有助于培养耐心和取得学业成功。

2018年5月发表于《心理科学》的一项研究进行了概念性重复实验：沿用同一研究思路，换用新的样本，并调整研究方法。研究者对来自不同背景的900多名儿童，在四岁半和15岁时分别进行了评估。未经调整的关联强度，大约只有早期研究结果的一半；进一步控制家庭背景和早期认知能力的影响后，这种关联又缩小了约三分之二，大多数估计结果都未达到统计显著性。

这项测验并非毫无价值，它或许仍能反映一些有实质意义的信息。2013年1月发表于《认知》的一项研究补充了背景：刚刚遭遇成年人失信的孩子，等待时间明显短于与守信成年人相处的同龄人。孩子能坚持多久，部分取决于他们觉得身边的大人有多可信。

“统计上不显著”不等于已经证明不存在关联；这里强调，计入背景差异后，早期结论的支持力度明显减弱。

核心词汇

先记这22项。连着下方的原文短语一起回想。

resist v.
抵挡；忍住
resist temptation

temptation n.
诱惑
resist temptation for about 15 minutes

impulsive adj.
冲动的
Impulsive children often surrender

surrender v.
放弃抵抗；屈服
often surrender instantly;

resolute adj.
意志坚定的
resolute ones hold out.

hold out phr. v.
坚持下去
resolute ones hold out.

legendary adj.
广为传颂的；著名的
Follow-ups made the test legendary.

restraint n.
克制；自制
restraint at four was glorified

glorify v.
过分美化；推崇
glorified as a predictor of destiny

predictor n.
预测因素；预测指标
a predictor of destiny.

precarious adj.
不稳固的；不可靠的
Yet the foundation was precarious.

homogeneous adj.
同质的；背景相似的
a homogeneous, privileged group

privileged adj.
享有优越条件的
a homogeneous, privileged group

foster v.
培养；促进
foster patience and academic success

replication n.
重复研究；重复实验
ran a conceptual replication:

diverse adj.
多种多样的
from diverse backgrounds

assess v.
评估
were assessed at four and a half

adjust for phr. v.
控制……的影响
Adjusting for family background

cognitive adj.
认知的
early cognitive ability

insignificant adj.
未达统计显著性的
statistically insignificant.

betray v.
辜负；背弃
trust an adult had just betrayed

trustworthy adj.
可信赖的
how trustworthy the adults around them

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

nursery school n.
幼儿园

campus n.
大学校园

in solitude phr.
独自；独处地

blunt adj.
直截了当的

bargain n.
交易；约定的条件

earn v.
挣得；赢得

instantly adv.
立即；马上

follow-up n.
后续追踪研究

adolescent n.
青少年

attain v.
达到；取得

trivial adj.
微不足道的

snack n.
零食

emblem n.
象征；标志

virtue n.
美德

destiny n.
命运；人生走向

foundation n.
（结论的）基础

a few dozen phr.
几十个

predominantly adv.
主要地；大多

faculty n.
大学全体教员

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

postgraduate adj.
研究生的

scarcely adv.
几乎不

ordinary adj.
普通的

prosperous adj.
富裕的

alike adv.
同样地；两者都

conceptual adj.
概念层面的

modified adj.
调整过的；改动过的

youngster n.
儿童；少年

link n.
关联；联系

statistically adv.
在统计学上

worthless adj.
毫无价值的

reflect v.
反映

substantive adj.
有实质意义的

context n.
背景；具体情境

markedly adv.
明显地

peer n.
同龄人

persevere v.
坚持；不放弃

depend v.
取决于（常与on连用）

on prep.
与depend连用，表示取决于

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Impulsive children often surrender instantly; _____ ones hold out.

02 Yet the foundation was _____.

03 Adjusting for family background and early cognitive ability shrank it by about two thirds, leaving most estimates statistically _____.

04 How long a child perseveres depends partly on how _____ the adults around them seem.

文章主线

- 选择 四五岁孩子可立刻吃一份零食，或等约15分钟换两份，用来观察延迟满足。
- 走红 早期追踪发现，等得久的孩子后来SAT（美国大学入学考试）成绩较高，测验遂被当作成功预测器。
- 局限 早期SAT分析仅数十名孩子，且多来自优裕家庭；家庭条件可能同时影响耐心与成绩。
- 复核 2018年复核900多名不同背景儿童；计入家庭背景与早期认知差异后，关联大幅减弱，多数估计不具统计显著性。
- 关键 另一实验发现，刚遇到不守信大人的孩子等得更短；等待也反映信任，不能据此断定一个人的命运。

参考答案　1 resolute 2 precarious 3 insignificant 4 trustworthy

两度摘得诺奖的女人

The Woman Who Won Two Nobels

居里夫人的两座诺奖从何而来?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Marie Curie was born Maria Skłodowska on 7 November 1867 in Warsaw, then under Russian dominion, where women were denied university admission. Eager to pursue formal study, she emigrated to Paris in 1891, enrolling at the Sorbonne. On 26 July 1895 she wed the physicist Pierre Curie, forging a research partnership.

Henri Becquerel found that uranium emitted rays; she named this property of certain elements radioactivity. In July 1898 the Curies announced polonium, a tribute to her motherland, and on 26 December, radium. In a leaky shed they painstakingly processed tonnes of pitchblende, a uranium ore; by 1902 she had recovered about a tenth of a gram of radium chloride.

On 10 December 1903 she jointly won the physics Nobel with Pierre and Becquerel, the first woman so recognised. The nomination had initially left her out; at Pierre's insistence, the omission was rectified. On 19 April 1906 a horse-drawn wagon fatally struck Pierre. On 5 November she gave her inaugural lecture as his successor, the Sorbonne's first woman professor.

In November 1911 a press scandal about her private life erupted; she was urged to stay away. Undeterred, she collected the chemistry Nobel, won outright, on 10 December. Her distinction in two sciences remains unmatched. In the First World War she mobilised X-ray vans and stationary posts that examined over a million casualties.

She died on 4 July 1934 of a blood disorder, probably caused by prolonged exposure to radiation. Her notebooks, still radioactive, are preserved in lead-lined boxes at the National Library of France; readers sign a liability waiver and don protective garments. The pages retain her handwriting and her radium.

中文译文

玛丽·居里原名玛丽亚·斯克沃多夫斯卡，1867年11月7日生于当时受俄国统治的华沙。那里的大学不招收女性。为了接受正规的高等教育，她于1891年移居巴黎，进入索邦大学。1895年7月26日，她与物理学家皮埃尔·居里结婚，两人也结成了科研搭档。

亨利·贝克勒尔发现铀会发出射线；玛丽把某些元素的这种性质命名为“放射性”。1898年7月，居里夫妇宣布发现钋，以此向她的祖国致敬；同年12月26日，他们又宣布发现镭。两人在一间漏雨的棚屋里，费尽心力处理了成吨的沥青铀矿。到1902年，她提取出了约十分之一克氯化镭。

1903年12月10日，她与皮埃尔、贝克勒尔共同获得诺贝尔物理学奖，成为首位女性诺奖得主。最初的提名没有她的名字，后来在皮埃尔的坚持下，这一遗漏才得到纠正。1906年4月19日，皮埃尔被一辆马拉货车撞倒身亡。同年11月5日，她接替丈夫的职位，发表就职演讲，成为索邦大学首位女教授。

1911年11月，媒体围绕她的私生活掀起一场风波，有人劝她不要前去领奖。她没有退缩，仍于12月10日领取了自己独自获得的诺贝尔化学奖。她在两门科学领域获得诺奖的成就，至今无人比肩。第一次世界大战期间，她组织调用X光车和固定检查站，为超过一百万名伤员做了检查。

1934年7月4日，她因一种血液疾病去世，很可能与长期接触辐射有关。她的笔记本至今仍有放射性，保存在法国国家图书馆内衬铅的盒子里；读者须签署免责协议，穿上防护服才能查阅。书页上既留着她的笔迹，也留着她的镭。

核心词汇

先记这22项。连着下方的原文短语一起回想。

dominion n.
统治；管辖
then under Russian dominion,

deny v.
被拒绝给予；被剥夺
were denied university admission.

admission n.
入学资格；准入
were denied university admission.

pursue v.
追求；从事
Eager to pursue formal study,

emigrate v.
移居国外
she emigrated to Paris in 1891,

enrol v.
注册入学
emigrated to Paris in 1891, enrolling

forge v.
建立；缔结
forging a research partnership.

partnership n.
合作关系
forging a research partnership.

property n.
性质；特性
she named this property of certain

radioactivity n.
放射性
of certain elements radioactivity.

painstakingly adv.
煞费苦心地；精细认真地
they painstakingly processed tonnes of

recover v.
提取；回收
by 1902 she had recovered about a

jointly adv.
共同地
10 December 1903 she jointly won the

insistence n.
坚持；坚决要求
Pierre's insistence, the omission was

omission n.
遗漏；疏漏
Pierre's insistence, the omission was

rectify v.
纠正；改正
the omission was rectified.

inaugural adj.
就职的；首次的
she gave her inaugural lecture as his

successor n.
继任者
inaugural lecture as his successor,

undeterred adj.
未被吓退的
Undeterred,

distinction n.
殊荣；卓越成就
Her distinction in two sciences

mobilise v.
组织调用；动员
World War she mobilised X-ray vans and

prolonged adj.
持续很久的
probably caused by prolonged exposure

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

eager adj.
渴望的；热切的

formal adj.
正规的；正式的

wed v.
与……结婚

ray n.
射线

element n.
化学元素

tribute to n.phr.
对……的致敬

motherland n.
祖国

radium n.
镭

leaky adj.
漏水的；漏雨的

tonne n.
公吨

ore n.
矿石

gram n.
克

chloride n.
氯化物

recognise v.
认可；表彰

initially adv.
最初

leave out phr.v.
与out连用：遗漏；排除

out adv.
与leave连用：排除在外

wagon n.
货运马车

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

fatally adv.
致命地

scandal n.
丑闻；引发非议的风波

erupt v.
突然爆发

outright adv.
独自地；不与他人分享地

unmatched adj.
无人能及的

van n.
厢式汽车

stationary adj.
固定不动的

casualty n.
伤亡人员；此处指伤员

disorder n.
疾病

radioactive adj.
有放射性的

preserve v.
保存

lead n.
铅

liability waiver n.phr.
免责协议

don v.
穿上

protective adj.
防护的

garment n.
衣服；服装

retain v.
保留；留存

handwriting n.
笔迹

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Eager to _____ formal study,

02 at Pierre's insistence, the omission was _____.

03 _____, she collected the chemistry Nobel, won outright, on 10 December.

04 probably caused by _____ exposure to radiation.

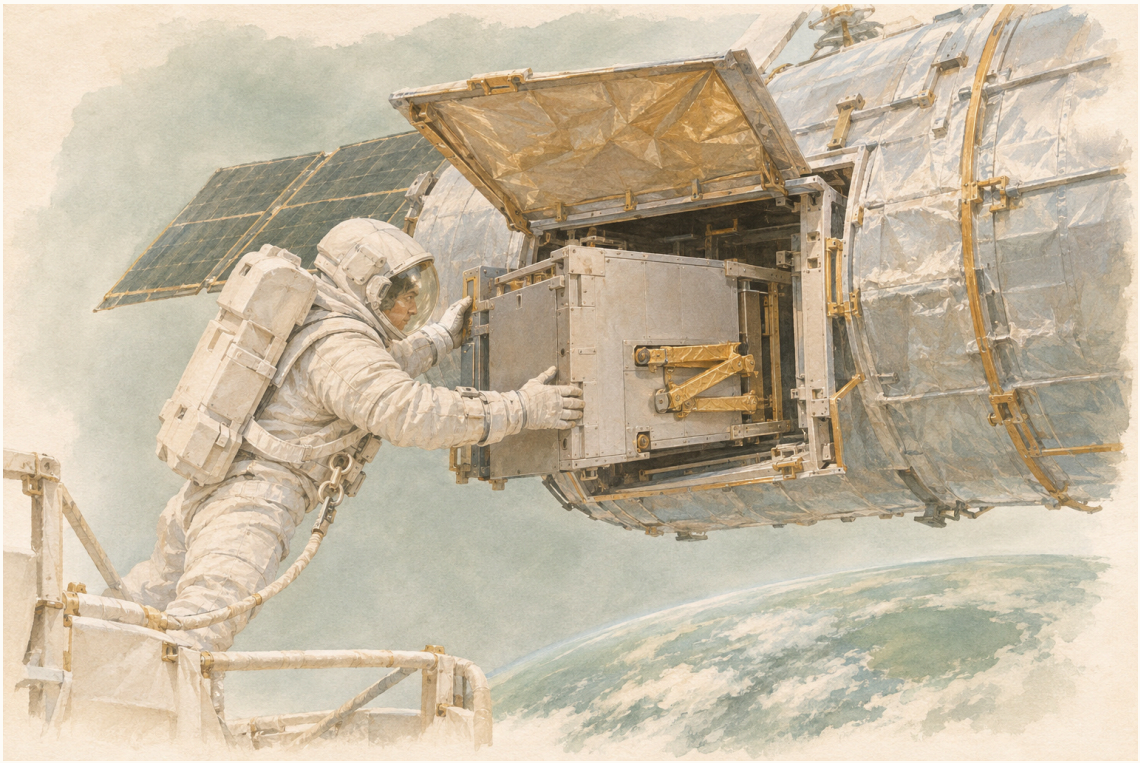
文章主线

- 求学 当时华沙的大学不收女性，玛丽·居里1891年赴巴黎求学，后与皮埃尔结婚合作研究。
- 发现 贝克勒尔发现铀会放出射线，她将这种性质称为放射性；居里夫妇发现钋、镭，从成吨矿石提得约0.1克氯化镭。
- 首奖 1903年，她与皮埃尔、贝克勒尔共享物理学奖，成为首位女性诺奖得主；最初漏提名经皮埃尔坚持纠正。
- 双奖 皮埃尔遇难后，她接任成为索邦首位女教授；1911年顶住私生活风波独获化学奖，仍是唯一获两门科学诺奖者。
- 救护代价 一战时她组织X光设施检查逾百万伤员；后来死于很可能与长期辐射有关的血液病，笔记至今仍有放射性。

参考答案　1 pursue 2 rectified 3 Undeterred 4 prolonged

需要戴眼镜的望远镜

The Telescope That Needed Glasses



主镜没有修好，影像怎样变清晰？

READ 纯净阅读 | CHECK 中文核对 | LEARN 词汇记忆 | RECALL 回想巩固

英文原文

On 24 April 1990 the space shuttle Discovery lofted the Hubble Space Telescope into orbit. The observatory cradled in the payload bay had cost about $1.5 billion, a lavish investment meant to fix its gaze on the cosmos and its galaxies. Money, it soon emerged, could not buy clarity.

Its first snapshots came back blurred and fuzzy. In June 1990 NASA announced a flaw in the main mirror: its outer rim had been flattened by 2.2 micrometres, roughly a fiftieth of the width of a hair. That deviation marred the optics, leaving light without a sharp focus.

The trouble began with calibration. A lens in the assembly that checked the mirror's shape had been offset by about 1.3 millimetres, so the mirror was ground to the wrong curve. A second, independent test by the contractor, Perkin-Elmer, gave a divergent result, but engineers and NASA managers judged it less reliable and left the contradictory reading unexplored.

The mirror could not be repaired in orbit, so engineers prescribed tailored glasses. COSTAR, a corrective cabinet the size of a refrigerator, carried small mirrors on adjustable arms that intercepted the light and compensated for the imperfect mirror's distortion before the light reached the instruments. The mirror itself would stay untouched, and a new main camera was prepared.

On 2 December 1993 Endeavour ascended with seven astronauts. After a rendezvous, they worked in tandem across five spacewalks, about 35 and a half hours of exertion, installing COSTAR and the new camera. The imagery turned crisp, Hubble's vision renewed. The mirror still carries that 2.2-micrometre flaw, and the light it gathers is corrected only after leaving it.

中文译文

1990年4月24日，发现号航天飞机将哈勃空间望远镜送入轨道。这座安放在载荷舱中的太空天文台耗资约15亿美元，是一笔用于凝视宇宙与星系的巨额投资。但人们很快发现，花钱并不一定能买来清晰的影像。

哈勃传回的首批照片模糊不清。1990年6月，美国航天局宣布主镜有缺陷：外缘磨得过平，与正确形状相差2.2微米，约为头发丝宽度的五十分之一。这点偏差损害了光学系统，使光线无法准确聚焦。

问题出在校准上。检测主镜形状的装置中，一块透镜错位了约1.3毫米，主镜因此被磨成了错误的曲面。承包商珀金-埃尔默公司另一项独立测试曾给出不同结果，但工程师和美国航天局管理人员认为该测试较不可靠，没有深入追查那组相互矛盾的读数。

主镜无法在轨道上修复，于是工程师为它设计了一副“定制眼镜”。矫正装置COSTAR的箱体约有冰箱大小，可调机械臂上装着小反射镜，在光线进入仪器前将其截住，抵消有缺陷的主镜造成的畸变。主镜本身保持不动，同时还准备了一台新的主相机。

1993年12月2日，奋进号载着七名宇航员升空。完成交会后，他们协同作业，通过五次太空行走，用约35个半小时安装了COSTAR和新相机。哈勃的影像恢复清晰，仿佛重获视力。但主镜那2.2微米的缺陷仍然存在，光线是在离开主镜之后才得到矫正的。

2.2微米是主镜形状误差，1.3毫米是检测装置透镜的错位；安装矫正装置并未修磨主镜。

核心词汇

先记这22项。连着下方的原文短语一起回想。

observatory n.
天文台
The observatory

payload n.
有效载荷
in the payload bay

lavish adj.
耗资巨大的
a lavish investment

clarity n.
清晰度
could not buy clarity

flaw n.
缺陷
a flaw in the main mirror

deviation n.
偏差
That deviation marred the optics

optics n.
光学系统
marred the optics

calibration n.
校准
The trouble began with calibration

assembly n.
装置；组件
A lens in the assembly

offset v.
使偏离位置
offset by about 1.3 millimetres

contractor n.
承包商
the contractor, Perkin-Elmer

divergent adj.
不同的；不一致的
a divergent result

reliable adj.
可靠的
judged it less reliable

contradictory adj.
相互矛盾的
the contradictory reading

corrective adj.
矫正的
a corrective cabinet

intercept v.
截住
intercepted the light

compensate for 动词短语
弥补；抵消
compensated for

distortion n.
畸变
the imperfect mirror's distortion

untouched adj.
未改动的
The mirror itself would stay untouched

rendezvous n.
交会
After a rendezvous

in tandem 介词短语
协同地
worked in tandem

imagery n.
影像
The imagery turned crisp

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

loft v.
向高处送出

telescope n.
望远镜

into prep.
进入；到……中

cradle v.
稳妥安放

investment n.
投资

gaze n.
凝视；目光

cosmos n.
宇宙

galaxy n.
星系

snapshot n.
照片；快照

blurred adj.
模糊的

fuzzy adj.
模糊不清的

outer adj.
外侧的

rim n.
边缘

flatten v.
磨平；使变平

width n.
宽度

mar v.
损害

sharp adj.
清晰锐利的

lens n.
透镜

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

millimetre n.
毫米

be ground to 动词短语
被磨成

curve n.
曲面；曲线

independent test n.
独立测试

unexplored adj.
未被深入调查的

prescribe v.
开出；提出（方案）

tailored adj.
量身定制的

cabinet n.
箱体

refrigerator n.
冰箱

adjustable adj.
可调节的

imperfect adj.
有缺陷的

ascend v.
升空；上升

spacewalk n.
太空行走

exertion n.
费力；辛苦作业

install v.
安装

crisp adj.
清晰鲜明的

vision n.
视力

renew v.
使恢复

gather v.
收集

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The trouble began with _____.

02 engineers and NASA managers judged it less _____ and left the contradictory reading unexplored.

03 The mirror itself would stay _____, and a new main camera was prepared.

04 The _____ turned crisp, Hubble's vision renewed.

文章主线

- 失焦 1990年，耗资约15亿美元的哈勃望远镜升空，照片却模糊；主镜外缘磨错2.2微米，无法准确聚光。
- 根源 主镜被磨错，是因检测装置的透镜错位约1.3毫米；另一独立测试读数不同，却被认为较不可靠而未追查。
- 方案 主镜无法在轨修复，工程师设计矫正装置COSTAR，用小反射镜在光线进入仪器前抵消畸变，并另备新相机。
- 修复 1993年，七名宇航员在五次太空行走中协同作业约35.5小时，装上COSTAR和新相机。
- 结果 影像恢复清晰，但主镜本身始终未动，2.2微米缺陷依旧；被矫正的是离开主镜后的光线。

参考答案　1 calibration 2 reliable 3 untouched 4 imagery

让蚂蚁变僵尸的真菌

The Fungus That Turns Ants into Zombies

蚂蚁为何会咬住叶片不松口?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Deep in a Thai rainforest, a carpenter ant breaks from its colony's column in the lofty canopy. Swaying on unsteady legs, it wanders downward, as if intoxicated, into leafy undergrowth. Around midday it clenches its jaws on a major vein underneath a leaf, about 25 centimetres above the ground.

By dusk the insect is lifeless, yet the bite holds rigidly. A fungus has destroyed its jaw muscles, so they cannot loosen. Two or three days later a fungal stalk pushes from its head and releases infectious spores on to the ground below, where foraging ants tread.

The invader, Ophiocordyceps unilateralis, is fussy about location. David Hughes and colleagues found the bites mostly on the north-west side of plants, where the air is humid and the temperature suits the fungus. When researchers moved infected ants higher, the fungus grew abnormally. Placed on the ground, the bodies disappeared.

In 2017 a Penn State team examined infected ants with three-dimensional microscope imaging. Fungal cells had formed an intertwined lattice around muscle fibres in the legs and jaw, a picture that evokes the strings of a puppet. Yet the fungus had not infiltrated the brain.

A Planet Earth episode, with David Attenborough's narration, showed one such ant. It helped inspire The Last of Us, a 2013 video game whose makers envisioned the parasite jumping to humans. The hypothetical scenario gave gamers a fictional nightmare.

The real parasite's roots run far deeper, into the geological record. A 48-million-year-old fossil leaf from Messel, in Germany, bears the same characteristic bite scars.

中文译文

在泰国雨林深处，一只木匠蚁脱离了高处树冠中的蚁群队列。它腿脚不稳，摇摇晃晃地向下游荡，仿佛喝醉一般，钻进枝叶茂密的林下植被。正午前后，它用双颚紧紧咬住一片叶子背面的主叶脉，那里离地面约25厘米。

到黄昏时，蚂蚁已经死去，却仍牢牢咬住叶子。真菌破坏了它的颚部肌肉，使其无法放松。两三天后，一根菌柄从蚂蚁头部伸出，向下方地面散播具有传染性的孢子，那里正有觅食的蚂蚁经过。

这种入侵者叫偏侧蛇虫草菌，对位置很挑剔。戴维·休斯及其同事发现，咬合点大多位于植物的西北侧，那里空气潮湿，温度也适合真菌。研究者将受感染的蚂蚁移到更高处后，真菌生长得不正常；放到地面的蚁尸则不见了。

2017年，宾夕法尼亚州立大学的团队通过三维显微成像检查受感染的蚂蚁。真菌细胞在腿部和颚部的肌肉纤维周围织成相互交错的网络，令人联想到操纵木偶的提线。然而，真菌并未侵入蚂蚁的大脑。

由戴维·阿滕伯勒解说的纪录片《地球脉动》曾在一集中展示过这样一只蚂蚁。这启发了2013年电子游戏《最后生还者》的创作者，他们设想这种寄生菌转而感染人类。这种假想情景给玩家带来了一场虚构的噩梦。

现实中，这种寄生现象的历史要久远得多，可以追溯到地质记录之中。德国梅塞尔出土的一片4800万年前的叶子化石上，留有同样特征的咬痕。

核心词汇

先记这22项。连着下方的原文短语一起回想。

colony n.
（蚂蚁等的）群体
its colony's column

canopy n.
森林的树冠层
in the lofty canopy

unsteady adj.
不稳的
on unsteady legs

wander v.
游荡；漫无目的地走
it wanders downward

intoxicated adj.
醉酒般的
as if intoxicated

undergrowth n.
林下灌丛；林下植被
into leafy undergrowth

clench v.
紧咬；紧闭
it clenches its jaws

vein n.
叶脉
a major vein underneath a leaf

dusk n.
黄昏
By dusk

rigidly adv.
僵硬地；牢牢地
the bite holds rigidly

infectious adj.
具有传染性的
releases infectious spores

tread v.
行走；踏过
where foraging ants tread

fussy adj.
挑剔的
is fussy about location

humid adj.
潮湿的
the air is humid

abnormally adv.
不正常地；反常地
the fungus grew abnormally

intertwined adj.
交织的；缠绕的
an intertwined lattice

lattice n.
网格；格状结构
an intertwined lattice

evoke v.
使人联想到
a picture that evokes

infiltrate v.
侵入；渗入
had not infiltrated the brain

hypothetical adj.
假想的；假设的
The hypothetical scenario

scenario n.
情景；设想
The hypothetical scenario

fossil n.
化石
fossil leaf from Messel

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

rainforest n.
雨林

carpenter n.
木匠（木匠蚁名称的一部分）

column n.
行进的队列

lofty adj.
高耸的；高处的

sway v.
摇晃；摇摆

downward adv.
向下

as if 连词短语
仿佛；好像

leafy adj.
枝叶茂密的

midday n.
正午

jaw n.
颚；口部咬合器官

lifeless adj.
死去的；无生命的

muscle n.
肌肉

loosen v.
放松；松开

fungal adj.
真菌的

invader n.
入侵者

location n.
位置；地点

bite n.
咬合点；咬痕

north-west adj.
西北侧的；西北方的

disappear v.
消失；不见

three-dimensional adj.
三维的

microscope n.
显微镜

puppet n.
木偶

narration n.
解说；旁白

inspire v.
启发；赋予灵感

maker n.
创作者；制作者

envision v.
设想；想象

gamer n.
电子游戏玩家

fictional adj.
虚构的

geological adj.
地质的

characteristic adj.
有代表性的；典型的

scar n.
疤痕；留下的痕迹

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Around midday it _____ its jaws on a major vein underneath a leaf, about 25 centimetres above the ground.

02 When researchers moved infected ants higher, the fungus grew _____.

03 Yet the fungus had not _____ the brain.

04 The _____ scenario gave gamers a fictional nightmare.

文章主线

- 感染 泰国木匠蚁感染偏侧蛇虫草菌后离群，正午咬住离地约25厘米的叶背主脉，黄昏死去。
- 传播 真菌破坏颚肌，使蚂蚁死后仍咬紧；两三天后菌柄顶出，孢子散向下方蚂蚁觅食的地面。
- 选址 咬痕多在温湿度适合的植物西北侧；感染蚂蚁移高后真菌生长反常，放到地面的蚁尸则消失。
- 成像 2017年三维成像发现，真菌在腿、颚肌肉周围织成网络，却没有侵入蚂蚁大脑。
- 边界 4800万年前的叶化石已有同类咬痕；纪录片启发了游戏《最后生还者》，但感染人类的情节是虚构。

参考答案　1 clenches 2 abnormally 3 infiltrated 4 hypothetical

禁区里的生机

Life in the Forbidden Zone

禁区里动物变多，就能说明辐射无害吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Chernobyl is synonymous with disaster, yet it holds a startling amount of wildlife. Early on 26 April 1986, one of its reactors exploded in Soviet Ukraine, dispersing radioactive fallout across Europe. On 27 April, nearby Pripyat was emptied of about 45,000 inhabitants within hours. A zone some 30 kilometres in radius became off-limits, and Belarus fenced off adjoining terrain.

In October 2015, a study in Current Biology combined aerial surveys with footprint counts on the Belarusian side. Elk, roe deer, red deer, and wild boar were comparable in density to those in four reserves free of contamination, and wolves were over seven times as numerous. Earlier helicopter counts showed elk, roe deer, and boar surging in the post-disaster decade.

Later fieldwork bolstered these findings. In April 2016, camera traps snapped wolves, boar, and raccoon dogs near the most hazardous spots, with no sign of suppressed numbers. The probable reason is mundane: the absence of agriculture, forestry, and hunting.

Smaller organisms fared worse in some studies. One, in March 2007, found fewer nesting birds, and fewer species, where radiation was more intense. Another, in March 2009, found pollinators and spiders less plentiful where exposure was greater. A third, in March 2014, found leaves decomposing about 40 per cent less at the most polluted sites, impeding the return of nutrients to the soil.

Counting heads cannot guarantee the health of individual animals, and how much damage low but continual doses inflict remains contentious, as studies clash over methods. The burgeoning herds may trumpet the dividend of human departure, but the sluggish rot of fallen leaves keeps a quieter ledger of radiation's cost.

中文译文

切尔诺贝利是灾难的代名词，却栖息着数量惊人的野生动物。1986年4月26日凌晨，苏联乌克兰境内这里的一座反应堆爆炸，放射性尘降物散布到欧洲各地。4月27日，附近普里皮亚季的约4.5万居民在数小时内撤离。半径约30公里的区域被列为禁区，白俄罗斯也将相邻地带围封起来。

2015年10月，《当代生物学》发表的一项研究结合白俄罗斯一侧的空中调查与足迹计数。当地驼鹿、狍、马鹿和野猪的密度，与四个未受污染保护区的同类相当；狼的数量则为对照保护区的七倍多。较早的直升机计数显示，灾后十年里，驼鹿、狍和野猪数量激增。

后来的实地研究为这些发现提供了支持。2016年4月，自动感应相机在危险程度最高的一些地点附近拍到狼、野猪和貉，没有发现数量受抑制的迹象。可能的原因其实很普通：农业、林业和狩猎活动消失了。

但一些研究发现，较小的生物境况更差。2007年3月的一项研究发现，辐射较强的地方，筑巢鸟类的数量和鸟种都较少。2009年3月的另一项研究发现，辐射暴露更高的地方，传粉动物和蜘蛛较少。2014年3月的第三项研究发现，污染最严重地点的落叶分解量少了约40%，妨碍养分回到土壤。

清点动物数量，不能保证每一个体都健康。持续低剂量辐射究竟造成多大伤害，仍有争议，各项研究在方法上也存在分歧。不断扩大的兽群，或许彰显了人类离开带来的好处；而落叶迟缓的腐烂，则默默记下了辐射的另一笔代价。

核心词汇

先记这22项。连着下方的原文短语一起回想。

synonymous with 形容词短语
是……的代名词
is synonymous with disaster

disperse v.
散布；扩散
dispersing radioactive fallout

radioactive adj.
放射性的
radioactive fallout across Europe

fallout n.
放射性尘降物
radioactive fallout across Europe

inhabitant n.
居民
about 45,000 inhabitants

comparable adj.
相当的；可比较的
comparable in density

density n.
密度
comparable in density

contamination n.
污染
free of contamination

fieldwork n.
实地研究
Later fieldwork bolstered

bolster v.
支持；加强证据
fieldwork bolstered these findings

hazardous adj.
危险的
the most hazardous spots

absence n.
不存在；缺少
the absence of agriculture

organism n.
生物
Smaller organisms fared worse

exposure n.
辐射暴露
where exposure was greater

decompose v.
分解；腐烂
leaves decomposing about 40 per cent

impede v.
阻碍
impeding the return of nutrients

guarantee v.
保证
guarantee the health

continual adj.
持续的
low but continual doses

inflict v.
造成伤害
low but continual doses inflict

contentious adj.
有争议的
remains contentious

dividend n.
好处；红利
the dividend of human departure

sluggish adj.
缓慢的
the sluggish rot of fallen leaves

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

disaster n.
灾难

startling adj.
惊人的

wildlife n.
野生动物

reactor n.
核反应堆

be emptied of 动词短语
被撤空；人员被全部迁走

radius n.
半径

off-limits adj.
禁止进入的

fence off 动词短语
围封；用围栏隔开

adjoining adj.
相邻的

terrain n.
地带；地形

Current Biology 专名
《当代生物学》

aerial adj.
空中的

footprint n.
足迹

elk n.
驼鹿

roe deer 名词短语
狍

boar n.
野猪

to prep.
与；引出比较对象

reserve n.
自然保护区

numerous adj.
数量众多的

helicopter n.
直升机

elk n.
驼鹿

surge v.
迅速增加

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

post-disaster adj.
灾后的

snap v.
拍下照片

suppress v.
抑制；压低

probable adj.
很可能的

mundane adj.
普通的；平常的

agriculture n.
农业

forestry n.
林业

nest v.
筑巢

intense adj.
强烈的

pollinator n.
传粉动物

plentiful adj.
数量多的；丰富的

polluted adj.
受污染的

individual adj.
个体的

how much damage 名词短语
多大伤害

clash over 动词短语
在……问题上存在分歧

burgeoning adj.
迅速增长的

herd n.
兽群

trumpet v.
大力宣扬；彰显

departure n.
离开

rot n.
腐烂

ledger n.
账本；此处比喻记录

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Chernobyl is _____ disaster, yet it holds a startling amount of wildlife.

02 Later fieldwork _____ these findings.

03 The probable reason is mundane: the _____ of agriculture, forestry, and hunting.

04 Counting heads cannot _____ the health of individual animals

文章主线

- 灾后 1986年切尔诺贝利核事故后，居民撤离、周边封禁，受污染地区却出现大量野生动物。
- 繁盛 白俄罗斯一侧多种兽类密度与四个未污染保护区相当，狼的数量为对照区的七倍多。
- 原因 可能的解释是农耕、伐木和狩猎消失，人类活动减少给野生动物带来了生存空间。
- 代价 部分研究发现高辐射处筑巢鸟数量及鸟种较少，传粉昆虫和蜘蛛也少；最重污染处落叶少分解约40%，阻碍养分回土。
- 限度 动物多不等于个体健康；长期低剂量辐射造成多大伤害仍有争议，不能据此说辐射无害。

参考答案　1 synonymous with 2 bolstered 3 absence 4 guarantee

十一天不睡觉的少年

Eleven Days Without Sleep

十一天不睡觉，身体会怎样？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 28 December 1963 Randy Gardner, a 17-year-old science enthusiast at a San Diego high school, set out to forgo slumber for a regional science fair. Two companions diligently jolted him awake whenever his alertness waned and logged his mood. The project, a voluntary endeavour, went on to take first place.

William Dement, a sleep researcher at Stanford, read press coverage of the attempt and was intrigued. He drove to San Diego and assisted John J. Ross, a Navy researcher, in monitoring the sleepless boy closely. The two investigators attentively noted the fluctuations in his state as the deprivation progressively deepened.

The consequences soon mounted. Gardner grew moody, lost concentration, and found his short-term memory impaired. He became suspicious, at times paranoid, and began to hallucinate. On the fourth day he was gripped by the delusion that he was a professional American football player.

Gardner ended the wearying attempt on 8 January 1964, after 264.4 hours, about 11 days and 25 minutes. He then slept soundly for about 14 hours and woke apparently refreshed. Researchers probed the case in a 1965 scholarly paper, which helped highlight microsleeps, involuntary lapses into sleep lasting a few seconds. Gardner may have had such momentary bouts without noticing.

In 1997 Guinness World Records discontinued monitoring such attempts because of the health perils, and others have since alleged longer stints awake. In a 2017 interview Gardner said severe insomnia had tormented him for about a decade, and he himself pinned the blame on the experiment. The boy who had seemed to bounce back from exhaustion with one long rest would, decades later, barely doze off.

中文译文

1963年12月28日，圣迭戈一所高中里，17岁的科学爱好者兰迪·加德纳为参加地区科学展，开始尝试不睡觉。每当他的清醒程度下降，两名同伴就尽力把他唤醒，并记录他的情绪。这项自愿开展的尝试后来获得了第一名。

斯坦福大学睡眠研究者威廉·德门特读到有关报道后产生了兴趣。他开车来到圣迭戈，协助海军研究员约翰·J.罗斯密切观察这名不眠的少年。随着缺觉程度逐步加重，两位研究者仔细记录着他状态的变化。

不良后果很快加剧。加德纳变得喜怒无常，注意力难以集中，短时记忆也受到损害。他开始多疑，有时甚至出现偏执妄想，还产生了幻觉。到第四天，他竟认定自己是一名职业美式橄榄球运动员。

1964年1月8日，加德纳结束了这场令人疲惫的挑战，历时264.4小时，约合11天零25分钟。之后，他酣睡约14小时，醒来时似乎恢复了精神。研究者在1965年的一篇学术论文中探讨了这个案例，使人们更加关注“微睡眠”：人会不由自主地短暂入睡，持续几秒钟。加德纳可能经历过这种短暂睡眠，自己却没有察觉。

1997年，吉尼斯世界纪录因健康风险停止监测此类挑战，此后仍有人声称自己保持清醒的时间更长。2017年接受采访时，加德纳说，严重失眠已折磨了自己约十年，他本人将其归咎于当年的实验。那个曾经看似睡个长觉便能从极度疲惫中恢复的少年，几十年后却几乎连打个盹都做不到。

晚年失眠与当年实验的因果关系，文中并未证实。

核心词汇

先记这22项。连着下方的原文短语一起回想。

enthusiast n.
爱好者
a 17-year-old science enthusiast

forgo v.
放弃；不享用
to forgo slumber

alertness n.
警觉；清醒程度
his alertness waned

wane v.
减弱；衰退
his alertness waned

voluntary adj.
自愿的
a voluntary endeavour

endeavour n.
尝试；努力
a voluntary endeavour

intrigue v.
使感兴趣
was intrigued

fluctuation n.
波动；变化
the fluctuations in his state

deprivation n.
剥夺；缺失
the deprivation progressively deepened

concentration n.
注意力；专注
lost concentration

impair v.
损害；使受损
short-term memory impaired

paranoid adj.
偏执多疑的；有妄想的
at times paranoid

hallucinate v.
产生幻觉
began to hallucinate

delusion n.
妄想；错误的确信
gripped by the delusion

involuntary adj.
不由自主的
involuntary lapses into sleep

discontinue v.
停止；终止
discontinued monitoring such attempts

peril n.
危险；风险
because of the health perils

insomnia n.
失眠
severe insomnia

torment v.
折磨；使痛苦
had tormented him for about a decade

pin the blame on 动词短语
把责任归咎于
pinned the blame on the experiment

bounce back from 动词短语
从……中迅速恢复
bounce back from exhaustion

exhaustion n.
精疲力竭
bounce back from exhaustion

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

slumber n.
睡眠

regional adj.
地区的；区域性的

companion n.
同伴

diligently adv.
尽心尽力地

jolt v.
使猛然惊醒

awake adj.
醒着的

mood n.
情绪

take first place 动词短语
获得第一名

coverage n.
新闻报道

assist v.
协助；帮助

in prep.
在……方面（assist…in）

sleepless adj.
不眠的；未睡的

closely adv.
密切地

investigator n.
研究者；调查人员

attentively adv.
专注地；仔细地

progressively adv.
逐步地；不断地

deepen v.
加深；加重

consequence n.
后果

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

mount v.
增加；加剧

moody adj.
喜怒无常的

short-term adj.
短期的；短时的

suspicious adj.
多疑的；猜疑的

be gripped by 动词短语
被强烈情绪或想法控制

wearying adj.
令人疲惫的

soundly adv.
（睡得）熟；酣畅地

refreshed adj.
恢复精神的

probe v.
深入探究

scholarly adj.
学术的

highlight v.
使受到关注；凸显

lapse into 动词短语
不由自主地陷入

momentary adj.
短暂的；瞬间的

bout n.
一阵；一段发作

allege v.
声称（尚未证实）

stint n.
一段时间

barely adv.
几乎不；勉强

doze off 动词短语
打瞌睡；短暂入睡

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Two companions diligently jolted him awake whenever his _____ waned and logged his mood.

02 The project, a voluntary _____, went on to take first place.

03 Gardner grew moody, lost concentration, and found his short-term memory _____.

04 He became suspicious, at times paranoid, and began to _____.

文章主线

- 挑战 1963年底，17岁加德纳为科学展尝试不睡觉；同伴帮他保持清醒，睡眠研究者随后加入监测。
- 损害 他逐渐情绪不稳，注意力和短时记忆受损，出现多疑和幻觉，甚至以为自己是职业橄榄球员。
- 结束 264.4小时（约11天）后挑战结束；酣睡约14小时后，他醒来似乎恢复了精神。
- 疑点 他可能经历过未察觉的数秒微睡眠；吉尼斯1997年因健康风险停止记录此类挑战。
- 晚年 2017年，他说严重失眠已折磨自己约十年，并归咎于当年实验；文章没有证明这种因果关系。

参考答案　1 alertness 2 endeavour 3 impaired 4 hallucinate

集装箱：重塑世界的铁盒子

The Steel Box That Remade the World

一个普通铁箱，怎样改变了全球货运？

英文原文

On 26 April 1956 a converted Second World War tanker, the Ideal X, slipped out of Port Newark, New Jersey, for Houston. Clamped to her deck were 58 containers, trailer bodies detached from their wheels and chassis. Few spectators would have taken so unremarkable a cargo for a landmark.

At the time, freight travelled loose. Gangs of twenty or more dockworkers carried sacks, barrels, and crates piece by piece. It was strenuous drudgery. Ships could spend up to half their time berthed, and theft was rampant. In 1956 loading loose cargo cost about $5.83 a ton; loading the Ideal X cost, by the company's reckoning, about 16 cents.

Containers were not new. The hurdle was consensus: rival firms' boxes had incompatible dimensions and latches. An international standards committee formed in 1961. In January 1963 Sea-Land, the Ideal X's renamed operator, let the committee use its patents free of royalties, including those on corner fittings for hoisting. Lifted by crane, a box could be transferred smoothly between ship, lorry, and train.

This efficiency came at the expense of livelihoods. New York's waterfront employment shrank by about three quarters between 1963 and 1976. Much of the cargo crossed the harbour to a new terminal at Elizabeth, inaugurated in August 1962. Most of London's historic docks closed one after another from 1967 to 1980, while Felixstowe, on the Suffolk coast, expanded.

The rewards were substantial. A study published in the Journal of International Economics in 2016 estimated that containerisation had lifted trade between industrialised countries by a cumulative 700 per cent over 20 years. Tireless cranes now lift identical containers into orderly, immense stacks.

中文译文

1956年4月26日，一艘由二战油轮改装而成的“理想X号”悄然驶出新泽西州纽瓦克港，前往休斯敦。甲板上固定着58个集装箱：它们原是挂车车厢，已从车轮和底盘上拆下。几乎没有旁观者会想到，这批平淡无奇的货物将成为一个里程碑。

当时，货物都是散装运输。每组二十人或更多的码头工人，一件件搬运麻袋、木桶和板条箱。这是繁重而枯燥的苦差事。船舶多达一半的时间可能耗在泊港上，偷窃也十分猖獗。1956年，装载散货每吨约需5.83美元；而据运营公司估算，“理想X号”的装船费用每吨仅约16美分。

集装箱并非新事物，障碍在于达成共识：竞争公司的箱子尺寸与锁扣互不兼容。1961年，一个国际标准委员会成立。1963年1月，“理想X号”的运营方——当时已更名为海陆公司——允许委员会免付专利使用费，使用其专利，其中包括用于吊装的角配件专利。吊机把箱子吊起后，便能在船舶、卡车和火车之间顺畅转运。

这种效率提升也以人们的生计为代价。1963至1976年，纽约滨水区的就业人数缩减了约四分之三。大量货运转移到港湾对面的伊丽莎白新码头；该码头于1962年8月启用。1967至1980年，伦敦大多数历史悠久的码头相继关闭，萨福克海岸的费利克斯托港却不断扩张。

回报同样可观。2016年发表于《国际经济学杂志》的一项研究估计，集装箱化在20年间使工业化国家间的贸易累计增长了约700%。如今，不知疲倦的吊机把规格一致的集装箱吊起，码成整齐而庞大的箱垛。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cargo n.
船运货物
so unremarkable a cargo

landmark n.
里程碑；重大转折
a cargo for a landmark

freight n.
货物；货运
freight travelled loose

drudgery n.
单调繁重的苦差事
strenuous drudgery

rampant adj.
猖獗的；泛滥的
theft was rampant

reckoning n.
估算
by the company's reckoning

hurdle n.
障碍；难关
The hurdle was consensus

consensus n.
共识
The hurdle was consensus

incompatible adj.
互不兼容的
incompatible dimensions and latches

dimension n.
尺寸
incompatible dimensions and latches

patent n.
专利
use its patents free of royalties

royalty n.
专利使用费
use its patents free of royalties

crane n.
起重机；吊机
Lifted by crane

transfer v.
转运；转移
could be transferred smoothly

efficiency n.
效率
This efficiency came

at the expense of 介词短语
以……为代价
at the expense of livelihoods

livelihood n.
生计
at the expense of livelihoods

employment n.
就业；就业人数
waterfront employment shrank

inaugurate v.
正式启用
inaugurated in August 1962

substantial adj.
可观的；大量的
The rewards were substantial

cumulative adj.
累计的
a cumulative 700 per cent

identical adj.
完全相同的
lift identical containers

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

convert v.
改装；改造

tanker n.
油轮

slip out of 动词短语
悄然驶离

clamp v.
夹紧；固定

container n.
集装箱

trailer n.
挂车

detach from 动词短语
从……拆下

wheel n.
车轮

chassis n.
车辆底盘

spectator n.
旁观者

unremarkable adj.
平淡无奇的

loose adj.
散装的；未捆装的

gang n.
一组工人

dockworker n.
码头工人

sack n.
大袋；麻袋

barrel n.
大桶

crate n.
板条箱

strenuous adj.
繁重费力的

up to 介词短语
多达

berth v.
停泊；靠泊

theft n.
偷窃

load v.
装载

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

by prep.
据；按照

rival adj.
竞争对手的

latch n.
锁扣；门闩

international standard 名词短语
国际标准

fitting n.
配件

hoist v.
吊起；吊装

smoothly adv.
顺畅地

lorry n.
卡车

waterfront n.
滨水区

shrink v.
缩减

terminal n.
货运码头

historic adj.
历史悠久的

dock n.
码头

expand v.
扩张

reward n.
回报

economics n.
经济学

industrialised adj.
工业化的

tireless adj.
不知疲倦的

orderly adj.
整齐有序的

immense adj.
巨大的

stack n.
垛；整齐堆叠的一摞

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The hurdle was _____:

02 and theft was _____.

03 rival firms' boxes had _____ dimensions and latches.

04 This efficiency came at the expense of _____.

文章主线

- 启航 过去码头逐件搬散货，船常久停；1956年，理想X号载58个集装箱启航。
- 降本 据运营方估算，该船每吨装船费约0.16美元，远低于当时散货的5.83美元。
- 互通 各家箱型原不兼容；运营方海陆公司免费开放专利，推动标准统一，箱子可在船、卡车和火车间转运。
- 转移 货运转向新港，老码头和工人生计受冲击；1963至1976年，纽约滨水区就业缩减约四分之三。
- 回报 2016年一项研究估计，集装箱化在20年间使工业化国家间贸易累计增加约700%。

参考答案　1 consensus 2 rampant 3 incompatible 4 livelihoods

永不沉没的神话

The Myth of the Unsinkable

守法的泰坦尼克号，为何仍救不了所有人？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 10 April 1912 the opulent liner Titanic left Southampton on her maiden voyage. A 1911 trade journal had dubbed her and her sister ship "practically unsinkable". At about 11.40pm on 14 April she collided with an iceberg; by 2.20am she had vanished beneath the waves, and the death toll would surpass 1,500.

Folklore blames arrogance. Yet the Titanic was compliant with the law. Under 1894 rules, Britain's maritime regulator pegged lifeboat numbers to gross tonnage (volume, not weight) on a scale that topped out at 16 boats for 10,000 tons or more. At about 46,000 tons she carried 20, exceeding the requirement. Their capacity was 1,178, roughly half the 2,200 aboard: sufficient in law, inadequate at sea.

Within radio range, the steamer Californian kept an equally lawful routine. Her sole wireless operator, Cyril Evans, was on the payroll of the Marconi company; when he slept, so did the radio. Around 11.30pm he switched off his set. Hindsight may call that negligence, yet no rule obliged him to stay vigilant, so the Titanic's frantic distress calls went unheard there.

Between April and July 1912, inquiries in Washington and London heard testimony from survivors and the Californian's crew. That July, American legislation made round-the-clock radio watch compulsory. In January 1914 delegates of 13 nations signed the first Safety of Life at Sea convention, stipulating lifeboat seats for all, continuous radio watch, and an ice patrol. The First World War stalled the treaty; the patrol went ahead regardless.

The doomed liner sank within three hours. The outdated rules she sailed under had stayed afloat since 1894; it took her dead to discredit them.

中文译文

1912年4月10日，豪华邮轮泰坦尼克号驶离南安普敦，开始首航。1911年，一本行业期刊曾将它和姊妹船称为“几乎不会沉没的船”。4月14日晚约11时40分，它撞上一座冰山；到次日凌晨2时20分，船已沉入海中，最终遇难人数超过1500。

坊间说法将灾难归咎于傲慢。然而，泰坦尼克号符合当时的法律。根据1894年的规定，英国海事监管机构按总吨位确定救生艇数量；总吨位衡量的是容积，并非重量。这套分级标准的上限，是对1万总吨及以上的船只要求配备16艘救生艇。泰坦尼克号约4.6万总吨，带了20艘，超过法定要求。但这些艇只能容纳1178人，约为船上2200人的一半：法律上足够，海上救生却不足。

在无线电通信范围内，加州人号轮船的值班惯例同样没有违规。船上唯一的报务员西里尔·埃文斯受雇于马可尼公司；他一休息，电台也随之无人值守。晚上约11时30分，他关掉了设备。事后看来，这或许可称为疏忽，但当时没有规定要求他保持值守，因此泰坦尼克号急切的求救信号在加州人号上无人听到。

1912年4月至7月，华盛顿和伦敦的调查听取了幸存者和加州人号船员的证词。同年7月，美国立法强制要求全天候无线电值守。1914年1月，13个国家的代表签署首部《海上人命安全公约》，规定救生艇座位必须足够全员使用、电台连续值守，并开展冰情巡逻。第一次世界大战使公约搁置，冰情巡逻却仍然开展起来。

这艘在劫难逃的邮轮不到三小时就沉没了。它所遵循的过时规定，却自1894年起一直照常施行；直到船上众多生命逝去，这些规定才失去了公信力。

总吨位衡量容积；“没有违规”并不表示法律要求报务员关机。

核心词汇

先记这22项。连着下方的原文短语一起回想。

opulent adj.
豪华的；奢华的
the opulent liner Titanic

maiden voyage 名词短语
首航
on her maiden voyage

dub v.
把……称为
had dubbed her and her sister ship

collide with 动词短语
与……相撞
she collided with an iceberg

death toll 名词短语
遇难人数
the death toll would surpass 1,500

arrogance n.
傲慢；自大
Folklore blames arrogance

compliant with 形容词短语
符合；遵守
was compliant with the law

regulator n.
监管机构
Britain's maritime regulator

gross tonnage 名词短语
总吨位（衡量船内容积）
gross tonnage (volume, not weight)

capacity n.
容纳人数；容量
Their capacity was 1,178

sufficient adj.
足够的
sufficient in law

inadequate adj.
不足的；不够的
inadequate at sea

hindsight n.
事后认识；后见之明
Hindsight may call that negligence

negligence n.
疏忽；失职
Hindsight may call that negligence

oblige v.
强制；要求
no rule obliged him to stay vigilant

vigilant adj.
警觉的；保持值守的
no rule obliged him to stay vigilant

testimony n.
证词；证言
heard testimony from survivors

legislation n.
立法；法律
American legislation

compulsory adj.
强制的；必须履行的
radio watch compulsory

convention n.
国际公约
Safety of Life at Sea convention

stipulate v.
规定；明确要求
stipulating lifeboat seats for all

discredit v.
使失去信誉；证明不可靠
to discredit them

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

liner n.
大型客轮；邮轮

trade journal 名词短语
行业期刊

practically adv.
几乎；实际上

unsinkable adj.
不会沉没的

iceberg n.
冰山

vanish v.
消失

surpass v.
超过

folklore n.
坊间流传的说法

maritime adj.
海事的；海上的

peg v.
使挂钩；按某标准确定

to prep.
与……（peg…to搭配）

scale n.
分级标准；等级表

top out at 动词短语
最高达到；上限为

exceed v.
超过

requirement n.
规定的要求

roughly adv.
大约；大致

equally adv.
同样地

lawful adj.
合法的；未违规的

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

sole adj.
唯一的

wireless adj.
无线电的

operator n.
操作员；报务员

on the payroll of 介词短语
受雇于；领取……的薪酬

switch off 动词短语
关掉（设备）

frantic adj.
慌乱而急切的

distress call 名词短语
求救呼叫；求救信号

unheard adj.
未被听见的

survivor n.
幸存者

round-the-clock adj.
全天候的；不间断的

delegate n.
代表

continuous adj.
连续的；不间断的

patrol n.
巡逻

stall v.
使暂缓；使停滞

treaty n.
条约

regardless adv.
尽管如此；仍然

doomed adj.
在劫难逃的

outdated adj.
过时的

stay afloat 动词短语
维持有效；继续存在（比喻）

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Yet the Titanic was _____ the law.

02 Their capacity was 1,178, roughly half the 2,200 aboard: sufficient in law, _____ at sea.

03 Hindsight may call that _____, yet no rule obliged him to stay vigilant, so the Titanic's frantic distress calls went unheard there.

04 That July, American legislation made round-the-clock radio watch _____.

文章主线

- 沉船 1912年，号称几乎不沉的泰坦尼克号首航撞冰山，不足三小时沉没，逾1500人遇难。
- 救生 旧规按总吨位（容积）定艇数，万吨以上只需16艇；它有20艇、1178座，仅够船上2200人约一半。
- 守听 附近加州人号的唯一报务员关机休息并未违规，电台没听到求救；当时没有全天值守的强制规定。
- 改革 1912年美国强制全天守听；1914年13国签约，要求全员救生艇座位、连续守听和冰情巡逻。
- 教训 一战使公约搁置，冰情巡逻仍启动；沿用自1894年的旧规暴露出：合法配置，并不等于安全足够。

参考答案　1 compliant with 2 inadequate 3 negligence 4 compulsory

法老的诅咒

The Curse of the Pharaoh

法老的诅咒，经得起证据检验吗？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 4 November 1922, in the Valley of the Kings, labourers engaged by the archaeologist Howard Carter uncovered a staircase carved into the rock. On 26 November, with his wealthy sponsor Lord Carnarvon beside him, Carter made a small hole in the sealed doorway and glimpsed the glittering gold of the boy king Tutankhamun. The spectacular find riveted the world.

Carnarvon sold exclusive rights to The Times, a monopoly that left competing reporters starved of news. Many resorted to speculation or sought outside commentators. In March 1923 the best-selling novelist Marie Corelli warned publicly that dire punishment awaited anyone who entered a sealed tomb. Within a fortnight, her words would seem uncanny.

That month a mosquito bit Carnarvon. He nicked the bite with a razor while shaving, it turned septic, and blood poisoning led to pneumonia. He died in Cairo on 5 April 1923. Newspapers reported that the city's lights failed at that instant and that his dog in England howled and died. Yet blackouts were almost a daily occurrence in Cairo.

No curse inscription was ever found in the tomb. In December 2002 a British Medical Journal study traced the fate of 44 Westerners in Egypt at the time. The 25 exposed to the tomb died at an average age of 70, versus 75 for the rest, a margin not statistically significant. The mortality figures lent the ominous rumour no credibility.

Carnarvon's untimely demise had fuelled the tale, giving it impetus. Carter, the excavator who had unsealed the burial place, died in March 1939, aged 64, almost sixteen years after his patron. The legend had left him conspicuously unharmed.

中文译文

1922年11月4日，在埃及帝王谷，考古学家霍华德·卡特雇用的工人发现了一段凿入岩石的阶梯。11月26日，富有的赞助人卡纳冯勋爵站在身旁，卡特在封住的入口上开了一个小孔，瞥见少年国王图坦卡蒙墓中闪耀的黄金。这一惊人发现吸引了全世界的目光。

卡纳冯将独家报道权卖给《泰晤士报》，使其他竞争记者严重缺少消息。许多人转而猜测，或向外界评论者寻求说法。1923年3月，畅销小说家玛丽·科雷利公开警告，任何进入封闭陵墓的人都会受到可怕的惩罚。不到两周，她的话就将显得诡异。

当月，一只蚊子叮了卡纳冯。他刮胡子时用剃刀划破叮咬处，伤口感染化脓，随后败血症引发肺炎。1923年4月5日，他在开罗去世。报纸声称，就在他去世的一刻，开罗全城停电，他在英国的狗也嚎叫着死去。然而，停电在当时的开罗几乎天天发生。

墓中从未发现过诅咒铭文。2002年12月，《英国医学杂志》的一项研究追溯了当时身在埃及的44名西方人的命运。接触过陵墓的25人平均享年70岁，其余人平均75岁，这一差距在统计上并不显著。死亡统计并未为那则不祥的传闻提供可信依据。

卡纳冯的过早离世助长了诅咒传说，使它越传越广。但亲自打开陵墓的发掘者卡特，直到1939年3月才去世，享年64岁，比赞助人的去世时间晚了近16年。这位开墓者却明显没有遭遇传说中的灾祸。

近16年指两人去世日期的间隔，并不是两人的寿命相差16年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sponsor n.
赞助人；资助者
his wealthy sponsor Lord Carnarvon

rivet v.
牢牢吸引注意力
The spectacular find riveted the world

exclusive adj.
独家的；专属的
sold exclusive rights

monopoly n.
垄断；独占
a monopoly

be starved of 动词短语
严重缺少
reporters starved of news

resort to 动词短语
转而采用；诉诸
resorted to speculation

speculation n.
猜测；推测
resorted to speculation

dire adj.
可怕的；极其严重的
dire punishment

fortnight n.
两周
Within a fortnight

uncanny adj.
诡异的；不可思议的
her words would seem uncanny

turn septic 动词短语
感染化脓
it turned septic

pneumonia n.
肺炎
blood poisoning led to pneumonia

inscription n.
铭文；刻文
No curse inscription

be exposed to 动词短语
接触到
The 25 exposed to the tomb

margin n.
差距；差额
a margin not statistically significant

significant adj.
（统计上）显著的
not statistically significant

mortality n.
死亡情况；死亡统计
The mortality figures

credibility n.
可信度
no credibility

demise n.
去世；死亡
Carnarvon's untimely demise

fuel v.
助长；促使加剧
had fuelled the tale

impetus n.
推动力
giving it impetus

conspicuously adv.
显眼地；格外醒目地
conspicuously unharmed

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

labourer n.
工人；体力劳动者

engage v.
雇用；聘请

uncover v.
发现；发掘

staircase n.
楼梯；阶梯

carve into 动词短语
雕凿于；刻入

wealthy adj.
富有的

doorway n.
门口；入口

glittering adj.
闪闪发光的

spectacular adj.
惊人的；壮观的

competing adj.
相互竞争的

commentator n.
评论者

best-selling adj.
畅销的

punishment n.
惩罚

tomb n.
陵墓

mosquito n.
蚊子

nick v.
轻轻划破；割破

razor n.
剃刀

shave v.
刮胡子

at that instant 介词短语
就在那一瞬间

howl v.
嚎叫

blackout n.
停电

occurrence n.
发生的事；事件

fate n.
命运；遭遇

versus prep.
与……相比

ominous adj.
不祥的

rumour n.
传闻；谣言

untimely adj.
过早的；不合时宜的

excavator n.
发掘者

unseal v.
启封；打开封口

burial n.
埋葬；墓葬

patron n.
资助人；赞助人

unharmed adj.
未受伤害的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Carnarvon sold exclusive rights to The Times, a _____ that left competing reporters starved of news.

02 Many resorted to _____ or sought outside commentators.

03 No curse _____ was ever found in the tomb.

04 The mortality figures lent the ominous rumour no _____.

文章主线

- 发现 1922年，卡特与赞助人卡纳冯发现图坦卡蒙墓，墓中珍宝震惊世界。
- 传言 《泰晤士报》买断报道，其他记者转向猜测；卡纳冯次年去世，诅咒故事随之盛行。
- 病因 卡纳冯剃破蚊叮处，感染后发生败血症和肺炎；报纸所述停电，在开罗本就几乎天天发生。
- 对照 研究追踪44人：接触墓的25人平均活70岁，其余75岁，这点差距仍可能来自偶然。
- 结论 墓中从未发现诅咒铭文；卡纳冯死后，亲自开墓的卡特又活了近16年，现有证据不支持诅咒。

参考答案　1 monopoly 2 speculation 3 inscription 4 credibility

孩子们发明的语言

The Language the Children Invented

没有共同手语的孩子，怎样创造出一门语言？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On a rattling bus in Managua, deaf pupils kept up a brisk, voiceless hubbub across the aisle, palms fluttering, wrists flicking. Their teachers could not follow a word.

Before the late 1970s Nicaragua had no deaf community. Most deaf people led secluded lives, sharing a homemade repertoire of gestures with parents and siblings. In 1977 a special-education centre opened in Managua's San Judas district with 50 deaf pupils. By 1983 the centre and a vocational school for teenage pupils held more than 400.

The curriculum hinged on oral Spanish and lipreading, and teachers signed only the alphabet. Pupils studied lips in vain, and most failed to grasp what a Spanish word was. At recess, in the street, and on the buses, they mingled gestures with home signs. A rough jumble ripened rapidly into a coherent language entirely their own.

The teachers found it unintelligible, and in June 1986 the Ministry of Education brought in Judy Kegl, an American Sign Language linguist from MIT. A 2001 study found that pupils who arrived younger, and later intakes, used grammatical devices more often and more uniformly than their forerunners. Each fresh wave of young signers added complexity.

In a 2004 study, gesturing Spanish speakers and the earliest signers described something rolling downhill by welding how it moved and where it went into one undivided movement. Later signers dismantled it into discrete segments in linear order, a framework found nowhere in the gestures around them.

Linguists had a rare chance to watch a language grow from infancy to maturity. Its builders were not teachers or experts but deaf playmates in playgrounds, streets, and buses.

中文译文

马那瓜一辆颠簸作响的公车上，聋人学生隔着过道热烈地“聊”着天，手掌挥动，手腕轻摆，热闹却无声。老师们一个字也看不懂。

20世纪70年代末以前，尼加拉瓜尚未形成聋人社群。多数聋人生活闭塞，只与父母、兄弟姐妹共用一套自创手势。1977年，一所特殊教育中心在马那瓜圣胡达斯区开办，招收了50名聋人学生。到1983年，这所中心与一所面向青少年学生的职业学校，共有400多名学生。

课程以西班牙语口语和唇读为中心，老师只用手势比划字母。学生努力观察说话时的嘴唇，却徒劳无功；多数人连一个西班牙语单词是什么都没能弄明白。但在课间、街头和公车上，他们把一般手势与各自家中的手势混合起来。一套粗糙杂乱的表达迅速发展为一门连贯、自成体系的语言。

老师们看不懂这种语言。1986年6月，教育部请来麻省理工学院研究美国手语的语言学家朱迪·凯格尔。2001年的一项研究发现，与较早的学生相比，入学时年龄更小的学生，以及后来几批入学者，使用语法手段更频繁，也更统一。每一批新的年幼手语使用者，都为这门语言增添了复杂性。

2004年的一项研究发现，打手势的西班牙语使用者和最早一批手语使用者，在描述某个东西滚下山坡时，会把“怎样运动”和“往哪里去”合成一个不可分的动作。后来的手语使用者则把它拆成彼此独立的片段，按先后顺序表达；这种结构在他们周围人的手势中无处可寻。

语言学家难得有机会亲眼看着一门语言从初创走向成熟。创造它的不是教师或专家，而是在操场、街头和公车上一起玩耍、交流的聋人孩子。

核心词汇

先记这22项。连着下方的原文短语一起回想。

community n.
社群
no deaf community

secluded adj.
与外界隔绝的
led secluded lives

repertoire n.
掌握的一整套表达手段
a homemade repertoire of gestures

curriculum n.
课程；课程设置
The curriculum hinged on oral Spanish

hinge on 动词短语
以……为中心；取决于
hinged on oral Spanish and lipreading

in vain 介词短语
徒劳地
Pupils studied lips in vain

grasp v.
理解；领会
failed to grasp

mingle v.
混合；融合
mingled gestures with home signs

coherent adj.
连贯的；自成体系的
a coherent language

unintelligible adj.
无法理解的
The teachers found it unintelligible

linguist n.
语言学家
an American Sign Language linguist

intake n.
新招收的几批学生
later intakes

grammatical adj.
语法的
used grammatical devices

uniformly adv.
一致地；统一地
more uniformly than their forerunners

complexity n.
复杂性
added complexity

dismantle v.
拆解
dismantled it into discrete segments

discrete adj.
分立的；彼此可区分的
discrete segments in linear order

segment n.
片段
discrete segments in linear order

linear order 名词短语
线性顺序；依次排列
segments in linear order

framework n.
结构；框架
a framework found nowhere

infancy n.
初创阶段；幼年期
from infancy to maturity

maturity n.
成熟阶段
from infancy to maturity

文中词汇与搭配

补充词汇共41项。保留每个表达在本文中的意思。

rattle v.（分词作定语）
颠簸作响

brisk adj.
轻快热烈的

voiceless adj.
无声的

hubbub n.
喧闹；热闹的交流

aisle n.
过道

palm n.
手掌

flutter v.
挥动；轻快拍动

wrist n.
手腕

flick v.
轻快地摆动

homemade adj.
自创的；自家制作的

sibling n.
兄弟姐妹

special education n.（作定语）
特殊教育

district n.
地区；区

pupil n.
学生

vocational school 名词短语
职业学校

teenage adj.
青少年的

oral adj.
口语的

lipreading n.
唇读；看口形辨话

alphabet n.
字母表

lip n.
嘴唇；此处涉及说话口形

文中词汇与搭配 续

补充词汇共41项。保留每个表达在本文中的意思。

at recess 介词短语
在课间休息时

with prep.
与；同

rough adj.
粗糙的；尚未完善的

jumble n.
杂乱的混合物

ripen v.
逐渐发展成熟

rapidly adv.
迅速地

into prep.
成为；用于引出变化结果

entirely adv.
完全地

Ministry of Education 专名
教育部

bring in 动词短语
请来；引入

American Sign Language 专名
美国手语

MIT 专名
麻省理工学院

forerunner n.
较早的一批人；前辈

signer n.
手语使用者

roll v.
滚动

downhill adv.
向坡下

weld v.
熔合；把两部分合为一体

undivided adj.
未分开的；不可分的

nowhere adv.
无处

builder n.
创造者

playmate n.
玩伴

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 A rough jumble ripened rapidly into a _____ language entirely their own.

02 The curriculum _____ oral Spanish and lipreading, and teachers signed only the alphabet.

03 Each fresh wave of young signers added _____.

04 Later signers dismantled it into discrete segments in _____, a framework found nowhere in the gestures around them.

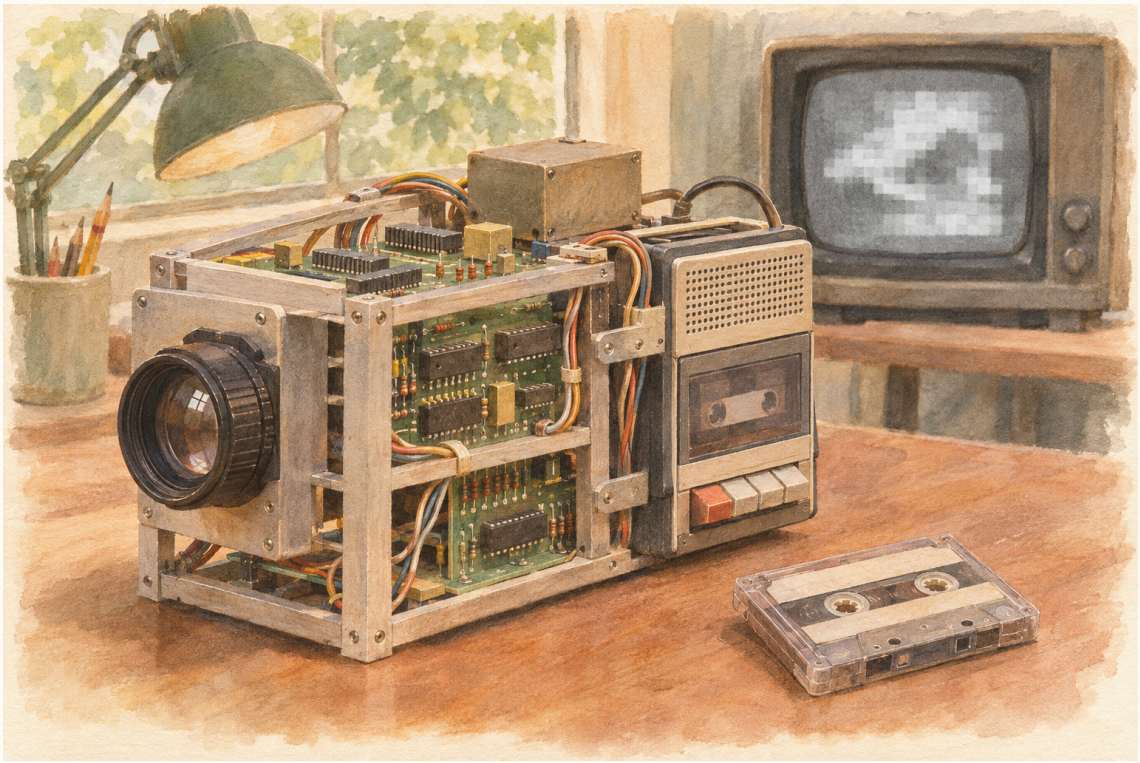
文章主线

- 相聚 1977年，尼加拉瓜学校把聋童聚在一起；此前他们多只与家人用自创手势交流。
- 受阻 学校教西班牙语口语和唇读，多数学生未能掌握；课外，孩子们开始用手势相互交流。
- 自创 孩子在课间、街头和公车上交流，把自家手势混合起来，逐渐形成了一门共同语言。
- 演变 入学年龄更小的孩子和后来几批新生使语法更统一复杂；后来的孩子还把同一手势里的怎么动、往哪去拆开，按先后顺序表达。
- 意义 这门语言出自孩子的同伴交流，而非专家传授；语言学家得以观察它逐渐成熟。

参考答案　1 coherent 2 hinged on 3 complexity 4 linear order

发明了数码相机，却被它打败

The Company That Invented Its Own Killer



造出数码相机的公司，为什么仍被数码时代冲击？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1975 Kodak engineer Steven Sasson built the first digital camera. The toaster-sized gadget, a cumbersome heavyweight of about eight pounds, was hardly portable, let alone handheld. Its first rudimentary picture held 10,000 black-and-white pixels. Recording it on a cassette took 23 seconds, and a playback unit showed it on a television.

Kodak's revenues showed its dependence on film, the profitable pillar of its business. Technical staffers were appreciative of the camera, but it needed no film. By Sasson's recollection, his superiors at headquarters called it cute and urged him to keep it hushed. A patent followed in 1978. Production never did.

The advent of digital photography proved disruptive. In June 2009 Kodak said it would stop making Kodachrome, its analogue film, citing lessened demand in the digital era. Kodak's manpower had reached an apex of about 145,000 in the 1980s. By 2012, layoffs since 2003 had slashed 47,000 jobs, alongside the shutdown of 13 factories and some 130 labs.

On 19 January 2012 the 131-year-old firm sought Chapter 11 protection from its creditors, sapped by pension costs and years of slowness in going digital. Citigroup, a lender, provided $950 million for its restructuring. A portfolio of 1,100 digital imaging patents fetched about $525 million from a consortium of twelve buyers, including the giants Apple and Google.

After 19 months of reorganisation, the once-mighty Kodak left bankruptcy in September 2013, downsized, streamlined, and refocused as a commercial printer for business clients. In November 2009 President Obama had decorated Sasson with national laurels, commending his digital camera. That medal's year also saw Kodak announce the end of Kodachrome.

中文译文

1975年，柯达工程师史蒂文·萨松造出第一台数码相机。这台烤面包机大小的装置是个约八磅重的笨重大块头，几乎谈不上便携，更不用说手持。它的第一张粗糙照片只有一万个黑白像素；把照片记录到盒式磁带上需要23秒，再由回放装置显示在电视上。

柯达的营收显示，公司依赖胶卷这项利润支柱。技术人员很欣赏这台相机，可它不需要胶卷。据萨松回忆，总部上司说它挺可爱，却让他别声张。1978年，相机取得专利，但这台原型机始终没有投产。

数码摄影的问世带来颠覆性冲击。2009年6月，柯达宣布将停产Kodachrome这一传统胶卷产品，理由是数码时代需求减少。公司员工规模曾在20世纪80年代达到约14.5万人的顶峰。到2012年，自2003年起的裁员已削减4.7万个岗位，同时关闭了13家工厂与约130个冲印实验室。

养老金成本与多年数码转型迟缓不断削弱公司。2012年1月19日，这家拥有131年历史的企业申请美国《破产法》第11章保护，以免债权人追讨。贷款方花旗集团提供9.5亿美元用于重组。包括苹果、谷歌等巨头在内的十二家买方组成财团，以约5.25亿美元买下1100项数字成像专利组成的专利组合。

经过19个月重组，昔日强大的柯达于2013年9月走出破产程序：缩小规模、精简业务，重新聚焦于为企业客户提供商业印刷服务。早在2009年11月，奥巴马总统就授予萨松国家级荣誉，表彰他的数码相机。而在这枚奖章颁发的同一年，柯达也宣布了Kodachrome胶卷的终结。

史实说明：这是2009年度的国家技术与创新奖章，实际颁发日为2010年11月17日（白宫典礼记录）。柯达宣布停产Kodachrome是在2009年，因此两件事并非发生于同一年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cumbersome adj.
笨重的
a cumbersome heavyweight

rudimentary adj.
初步的；粗陋的
Its first rudimentary picture

revenue n.
营收
Kodak's revenues

dependence on 名词短语
对……的依赖
its dependence on film

profitable adj.
盈利的；赚钱的
the profitable pillar of its business

appreciative of 形容词短语
欣赏；赞赏
appreciative of the camera

recollection n.
回忆
By Sasson's recollection

keep … hushed 动词短语
对此保密；别声张
urged him to keep it hushed

patent n.
专利
A patent followed in 1978

advent n.
出现；问世
The advent of digital photography

disruptive adj.
颠覆性的
proved disruptive

layoff n.
裁员
layoffs since 2003

creditor n.
债权人
protection from its creditors

sap v.
逐渐削弱
sapped by pension costs

restructuring n.
重组
for its restructuring

portfolio n.
一组持有的资产；此指专利组合
A portfolio of 1,100

consortium n.
财团；联合体
a consortium of twelve buyers

reorganisation n.
重组；改组
19 months of reorganisation

downsize v.
缩小规模
downsized, streamlined

streamline v.
精简
downsized, streamlined

refocus as 动词短语
重新定位为
refocused as a commercial printer

commend v.
表彰
commending his digital camera

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

digital camera 名词短语
数码相机

toaster n.
烤面包机

gadget n.
小装置

heavyweight n.
重家伙；大块头

portable adj.
便携的

let alone 固定短语
更不用说

handheld adj.
手持的

pixel n.
像素

cassette n.
盒式磁带

playback n.
回放

pillar n.
支柱

staffer n.
员工

superior n.
上司

headquarters n.
总部

analogue adj.
模拟式的；此处指传统胶片

lessen v.
减少

era n.
时代

manpower n.
员工规模；人力

apex n.
顶峰

slash v.
大幅削减

shutdown n.
关闭

pension n.
养老金

slowness n.
迟缓

lender n.
贷款方

buyer n.
买方

giant n.
行业巨头

once-mighty adj.
昔日强大的

printer n.
印刷企业；不是打印机

client n.
客户

decorate v.
授予荣誉；授勋

laurels n.
荣誉

medal n.
奖章

the end of 名词短语
……的终结

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ of digital photography proved disruptive.

02 By Sasson's _____, his superiors at headquarters called it cute and urged him to keep it hushed.

03 Citigroup, a lender, provided $950 million for its _____.

04 downsized, _____, and refocused as a commercial printer for business clients.

文章主线

- 先机 1975年，柯达工程师萨松造出第一台数码相机：约八磅重、一万黑白像素，存一张照片需23秒。
- 牵制 胶卷是利润支柱；据萨松回忆，上司让他别声张。原型机1978年获专利，却未投产。
- 衰退 数码摄影冲击胶卷需求，转型迟缓与养老金负担削弱公司，柯达于2012年申请破产保护。
- 重组 柯达获9.5亿美元贷款，并以约5.25亿美元出售1100项专利；2013年完成重组，转做企业印刷。
- 反差 柯达2009年宣布停产Kodachrome胶卷；萨松获2009年度美国国家技术与创新奖章，实际于2010年授勋。

参考答案　1 advent 2 recollection 3 restructuring 4 streamlined

青蒿一握，济世千万

A Handful of Wormwood That Saved Millions

古方里没有煮沸这一步，怎样启发了青蒿素研究？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Project 523, a covert military mission involving many institutes, was named after its launch date, 23 May 1967. Its urgent target was the chloroquine-resistant malaria weakening troops in the Vietnam War. In January 1969 Tu Youyou was entrusted with leading her institute's research group, despite having no doctorate or foreign training.

In an exhaustive search, her team combed over 2,000 folk recipes, compiled 640 prescriptions, and evaluated about 380 extracts from some 200 medicinal herbs. Qinghao, or sweet wormwood, showed flashes of promise, yet its results were erratic. Seeking a clue, Tu revisited the classical texts.

Around AD 340, the physician Ge Hong's handbook advised taking "a handful of qinghao immersed in two litres of water". This bypassed boiling. Tu suspected heat during extraction was degrading a delicate component, so she switched to ether, a volatile solvent boiling at about 35 degrees Celsius. On 4 October 1971 sample 191 completely inhibited the parasite in infected mice.

Efficacy in mice indicated nothing about human safety. In July 1972, to assess its toxicity, Tu and two colleagues voluntarily took the extract. Thereafter, from August to October, it was administered to 21 malaria patients in Hainan, and on 8 November her team obtained pure crystals of the substance later named artemisinin.

In September 2011 Tu was named a Lasker Award recipient; its jury credited the drug with saving millions. On 5 October 2015 she won half the Nobel Prize in Physiology or Medicine, the first Chinese woman Nobel laureate. Since 2006 WHO guidelines have endorsed artemisinin-based combination therapies against malaria's deadliest form. The essence of Ge Hong's line lay in the boiling it skipped.

中文译文

“523项目”是一项多家研究机构参与的秘密军事任务，名称来自启动日期：1967年5月23日。项目急需解决的是对氯喹产生耐药性的疟疾，当时这种疾病正削弱越南战争中的部队。1969年1月，尽管没有博士学位，也未在国外受过训练，屠呦呦仍受命领导所在研究所的研究组。

团队进行了全面搜寻，仔细翻检2000多份民间药方，汇编640个方剂，评估了来自约200种药用植物的约380种提取物。青蒿，也就是黄花蒿，时而显出效果，结果却不稳定。为寻找线索，屠呦呦重新翻阅古代文献。

公元340年前后，医师葛洪的手册建议取“一把浸在两升水中的青蒿”。这种做法不经煮沸。屠呦呦怀疑，提取过程中的高温会破坏某种不耐热的成分，于是改用乙醚：一种沸点约35摄氏度、容易挥发的溶剂。1971年10月4日，第191号样品完全抑制了感染小鼠体内的疟原虫。

小鼠身上有效，并不能说明对人体是否安全。1972年7月，为评估毒性，屠呦呦与两名同事自愿服下提取物。随后，8月至10月间，提取物用于海南21名疟疾患者；11月8日，团队取得了这种物质的纯净晶体，它后来被命名为青蒿素。

2011年9月，屠呦呦获拉斯克奖，评审团认为这种药物挽救了数百万人的生命。2015年10月5日，她获得当年诺贝尔生理学或医学奖的一半份额，成为首位中国女性诺贝尔奖得主。自2006年起，世卫组织指南就推荐以青蒿素为基础的多药联合疗法，对抗最致命的一类疟疾。葛洪那句话的关键，恰在于没有煮沸。

核心词汇

先记这22项。连着下方的原文短语一起回想。

resistant adj.
有抗药性的
chloroquine-resistant malaria

be entrusted with 动词短语
受托承担；受命负责
entrusted with leading

exhaustive adj.
全面彻底的
In an exhaustive search

comb v.
仔细搜寻；翻检
combed over 2,000 folk recipes

compile v.
汇编
compiled 640 prescriptions

evaluate v.
评估
evaluated about 380 extracts

extract n.
提取物
evaluated about 380 extracts

erratic adj.
不稳定的；时好时坏的
its results were erratic

extraction n.
提取
heat during extraction

degrade v.
降解；使成分受破坏
was degrading a delicate component

component n.
成分
a delicate component

volatile adj.
易挥发的
a volatile solvent

solvent n.
溶剂
a volatile solvent

inhibit v.
抑制
completely inhibited the parasite

efficacy n.
药效；效力
Efficacy in mice

assess v.
评估
to assess its toxicity

toxicity n.
毒性
to assess its toxicity

administer … to … 动词短语
给……用药
administered to 21 malaria patients

substance n.
物质
pure crystals of the substance

endorse v.
认可；推荐
have endorsed artemisinin-based

combination n.
联合；合用
combination therapies

essence n.
关键；精髓
The essence of Ge Hong's line

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

covert adj.
秘密的；隐蔽的

military adj.
军事的

mission n.
任务

urgent adj.
紧迫的

malaria n.
疟疾

weaken v.
削弱

troop n.
部队

Vietnam War 专名
越南战争

despite prep.
尽管

doctorate n.
博士学位

folk adj.
民间的

prescription n.
药方

medicinal adj.
药用的

herb n.
草药；药用植物

sweet wormwood 名词短语
黄花蒿

flash n.
短暂显现

promise n.
有效的希望；前景

clue n.
线索

revisit v.
重新审阅

classical adj.
古代经典的

handbook n.
手册

be immersed in 动词短语
浸在……中

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

litre n.
升

bypass v.
避开；略过

delicate adj.
易受损的；此处指不耐热的

switch to 动词短语
改用

ether n.
乙醚

degree Celsius 名词短语
摄氏度

parasite n.
寄生虫；此处指疟原虫

infected adj.
受感染的

indicate v.
表明；说明

voluntarily adv.
自愿地

thereafter adv.
此后

patient n.
患者

obtain v.
获得

crystal n.
晶体

recipient n.
获奖者

jury n.
评审团

physiology n.
生理学

laureate n.
获奖者；此处指诺奖得主

guideline n.
指南

therapy n.
疗法

deadly adj.
最致命的

skip v.
略过；省去

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 This _____ boiling.

02 Efficacy in mice indicated nothing about human _____.

03 In July 1972, to assess its _____, Tu and two colleagues voluntarily took the extract.

04 The _____ of Ge Hong's line lay in the boiling it skipped.

文章主线

- 任务 523项目联合攻关，目标是对旧药氯喹有抗药性的疟疾；屠呦呦带队查2000多药方，试约380种提取物。
- 线索 青蒿药效不稳；葛洪古方浸泡青蒿而不煮沸，启发屠呦呦怀疑高温会破坏有效成分。
- 突破 改用沸点约35°C的乙醚低温提取，1971年第191号样品完全抑制小鼠疟原虫；人体安全仍待验证。
- 验证 1972年，她与两名同事试服提取物，再用于21名疟疾患者，随后得到青蒿素晶体。
- 影响 以青蒿素为基础的多药合用疗法获世卫推荐，挽救众多生命；屠呦呦2015年获诺奖。

参考答案　1 bypassed 2 safety 3 toxicity 4 essence

洞穴里的十三个人

Thirteen in the Cave

找到十三人之后，为什么救他们出洞仍如此困难？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 23 June 2018, twelve boys aged 11 to 16 from a junior football squad went on an outing with their 25-year-old assistant coach. They entered Tham Luang, a sprawling maze of tunnels in northern Thailand, to penetrate its gloom. Monsoon rain then flooded the passages, and the rising water trapped all of them.

For nine days their whereabouts were unknown. On 2 July two elite British divers, threading through submerged passages, reached an inundated chamber about 4 km from the entrance. All 13 were there, huddled on a muddy mound and perched just above the water, every one of them alive.

The only exit ran through kilometres of constricted, treacherous passages, many of them navigable only with scuba gear. Rescuers began stockpiling oxygen cylinders along the route. On 6 July Saman Kunan, a former Thai Navy SEAL, fell unconscious while laying them and died.

The evacuation began on 8 July, and the boys came out in batches. Each was given doses of sedative, wrapped motionless in a stretcher, and passed from diver to diver, with medics stationed along the route. On 10 July the last four footballers and their coach were escorted out after their teammates.

The multinational mobilisation was a feat of logistics. Some 10,000 people converged on the site, including a contingent of more than 100 divers, plus 2,000 soldiers and 900 policemen, with teams deployed from abroad. The rescue had a sobering postscript: in December 2019 another Navy SEAL died of a blood infection caught during the operation. Every boy came home alive, and two rescuers were mourned.

中文译文

2018年6月23日，一支少年足球队的十二名男孩与25岁的助理教练外出游玩；孩子们年龄在11至16岁。他们进入泰国北部的睡美人洞，深入这个隧道交错、向四方延伸的幽暗迷宫。随后季风雨淹没通道，上涨的水将所有人困在洞内。

整整九天，没有人知道他们在哪里。7月2日，两名英国精英潜水员穿过淹没的通道，到达距入口约4公里的一处进水洞室。十三人全在那里，挤在一处刚高出水面的泥土堆上，每个人都还活着。

唯一的出路要经过数公里狭窄、暗藏危险的通道，其中许多路段只有借助水肺装备才能通行。救援人员开始沿途布放、储备氧气瓶。7月6日，前泰国海军海豹突击队员萨曼·古南在布放气瓶时失去知觉，随后死亡。

撤离于7月8日开始，男孩们分批出洞。他们分别接受镇静药，被固定包裹在担架中，保持不动，由潜水员逐段接力运送，沿途有医护人员驻守。7月10日，最后四名小球员与教练继其他队友之后，被护送出洞。

这场跨国动员堪称后勤协调的壮举。约一万人汇集现场，包括100多名潜水员组成的队伍，以及2000名士兵和900名警察，还有从国外调派来的救援团队。但救援还有一段令人沉痛的后续：2019年12月，另一名海军海豹突击队员死于救援行动中染上的血液感染。每个男孩都活着回家，两名救援者却长留在人们的悼念中。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sprawling adj.
向四周延伸的
a sprawling maze of tunnels

maze n.
迷宫
a sprawling maze of tunnels

whereabouts n.
下落；所在之处
their whereabouts were unknown

thread through 动词短语
穿行于；小心穿过
threading through submerged passages

submerged adj.
淹没在水下的
submerged passages

inundated adj.
被水淹的
an inundated chamber

constricted adj.
狭窄的
constricted, treacherous passages

treacherous adj.
暗藏危险的
constricted, treacherous passages

navigable adj.
可以通行的
navigable only with scuba gear

stockpile v.
储备
stockpiling oxygen cylinders

unconscious adj.
失去知觉的
fell unconscious

evacuation n.
撤离
The evacuation began on 8 July

in batches 介词短语
分批地
the boys came out in batches

sedative n.
镇静药
doses of sedative

motionless adj.
一动不动的
wrapped motionless in a stretcher

escort v.
护送
were escorted out

multinational adj.
多国参与的
The multinational mobilisation

mobilisation n.
动员
The multinational mobilisation

logistics n.
后勤；运输与协调
a feat of logistics

deploy v.
调派；部署
teams deployed from abroad

sobering adj.
令人警醒或沉痛的
a sobering postscript

mourn v.
悼念
two rescuers were mourned

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

junior adj.
青少年的

squad n.
球队；小队

go on an outing 动词短语
外出游玩

assistant coach 名词短语
助理教练

enter v.
进入

tunnel n.
隧道

penetrate v.
深入；进入深处

gloom n.
幽暗

monsoon n.
季风

flood v.
淹没

passage n.
通道

elite adj.
精英的；顶尖的

diver n.
潜水员

entrance n.
入口

huddle v.
挤在一起

muddy adj.
泥泞的

mound n.
土堆

perch v.
栖身在狭小高处

alive adj.
活着的

exit n.
出口

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

run through 动词短语
经过；贯穿

kilometre n.
公里

scuba n.
水肺；自携式水下呼吸器

rescuer n.
救援人员

cylinder n.
气瓶

along the route 介词短语
沿途

former adj.
以前的；前任的

dose n.
剂量；一次用药量

wrap v.
包裹

stretcher n.
担架

medic n.
医护人员

station v.
派驻；安排驻守

footballer n.
足球队员

teammate n.
队友

converge on 动词短语
汇集到

contingent n.
分队；一支队伍

policeman n.
警察

postscript n.
后续；事后补记

blood infection 名词短语
血液感染

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 For nine days their _____ were unknown.

02 Rescuers began _____ oxygen cylinders along the route.

03 The evacuation began on 8 July, and the boys came out _____.

04 The multinational mobilisation was a feat of _____.

文章主线

- 被困 2018年，泰国12名少年球员和助理教练进入睡美人洞，因季风雨淹没通道而受困。
- 发现 失联九天后，两名英国潜水员在距洞口约4公里处找到他们，十三人全部活着。
- 难题 救援需穿过数公里狭窄水道；约万人的多国队伍沿途布放气瓶，协调百余名潜水员。
- 救出 7月8日至10日，男孩用镇静药后固定在担架中，由潜水员接力带出，沿途有医护协助。
- 代价 十三人全获救；救援者萨曼・古南在前期布气瓶时遇难，另一名救援队员2019年死于救援时染上的血液感染。

参考答案　1 whereabouts 2 stockpiling 3 in batches 4 logistics

几个少年和一只狗发现的洞穴

The Cave Found by a Dog

谁发现了壁画，为什么今天只能看复制洞穴?

英文原文

Marcel Ravidat, 18, was on a woodland stroll near Montignac with Robot, his canine companion, when it began sniffing at a hollow left by a toppled tree. On 12 September 1940 he came back with three other boys for an adventurous excursion, and the four explorers climbed down into the darkness.

In later retellings, Ravidat embroidered his anecdote with a dramatic twist. Robot, he said, had fallen into the cave. That fanciful mishap is a fabrication. The animal only nosed about the hole, and the pivotal initiative was the boys' own. Ironically, the invented fall made the dog the hero of the story.

Some 680 painted figures and 1,500 engravings cover the rock. The artwork is usually dated to between 17,000 and 22,000 years ago; its creators, our distant ancestors, lived in antiquity. Horses, bulls, and stags are among the animals depicted in pigments on the murals. The paintings' fragility soon became clear.

Lascaux opened to the public on 14 July 1948, and tourism brought an influx of about 1,200 visitors a day. The throng exhaled carbon dioxide and added heat, humidity, and contaminants, and the deterioration of the paintings became plain to see. In 1963 the site closed to the public, followed by restoration and daily inspection.

Lascaux joined UNESCO's World Heritage List in 1979, and Lascaux II, a meticulously made partial facsimile, opened in 1983. A white fungus invaded the walls in a 2001 outbreak, and black stains spread after chemical treatment. Lascaux IV opened in December 2016. Admirers now meet the bulls only in a recreation, the originals being inaccessible to them.

中文译文

18岁的马塞尔·拉维达带着爱犬Robot，在蒙蒂尼亚克附近的林地散步时，Robot开始在一棵倒树留下的地洞旁嗅闻。1940年9月12日，拉维达带着另外三个男孩回来探险，四人爬进了黑暗中。

后来讲述这段经历时，拉维达添油加醋，加了一个戏剧性的转折：他说Robot掉进了洞穴。这个离奇的意外其实是编造的。狗只是在洞口嗅闻，作出关键探索决定的是少年们自己。讽刺的是，虚构的那一跤反而让狗成了故事里的英雄。

洞壁上布满约680个彩绘形象和1500幅雕刻图案。这些作品通常被断代为距今1.7万至2.2万年；创作者是生活在远古的先祖。壁画用颜料描绘了马、公牛、雄鹿等动物。然而，这些画作的脆弱很快显现出来。

拉斯科于1948年7月14日向公众开放，游客每天约有1200人。人群呼出二氧化碳，带来热量、湿气和污染物，壁画的退化变得清晰可见。1963年，原洞停止向公众开放，随后展开修复和每日检查。

1979年，拉斯科列入联合国教科文组织《世界遗产名录》。1983年，精心制作的局部复制洞穴“拉斯科二号”开放。2001年，一种白色真菌暴发并侵入洞壁；化学处理后，黑色污斑又蔓延开来。“拉斯科四号”于2016年12月开放。今天，参观者只能在复原品中欣赏那些公牛，原作所在的洞穴已不对他们开放。

核心词汇

先记这22项。连着下方的原文短语一起回想。

companion n.
伙伴
his canine companion

hollow n.
地面凹洞
a hollow left by a toppled tree

excursion n.
短途出游；探险
an adventurous excursion

embroider v.
添油加醋地讲述
embroidered his anecdote

anecdote n.
轶事；小故事
embroidered his anecdote

fabrication n.
编造；虚构
That fanciful mishap is a fabrication

pivotal adj.
关键的
the pivotal initiative

initiative n.
主动行动；主动性
the pivotal initiative

painted figure 名词短语
彩绘形象
680 painted figures

engraving n.
雕刻图案
1,500 engravings cover the rock

ancestor n.
祖先
our distant ancestors

depict v.
描绘
the animals depicted in pigments

pigment n.
颜料
depicted in pigments on the murals

mural n.
壁画
depicted in pigments on the murals

fragility n.
脆弱性
The paintings' fragility

influx n.
大量涌入
an influx of about 1,200 visitors

contaminant n.
污染物
heat, humidity, and contaminants

deterioration n.
退化；恶化
the deterioration of the paintings

restoration n.
修复
followed by restoration

facsimile n.
复制品
a meticulously made partial facsimile

recreation n.
复原品；再现
only in a recreation

inaccessible adj.
无法进入或接近的
the originals being inaccessible

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

woodland n.
林地

stroll n.
散步

canine adj.
犬类的

sniff v.
嗅闻

topple v.
倒下；倾倒

adventurous adj.
富于冒险的

explorer n.
探险者

darkness n.
黑暗

retelling n.
重新讲述

dramatic adj.
戏剧性的

fall into 动词短语
掉进

fanciful adj.
想象出来的；不真实的

mishap n.
意外；小事故

ironically adv.
讽刺的是

invent v.
编造

cover v.
布满；覆盖

artwork n.
艺术作品

creator n.
创作者

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

antiquity n.
远古；古代

bull n.
公牛

open to the public 动词短语
向公众开放

tourism n.
旅游；旅游业

throng n.
密集的人群

exhale v.
呼出

carbon dioxide 名词短语
二氧化碳

humidity n.
湿度；湿气

plain to see 形容词短语
显而易见的

inspection n.
检查

meticulously adv.
精细地；一丝不苟地

partial adj.
局部的；部分的

fungus n.
真菌

invade v.
侵入

outbreak n.
暴发

stain n.
污斑

chemical treatment 名词短语
化学处理

admirer n.
欣赏者；仰慕者

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That fanciful mishap is a _____.

02 the pivotal _____ was the boys' own.

03 The paintings' _____ soon became clear.

04 the originals being _____ to them.

文章主线

- 发现 1940年，狗在地洞旁嗅闻引起拉维达注意；他随后与另外三名少年下洞，发现拉斯科史前壁画。
- 澄清 后来拉维达把故事说成狗掉进洞里；其实狗只在洞口嗅闻，下洞探险是少年们主动决定的。
- 瑰宝 洞内约有680个彩绘形象和1500幅雕刻图案，常被断代为距今约1.7万至2.2万年，描绘马、牛、鹿等动物。
- 受损 1948年开放后，游客带来二氧化碳、热量、湿气和污染，壁画退化，1963年闭洞。
- 保护 闭洞后仍出现真菌、黑斑；1983年和2016年先后开放复制洞穴，公众如今只能看复制品。

参考答案　1 fabrication 2 initiative 3 fragility 4 inaccessible

黄石公园狼影再现

Shadows of Wolves Over Yellowstone

狼回来了，河谷恢复就都该归功于狼吗?

英文原文

Yellowstone's iconic grey wolves had been exterminated by 1926. On 12 January 1995 eight wolves captured in Alberta, Canada, were hauled to the park in crates; six more followed. Released from enclosures in late March, they roamed the wilderness.

In February 2014 a viral video narrated by writer George Monbiot spun a compelling narrative: elk, thinned by wolves and wary of them, retreated from valleys, letting willows flourish and riverbanks stabilise. Ecologists call this a trophic cascade: predators eat or deter plant-eaters, and vegetation rebounds.

Yet the census undermined the thinning claim: elk on the park's northern range peaked at about 19,000 in January 1994 and were dwindling before wolves arrived, depleted by hunters, drought, and bears. Fear looked weak too: in June 2013 Arthur Middleton's University of Wyoming team found that collared elk east of the park kept grazing unless wolves ventured within half a mile.

Trees fared no better: in November 2021 Utah State University's Elaine Brice showed that sampling only the tallest aspens overstated regrowth four- to sevenfold. The deeper flaw was water. Colorado State University's Tom Hobbs and David Cooper had excluded elk with fences and built artificial beaver dams. In January 2024 they reported: eroding streams had cut so deep that stunted willows no longer reached water.

In January 2025 Oregon State University's William Ripple, an advocate of the hypothesis, reported one of the world's strongest cascades; critics disputed his methods. The question is unresolved, but the wolves' shadow looks fainter than the fable suggested, and the science messier. A discipline willing to revise its cherished example may deserve more trust than the fable ever did.

中文译文

到1926年，黄石公园标志性的灰狼已被消灭。1995年1月12日，八只在加拿大艾伯塔省捕获的狼被装进板条箱运到公园，随后又来了六只。3月下旬，它们离开围栏，开始在荒野中活动。

2014年2月，一段由作家乔治·蒙比奥特解说的热门视频讲了一个动人的故事：狼捕杀北美马鹿，让鹿群数量减少，又因使它们害怕而迫使其离开河谷；柳树于是繁盛起来，河岸也更稳固。生态学家把这类连锁影响称为“营养级联”：捕食者吃掉或吓阻食草动物，继而让植被恢复。

然而，普查数据削弱了鹿群因狼而减少的说法。公园北部草场的北美马鹿在1994年1月达到约1.9万头的高峰，狼还没来，鹿群就已因猎捕、干旱和熊而减少。恐惧导致行为改变的证据也不强：2013年6月，阿瑟·米德尔顿领导的怀俄明大学团队发现，公园以东戴追踪项圈的鹿仍照常吃草，除非狼进入半英里以内。

树木方面的证据也不那么理想。2021年11月，犹他州立大学的伊莱恩·布赖斯指出，只抽样测量最高的山杨，会把再生程度夸大四到七倍。更深层的问题在水上。科罗拉多州立大学的汤姆·霍布斯和戴维·库珀用围栏排除鹿群，并修建人工海狸坝。2024年1月，他们报告说，溪流侵蚀使河床下切得太深，生长受阻的柳树已无法从根部够到水。

2025年1月，支持营养级联假说的俄勒冈州立大学学者威廉·里普尔发表研究，称黄石的级联效应是世界上最强的案例之一；批评者则质疑其方法。问题仍未解决，但狼对植被恢复的影响似乎不像那个简单故事所讲的那么强，科学证据也更复杂。一门愿意修正自己珍视案例的学科，也许比那个寓言般的故事更值得信任。

本文的elk指北美马鹿，不是中国物种名“麋鹿”所指的动物。争论的是狼促成植被恢复的程度，不是狼是否具有任何生态作用。

核心词汇

先记这22项。连着下方的原文短语一起回想。

iconic adj.
标志性的
Yellowstone's iconic grey wolves

exterminate v.
彻底消灭
had been exterminated by 1926

enclosure n.
围栏区域
Released from enclosures

wilderness n.
荒野
they roamed the wilderness

compelling adj.
引人入胜的
a compelling narrative

narrative n.
故事；叙事
a compelling narrative

wary of 形容词短语
提防；警惕
wary of them

trophic cascade n.
营养级联
a trophic cascade

predator n.
捕食者
predators eat or deter plant-eaters

deter v.
吓阻
deter plant-eaters

vegetation n.
植被
vegetation rebounds

census n.
普查
the census

undermine v.
削弱；动摇
undermined the thinning claim

dwindle v.
逐渐减少
dwindling before wolves arrived

deplete v.
使减少；消耗
depleted by hunters

sample v.
抽样测量
sampling only the tallest aspens

overstate v.
夸大
overstated regrowth four- to sevenfold

regrowth n.
再生
overstated regrowth four- to sevenfold

hypothesis n.
假说
an advocate of the hypothesis

dispute v.
质疑；对……提出异议
critics disputed his methods

unresolved adj.
尚未解决的
The question is unresolved

revise v.
修正
revise its cherished example

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

capture v.
捕获

haul v.
拖运

crate n.
板条箱

roam v.
漫游；四处活动

viral video n.
迅速走红的视频

be narrated by 动词短语
由……解说

spin v.
编织（故事）

retreat from 动词短语
从……撤离

willow n.
柳树

flourish v.
繁盛

stabilise v.
变得稳固；稳定

rebound v.
恢复；回升

peak at 动词短语
达到……峰值

hunter n.
猎人

drought n.
干旱

collared adj.
戴有追踪项圈的

graze v.
吃草

venture v.
冒险进入

fare no better 动词短语
情况也没有更好

aspen n.
山杨

flaw n.
问题；缺陷

exclude v.
排除；挡在外面

artificial adj.
人工的

dam n.
坝

erode v.
侵蚀

stunted adj.
生长受阻的

advocate n.
倡导者

faint adj.
较弱的；不明显的

fable n.
寓言式的故事

messy adj.
更错综复杂的

discipline n.
学科

cherished adj.
珍视的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a viral video narrated by writer George Monbiot spun a _____ narrative

02 elk on the park's northern range peaked at about 19,000 in January 1994 and were _____ before wolves arrived, depleted by hunters, drought, and bears.

03 Trees fared no better: in November 2021 Utah State University's Elaine Brice showed that sampling only the tallest aspens _____ regrowth four- to sevenfold.

04 In January 2025 Oregon State University's William Ripple, an advocate of the _____, reported one of the world's strongest cascades; critics disputed his methods.

文章主线

- 回归 黄石公园的灰狼到1926年已被消灭，1995年重新引入；流行视频称狼捕杀并吓退北美马鹿，让柳树复苏、河岸稳固。
- 质疑 这类捕食者经食草动物影响植被的连锁效应，叫“营养级联”；但鹿群在狼来前已因猎捕、干旱和熊而减少。
- 证据 除非狼接近至半英里内，公园以东被追踪的北美马鹿仍照常吃草；只测最高山杨，还会把再生夸大四至七倍。
- 瓶颈 围栏与人工海狸坝实验揭示，河床下切让部分柳树够不到水；减少北美马鹿啃食本身，不足以保证柳树恢复。
- 争议 2025年支持者仍称级联很强，批评者则质疑方法；狼促成植被恢复的程度仍有争议，应随证据修正这个经典案例。

参考答案　1 compelling 2 dwindling 3 overstated 4 hypothesis

走出苏联的俄罗斯方块

The Game That Escaped the Soviet Union

游戏风靡世界后，为什么作者多年未分到游戏收益?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In June 1984 Alexey Pajitnov, a 29-year-old Moscow coder, grew engrossed in a diversion he had built. His primitive Elektronika 60 had a purely textual display, so every geometric piece of four tiles was assembled from characters. The result proved addictive. He called it Tetris, a fusion of the Greek tetra- and tennis, his favourite pastime.

In the Soviet Union, the state, not its salaried designer, was the proprietor of this intellectual property, and Pajitnov held no entitlement to it. Copies mushroomed freely on floppy disks across the Eastern bloc.

In 1986 Robert Stein, an enterprising British software vendor, stumbled upon Tetris in Hungary. In his ignorance of Elorg, the state bureau regulating Soviet software exports, he hastily hawked rights he did not yet legitimately hold. The rights were shuffled onward through a chaotic muddle of overlapping deals among Western firms, one owned by the magnate Robert Maxwell.

In February 1989 Henk Rogers, a resourceful Dutch-born publisher, reached Moscow on a tourist visa and found his way to Elorg. Stein and Kevin Maxwell, the magnate's son, were in Moscow that same hectic week of negotiations, each a bidder. Pajitnov felt a fondness for Rogers, and Nintendo scooped the handheld rights, then worldwide console rights.

Nintendo bundled Tetris with its pocket-sized Game Boy, and that version shipped over 35 million copies. Pajitnov reaped none of the spoils for years. His earnings began only after his arrangement's expiry around 1995. In 1996 he and Rogers founded The Tetris Company to manage the rights.

The man whose puzzle once spread for nothing later helped decide who could sell it.

中文译文

1984年6月，29岁的莫斯科程序员阿列克谢·帕基特诺夫，沉迷于自己编写的一款小游戏。他那台简陋的Elektronika 60电脑只能显示文字，于是每个由四个小格构成的几何块，都用字符拼成。游戏让人欲罢不能。他把希腊语前缀tetra-与自己最爱的消遣——网球的英文tennis——结合起来，给它取名Tetris，也就是俄罗斯方块。

在苏联，这项知识产权归国家所有，并不属于领取工资的设计者，帕基特诺夫没有支配它的权利。游戏复制品通过软盘，在东欧集团各地自由传播、迅速增多。

1986年，富有进取心的英国软件商罗伯特·斯坦因在匈牙利偶然发现俄罗斯方块。他不知道Elorg是监管苏联软件出口的国家机构，尚未合法取得权利，便草率地兜售版权。权利在西方公司之间不断转手，各种相互重叠的交易混成一团；其中一家公司属于大亨罗伯特·马克斯韦尔。

1989年2月，出生于荷兰、足智多谋的发行商亨克·罗杰斯持旅游签证来到莫斯科，设法找到Elorg。同一个忙乱的谈判周里，斯坦因与大亨之子凯文·马克斯韦尔也在莫斯科，分别参与竞标。帕基特诺夫对罗杰斯颇有好感。任天堂最终抢得掌机版权，随后又拿到全球游戏主机版权。

任天堂把俄罗斯方块与口袋大小的Game Boy捆绑销售，这个版本出货超过3500万份。帕基特诺夫多年未能分享游戏带来的商业收益；直到原有协议于1995年前后到期，才开始从游戏中获得收益。1996年，他与罗杰斯创办俄罗斯方块公司，管理相关权利。

当年，他的益智游戏曾免费流传；后来，他也参与决定谁可以出售它。

核心词汇

先记这22项。连着下方的原文短语一起回想。

engrossed in 形容词短语
全神贯注于；沉迷于
grew engrossed in a diversion

textual adj.
文字的；文本的
a purely textual display

be assembled from 动词短语
由……拼成
assembled from characters

fusion n.
融合
a fusion of the Greek tetra-

proprietor n.
所有者
the proprietor

intellectual property 名词短语
知识产权
this intellectual property

entitlement n.
拥有某物的权利
held no entitlement to it

mushroom v.
迅速增多
Copies mushroomed freely

enterprising adj.
富有进取心的
enterprising British software vendor

vendor n.
销售商
enterprising British software vendor

stumble upon 动词短语
偶然发现
stumbled upon Tetris in Hungary

regulate v.
监管
regulating Soviet software exports

legitimately adv.
合法地
did not yet legitimately hold

overlap v.（分词作定语）
相互重叠
overlapping deals among Western firms

resourceful adj.
足智多谋的
a resourceful Dutch-born publisher

negotiation n.
谈判
hectic week of negotiations

bidder n.
竞标者
each a bidder

bundle v.
捆绑销售
Nintendo bundled Tetris

reap v.
收获；获得
reaped none of the spoils

earnings n.
收入；此处指游戏收益
His earnings began only after

arrangement n.
协议；安排
his arrangement's expiry

expiry n.
到期
his arrangement's expiry around 1995

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

Moscow 专名
莫斯科

coder n.
程序员

diversion n.
消遣；游戏

primitive adj.
简陋的

purely adv.
完全地；纯粹地

geometric adj.
几何的

tile n.
小格；方块

character n.
字符

addictive adj.
让人上瘾的

Tetris 专名
俄罗斯方块

Greek adj.
希腊语的

pastime n.
消遣

salaried adj.
领取工资的

designer n.
设计者

freely adv.
自由地；不受限制地

floppy disk 名词短语
软盘

Eastern bloc 专名
东欧集团

software n.
软件

Hungary 专名
匈牙利

in prep.
处于；与ignorance of连用

ignorance n.
不知情；缺乏了解

of prep.
对；引出不了解的对象

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

state bureau 名词短语
国家机构

hastily adv.
草率地；仓促地

hawk v.
兜售

shuffle v.
不断倒手；来回转移

onward adv.
继续向前；此处指继续转手

chaotic adj.
混乱的

muddle n.
混乱局面

magnate n.
商业大亨

publisher n.
发行商

visa n.
签证

find one's way to 动词短语
设法找到；来到

hectic adj.
忙乱的

fondness n.
好感；喜爱

Nintendo 专名
任天堂

scoop v.
抢先获得

console n.
游戏主机

pocket-sized adj.
口袋大小的

version n.
版本

spoils n.
获利；收益

found v.
创办

puzzle n.
益智游戏

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the state, not its salaried designer, was the proprietor of this _____

02 Copies _____ freely on floppy disks across the Eastern bloc.

03 Nintendo _____ Tetris with its pocket-sized Game Boy

04 His earnings began only after his arrangement's _____ around 1995.

文章主线

- 诞生 1984年，苏联程序员帕基特诺夫在只能显示文字的电脑上，用字符做出了俄罗斯方块。
- 归属 游戏版权属于苏联国家，作者没有支配权，复制品则通过软盘在东欧集团迅速流传。
- 混战 英国软件商斯坦因不知Elorg掌管苏联软件出口，未获授权就转卖版权，造成西方公司权利重叠。
- 突破 1989年，发行商罗杰斯找到Elorg谈判，任天堂获掌机及全球主机版权，Game Boy版出货3500多万份。
- 获益 帕基特诺夫到约1995年旧协议到期才开始获得游戏收益；1996年与罗杰斯成立公司，参与管理版权。

参考答案　1 intellectual property 2 mushroomed 3 bundled 4 expiry

多莉：一只羊的复制

Dolly, the Copied Sheep

白脸小羊为何由黑脸母羊生下?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Dolly was born at Roslin, Scotland, on 5 July 1996. Her genome came from a Finn Dorset ewe's mammary cell, fused with a Blackface egg stripped of its nucleus. A Blackface surrogate carried the embryo through pregnancy, yet her white face was proof of a genetically different lineage. She alone survived from 277 attempts, an extraordinary failure rate.

Dolly is popularly called the first clone, a misconception. Her predecessors included frogs cloned from transplanted nuclei in 1962 and Roslin's embryo-derived duo, Megan and Morag, in 1995. She was genuinely the first mammal cloned from an adult cell. Roslin's announcement on 22 February 1997 coincided with the paper, and her name, a mammary pun, honoured Dolly Parton.

The news provoked uproar and apprehension about the ethics of human cloning. In March 1997 Clinton prohibited federal funding for such experiments. In 1999 her telomeres, or chromosome tips, were reported shortened by about 20% for her age. That fed anxiety; some speculated she was ageing prematurely.

Dolly was put down, aged six, on 14 February 2003, with arthritis and a progressive lung disease caused by a sheep virus. Indoor confinement probably contributed. In July 2016, 13 clones, four from her cell line, were ageing normally, without hypertension or diabetes. In November 2017, her skeletal X-rays showed ordinary arthritis, and fears of early onset were judged unfounded.

She bore six lambs by natural mating; Bonnie, first of these descendants, arrived in April 1998. Since 2003 her stuffed body, now famously an icon and memorial, has been encased in the National Museum of Scotland. Behind glass, her white face plainly testifies to her Finn Dorset ancestry.

中文译文

多莉于1996年7月5日在苏格兰罗斯林出生。她的基因组来自一只芬兰多塞特母羊的乳腺细胞；研究人员把这个细胞与一枚已去除细胞核的黑脸羊卵细胞融合。胚胎由一只黑脸羊代孕并孕育至出生，但多莉的白脸显示，她的遗传血统与代孕母羊不同。277次尝试中，只有她存活下来，失败率高得惊人。

人们常把多莉称为第一个克隆体，这是误解。在她之前，已有1962年通过移植细胞核克隆的青蛙，以及罗斯林1995年用胚胎细胞克隆出的梅根和莫拉格两只羊。她真正创下的纪录，是成为首只由成年细胞克隆的哺乳动物。罗斯林于1997年2月22日公布消息，相关论文也同时发表。她的名字带有乳腺相关的双关意味，向多莉·帕顿致意。

消息引起轩然大波，也引发人们对克隆人伦理的忧虑。1997年3月，克林顿禁止以联邦资金资助这类实验。1999年有报告称，多莉的端粒，即染色体末端，比她这个年龄应有的长度短约20%。这加重了焦虑，有人推测她正在过早衰老。

2003年2月14日，六岁的多莉被实施安乐死。她当时患有关节炎，以及由一种绵羊病毒引起、不断恶化的肺病；长期圈养在室内可能也是促成病情的因素。2016年7月，一项研究中的13只克隆羊都在正常衰老，没有高血压或糖尿病，其中4只来自与多莉相同的细胞系。2017年11月，对多莉骨骼X光片的检查显示，她患的是普通关节炎，对过早发病的担忧被认为缺乏根据。

多莉通过自然交配生下六只小羊。第一个后代邦妮于1998年4月出生。自2003年起，她的填充标本一直陈列在苏格兰国家博物馆的玻璃展柜中，如今已是广为人知的象征和纪念物。隔着玻璃，她的白脸依然清楚地表明其芬兰多塞特祖源。

“基因组来自乳腺细胞”主要指细胞核中的遗传物质；去核卵细胞仍保留自身的线粒体遗传物质，代孕母羊则负责孕育胚胎。

核心词汇

先记这22项。连着下方的原文短语一起回想。

genome n.
基因组
Her genome came

fuse with 动词短语
与……融合
fused with a Blackface egg

strip of 动词短语
去除；剥去
stripped of its nucleus

nucleus n.
细胞核
stripped of its nucleus

surrogate n.
代孕母羊；代孕者
Blackface surrogate

embryo n.
胚胎
carried the embryo through pregnancy

lineage n.
血统；世系
a genetically different lineage

clone n.
克隆体
the first clone

misconception n.
误解；错误观念
a misconception

mammal n.
哺乳动物
the first mammal cloned

coincide with 动词短语
与……同时发生
coincided with the paper

provoke v.
引起；激起
The news provoked uproar

apprehension n.
忧虑；担心
apprehension about the ethics

ethics n.
伦理；道德规范
the ethics of human cloning

prohibit v.
禁止
Clinton prohibited federal funding

speculate v.
推测
some speculated

prematurely adv.
过早地
she was ageing prematurely

progressive adj.
病情逐渐加重的
a progressive lung disease

confinement n.
限制活动；圈养
Indoor confinement

onset n.
发病；开始出现
fears of early onset

unfounded adj.
没有根据的
were judged unfounded

ancestry n.
祖源；世系
her Finn Dorset ancestry

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

Scotland 专有名词
苏格兰

pregnancy n.
孕期；怀孕

proof of 名词短语
……的证据

genetically adv.
从遗传上说

extraordinary adj.
异乎寻常的

popularly adv.
通常；普遍地

predecessor n.
先前的同类；前身

transplant v.
移植

duo n.
一对；两个一组

genuinely adv.
真正地；确实

announcement n.
公告；宣布

pun n.
双关语

honour v.
向……致意

uproar n.
轩然大波

chromosome n.
染色体

shorten v.
缩短

anxiety n.
焦虑

put down 动词短语
对动物实施安乐死

arthritis n.
关节炎

lung n.
肺

virus n.
病毒

contribute v.
是促成因素之一

hypertension n.
高血压

diabetes n.
糖尿病

skeletal adj.
骨骼的

mating n.
交配

descendant n.
后代

stuffed adj.
制成填充标本

famously adv.
众所周知地

icon n.
象征；标志性事物

memorial n.
纪念物

encase in 动词短语
封装在……里面

plainly adv.
清楚地

testify to 动词短语
表明；证明

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Dolly is popularly called the first clone, a _____.

02 Her _____ came from a Finn Dorset ewe's mammary cell

03 some speculated she was ageing _____.

04 fears of early onset were judged _____.

文章主线

- 诞生 研究者将去掉细胞核的卵细胞与成年母羊乳腺细胞融合，培育胚胎，1996年由代孕母羊产下多莉。
- 突破 277次尝试仅多莉存活；她是首只由成年细胞克隆的哺乳动物，此前已能用胚胎细胞克隆羊。
- 争议 成果引发人类克隆伦理争论；多莉的端粒（染色体末端）比同龄羊短约20%，又引起早衰猜测。
- 离世 多莉自然交配育下六只小羊；患有关节炎和病毒性肺病，六岁时被实施安乐死，不能据此认定克隆导致早衰。
- 复核 后续研究中13只克隆羊正常衰老；多莉骨骼复查也未支持关节炎过早出现的担忧。

参考答案　1 misconception 2 genome 3 prematurely 4 unfounded

不合拍的风：塔科马大桥的陨落

The Wind That Kept No Time: The Fall of Galloping Gertie

风没按节拍吹，大桥为何还是倒了？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On the morning of 7 November 1940, a sustained wind of about 40 miles an hour set the Tacoma Narrows Bridge twisting. Its deck tilted from side to side in ever more violent oscillations until most of the main span tore apart and tumbled into the water. No one died. Footage of the spectacle would later fascinate students worldwide.

The bridge had opened in July 1940 as the third-longest suspension span in the world at the time, its roadway hung from steel cables. Girders only eight feet deep stiffened the deck, making it slender, elegant, and economical, but far from rigid. Even faint breezes made it heave, earning the nickname Galloping Gertie.

Textbooks have long invoked resonance: a periodic push that matches an object's natural frequency can amplify its motion, as when well-timed shoves propel a child on a swing. The analogy is intuitive but deceptive. That morning's wind did not pulse in time with the bridge; it blew fairly steadily.

The mechanism was subtler. Each twist altered the passing airflow, which reinforced the next twist: a feedback loop that kept tapping the wind for energy. In March 1941 a federal panel blamed the deck's flexibility and wing-like lift. Later analysis called the mechanism flutter; in February 1991 two engineers asserted in a physics journal that textbooks had erred.

The panel urged wind-tunnel tests, and trials on scaled-down replicas gradually became customary for suspension bridges. A sturdier replacement opened in October 1950. Gertie's fall may still instruct engineers, if read accurately: the wind never kept time with the bridge, the bridge's own motion turned a sustained wind destructive.

中文译文

1940年11月7日上午，时速约40英里的持续大风使塔科马海峡大桥开始扭转。桥面左右倾斜，振荡越来越剧烈，最终主跨的大部分断裂，坠入水中。无人遇难。这场景象的影像，后来吸引了世界各地的学生。

大桥于1940年7月通车，其悬索桥跨径当时位居世界第三，路面由钢缆悬吊。桥面只靠高八英尺的钢梁加强，因此显得纤细、优美又经济，却远不够刚硬。即使微风也会让它起伏，因而得了“奔腾的格蒂”这个绰号。

教科书长期用共振来解释这起事故：如果周期性的推力与物体的固有频率相合，就能放大运动，就像踩准节拍推秋千上的孩子。这个类比直观，却会误导人。那天上午的风并没有跟着大桥的节拍一阵阵吹来，而是相当平稳。

实际机制更微妙。桥面每一次扭转都会改变掠过的气流，气流又加强下一次扭转，形成不断从风中获取能量的反馈回路。1941年3月，一个联邦调查委员会把事故归因于桥面的柔性和类似机翼的升力。后来的分析把这一机制称为颤振；1991年2月，两位工程师在一份物理学期刊上指出，教科书的解释错了。

委员会力主做风洞试验，此后用缩尺模型测试悬索桥逐渐成为惯例。更坚固的新桥于1950年10月通车。只要读懂真正的机制，格蒂的倒塌仍能给工程师以启示：风从未与桥合拍，是桥自身的运动，让一股持续的风变得具有破坏力。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sustained adj.
持续的
a sustained wind of about 40 miles an

oscillation n.
振荡
side in ever more violent oscillations

tear apart phr.v.
撕裂；断裂
the main span tore apart and tumbled

footage n.
影片片段；影像
Footage of the spectacle would later

suspension n.
悬吊；悬索结构
as the third-longest suspension span

span n.
桥跨；跨径
as the third-longest suspension span

rigid adj.
刚硬的；不易变形的
and economical, but far from rigid.

heave v.
上下起伏
Even faint breezes made it heave,

invoke v.
援引；用作解释
Textbooks have long invoked resonance:

resonance n.
共振
Textbooks have long invoked resonance:

periodic adj.
周期性的
periodic push that matches an object's

frequency n.
频率
object's natural frequency can amplify

amplify v.
放大；增强
object's natural frequency can amplify

analogy n.
类比
The analogy is intuitive but

intuitive adj.
直观易懂的
analogy is intuitive but deceptive.

deceptive adj.
具有误导性的
analogy is intuitive but deceptive.

mechanism n.
机制；机理
The mechanism was subtler.

feedback n.
反馈
a feedback loop that kept tapping the

flexibility n.
柔性；易弯曲性
panel blamed the deck's flexibility

scaled-down adj.
按比例缩小的
scaled-down replicas gradually became

customary adj.
惯常的
replicas gradually became customary

destructive adj.
具有破坏力的
turned a sustained wind destructive.

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

twist v.
扭转

deck n.
桥面

tilt v.
倾斜

violent adj.
剧烈的

tumble v.
翻落；坠落

spectacle n.
壮观的景象

fascinate v.
使着迷；吸引

hang from phr.v.
悬挂于

girder n.
大梁；钢梁

stiffen v.
使更坚硬；加固

slender adj.
纤细的

elegant adj.
优美的

economical adj.
经济实惠的

faint adj.
微弱的

breeze n.
微风

nickname n.
绰号

textbook n.
教科书

well-timed adj.
时机恰当的

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

shove n.
推；猛推

propel v.
推动

pulse v.
有节奏地起伏

steadily adv.
平稳持续地

subtle adj.
更微妙的

alter v.
改变

reinforce v.
加强

tap v.
汲取；利用

panel n.
专家小组；调查委员会

assert v.
断言；明确提出

err v.
犯错

wind-tunnel adj.
风洞的

replica n.
复制模型

gradually adv.
逐渐地

sturdy adj.
更坚固的

replacement n.
替代物；此处指新桥

instruct v.
指导；给以启示

accurately adv.
准确地

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a _____ wind of about 40 miles an hour

02 can _____ its motion,

03 a _____ loop that kept tapping the wind for energy.

04 gradually became _____ for suspension bridges.

文章主线

- 事故 1940年，刚通车数月的塔科马大桥在持续大风中剧烈扭转、断裂坍塌，无人遇难。
- 隐患 轻巧经济的桥面刚度不足，连微风都能使其起伏，早已有“奔腾的格蒂”之名。
- 纠误 课本常用共振解释事故，但当天的风相当平稳，并没有按桥的节拍反复推它。
- 机制 桥的扭转改变气流，气流又加强下一次扭转，不断从风中取能；这一机制称为颤振。
- 后续 事故推动悬索桥模型风洞试验成为惯例；更坚固的新桥于1950年通车。

参考答案　1 sustained 2 amplify 3 feedback 4 customary

改变世界的科学巨匠

The Scientific Giant Who Changed the World

爱因斯坦因为什么获得诺贝尔奖?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 29 May 1919, as a total eclipse obscured the sun, two British expeditions photographed stars in its apparent vicinity. Arthur Eddington observed from Príncipe, an island off West Africa; Crommelin and Davidson from Sobral, in north-eastern Brazil. They were testing general relativity's bold prediction that the sun's gravity would deflect starlight.

On 6 November, at a joint meeting of the Royal Society and the Royal Astronomical Society, the findings were hailed as confirming it. The Times proclaimed a "Revolution in Science"; a New York Times headline read "Lights All Askew in the Heavens". Almost overnight, Einstein, eminent among physicists, became a global celebrity.

Stockholm proved harder to dazzle. The Nobel physics committee valued rigorous experimental confirmation, and committee member Allvar Gullstrand, a Nobel-winning eye specialist, deemed the empirical support for relativity insufficient. Since no nomination fulfilled Nobel's criteria, the 1921 prize was reserved for a year, as the statutes permit.

On 9 November 1922 the Royal Swedish Academy of Sciences conferred it on Einstein, citing his services to theoretical physics, especially his discovery of the law of the photoelectric effect. That law, which governs how light ejects electrons from metal, helped underpin quantum theory. A letter the next day clarified that the prize took no account of relativity. Einstein, then aboard a steamer bound for Japan, would miss the December ceremony.

On 11 July 1923, in Gothenburg, Sweden, he delivered his Nobel lecture before a congress of Nordic scientists, devoting it to the fundamental ideas and problems of relativity. The theory the committee looked past reached a Nobel podium after all, carried there by the man the prize had honoured.

中文译文

1919年5月29日，日全食遮住太阳时，两支英国考察队拍摄了天空中看似靠近太阳的恒星。阿瑟·爱丁顿在西非外海的普林西比岛观测，克罗姆林和戴维森则在巴西东北部的索布拉尔观测。他们要检验广义相对论的一个大胆预言：太阳的引力会让星光发生偏折。

同年11月6日，在英国皇家学会和皇家天文学会的联席会议上，这些观测结果被盛赞为证实了该预言。《泰晤士报》宣告“科学革命”；《纽约时报》则以“天上的光全歪了”为标题。几乎一夜之间，原本已在物理学界享有盛名的爱因斯坦，变成了全球名人。

斯德哥尔摩的诺奖评审却没那么容易被打动。诺贝尔物理学委员会重视严格的实验证实，而委员阿尔瓦·古尔斯特兰德——一位获得过诺奖的眼科专家——认为相对论的实证支持不足。由于没有一项提名符合诺贝尔规定的标准，1921年的物理学奖按照章程保留一年。

1922年11月9日，瑞典皇家科学院把这个奖授予爱因斯坦，表彰他对理论物理学的贡献，尤其是发现光电效应定律。这一定律揭示光如何使金属释放电子，并帮助奠定了量子理论的基础。授奖决定次日的一封信明确说明，奖项并未考虑相对论。爱因斯坦当时乘轮船前往日本，因此未能参加12月的颁奖典礼。

1923年7月11日，爱因斯坦在瑞典哥德堡的北欧科学家大会上发表诺贝尔演讲，专讲相对论的基本思想与问题。那个被委员会绕开的理论，终究登上了诺贝尔讲台；把它带上去的，正是获得这个奖的人。

核心词汇

先记这22项。连着下方的原文短语一起回想。

eclipse n.
日食；月食
as a total eclipse obscured the sun,

obscure v.
遮住；使模糊
as a total eclipse obscured the sun,

expedition n.
考察队；远征
British expeditions photographed stars

vicinity n.
附近
stars in its apparent vicinity.

deflect v.
使偏折
sun's gravity would deflect starlight.

findings n.
研究或观测结果
the findings were hailed as confirming

hail as v.phr.
被盛赞为
the findings were hailed as confirming

eminent adj.
卓越的；著名的
Almost overnight, Einstein, eminent

rigorous adj.
严格缜密的
rigorous experimental confirmation,

confirmation n.
证实
rigorous experimental confirmation,

deem v.
认定；认为
deemed the empirical support for

empirical adj.
实证的；以观察实验为依据的
deemed the empirical support for

insufficient adj.
不足的
support for relativity insufficient.

criterion n.
标准
nomination fulfilled Nobel's criteria,

reserve v.
保留；暂缓颁发
the 1921 prize was reserved for a

confer on phr.v.
与on连用：授予
Swedish Academy of Sciences conferred

underpin v.
为……奠定基础
helped underpin quantum theory.

clarify v.
澄清；明确说明
letter the next day clarified that the

take no account of v.phr.
不予考虑
prize took no account of relativity.

bound for adj.phr.
驶往；开往
then aboard a steamer bound for Japan,

fundamental adj.
基本的；根本的
to the fundamental ideas and problems

honour v.
表彰；授予荣誉
by the man the prize had honoured.

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

observe v.
观测

general relativity n.phr.
广义相对论

bold adj.
大胆的

gravity n.
引力

joint adj.
联合的；联席的

proclaim v.
宣告

headline n.
新闻标题

overnight adv.
一夜之间

physicist n.
物理学家

celebrity n.
名人

dazzle v.
使赞叹；使目眩

experimental adj.
实验的

committee n.
委员会

specialist n.
专家

nomination n.
提名

fulfil v.
符合；满足

statute n.
章程；成文规定

permit v.
允许

on prep.
标示获授予的人

cite v.
列举；引作表彰理由

govern v.
支配；决定

eject v.
使喷出；释放出

electron n.
电子

quantum adj.
量子的

aboard prep.
在船上；乘船

steamer n.
轮船

ceremony n.
典礼

deliver v.
发表演讲

lecture n.
学术演讲；讲座

congress n.
大会

devote to phr.v.
与to连用：专门用于

to prep.
标示投入的对象

podium n.
讲台

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the sun's gravity would _____ starlight.

02 valued _____ experimental confirmation,

03 helped _____ quantum theory.

04 the prize _____ relativity.

文章主线

- 成名 1919年日食观测被誉为证实太阳引力会使星光偏折，这一相对论预言让爱因斯坦迅速举世闻名。
- 分歧 委员古尔斯特兰德认为相对论证据不足；1921年因无提名达标，物理学奖按章程保留一年。
- 授奖 1922年，保留的1921年度物理学奖授予爱因斯坦，表彰理论物理贡献，尤其是发现光电效应定律。
- 区别 光电效应指光使金属释放电子，这一定律帮助奠定量子理论基础；授奖决定次日的信件明确：该奖不考虑相对论。
- 演讲 爱因斯坦赴日错过颁奖典礼；1923年，他的诺贝尔演讲却专讲相对论，让被奖项绕开的理论登上讲台。

参考答案　1 deflect 2 rigorous 3 underpin 4 took no account of

末日种子库的第一次取种

The Day the Doomsday Vault Opened

全球种子保险箱第一次为谁打开?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On Spitsbergen, in Norway's Svalbard archipelago, a corridor has been drilled deep into a frosted hillside. At its end, cartons of precious seed sit silently in a sub-zero repository, insulated by permanently frozen ground.

The Svalbard Global Seed Vault opened near Longyearbyen in February 2008 to insure crops such as beans, rice, and wheat against the horrors people dread, from atomic war to disease. It has ample room for about four and a half million samples. Essentially the planet's ultimate backup, it is run like a bank's safe-deposit box. Depositors keep ownership, and only they can take their seeds out.

The first retrieval came not after an asteroid strike but amid strife on the ground. ICARDA, a research centre whose gene bank stood near Aleppo, had moved its headquarters to Beirut during Syria's civil war. In September 2015 it asked for about 116,000 samples, 130 of the 325 boxes it had sent to the Arctic before the hostilities. They included robust strains of wheat and barley tolerant of drought and heat.

In Lebanon and Morocco, ICARDA sowed the seeds, and vigorous plants sprouted to regenerate its collection. Once multiplication had gone far enough, a homecoming began in 2017, consignment by consignment, restocking the Arctic stronghold.

That same year brought a scare. After record-breaking warmth, meltwater ran into the roughly 100-metre entrance tunnel and solidified. The intrusion stopped short of the seeds, which stayed safely at minus 18 degrees Celsius. Norway waterproofed the tunnel and dug drainage ditches.

A haven built against the apocalypse had a weakness at its own entrance, and warmer weather found it.

中文译文

在挪威斯瓦尔巴群岛的斯匹次卑尔根岛，一条通道深深钻入结霜的山坡。通道尽头，一箱箱珍贵种子静静存放在零度以下的储藏库里，永久冻土为它们提供隔热保护。

斯瓦尔巴全球种子库于2008年2月在朗伊尔城附近启用，为豆类、水稻、小麦等作物提供保障，以防人们畏惧的种种灾难，从核战争到病害。库内空间充足，可容纳约450万份样本。它实质上是地球最后一道种子备份，运作方式类似银行保险箱：存放者保留种子的所有权，也只有他们有权取回。

这座种子库的首次取种，并非发生在小行星撞击之后，而是在地面战乱之中。研究中心ICARDA的基因库原设在阿勒颇附近，叙利亚内战期间，其总部迁往贝鲁特。2015年9月，该中心申请取回约11.6万份样本，即战事发生前由它送往北极的325箱中的130箱。其中包括耐旱、耐热的健壮小麦和大麦品系。

ICARDA在黎巴嫩和摩洛哥播下这些种子，茁壮的植株萌发生长，重建了该中心自己的种子样本库。繁育数量足够多后，一场回归之旅于2017年开始：种子一批接一批送回，重新补充这座北极堡垒。

同年，种子库也经历了一场虚惊。创纪录的暖天气过后，融水流入约100米长的入口隧道，并凝结成冰。但水和冰没有抵达种子存放处，种子始终安全保存在零下18摄氏度。挪威随后给隧道加做防水，并挖出排水沟。

这座为抵御末日灾难而建的庇护所，弱点却在自己的入口，变暖的天气把它暴露了出来。

核心词汇

先记这22项。连着下方的原文短语一起回想。

corridor n.
通道；走廊
a corridor has been drilled

repository n.
储藏库
a sub-zero repository

insulate v.
隔热保护
insulated by permanently frozen ground

seed vault 名词短语
种子储藏库
The Svalbard Global Seed Vault

insure v.
为……提供保障
insure crops such as beans

ample adj.
充足的
ample room

backup n.
备份；后备
the planet's ultimate backup

ownership n.
所有权
Depositors keep ownership

retrieval n.
取回
The first retrieval

strife n.
冲突；战乱
amid strife on the ground

gene bank 名词短语
基因库；种质资源库
a research centre whose gene bank

headquarters n.
总部
moved its headquarters to Beirut

robust adj.
健壮的；适应力强的
robust strains of wheat and barley

strain n.
作物品系
robust strains of wheat and barley

tolerant of 形容词短语
能耐受……的
tolerant of drought and heat

sow v.
播种
ICARDA sowed the seeds

regenerate v.
重新繁育；恢复
to regenerate its collection

consignment n.
一批运送的货物
consignment by consignment

restock v.
重新补充库存
restocking the Arctic stronghold

intrusion n.
侵入
The intrusion stopped short

stop short of 动词短语
在到达……前停住
stopped short of the seeds

drainage n.
排水
dug drainage ditches

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

archipelago n.
群岛

drill v.
钻凿

frosted adj.
覆着霜的

hillside n.
山坡

carton n.
纸箱

precious adj.
珍贵的

silently adv.
静静地

sub-zero adj.
零度以下的

permanently adv.
永久地

bean n.
豆类

against prep.
防范；抵御

horror n.
可怕的灾难

dread v.
畏惧

atomic adj.
原子能的；核的

essentially adv.
实质上；基本上

ultimate adj.
最终的；最后一道的

safe-deposit box 名词短语
银行保险箱

depositor n.
存放者

asteroid n.
小行星

civil war 名词短语
内战

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

ask for 动词短语
申请取回；请求

Arctic 专有名词
北极地区

hostilities 复数名词
战事；敌对行动

barley n.
大麦

vigorous adj.
茁壮的；旺盛的

sprout v.
发芽

multiplication n.
繁殖；扩繁

homecoming n.
回归

stronghold n.
堡垒；坚固据点

scare n.
虚惊；惊慌

record-breaking adj.
破纪录的

tunnel n.
隧道

solidify v.
凝固

safely adv.
安全地

waterproof v.
做防水处理

ditch n.
沟渠

haven n.
庇护所；安全之地

apocalypse n.
末日灾难

weakness n.
弱点

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Depositors keep _____, and only they can take their seeds out.

02 The first _____ came not after an asteroid strike but amid strife on the ground.

03 robust strains of wheat and barley _____ drought and heat.

04 The _____ stopped short of the seeds, which stayed safely at minus 18 degrees Celsius.

文章主线

- 备份 2008年启用的斯瓦尔巴种子库，为防战争、病害保存农作物种子备份；种子归存放者，只有他们能取回。
- 取种 叙利亚内战中，农业研究中心ICARDA迁移总部；2015年取回自存的部分备份约11.6万份，这是种子库首次出库。
- 重建 该中心在黎巴嫩、摩洛哥繁育种子，恢复自己的种子样本库，并从2017年起分批将种子存回北极。
- 虚惊 同年，高温后的融水进入入口隧道并结冰，未触及种子；种子仍安全保存在零下18摄氏度。
- 加固 挪威给隧道加做防水并开挖排水沟，修补这座全球种子保险箱因天气变暖而暴露的入口弱点。

参考答案　1 ownership 2 retrieval 3 tolerant of 4 intrusion

郁金香狂热：泡沫与传说

Tulip Mania: The Bubble and the Legend

郁金香真的让整个荷兰破产了吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In the winter of 1636–37, Dutch merchants and brokers met in taverns to trade contracts for tulip bulbs still in the ground, a speculative business they called windhandel, "trading in the wind". Prices of prized "broken" tulips, with streaked petals, soared.

In early February 1637, at a sale in Haarlem, the hub of the trade, no one would bid. Days later, at an auction in Alkmaar, one bulb still fetched 5,200 guilders, the Dutch currency of the day, more than a decade's wages for a skilled craftsman. But the frenzy was over. Prices collapsed, many buyers defaulted, and unsettled contracts piled up. Why the market turned so swiftly is still contested.

Two centuries later, Charles Mackay, a Scottish journalist, recounted the episode in a book on popular delusions, casting it as national hysteria. In his telling, rich and poor gambled, great merchants were reduced almost to begging, and Dutch commerce reeled. His vivid account made tulip mania shorthand for bubbles fuelled by reckless buying.

Yet the archives tell a smaller story. Mackay's story relied largely on satirical pamphlets of 1637 that preached against greed. Anne Goldgar, a historian at King's College London, scoured Dutch records and counted only a few hundred traders, mostly merchants and craftsmen. She found no bankruptcy caused by tulips and no measurable harm to the economy. Many buyers were later allowed to settle their contracts by paying a small fraction of the price, and few were ruined.

The boom was real but fleeting. What spread, and still circulates nearly four centuries on, was the moral: a sermon drawn from those pamphlets and retold by Mackay as history.

中文译文

1636至1637年冬天，荷兰商人和经纪人聚在小酒馆里，交易还埋在土中的郁金香球茎合约。他们把这门投机生意称为windhandel，即“风中交易”。名贵的“碎色”郁金香花瓣带有条纹，价格一路飙升。

1637年2月初，在交易中心哈勒姆的一场买卖中，竟无人愿意出价。几天后，阿尔克马尔的一次拍卖仍有一只球茎售得5200荷兰盾——那是当时荷兰的货币，金额超过熟练工匠十年的工资。但狂热已经结束：价格暴跌，许多买家违约，未结清的合约越积越多。市场为何如此迅速转向，至今仍有争议。

两个世纪后，苏格兰记者查尔斯·麦凯在一本讨论群体性错觉的书中重述此事，将它描绘成全国性的狂热。在他的笔下，穷人富人都参与赌博，大商人几乎沦为乞丐，荷兰商业遭到重创。他生动的叙述使“郁金香狂热”成为不计后果的买入所催生的投机泡沫的代名词。

然而，档案呈现的规模要小得多。麦凯主要依据1637年那些讽刺贪婪、劝人戒贪的小册子来讲故事。伦敦国王学院历史学家安妮·戈德加仔细搜遍荷兰档案，只找到数百名交易者，大多是商人和工匠。她未找到因郁金香而破产的案例，也未发现经济遭受可衡量损害的证据。许多买家后来获准只支付原价的一小部分便结清合约，倾家荡产者寥寥无几。

这场热潮真实存在，却很短暂。真正广为传播、近四个世纪后仍在流传的，是其中的道德教训：一篇源于那些小册子的训诫，被麦凯当作历史重新讲述。

“未找到破产或经济损害的证据”是档案研究的发现，不等于断言无人蒙受损失。

核心词汇

先记这22项。连着下方的原文短语一起回想。

broker n.
经纪人；交易中介
merchants and brokers

contract n.
合约；合同
contracts for tulip bulbs

speculative adj.
投机性的
a speculative business

soar v.
飙升；猛涨
soared

bid v.
出价；竞价
no one would bid

fetch v.
售得；拍出某价格
one bulb still fetched 5,200 guilders

frenzy n.
狂热
the frenzy was over

collapse v.
暴跌；崩落
Prices collapsed

default v.
违约；未履行付款义务
many buyers defaulted

contest v.
受到争议或质疑
is still contested

hysteria n.
群体性狂热
national hysteria

shorthand for 名词短语
……的代名词
shorthand for bubbles

reckless adj.
不计后果的；鲁莽的
reckless buying

archive n.
档案；档案资料
the archives tell a smaller story

satirical adj.
讽刺性的
satirical pamphlets

pamphlet n.
小册子
satirical pamphlets

scour v.
彻底搜寻；仔细查遍
scoured Dutch records

bankruptcy n.
破产
no bankruptcy caused by tulips

measurable adj.
可以衡量的
no measurable harm to the economy

settle v.
结清；了结合约
settle their contracts

fraction n.
一小部分；分数
a small fraction of the price

fleeting adj.
短暂的；转瞬即逝的
real but fleeting

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

merchant n.
商人

tavern n.
小酒馆

tulip n.
郁金香

streaked adj.
带条纹的

petal n.
花瓣

hub n.
中心；枢纽

auction n.
拍卖会

currency n.
货币

wage n.
工资；工钱

unsettled adj.
未结清的

pile up 动词短语
堆积；越积越多

swiftly adv.
迅速地

journalist n.
记者

recount v.
讲述；叙述

episode n.
事件；一段往事

popular delusion 名词短语
群体性错觉；大众妄想

gamble v.
赌博

commerce n.
商业；贸易

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

reel v.
遭受冲击；摇摇欲坠

vivid adj.
生动的；鲜明的

mania n.
狂热

bubble n.
投机泡沫

rely v.
依赖；依据

on prep.
表示依赖或依据的对象

preach v.
宣扬；说教

against prep.
反对；抵制

greed n.
贪婪

historian n.
历史学家

harm n.
损害

economy n.
经济

a art.
不定冠词，表示一份或一部分

of prep.
……的

boom n.
繁荣；热潮

circulate v.
流传；传播

moral n.
寓意；道德教训

sermon n.
训诫；说教

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Prices collapsed, many buyers _____, and unsettled contracts piled up.

02 Why the market turned so swiftly is still _____.

03 Mackay's story relied largely on _____ pamphlets of 1637 that preached against greed.

04 She found no bankruptcy caused by tulips and no _____ harm to the economy.

文章主线

- 热潮 1636至1637年，荷兰商人和经纪人交易尚在土里的郁金香球茎合约，稀有品种价格暴涨。
- 破裂 1637年2月买盘骤冷，仍有球茎卖5200盾、抵工匠十余年收入；随后价格崩落、违约堆积，原因仍有争议。
- 传说 两个世纪后，麦凯主要依据讽刺贪婪的小册子，把它写成全民投机、商人破产、经济重创的故事。
- 档案 戈德加查档只找到数百名交易者，未找到郁金香引发破产或经济遭受可衡量损害的证据。
- 辨别 许多买家只付原价一小部分便结清合约，鲜有人倾家荡产；热潮真实，广传的全国灾难故事却被放大。

参考答案　1 defaulted 2 contested 3 satirical 4 measurable

一座火山，造出了《弗兰肯斯坦》

The Volcano That Gave Us Frankenstein

远在热带的火山，怎样与《弗兰肯斯坦》相连?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In June 1816 a ceaseless downpour kept a party besieged at the Villa Diodati, a mansion the poet Lord Byron had leased on the outskirts of Geneva. After reading an anthology of German ghost stories in French translation, Byron proposed that each guest write a supernatural tale.

The melancholy weather was attributable to a distant blast. On 10 April 1815 Mount Tambora, on the Indonesian island of Sumbawa, blew itself apart in one of the most potent eruptions in recorded history. About 10,000 people died immediately, and tens of thousands more of starvation and disease in the aftermath.

The volcano pumped prodigious quantities of sulphur dioxide from its crater into the stratosphere, where it formed sulphate aerosols that blanketed the globe, reflecting sunshine and thereby lowering temperatures world-wide. Europe and North America came to know 1816 as the Year Without a Summer.

Europe's summer was chilly and relentlessly wet. Crops failed or lay rotten in saturated fields, cereal and bread prices escalated, and scarcity, disease, and unrest followed.

The eighteen-year-old Mary Godwin, later Mary Shelley, had set out in May alongside Percy Bysshe Shelley and Claire Clairmont. In her 1831 preface she would recollect a damp, gloomy summer whose torrents often kept the group indoors for days.

Byron's challenge proved fruitful. His physician John Polidori produced The Vampyre, whilst Byron's poetic output included Darkness. Mary began the ghastly yarn that became Frankenstein, which appeared anonymously in 1818 and was later reissued under her name. It is often called the founding work of the science-fiction genre.

A tropical volcano had supplied the rain, and a teenager the monster.

中文译文

1816年6月，连绵的倾盆大雨把一群人困在迪奥达蒂别墅。这座宅邸位于日内瓦郊外，由诗人拜伦勋爵租住。读过一本德国鬼故事选集的法文译本后，拜伦提议每位客人各写一个超自然故事。

这阴郁的天气，源于遥远的一次猛烈喷发。1815年4月10日，印度尼西亚松巴哇岛上的坦博拉火山炸裂，成为有记录以来最猛烈的火山喷发之一。约一万人当场死亡，随后又有数万人死于饥饿和疾病。

火山把大量二氧化硫从火山口送进平流层。这些气体在那里形成硫酸盐气溶胶，覆盖全球、反射阳光，从而使世界各地降温。欧洲和北美后来将1816年称为“无夏之年”。

欧洲的夏天寒冷而持续多雨。庄稼不是歉收，就是烂在浸透水的田里；谷物和面包价格飞涨，物资短缺、疾病与社会动荡接踵而来。

18岁的玛丽·戈德温，也就是后来的玛丽·雪莱，早在5月便与珀西·比希·雪莱、克莱尔·克莱蒙特一同启程。她在1831年的序言中回忆，那是一个潮湿阴沉的夏天，大雨常让一行人接连几天无法出门。

拜伦的提议结出了丰硕成果。他的医生约翰·波利多里写出《吸血鬼》，拜伦创作的诗歌中则有《黑暗》。玛丽开始写那个恐怖故事，它后来成为《弗兰肯斯坦》：1818年匿名出版，之后以她的名字再版。这部作品常被称为科幻文学的开山之作。

热带的一座火山带来了大雨，一位少女则创造了怪物。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ceaseless adj.
无休止的；持续不断的
a ceaseless downpour

besiege v.
围困；使无法离开
kept a party besieged

anthology n.
选集
an anthology of German ghost stories

attributable to 形容词短语
可归因于
attributable to a distant blast

potent adj.
强大的；猛烈的
one of the most potent eruptions

aftermath n.
灾难之后；后果
in the aftermath

prodigious adj.
数量庞大的
prodigious quantities

sulphur dioxide 名词短语
二氧化硫
quantities of sulphur dioxide

aerosol n.
气溶胶；悬浮的微小颗粒
sulphate aerosols

thereby adv.
从而
thereby lowering temperatures

relentlessly adv.
持续不断地
chilly and relentlessly wet

saturated adj.
浸透水的
saturated fields

escalate v.
迅速上涨
bread prices escalated

scarcity n.
短缺
scarcity, disease, and unrest

unrest n.
社会动荡
scarcity, disease, and unrest

preface n.
序言
In her 1831 preface

recollect v.
回忆
she would recollect

fruitful adj.
成果丰硕的
Byron's challenge proved fruitful

anonymously adv.
匿名地
appeared anonymously in 1818

reissue v.
重新出版
reissued under her name

founding work 名词短语
开山之作
the founding work

genre n.
文学体裁；类型
the science-fiction genre

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

downpour n.
倾盆大雨

mansion n.
大宅；宅邸

poet n.
诗人

lease v.
租用

outskirts n.
郊外

ghost story 名词短语
鬼故事

translation n.
译本

supernatural adj.
超自然的

melancholy adj.
阴郁的

Indonesian adj.
印度尼西亚的

blow v.
爆炸；与apart连用指炸裂

apart adv.
成碎片；与blow连用指炸裂

eruption n.
火山喷发

tens of thousands 数量短语
数万

starvation n.
饥饿；饿死

volcano n.
火山

pump v.
大量送入；喷出

crater n.
火山口

blanket v.
覆盖；笼罩

sunshine n.
阳光

lower v.
降低

world-wide adv.
在全世界范围内

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

know v.
把……称为；与as连用

as prep.
作为；引出称呼

chilly adj.
寒冷的

crop n.
庄稼

fail v.
歉收；收成不好

rotten adj.
腐烂的

cereal n.
谷物

eighteen-year-old adj.
十八岁的

set out 动词短语
启程

alongside prep.
与……一起

damp adj.
潮湿的

gloomy adj.
阴沉的

torrent n.
倾盆大雨

whilst conj.
而；与此同时

poetic adj.
诗歌的

output n.
创作成果

ghastly adj.
恐怖的

yarn n.
故事

Frankenstein 专名
《弗兰肯斯坦》

science fiction n.（作定语）
科幻文学

tropical volcano 名词短语
热带火山

monster n.
怪物

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The melancholy weather was _____ a distant blast.

02 Byron's challenge proved _____.

03 which appeared _____ in 1818 and was later reissued under her name.

04 It is often called the founding work of the science-fiction _____.

文章主线

- 喷发 1815年，印尼坦博拉火山喷发，当场约一万人死亡，随后又有数万人死于饥饿和疾病。
- 降温 火山喷出的二氧化硫在高空形成硫酸盐微粒，反射阳光降温，使1816年成为欧美无夏之年。
- 灾情 欧洲夏季寒冷多雨，庄稼歉收、粮价上涨，饥饿、疾病与社会动荡接踵而来。
- 契机 大雨把拜伦、玛丽·雪莱等人困在日内瓦郊外的别墅，拜伦提议每人写一篇恐怖故事。
- 成果 18岁的玛丽开始写《弗兰肯斯坦》，1818年匿名出版，这部作品后来常被视为科幻小说的开山之作。

参考答案　1 attributable to 2 fruitful 3 anonymously 4 genre

从车库到世界之巅

From a Garage to the Top of the World

离开苹果后，乔布斯怎样走回苹果？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In May 1985, after a bruising confrontation, Apple's board ultimately backed chief executive John Sculley. Steve Jobs, 30, was stripped of his managerial duties at the company he had co-founded. That September, sidelined and alienated, he resigned.

Rewind three decades. Born in San Francisco on 24 February 1955, Jobs was adopted as a newborn. On 1 April 1976 he founded Apple with Steve Wozniak, the electronics engineer behind its first prototypes. Apple's garage origins are largely symbolic; Wozniak calls their significance inflated. Apple floated on 12 December 1980; the flagship Macintosh debuted on 24 January 1984.

Exile proved productive. Later in 1985 Jobs founded NeXT, a computer start-up. In February 1986 he acquired Lucasfilm's computer-graphics outfit, which became Pixar. It began as a manufacturer of specialised hardware, not chiefly an animation studio. Toy Story, released on 22 November 1995, was the first feature-length computer-animated film.

The reversal came on 20 December 1996: Apple agreed to buy NeXT, gaining the core software of the eventual Mac OS X and reuniting with its estranged co-founder. In August 1997 Microsoft invested $150 million in its distressed competitor; that September Jobs took the reins as interim chief. Apple, he later said, had been weeks from ruin. The iMac, iPod, and iPhone followed, the iPhone showcased on 9 January 2007.

In October 2003 Jobs received a diagnosis: an uncommon, cancerous pancreatic tumour. In August 2011 Apple briefly overtook Exxon Mobil as the world's most valuable company. His tenure ended on 24 August 2011; he died on 5 October 2011, aged 56. He had spent nearly fourteen years at the helm of the company that ousted him.

中文译文

1985年5月，经历一场激烈交锋后，苹果董事会最终支持了首席执行官约翰·斯卡利。30岁的史蒂夫·乔布斯，在自己参与创办的公司里被解除管理职务。同年9月，遭到架空和疏远的他辞职了。

把时间倒回三十年前。乔布斯于1955年2月24日生于旧金山，出生不久便被收养。1976年4月1日，他和史蒂夫·沃兹尼亚克创办苹果；首批样机由电子工程师沃兹尼亚克研制。苹果的“车库起源”很大程度上具有象征意义，沃兹尼亚克认为它的重要性被夸大了。苹果于1980年12月12日上市，旗舰产品Macintosh于1984年1月24日首次推出。

离开苹果后的岁月颇有成果。1985年晚些时候，乔布斯创办电脑初创公司NeXT。1986年2月，他收购卢卡斯影业的电脑图形团队，后来发展为皮克斯。皮克斯起初制造专用硬件，主要还不是动画工作室。1995年11月22日上映的《玩具总动员》，成为首部长篇电脑动画电影。

转折发生在1996年12月20日：苹果同意收购NeXT，获得后来Mac OS X所用的核心软件，也让苹果与关系疏远的联合创始人乔布斯重新走到一起。1997年8月，微软向陷入困境的竞争对手苹果投资1.5亿美元；同年9月，乔布斯接掌大权，出任临时负责人。他后来回忆，当时苹果离破产只差几周。此后，iMac、iPod和iPhone相继问世，其中iPhone于2007年1月9日公开亮相。

2003年10月，乔布斯被诊断患有一种罕见的恶性胰腺肿瘤。2011年8月，苹果一度超过埃克森美孚，成为全球市值最高的公司。乔布斯于2011年8月24日卸任，10月5日去世，终年56岁。在那家曾经将他逐出的公司里，他重新掌舵了将近十四年。

eventual Mac OS X指后来出现的系统，不是某个“最终版本”；微软投资的对象是苹果。

核心词汇

先记这22项。连着下方的原文短语一起回想。

confrontation n.
对抗；交锋
a bruising confrontation

board n.
董事会
Apple's board

executive n.
高管；执行官
chief executive John Sculley

be stripped of 动词短语
被剥夺
was stripped of his managerial duties

managerial adj.
管理的
his managerial duties

sideline v.
使失去实权；架空
sidelined and alienated

resign v.
辞职
he resigned

prototype n.
样机
its first prototypes

symbolic adj.
象征性的
largely symbolic

significance n.
重要性；意义
their significance inflated

productive adj.
富有成果的
Exile proved productive

acquire v.
收购
acquired Lucasfilm's

manufacturer n.
制造商
a manufacturer of specialised hardware

reversal n.
局面逆转
The reversal came

eventual adj.
后来出现的；最终的
the eventual Mac OS X

estranged adj.
关系疏远的
its estranged co-founder

distressed adj.
陷入财务困境的
its distressed competitor

competitor n.
竞争对手
its distressed competitor

interim adj.
临时的
as interim chief

diagnosis n.
诊断
received a diagnosis

tenure n.
任期
His tenure ended

at the helm of 介词短语
掌管；领导
at the helm of the company

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

bruising adj.
激烈而令人受挫的

ultimately adv.
最终

co-found v.
共同创办

alienate v.
使疏远

rewind v.
倒回；回溯

San Francisco n.
旧金山

adopt v.
收养

newborn n.
新生儿

found v.
创办

electronics n.
电子技术

origin n.
起源

inflate v.
夸大

float v.
上市；公开发行股票

flagship adj.
旗舰的；主打的

debut v.
首次推出

exile n.
被迫离开的时期

start-up n.
初创公司

graphics n.
电脑图形

outfit n.
团队；机构

Pixar n.
皮克斯

specialised adj.
专用的

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

hardware n.
硬件

animation n.
动画

studio n.
工作室

release v.
上映；发行

feature-length adj.
长篇电影长度的

computer-animated adj.
电脑动画制作的

core adj.
核心的

reunite with 动词短语
与……重聚

invest v.
投资

in prep.
向；对（投资对象）

take the reins 动词短语
接掌大权

ruin n.
破产；毁灭

showcase v.
公开展示

uncommon adj.
罕见的

cancerous adj.
恶性的

pancreatic adj.
胰腺的

tumour n.
肿瘤

briefly adv.
短暂地

overtake v.
超过

valuable adj.
最有价值的；此处指市值最高

oust v.
逐出

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In May 1985, after a bruising _____, Apple's board ultimately backed chief executive John Sculley.

02 On 1 April 1976 he founded Apple with Steve Wozniak, the electronics engineer behind its first _____.

03 In February 1986 he _____ Lucasfilm's computer-graphics outfit, which became Pixar.

04 In August 1997 Microsoft invested $150 million in its distressed competitor; that September Jobs took the reins as _____ chief.

文章主线

- 创业 1976年，乔布斯与沃兹尼亚克创办苹果，后者研制首批样机；苹果1980年上市，车库起源的意义被夸大。
- 离开 1985年，董事会在冲突中支持首席执行官斯卡利，解除乔布斯管理职务；被架空的乔布斯随后辞职。
- 另起 他创办NeXT，收购皮克斯前身；皮克斯原造专用硬件，1995年推出全球首部长篇电脑动画《玩具总动员》。
- 回归 1996年苹果同意收购NeXT，获得后来Mac OS X的核心软件；次年微软向苹果注资1.5亿美元，乔布斯回归掌舵。
- 重振 iMac、iPod、iPhone相继推出，苹果2011年一度市值全球第一；患胰腺恶性肿瘤的乔布斯同年卸任、去世。

参考答案　1 confrontation 2 prototypes 3 acquired 4 interim

夺命毒雾：伦敦1952

The Killer Smog of 1952

温暖的炉火为何变成致命烟雾？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On Friday 5 December 1952, cold, stagnant air descended on London. Overhead, warmer air pressed down like a lid, holding the chill captive beneath. Londoners were accustomed to murky, dreary winters. This time, though, the capital's smoke was trapped too, thickening into a choking smog, a mix of smoke and fog.

That smoke, from coal burned in domestic fireplaces and power stations, was laden with soot and noxious sulphur dioxide. In places, visibility fell to a few metres. Ambulance services were suspended, an opera at the Sadler's Wells theatre was abandoned at the interval, and prize cattle at the Smithfield agricultural show died or had to be slaughtered.

The smog dissipated on Tuesday 9 December. Officials at the time put the excess deaths at about 4,000, but a 1954 Ministry of Health report blamed later deaths on influenza. A study published in Environmental Health Perspectives in 2001 reassessed the evidence and cast doubt on that explanation, estimating about 12,000 excess deaths between December 1952 and February 1953.

For decades Londoners had tolerated the smoke as a nuisance. One villain flickered in their own grates, a source of warmth and comfort so habitual that few thought to accuse it. Such complacency may help to explain the government's grudging response. It withstood pressure for an inquiry until May 1953.

Reform proved protracted, slowed by official inertia. The Clean Air Act became law on 5 July 1956, three and a half years after the smog. It empowered municipal councils to create smokeless zones, combating smoke at its source. The cosy hearth, beloved of British families, had, all along, been among the chief offenders.

中文译文

1952年12月5日星期五，寒冷而凝滞的空气笼罩了伦敦。上方较暖的空气像盖子一样压住下层，把冷空气困在下面。伦敦人早已习惯冬日的昏暗阴沉，但这一次，城里的煤烟也无法散去，越积越浓，与雾混合成令人窒息的烟雾。

煤烟来自家庭壁炉和发电厂燃烧的煤，其中含有大量烟尘和有毒的二氧化硫。有些地方的能见度降到只有几米。救护车服务暂停，萨德勒之井剧院的一场歌剧在幕间休息时中止，史密斯菲尔德农业展上的获奖牛也有的死亡，有的不得不被宰杀。

烟雾在12月9日星期二消散。当时官方估计，死亡人数比平常多出约4000人，但卫生部1954年的报告把后续死亡归咎于流感。2001年发表于《环境健康展望》的一项研究重新审查证据，对这种解释提出质疑，估计1952年12月至1953年2月期间的超额死亡约为1.2万人。

几十年来，伦敦人容忍着煤烟，只把它视为一种烦恼。其中一个元凶，就在自家的炉栅里闪烁：那是人们习以为常的温暖与舒适之源，很少有人想到要责怪它。这种安于现状的心态，或许能部分解释政府为什么不情愿作出回应。政府顶住要求调查的压力，一直拖到1953年5月。

改革旷日持久，又被官方的惰性拖慢。《清洁空气法》直到1956年7月5日才成为法律，距烟雾事件已过三年半。它授权市镇议会划设无烟区，从源头治理煤烟。英国家庭钟爱的温暖炉膛，原来始终都是主要的污染元凶之一。

核心词汇

先记这22项。连着下方的原文短语一起回想。

stagnant adj.
凝滞的；不流动的
5 December 1952, cold, stagnant air

descend on phr.v.
笼罩；降临于
stagnant air descended on London.

accustomed to adj.phr.
习惯于
accustomed to murky, dreary winters.

murky adj.
昏暗的；污浊的
accustomed to murky, dreary winters.

choking adj.
令人窒息的
thickening into a choking smog, a mix

smog n.
烟雾
thickening into a choking smog, a mix

laden with adj.phr.
充满；载有大量
was laden with soot and noxious

noxious adj.
有毒有害的
with soot and noxious sulphur dioxide.

visibility n.
能见度
In places, visibility fell to a few

suspend v.
暂停；中止
Ambulance services were suspended,

dissipate v.
消散
The smog dissipated on Tuesday 9

excess adj.
超出正常数量的
at the time put the excess deaths at

reassess v.
重新评估
reassessed the evidence

cast doubt on v.phr.
对……提出质疑
and cast doubt on that explanation,

nuisance n.
令人烦恼的事物
had tolerated the smoke as a nuisance.

habitual adj.
习惯性的；习以为常的
of warmth and comfort so habitual

complacency n.
自满；安于现状
Such complacency may help to explain

grudging adj.
不情愿的
the government's grudging response.

inquiry n.
调查
for an inquiry until May 1953.

protracted adj.
拖延很久的
Reform proved protracted, slowed by

inertia n.
惰性；因循不变
slowed by official inertia.

empower v.
授权
It empowered municipal councils to

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

overhead adv.
在头顶上方

lid n.
盖子

chill n.
寒气

captive adj.
被困住的

dreary adj.
阴沉乏味的

smoke n.
烟；煤烟

fog n.
雾

domestic adj.
家庭的

fireplace n.
壁炉

power station n.phr.
发电厂

ambulance n.
救护车

interval n.
幕间休息

cattle n.
牛；牛群

agricultural adj.
农业的

slaughter v.
宰杀

influenza n.
流感

explanation n.
解释

tolerate v.
容忍

villain n.
元凶；祸首

flicker v.
闪烁；摇曳

warmth n.
温暖

comfort n.
舒适

accuse v.
指责；归罪于

response n.
回应

withstand v.
顶住；抵挡

act n.
法案；法令

municipal adj.
市镇的

smokeless adj.
无烟的

combat v.
治理；对抗

source n.
源头；来源

cosy adj.
温暖舒适的

hearth n.
炉膛；壁炉前地面

beloved of adj.phr.
深受……喜爱的

offender n.
肇事者；此处指污染元凶

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 cold, _____ air descended on London.

02 was laden with soot and _____ sulphur dioxide.

03 reassessed the evidence and _____ that explanation,

04 Reform proved protracted, slowed by official _____.

文章主线

- 困住 1952年12月5至9日，伦敦上方暖空气像盖子压住冷空气，煤烟无法散去，形成致命烟雾。
- 源头 家庭壁炉和发电厂烧煤，释放烟尘及有毒二氧化硫；熟悉的温暖炉火也是污染元凶。
- 瘫痪 局部能见度仅几米，救护车停运、演出中止，农业展的牛也死亡或被迫宰杀。
- 复核 官方初估比常态多死亡约4000人，后续死亡曾被归于流感；2001年研究重估1952年12月至1953年2月，多死亡约1.2万人。
- 改革 政府拖延调查与改革，直到三年半后的1956年才制定《清洁空气法》，授权设无烟区，从源头控烟。

参考答案　1 stagnant 2 noxious 3 cast doubt on 4 inertia

稻济天下

Rice for the World

杂交稻的丰收，怎样由许多人共同促成？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In July 1964, inspecting rice ear by ear in a paddy at Anjiang Agricultural School in Hunan province, the instructor Yuan Longping spotted a peculiar plant. It was male-sterile, shedding no viable pollen. Rice pollinates itself, so breeding hybrid seed in bulk depends on such plants. In February 1966 he detailed the anomaly in the journal Chinese Science Bulletin.

The specimen launched a six-year hunt: Yuan and his colleagues could not breed a sterile line whose offspring reliably inherited the trait. The breakthrough came on 23 November 1970, in a marsh near Nanhong Farm in Sanya, Hainan. Yuan's assistant, Li Bihu, and Feng Keshan, a farm technician, found wild rice whose pollen was abortive.

Then came Yuan's decisive move: rather than withhold the strain, he distributed it to collaborators nationwide. Hybrids need three distinct lines: a sterile line, one to maintain its sterility, and one to restore fertility in the hybrids. Yan Longan's team in Jiangxi bred sterile lines and Zhang Xiancheng in Guangxi first identified a restorer line. In October 1973, at a national conference in Suzhou, Yuan declared the three-line system complete.

By 1976 hybrids were under cultivation on about 140,000 hectares. The World Food Prize Foundation estimates that hybrid rice can yield up to a fifth more grain than conventional varieties. By the early 2000s it covered almost half of China's rice fields.

Yuan, who shared the 2004 World Food Prize with Sierra Leone's Monty Jones, died in Changsha on 22 May 2021, aged 91. Yet popular myth portrays him as a solitary genius. His eye began the search; his generosity let many hands bring in the harvest.

中文译文

1964年7月，湖南安江农校教员袁隆平在稻田里逐穗检查，发现一株不同寻常的水稻：它雄性不育，无法散出有活性的花粉。水稻通常自花授粉，因此大批量培育杂交种子，需要这样的植株。1966年2月，他在《科学通报》详细报告了这一反常现象。

这株标本开启了长达六年的搜寻。袁隆平与同事一直无法育成一个不育系，使其后代稳定继承雄性不育的性状。突破出现在1970年11月23日：在海南三亚南红农场附近的沼泽中，袁隆平的助手李必湖和农场技术员冯克珊找到了花粉败育的野生稻。

袁隆平随后作出了关键决定：他没有将这个品系留在自己手中，而是把材料分给全国各地的协作者。杂交稻需要三个不同的系：不育系、保持其不育性的保持系，以及让杂交后代恢复育性的恢复系。江西颜龙安团队育成了不育系，广西张先程率先找到了恢复系。1973年10月，袁隆平在苏州的一次全国会议上宣布，三系已经配套。

到1976年，杂交稻种植面积约为14万公顷。世界粮食奖基金会估计，它比常规品种最多可多产五分之一的稻谷。到21世纪初，杂交稻已覆盖中国近一半稻田。

袁隆平与塞拉利昂的蒙蒂·琼斯共同获得2004年世界粮食奖。他于2021年5月22日在长沙去世，享年91岁。然而，流行的传奇常把他描绘成单枪匹马的天才。他的敏锐观察开启了搜寻，他愿意分享材料，才让许多人的共同努力促成丰收。

雄性不育指缺少有效花粉；植株仍能作母本接受其他植株的花粉。文中的增产两成是基金会估计的最高幅度。

核心词汇

先记这22项。连着下方的原文短语一起回想。

inspect v.
仔细检查
inspecting rice ear by ear

peculiar adj.
不同寻常的
a peculiar plant

sterile adj.
不育的；本文指雄性不育
It was male-sterile

viable adj.
有活性的；能正常发育的
no viable pollen

pollen n.
花粉
no viable pollen

hybrid adj.
杂交的
breeding hybrid seed in bulk

in bulk 介词短语
大批量地
breeding hybrid seed in bulk

anomaly n.
反常现象
detailed the anomaly

specimen n.
标本
The specimen launched a six-year hunt

breed v.
培育；繁育
breed a sterile line

inherit v.
遗传获得
reliably inherited the trait

trait n.
性状；特征
reliably inherited the trait

breakthrough n.
突破
The breakthrough came

abortive adj.
（花粉）败育的
whose pollen was abortive

decisive adj.
关键的；决定性的
Yuan's decisive move

withhold v.
不予提供；保留
rather than withhold the strain

collaborator n.
协作者
collaborators nationwide

distinct adj.
不同的
three distinct lines

restore v.
恢复
restore fertility in the hybrids

fertility n.
育性；繁殖能力
restore fertility in the hybrids

yield v.
产出
yield up to a fifth more grain

generosity n.
慷慨；乐于分享
his generosity

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

paddy n.
稻田

instructor n.
教员

shed v.
散出；释放

pollinate itself 动词短语
自花授粉

detail v.
详细描述

journal n.
学术期刊

launch v.
开启；发起

hunt n.
搜寻

colleague n.
同事

offspring n.
后代

reliably adv.
稳定可靠地

marsh n.
沼泽

technician n.
技术员

strain n.
品系

distribute v.
分发

to prep.
给；向

nationwide adv.
在全国范围

maintain v.
保持

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

sterility n.
不育性

identify v.
找出；确认

national conference n.
全国性会议

declare v.
宣布

under cultivation 介词短语
处于种植中

hectare n.
公顷

estimate v.
估计

grain n.
谷物；此处指稻谷

conventional adj.
常规的

variety n.
品种

cover v.
覆盖；占据

rice field n.
稻田

World Food Prize n.
世界粮食奖

myth n.
流传的错误观念

portray v.
描绘

as prep.
作为；描绘成

solitary adj.
独自一人的

genius n.
天才

bring in the harvest 动词短语
收获庄稼；促成丰收

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In July 1964, inspecting rice ear by ear in a paddy at Anjiang Agricultural School in Hunan province, the instructor Yuan Longping spotted a _____ plant.

02 It was male-sterile, shedding no _____ pollen.

03 Then came Yuan's decisive move: rather than _____ the strain, he distributed it to collaborators nationwide.

04 The World Food Prize Foundation estimates that hybrid rice can _____ up to a fifth more grain than conventional varieties.

文章主线

- 难题 水稻通常自花授粉，大批杂交制种要用不产有效花粉的母株；袁隆平1964年发现这样的稻株，却难让后代稳定保持不育性。
- 协作 1970年李必湖、冯克珊找到花粉败育野生稻；袁隆平向全国团队共享材料，颜龙安团队育成不育系，张先程率先找到恢复系。
- 机制 1973年三系配套：不育系作母本；保持系为母本授粉，繁殖仍不育的母株；恢复系为母本授粉，产生可结实的杂交后代。
- 推广 1976年已种约14万公顷，21世纪初覆盖中国近半稻田；世界粮食奖基金会估计，最多比常规稻增产两成。
- 贡献 袁隆平的发现、材料共享与全国协作共同促成丰收，并非一人独创；他2004年共获世界粮食奖，2021年去世。

参考答案　1 peculiar 2 viable 3 withhold 4 yield

炸药大王的救心药

Dynamite for the Heart

同一种液体，怎样既能炸山又能救心？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Around 1847 the Italian chemist Sobrero first made nitroglycerin, a colourless, oily liquid of acute instability. A mere droplet on the tongue, he reported, left his head throbbing violently for hours. On 3 September 1864 a nitroglycerin shed in Stockholm exploded. The tragic blast killed five, including Alfred Nobel's younger brother Emil.

Nobel tamed the liquid by blending it with kieselguhr, a porous earth, into a paste shaped into rods. Patented in 1867 as dynamite, it was harnessed for mining, quarrying, and demolition, enriching him immensely. His 1895 testament made the fortune an endowment for prizes, first awarded in 1901.

Nitroglycerin had side-effects. Factory operatives suffered racking headaches, and their tolerance, built over the week, ebbed at weekends. Mondays brought a recurrence, with dizziness and palpitations, hence the name Monday disease. In 1879 William Murrell reported in The Lancet on its therapeutic use against angina pectoris, attacks of chest pain.

On 25 October 1896 Nobel, who had angina, wrote to his aide that by an irony of fate doctors had prescribed him nitroglycerin to take internally. Its alias, Trinitrin, was chosen, he explained, to avoid frightening pharmacists or the public. He did not comply. Partly paralysed by a cerebral haemorrhage, he died at his San Remo villa on 10 December 1896.

In 1977 Murad ascertained that nitroglycerin releases nitric oxide, which widens blood vessels. Furchgott found an unidentified relaxing factor in vascular linings in 1980, and Ignarro recognised it as nitric oxide in 1986. The trio shared the 1998 Nobel Prize in Physiology or Medicine. Its presentation speech, on 10 December, the anniversary of Nobel's death, quoted his ironic letter.

中文译文

1847年前后，意大利化学家索布雷罗首次制得硝酸甘油。这是一种无色、油状、极不稳定的液体。他报告说，仅在舌头上滴一小滴，就会使头部剧烈抽痛数小时。1864年9月3日，斯德哥尔摩一间存放硝酸甘油的棚屋爆炸，造成五人死亡，其中包括阿尔弗雷德·诺贝尔的弟弟埃米尔。

诺贝尔将这种液体与多孔的硅藻土混合，制成糊状物，再塑成棒状，使它变得更可控。这种炸药于1867年获得专利，用于采矿、采石和拆除工程，让他积累了巨额财富。他在1895年立下遗嘱，以这笔财产设立奖项基金，奖项于1901年首次颁发。

硝酸甘油也有副作用。工厂工人饱受剧烈头痛之苦；他们在一周工作中逐渐产生的耐受性，到周末又会减退。每到星期一，头痛便再次发作，还伴有眩晕和心悸，因此被称为“星期一病”。1879年，威廉·默雷尔在《柳叶刀》报告了硝酸甘油治疗心绞痛的用途。心绞痛就是阵发性的胸痛。

1896年10月25日，患有心绞痛的诺贝尔写信给助手，感叹命运的讽刺：医生竟开了硝酸甘油让他内服。他解释说，这种药另取名为Trinitrin，是为了不吓到药剂师或公众。他没有遵从医嘱。后来，他因脑出血而部分瘫痪，于1896年12月10日在圣雷莫的别墅中去世。

1977年，穆拉德查明，硝酸甘油会释放一氧化氮，使血管扩张。1980年，弗奇戈特在血管内壁发现了一种身份尚不明确的舒张因子；1986年，伊格纳罗认定它就是一氧化氮。三人共同获得1998年诺贝尔生理学或医学奖。12月10日颁奖时，致辞引用了诺贝尔那封感叹命运讽刺的信；那天也正是他的逝世纪念日。

未遵医嘱与死亡不应混作因果；文章明确交代诺贝尔脑出血。

核心词汇

先记这22项。连着下方的原文短语一起回想。

chemist n.
化学家
the Italian chemist Sobrero

nitroglycerin n.
硝酸甘油
first made nitroglycerin

acute adj.
极度的；严重的
of acute instability

instability n.
不稳定性
of acute instability

tame v.
使受控；驯服
Nobel tamed the liquid

blend v.
混合
blending it with kieselguhr

porous adj.
多孔的
a porous earth

harness v.
控制并利用
harnessed for mining

demolition n.
拆除工程
mining, quarrying, and demolition

enrich v.
使富有
enriching him immensely

testament n.
遗嘱
His 1895 testament

fortune n.
巨额财富
the fortune an endowment

endowment n.
捐赠基金
an endowment for prizes

tolerance n.
耐受性
their tolerance

recurrence n.
复发
a recurrence, with dizziness

therapeutic adj.
治疗的；有疗效的
its therapeutic use

prescribe v.
开具（药物）处方
doctors had prescribed him

comply v.
遵从
He did not comply

ascertain v.
查明
Murad ascertained

unidentified adj.
尚未确认身份的
an unidentified relaxing factor

vascular adj.
血管的
vascular linings

anniversary n.
周年纪念日
the anniversary of Nobel's death

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

Italian adj.
意大利的

colourless adj.
无色的

oily adj.
油状的

liquid n.
液体

mere adj.
仅仅的；不过是

droplet n.
小滴

throb v.
一阵阵抽痛

violently adv.
剧烈地

shed n.
棚屋

tragic adj.
悲惨的

paste n.
糊状物

rod n.
棒；条

mining n.
采矿

quarrying n.
采石

immensely adv.
极其

nitroglycerin n.
硝酸甘油

side effect n.
副作用

factory operative n.
工厂工人

racking adj.
剧烈折磨人的

ebb v.
减退

dizziness n.
眩晕

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

palpitation n.
心悸

hence adv.
因此

aide n.
助手

by an irony of fate 介词短语
命运弄人地

internally adv.
通过内服

alias n.
别名

frighten v.
使害怕

pharmacist n.
药剂师

partly adv.
部分地

paralyse v.
使瘫痪

cerebral haemorrhage n.
脑出血

villa n.
别墅

nitric oxide n.
一氧化氮

widen v.
使扩张

factor n.
因子

lining n.
内壁；内层

trio n.
三人组

Nobel Prize n.
诺贝尔奖

presentation n.
颁授；颁奖

quote v.
引用

ironic adj.
具有讽刺意味的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Around 1847 the Italian chemist Sobrero first made nitroglycerin, a colourless, oily liquid of acute _____.

02 Factory operatives suffered racking headaches, and their _____, built over the week, ebbed at weekends.

03 He did not _____.

04 In 1977 Murad _____ that nitroglycerin releases nitric oxide, which widens blood vessels.

文章主线

- 危险 1847年前后，索布雷罗制得极不稳定的硝酸甘油；1864年的爆炸夺走五人生命，包括诺贝尔的弟弟。
- 炸药 诺贝尔将硝酸甘油与硅藻土混合制成炸药，1867年获专利，凭此致富，后来遗赠财产设立奖项。
- 药用 工人接触它会头痛，周末耐受下降、周一复发；1879年，默雷尔报告它能治疗心绞痛。
- 讽刺 诺贝尔晚年患心绞痛，也被开了硝酸甘油；他写信感叹命运讽刺，却未遵医嘱。他于1896年死于脑出血。
- 揭谜 后来查明它释放一氧化氮、扩张血管；三位相关研究者获1998年生理学或医学诺奖，颁奖致辞在诺贝尔忌日引用了这封信。

参考答案　1 instability 2 tolerance 3 comply 4 ascertained

被拒绝了十二次的哈利·波特

The Book Twelve Publishers Turned Down

十二次拒绝后，一位八岁读者带来了什么转机？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1990, on a delayed train from Manchester to London's King's Cross, Joanne Rowling, a would-be novelist, was unexpectedly struck by the inspiration for Harry Potter, an orphaned young wizard. Grief soon overshadowed that spark: her mother died on 30 December 1990, aged 45. Rowling then relocated to Porto to teach English.

Rowling married in Porto in October 1992, and a daughter's arrival in July 1993 ushered in motherhood. At the end of 1993 she resettled in Edinburgh, a lone mother reliant on state welfare. She habitually wrote in cafés, patiently polishing her unfinished novel.

A literary agent, Christopher Little, took on the manuscript and pitched it repeatedly to prospective buyers. It was reportedly rejected by twelve publishers in succession, a relentless run of refusals that offered the newcomer little encouragement.

At Bloomsbury, the chairman Nigel Newton gave the first chapter to his eight-year-old daughter, Alice, who relished it and eagerly asked for the next instalment. The firm paid a sum of 1,500 pounds as an advance. Fearful that boys might disregard a book by a woman, the publishers opted for initials to mask her gender. Having no middle name, Rowling took the K for her pseudonym from her grandmother Kathleen.

Harry Potter and the Philosopher's Stone was published on 26 June 1997, in a cautious first print run of 500 hardbacks, 300 allocated to libraries. The American publisher Scholastic paid $105,000 for the US rights, and the renamed Sorcerer's Stone appeared there on 1 September 1998. The series and its spin-off volumes have collectively surpassed 500 million copies sold, in 80 languages. Alice's appetite was evidently shared by millions.

中文译文

1990年，一列从曼彻斯特驶往伦敦国王十字车站的火车晚点了。车上，想成为小说家的乔安妮·罗琳意外得到灵感，构思出孤儿少年巫师哈利·波特。然而悲痛很快掩盖了这点火花：母亲于1990年12月30日去世，年仅45岁。罗琳随后移居波尔图，教授英语。

罗琳于1992年10月在波尔图结婚，1993年7月女儿降生，她开始为人母。同年年底，她在爱丁堡重新安家，成为依靠国家福利生活的单亲母亲。她常在咖啡馆写作，耐心打磨那部尚未完成的小说。

文学经纪人克里斯托弗·利特尔接下书稿，多次向潜在买家推介。据报道，书稿接连被十二家出版社拒绝。这一连串毫不留情的回绝，几乎没有给这位新人带来鼓励。

布卢姆斯伯里出版社董事长奈杰尔·牛顿把第一章给八岁的女儿爱丽丝看。她读得津津有味，急着要下一部分。出版社支付了1500英镑预付金。担心男孩会忽略女作者的书，出版方选择用名字首字母缩写掩去作者性别。罗琳没有中间名，便从祖母凯瑟琳的名字中取来K，组成笔名的一部分。

《哈利·波特与魔法石》于1997年6月26日出版，第一批精装本谨慎地只印了500册，其中300册分配给图书馆。美国学乐出版社花10.5万美元买下美国版权；更换英文书名后的美版于1998年9月1日问世。这一系列及衍生图书后来已有80种语言版本，合计销量超过五亿册。显然，数百万读者和爱丽丝有着同样的阅读胃口。

核心词汇

先记这22项。连着下方的原文短语一起回想。

inspiration n.
灵感
the inspiration for Harry Potter

grief n.
悲痛
Grief soon overshadowed that spark

overshadow v.
使黯然失色；盖过
overshadowed that spark

relocate v.
移居
relocated to Porto

usher in 动词短语
开启；带来
ushered in motherhood

reliant on 形容词短语
依靠
reliant on state welfare

welfare n.
政府福利救济
state welfare

polish v.
润色；打磨
polishing her unfinished novel

literary adj.
文学的
A literary agent

pitch v.
推介；推销
pitched it repeatedly

prospective adj.
潜在的；可能成为的
prospective buyers

reject v.
拒绝
rejected by twelve publishers

in succession 介词短语
接连；连续
twelve publishers in succession

relish v.
十分享受；津津有味地读
who relished it

disregard v.
忽视
disregard a book by a woman

opt for 动词短语
选择
opted for initials

initial n.
姓名首字母缩写
initials to mask her gender

pseudonym n.
笔名
the K for her pseudonym

hardback n.
精装本
first print run of 500 hardbacks

allocate v.
分配
300 allocated to libraries

collectively adv.
合计；共同地
have collectively surpassed

surpass v.
超过
surpassed 500 million copies sold

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

delay v.（分词作定语）
延误；晚点

Manchester 专名
曼彻斯特

would-be adj.
有志成为的

novelist n.
小说家

unexpectedly adv.
出乎意料地

orphaned adj.
失去父母的；成为孤儿的

wizard n.
巫师

spark n.
火花；灵感的萌芽

arrival n.
降生；到来

motherhood n.
为人母的生活

resettle v.
重新安家

Edinburgh 专名
爱丁堡

lone adj.
独自的；单亲的

habitually adv.
习惯性地；经常

patiently adv.
耐心地

unfinished adj.
尚未完成的

take on 动词短语
接手；接下

repeatedly adv.
多次地

relentless adj.
毫不留情的；持续不断的

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

refusal n.
回绝

newcomer n.
新人

encouragement n.
鼓励

chairman n.
董事长

chapter n.
章

eagerly adv.
急切地

instalment n.
分期的一部分；此指后续章节

a sum of 名词短语
一笔……的款项

as an advance 介词短语
作为预付金

fearful adj.
担心的

mask v.
掩饰；隐瞒

gender n.
性别

cautious adj.
谨慎的

rename v.
更名

series n.
系列

spin-off adj.
衍生的

volume n.
书；卷

appetite n.
阅读欲望；兴趣

evidently adv.
显然

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 It was reportedly _____ by twelve publishers in succession

02 the publishers _____ initials to mask her gender.

03 in a cautious first print run of 500 _____

04 The series and its spin-off volumes have collectively _____ 500 million copies sold, in 80 languages.

文章主线

- 起笔 1990年，罗琳在一列晚点火车上得到哈利·波特的灵感，开始构思孤儿巫师的故事。
- 坚持 经历丧母后，她在爱丁堡独自抚养女儿，依靠福利生活，仍常在咖啡馆打磨小说。
- 转机 书稿据报遭十二家出版社拒绝；布卢姆斯伯里董事长的八岁女儿爱丽丝，读首章后急着看后续。
- 问世 该社接纳书稿，因担心男孩忽略女作者的书，采用J.K.缩写署名；1997年首册《魔法石》在英国首印500册精装本。
- 回响 美国版于1998年问世；系列及衍生图书后来已有80种语言版本，合计销量超过五亿册。

参考答案　1 rejected 2 opted for 3 hardbacks 4 surpassed

从空气中造面包

The Bread Made from Air

同一种化学工艺，为何既能养人也能助战?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In July 1909, BASF envoys gathered round a squat machine on a bench in Fritz Haber's Karlsruhe laboratory. Behind its gleaming, bolted valves, nitrogen and hydrogen met at crushing pressure and furnace heat, and transparent beads of ammonia began to drip slowly. Haber had pried nourishment from the air.

BASF's Carl Bosch scaled the method up, and from 1913 a gigantic plant at Oppau mass-produced ammonia. Hailed for making bread from air, Haber won the coveted 1918 Nobel Prize. Ploughed into stony soils as fertiliser, ammonia wards off famine as no manure heap could, bringing abundance. By 2008 it was feeding nearly half of mankind.

Haber's ambition had a sinister side. Ammonia stocked Germany's arsenal with nitrates for munitions despite a British blockade, and Captain Haber marshalled chemical warfare. On 22 April 1915 at Ypres, troops vented about 168 tonnes of greenish, poisonous chlorine towards the Allied trenches, the first large-scale gas onslaught.

Haber's wife, Clara Immerwahr, Germany's first woman with a doctoral degree in chemistry, voiced objections to his work. On 2 May 1915, after festivities in his honour, she shot herself in the garden with his service pistol. The next day the widower left for the Eastern Front.

Prestige offered no sanctuary. Of Jewish origin, Haber left Germany an outcast after Hitler's regime seized power, dying in Basel in 1934. His institute's pesticide research had produced Zyklon. A later variant, Zyklon B, was used in Nazi death camps, where relatives of his died.

About half the nitrogen in our proteins and DNA has passed through a Haber-Bosch factory, so each of us carries a little of Haber's chemistry.

中文译文

1909年7月，BASF公司的代表围在弗里茨·哈伯位于卡尔斯鲁厄的实验室工作台旁，看着一台矮壮的机器。在闪亮、用螺栓紧固的阀门后，氮与氢在极高压力和熔炉般的高温下反应，透明的氨液滴开始缓缓滴落。哈伯从空气中撬出了养分。

BASF的卡尔·博施将这一方法扩大到工业规模。1913年起，奥保的一座大型工厂开始批量生产氨。哈伯因“从空气中造面包”而广受赞誉，获得了令人向往的1918年诺贝尔奖。氨作为化肥被施入多石的土壤，以粪肥远不能及的能力抵御饥荒，带来丰收。到2008年，它已养活近半数人类。

哈伯的雄心也有阴暗的一面。尽管英国实施封锁，氨仍为德国军火库提供制造弹药所需的硝酸盐；而哈伯上尉则组织化学战。1915年4月22日，在伊普尔，德军向协约国战壕排放了约168吨黄绿色的有毒氯气，发动了首次大规模毒气攻击。

哈伯的妻子克拉拉·伊梅瓦尔是德国首位获得化学博士学位的女性，她曾反对他的工作。1915年5月2日，在为哈伯举行的庆祝活动之后，她在花园中用他的军用手枪自杀。次日，这位刚刚丧妻的人便前往东线。

声望并未给哈伯提供庇护。身为犹太裔，他在希特勒政权掌权后受到排斥，离开德国，1934年死于巴塞尔。他所在研究所的杀虫剂研究曾催生齐克隆；后来的变体齐克隆B被用于纳粹死亡营，他的一些亲属也死在这些营中。

我们体内蛋白质和DNA所含的氮，约有一半曾经过哈伯—博施工厂。因此，每个人的身体里都带着一点哈伯化学的痕迹。

齐克隆B是后来的变体；文中未认定哈伯亲属的具体死因。

核心词汇

先记这22项。连着下方的原文短语一起回想。

valve n.
阀门
its gleaming, bolted valves

ammonia n.
氨
transparent beads of ammonia

nourishment n.
养分；营养
pried nourishment from the air

scale up 动词短语
扩大规模
scaled the method up

mass-produce v.
批量生产
mass-produced ammonia

coveted adj.
人人渴望得到的
the coveted 1918 Nobel Prize

fertiliser n.
化肥；肥料
as fertiliser

ward off 动词短语
抵御；防止
ammonia wards off famine

famine n.
饥荒
ammonia wards off famine

abundance n.
丰饶；充裕
bringing abundance

sinister adj.
险恶的；阴暗的
a sinister side

arsenal n.
军火库
Germany's arsenal

munitions n.
弹药；军需品
nitrates for munitions

blockade n.
封锁
despite a British blockade

marshal v.
统筹；组织调度
marshalled chemical warfare

onslaught n.
猛烈攻击
the first large-scale gas onslaught

objection to 名词短语
对……的反对意见
voiced objections to his work

sanctuary n.
庇护；避难所
Prestige offered no sanctuary

outcast n.
被排斥的人
left Germany an outcast

pesticide n.
杀虫剂；农药
His institute's pesticide research

variant n.
变体
A later variant, Zyklon B

nitrogen n.
氮
nitrogen and hydrogen met

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

envoy n.
代表；使者

squat adj.
矮壮的；矮而宽的

bench n.
实验工作台

gleaming adj.
闪亮的

bolted adj.
用螺栓紧固的

crushing adj.
极高的；压倒性的

furnace n.
熔炉；高温炉

transparent adj.
透明的

bead n.
液体小滴

drip v.
滴落

slowly adv.
缓慢地

pry v.
撬出；费力取得

from prep.
从；由

up adv.
表示规模扩大

gigantic adj.
巨大的

hail for 动词短语
因……而受到赞誉

plough into 动词短语
翻耕埋入；施入土壤

stony adj.
多石的

soil n.
土壤

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

manure n.
粪肥

heap n.
堆；一堆

mankind n.
人类

ambition n.
雄心；抱负

stock v.
为……储备；充实

with prep.
用；以（stock…with搭配）

vent v.
排放；释放气体

greenish adj.
略带绿色的

poisonous adj.
有毒的

trench n.
战壕

doctoral adj.
博士学位的

festivity n.
庆祝活动

pistol n.
手枪

widower n.
鳏夫；丧妻的人

prestige n.
声望；威望

regime n.
政权

seize v.
夺取；攫取

protein n.
蛋白质

pass through 动词短语
经过；通过

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Behind its gleaming, bolted valves, nitrogen and hydrogen met at crushing pressure and furnace heat, and transparent beads of _____ began to drip slowly.

02 Ploughed into stony soils as _____, ammonia wards off famine as no manure heap could, bringing abundance.

03 Haber's ambition had a _____ side.

04 Prestige offered no _____.

文章主线

- 突破 1909年哈伯用空气里的氮，在高温高压下与氢合成氨；博施将其放大，1913年实现工业量产。
- 养育 氨制成化肥，提高粮食产量，到2008年已支撑近半人类；人体蛋白质和DNA中约一半的氮也经此工艺。
- 战争 合成氨也供应弹药原料；哈伯1915年组织首次大规模毒气攻击，妻子反对其战争工作。
- 命运 妻子后来自杀，他次日仍赴前线；虽获1918年诺奖，犹太裔的他后来仍遭纳粹排斥，1934年死于流亡中。
- 余波 他研究所的杀虫剂齐克隆，后来衍生出齐克隆B；纳粹将其用于死亡营，他的一些亲属也死在营中。

参考答案　1 ammonia 2 fertiliser 3 sinister 4 sanctuary

失败的胶水，成功的便利贴

The Glue That Failed Its Way to Success

黏得不牢的胶，为什么成了畅销文具？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1968 Spencer Silver, a scientist at 3M, set out to make a super-strong adhesive. The polymer he produced did something else. Its tiny spheres formed a layer with little tack, giving paper a temporary hold, so it would adhere, peel off, and be reapplied. To an inventor chasing strength, the formulation was a setback.

For five years Silver acted as its promoter within 3M's corporate ranks, in informal chats and at seminars. In his own summary, it was a solution without a problem. His perseverance met indifference. The unconventional material found no use, and the pursuit proved fruitless.

In 1974 Art Fry, a colleague who had heard one of those talks, faced a small annoyance at church. The paper bookmarks wedged in his choir's hymn book kept getting dislodged and falling out. The frustration gave Fry an idea. The unwanted adhesive could make a marker that would cling to a page, stay put, and leave it undamaged.

In 1977 the notes were test-marketed as Press 'n Peel in four American cities, where purchasers showed little enthusiasm. In 1978 3M's marketers staged a blitz in Boise, Idaho, and dispensed complimentary packets of notes to offices. More than 90 per cent of respondents who tried them said they would buy. After the coolness came a turnaround.

On 6 April 1980 the pads, rebranded as Post-it Notes, went on sale nationally in the US, and Canada and Europe followed in 1981. The stationery staple owes its signature canary tone to an unplanned twist. The laboratory next door had only leftover yellow paper.

中文译文

1968年，3M公司的科学家斯潘塞·西尔弗着手研制超强黏合剂，做出的聚合物却有另一种性能：微小球粒形成黏性较弱的涂层，让纸张暂时贴住，揭下后还能重新贴上。对追求黏合强度的发明者来说，这个配方却是一次挫折。

此后五年，西尔弗在3M内部不断推广它，在闲聊和研讨会上反复介绍。用他自己的话说，这是一个还没找到问题的解决方案。他的坚持换来的却是冷淡。这种不同寻常的材料始终找不到用途，探索似乎毫无结果。

1974年，听过他一次讲座的同事阿特·弗莱在教堂遇到一件烦心事：夹在唱诗班赞美诗集里的纸书签总是移位、掉落。这种挫败感让他想到，没人要的弱黏胶，也许能做出贴在书页上、不易移动、揭下又不伤纸的书签。

1977年，这种便条以Press’n Peel之名在美国四个城市试销，购买者反应冷淡。1978年，3M的营销人员在爱达荷州博伊西发动密集推广，向办公室免费派发一包包便条。试用过的受访者中，超过90%表示愿意购买。产品终于由冷遇迎来转机。

1980年4月6日，产品以“便利贴”的新名字在美国全国销售，1981年又进入加拿大和欧洲。如今这种文具已是常用品；它标志性的金丝雀黄色却纯属偶然：当时隔壁实验室只有剩下的黄纸。

超过90%说愿意购买，指试用后的受访者，不是所有消费者，也不是实际购买率。

核心词汇

先记这22项。连着下方的原文短语一起回想。

adhesive n.
黏合剂
a super-strong adhesive

polymer n.
聚合物
The polymer he produced

tack n.
黏着力
with little tack

temporary adj.
暂时的
a temporary hold

adhere v.
黏附
adhere, peel off, and be reapplied

peel off 动词短语
揭下；剥离
adhere, peel off, and be reapplied

formulation n.
配方
the formulation was a setback

setback n.
挫折
the formulation was a setback

perseverance n.
坚持不懈
His perseverance met indifference

indifference n.
冷淡；漠不关心
His perseverance met indifference

unconventional adj.
不同寻常的
The unconventional material

fruitless adj.
无结果的
the pursuit proved fruitless

dislodge v.
使移位
getting dislodged and falling out

frustration n.
挫败；懊恼
The frustration gave Fry an idea

cling to 动词短语
黏在；紧附
cling to a page

test-market v.
试销
the notes were test-marketed

enthusiasm n.
热情
purchasers showed little enthusiasm

dispense v.
分发
dispensed complimentary packets

complimentary adj.
免费赠送的
complimentary packets of notes

respondent n.
受访者
respondents who tried them

turnaround n.
转机；好转
After the coolness came a turnaround

staple n.
常用品
The stationery staple

文中词汇与搭配

补充词汇共41项。保留每个表达在本文中的意思。

set out to 动词短语
着手；开始做

super-strong adj.
超强的

something else 代词短语
别的东西；另一回事

sphere n.
球状颗粒

reapply v.
重新贴上

inventor n.
发明者

act as 动词短语
充当

promoter n.
推广者

corporate adj.
公司的

informal adj.
非正式的

seminar n.
研讨会

summary n.
概括；总结

solution n.
解决方案

find no use 动词短语
找不到用途

pursuit n.
探索；追求

annoyance n.
烦心事

bookmark n.
书签

wedge v.
夹住；塞入

choir n.
唱诗班

hymn n.
赞美诗

文中词汇与搭配 续

补充词汇共41项。保留每个表达在本文中的意思。

fall out 动词短语
掉出来

unwanted adj.
没人要的

stay put 动词短语
留在原处

undamaged adj.
完好无损的

note n.
便条

purchaser n.
购买者

marketer n.
营销人员

stage v.
组织；发起

blitz n.
密集攻势

packet n.
小包

coolness n.
冷淡

pad n.
便笺本

rebrand as 动词短语
更名为；改品牌为

Post-it Notes n.
便利贴（产品名）

nationally adv.
在全国

stationery n.
文具

signature adj.
标志性的

canary adj.
金丝雀黄色的

tone n.
色调

unplanned adj.
计划外的

leftover adj.
剩余的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In 1968 Spencer Silver, a scientist at 3M, set out to make a super-strong _____.

02 His _____ met indifference.

03 In 1978 3M's marketers staged a blitz in Boise, Idaho, and dispensed _____ packets of notes to offices.

04 After the coolness came a _____.

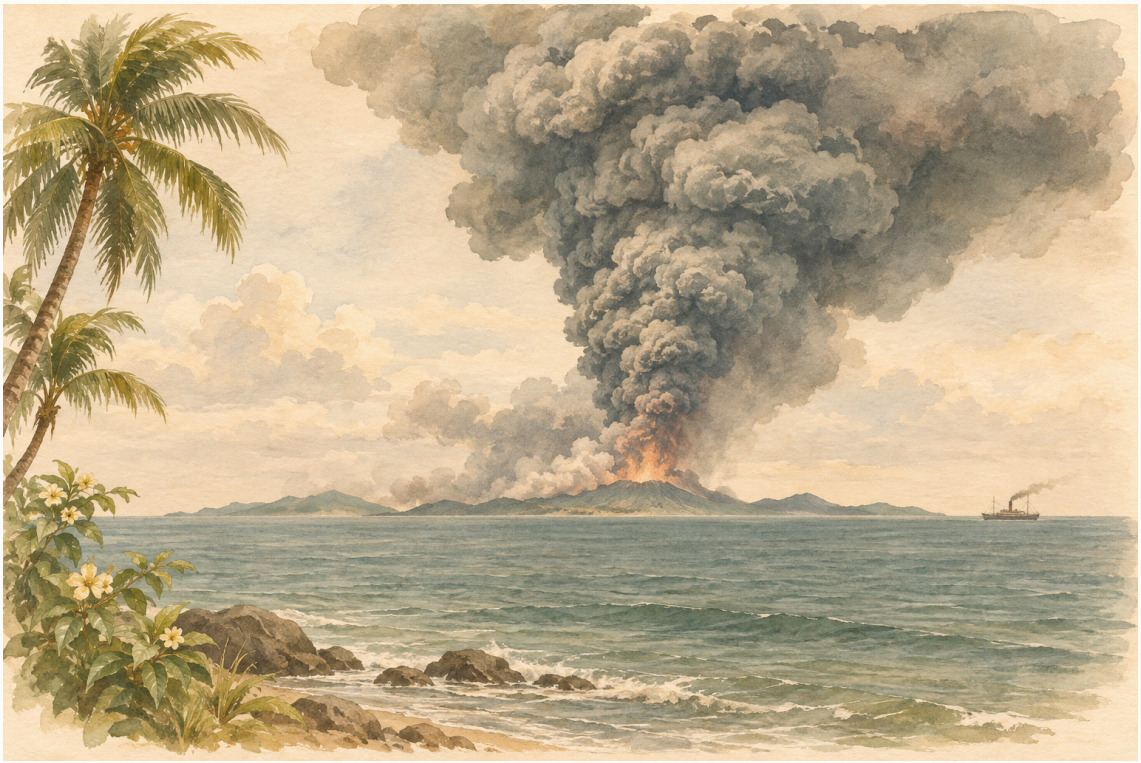
文章主线

- 意外 1968年，3M的西尔弗想造超强胶，却做出由微小球粒组成的弱黏胶，纸张能贴住、揭下再贴。
- 用途 他在公司推广多年后，同事弗莱于1974年想到，用这种胶固定总从赞美诗集中滑落的书签，且不伤书页。
- 冷遇 1977年，这种便条在美国四个城市试销，消费者反应冷淡。
- 转机 1978年，3M在博伊西向办公室免费发试用品；试用过的受访者中，超过九成表示愿意购买。
- 上市 1980年，产品改名“便利贴”在全美上市，次年进入加拿大和欧洲；标志性的黄色，仅因隔壁实验室剩下黄纸。

参考答案　1 adhesive 2 perseverance 3 complimentary 4 turnaround

震动世界的一声巨响

The Sound That Shook the World



火山的一声巨响，如何传遍世界？

英文原文

Shortly after ten in the morning on 27 August 1883, Krakatoa, a volcanic island in the Sunda Strait, unleashed its most thunderous explosion. About four hours later, islanders on Rodrigues, nearly 4,800 kilometres across the Indian Ocean, took the roar for distant gunfire. Barometers worldwide recorded the atmospheric pressure wave, some of them seven times in about five days.

Restless since 20 May, the volcano reached a catastrophic climax on 26–27 August. About two-thirds of the island collapsed into its emptied magma chamber, vanishing beneath the sea. Tsunamis, reportedly over 30 metres high in places, swamped coastal settlements on Java and Sumatra. Dutch colonial administrators counted 36,417 dead, mostly drowned, perhaps underestimating the toll.

Telegrams from Batavia (now Jakarta), relayed by submarine cable, reached London within hours. Western newspapers soon began chronicling what is often called the first global news event. Yet early dispatches were terse; the magnitude of the devastation emerged piecemeal over days.

Ash and sulphur dioxide, hurled tens of kilometres high, formed a haze that circled the equator in about two weeks. In November 1883, crimson sunsets glowed over Europe and North America, summoning firefighters in New York and Connecticut to phantom blazes. Later climate reconstructions suggest global temperatures dipped a few tenths of a degree the following year.

Some experts discounted the volcanic explanation, blaming the sunsets on moisture at high altitude or cosmic dust. In February 1884, Britain's Royal Society appealed in The Times for dated observations, eliciting worldwide replies; the Society's 1888 report tied the glow to Krakatoa. Through barometers, cables, and crimson skies, far-flung observers began to glimpse how interconnected the planet already was.

中文译文

1883年8月27日上午10点刚过，巽他海峡的火山岛喀拉喀托发生了最震耳欲聋的一次爆炸。约四小时后，远隔印度洋、距其近4800公里的罗德里格斯岛居民，把轰鸣误当成远处的炮声。世界各地的气压计都记录到了大气压力波，一些仪器在大约五天内录到了七次。

火山自5月20日起便躁动不安，到了8月26日至27日，活动达到灾难性的高潮。约三分之二的岛体塌入已被排空的岩浆房，没入海中。据报道，局部高逾30米的海啸淹没了爪哇和苏门答腊的沿海居民点。荷兰殖民当局统计死亡36417人，大多为溺亡，而这一数字还可能偏低。

从巴达维亚，也就是今天的雅加达发出的电报，经海底电缆转发，数小时内便抵达伦敦。西方报纸很快开始报道这起常被称为首个全球新闻事件的灾难。然而，早期电讯十分简短，灾情究竟有多严重，直到几天后才逐渐清楚。

火山灰和二氧化硫被抛向数十公里高空，形成的薄雾约用两周便绕赤道一圈。1883年11月，深红的晚霞照亮欧洲和北美；纽约州和康涅狄格州的消防员甚至被引去扑救并不存在的大火。后来的气候重建研究推测，次年全球气温下降了零点几度。

一些专家不接受火山解释，认为晚霞来自高空水汽或宇宙尘埃。1884年2月，英国皇家学会通过《泰晤士报》征集注明日期的观测记录，得到世界各地的回应；学会1888年的报告将红霞与喀拉喀托联系起来。借助气压计、电缆和绯红的天空，相隔遥远的观测者开始看见：地球各处原来早已相互关联。

核心词汇

先记这22项。连着下方的原文短语一起回想。

unleash v.
释放；迸发出
unleashed its most thunderous

thunderous adj.
雷鸣般响亮的
its most thunderous explosion.

barometer n.
气压计
Barometers worldwide recorded the

atmospheric adj.
大气的
the atmospheric pressure wave,

catastrophic adj.
灾难性的
a catastrophic climax on 26–27 August.

climax n.
高潮；顶点
a catastrophic climax on 26–27 August.

collapse into phr.v.
塌入
the island collapsed into its emptied

reportedly adv.
据报道
Tsunamis, reportedly over 30 metres

swamp v.
淹没
swamped coastal settlements on Java

underestimate v.
低估
perhaps underestimating the toll.

toll n.
死亡人数；伤亡总数
perhaps underestimating the toll.

relay v.
转发；传递
relayed by submarine cable,

terse adj.
简短的
Yet early dispatches were terse;

magnitude n.
规模；程度
the magnitude of the devastation

devastation n.
严重破坏；浩劫
the devastation emerged piecemeal over

piecemeal adv.
一点一点地
the devastation emerged piecemeal over

haze n.
薄雾；霾
haze that circled the equator in about

reconstruction n.
重建；历史状况复原
Later climate reconstructions suggest

discount v.
不予采信；低估
Some experts discounted the volcanic

elicit v.
引起；引出
eliciting worldwide replies;

far-flung adj.
相隔遥远的
far-flung observers began to glimpse

interconnected adj.
相互关联的
how interconnected the planet already

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

volcanic adj.
火山的

strait n.
海峡

islander n.
岛民

roar n.
轰鸣；咆哮声

distant adj.
远处的

gunfire n.
炮火声；枪炮射击

restless adj.
躁动不安的

emptied adj.
被排空的

magma n.
岩浆

vanish v.
消失

tsunami n.
海啸

coastal adj.
沿海的

settlement n.
居民点；聚居地

colonial adj.
殖民地的

administrator n.
行政官员

drown v.
溺亡

telegram n.
电报

submarine adj.
海底的

cable n.
电缆

chronicle v.
记述；报道

dispatch n.
电讯；发回的报道

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

ash n.
灰；此处指火山灰

sulphur n.
硫

hurl v.
猛抛；抛射

equator n.
赤道

crimson adj.
深红色的

glow v.
发光；泛出光彩

summon v.
召来

firefighter n.
消防员

phantom adj.
虚幻的；并不存在的

blaze n.
大火

dip v.
小幅下降

sunset n.
晚霞；日落

moisture n.
水汽；水分

at high altitude prep.phr.
在高空

cosmic adj.
宇宙的

Royal Society proper n.
英国皇家学会

appeal v.
呼吁

tie to phr.v.
与to连用：把……联系起来

to prep.
标示联系到的对象

observer n.
观察者；观测者

glimpse v.
初步认识到；窥见

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 _____ its most thunderous explosion.

02 Tsunamis, _____ over 30 metres high in places,

03 Yet early dispatches were _____;

04 how _____ the planet already was.

文章主线

- 巨响 1883年喀拉喀托火山喷发，声传近4800公里；全球气压计录到压力波，有些约5天录到7次。
- 灾难 约三分之二岛体塌入已排空的岩浆房并没入海中；海啸据报局部逾30米，官方记36417人死亡，多数溺亡，或仍低估。
- 快讯 海底电缆让电报数小时到伦敦，消息迅速全球传播；但早期报道简短，灾情规模数日后才逐渐清楚。
- 余波 火山灰和二氧化硫形成高空薄雾，远方出现红霞；后来的气候重建提示，1884年全球降温零点几度。
- 考证 有人把红霞归于高空水汽或宇宙尘埃；皇家学会汇集全球观测，1888年报告将红霞与喷发联系起来。

参考答案　1 unleashed 2 reportedly 3 terse 4 interconnected

光之殇

Lost in the Glare

昆虫绕着灯飞，真只是因为喜欢光吗？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

A moth skims over an ultraviolet lamp and flips upside down, belly to the sky. In Costa Rica's misty Monteverde cloud-forest, stereo cameras caught such acrobatics for a 2024 Imperial College study that unravelled an age-old puzzle, the helpless swirl of insects round porch lights.

Folk wisdom blames the circling on attraction. In fact, insects instinctively keep their backs to whatever shines brightest, normally the sky, to stay level. A lamp, a counterfeit sky, hijacks this built-in habit, leaving insects eddying in its halo or inverted above it. In 477 field clips, precious few darted straight at the light.

Many fireflies flirt with their lanterns. A male blinks an overture, a female winks back, and bright nights smother the banter. At Tufts University, suitors under artificial light mustered about five flash patterns a minute instead of ten. Females reciprocated about half as often under red or blue light, scarcely at all under white, and never under amber.

The harm spills into the ecosystem. Along 55 Thames Valley roadsides, lamp-lit hedges held 47% fewer moth caterpillars than dark ones, with harsh white LEDs worse than mellower sodium lamps. Street lamps rigged up in seven of 14 Swiss meadows cut nocturnal pollinator visits by 62%. The plants set fewer fruit, and daytime pollinators proved poor understudies.

Over 80% of humanity lives beneath a luminous veil, which hides the pale ribbon of the Milky Way from more than a third. Citizen scientists' star counts from 2011 to 2022 show the sky brightening about 9.6% a year. At that pace, a child born where 250 stars twinkle would see about 100 by their eighteenth birthday.

中文译文

一只飞蛾掠过紫外灯，突然翻转，肚皮朝向天空。2024年伦敦帝国理工学院的一项研究，在哥斯达黎加蒙特韦尔德雾气弥漫的云雾林中，用立体摄像机捕捉到这样的飞行动作，解开了一个古老谜团：昆虫为什么会身不由己地绕着门廊灯盘旋。

人们通常认为，昆虫绕灯飞是受到灯光吸引。实际上，为保持身体平稳，昆虫会本能地让背部朝向最亮的方向，通常就是天空。人造灯冒充了天空，扰乱这种天生习性，使昆虫在光晕中打转，或在灯上方翻倒。477段野外录像中，真正径直冲向光源的昆虫寥寥无几。

许多萤火虫用发光来求偶：雄虫先闪光，雌虫再回应，太亮的夜晚却会淹没这番交流。在塔夫茨大学的实验中，人造光下的雄虫每分钟只能发出约五组闪光，而原本约有十组。实验中的雌虫在红光或蓝光下回应的次数约减半，在白光下几乎没有回应，在琥珀色光下则完全没有回应。

影响还延伸到生态系统。对泰晤士河谷55处路边的调查发现，有灯照明的树篱里，蛾类毛虫比无灯处少47%；刺眼的白色LED处，情况比柔和的钠灯处更严重。在瑞士14片草甸中的7片临时架起路灯后，夜间传粉者的造访减少62%。植物结的果实也更少，白天的传粉者未能补足夜间的损失。

全球超过80%的人口生活在发光的夜空帷幕下，超过三分之一的人看不到银河那条淡淡的光带。参与观星的公众在2011年至2022年间记录的星数表明，夜空每年约变亮9.6%。按这一速度，一个孩子出生时若能在当地看到250颗星，到18岁时便大约只能看到100颗。

树篱毛虫少47%来自路边调查；草甸的62%来自架灯实验，不应把两种证据都写成同一种因果证明。

核心词汇

先记这22项。连着下方的原文短语一起回想。

skim over 动词短语
掠过
A moth skims over an ultraviolet lamp

unravel v.
解开（谜团）
unravelled an age-old puzzle

age-old adj.
由来已久的
an age-old puzzle

folk wisdom n.
民间流传的观念
Folk wisdom blames the circling

attraction n.
吸引
blames the circling on attraction

instinctively adv.
本能地
insects instinctively keep their backs

counterfeit adj.
假冒的
a counterfeit sky

hijack v.
劫持；此处指扰乱
hijacks this built-in habit

built-in adj.
与生俱来的
this built-in habit

inverted adj.
翻转倒置的
inverted above it

precious few 数量短语
少之又少
precious few darted straight

reciprocate v.
回应
Females reciprocated

ecosystem n.
生态系统
The harm spills into the ecosystem

caterpillar n.
毛虫；蛾蝶幼虫
47% fewer moth caterpillars

harsh adj.
刺眼的；强烈的
harsh white LEDs

sodium lamp n.
钠灯
mellower sodium lamps

rig up 动词短语
临时架设
Street lamps rigged up

nocturnal adj.
夜间的
nocturnal pollinator visits

understudy n.
替补
proved poor understudies

luminous adj.
发光的
a luminous veil

citizen n.
公民；此处指公众参与者
Citizen scientists' star counts

pace n.
速度
At that pace

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

belly n.
腹部

misty adj.
雾气弥漫的

cloud forest n.
云雾林

stereo adj.
立体成像的

acrobatics n.
杂技般的动作

helpless adj.
身不由己的

swirl n.
盘旋；打转

porch n.
门廊

eddy v.
打旋

halo n.
光晕

clip n.
视频片段

dart v.
猛冲；疾驰

flirt v.
求偶；调情

lantern n.
灯笼；此处喻发光器官

blink v.
闪烁

overture n.
序曲；此处喻求偶信号

wink v.
眨眼；此处喻闪光回应

smother v.
压制；淹没

banter n.
打趣；此处喻求偶交流

suitor n.
求偶者

muster v.
勉强发出（闪光）

amber n.
琥珀色

spill into 动词短语
蔓延到

roadside n.
路边

lamp-lit adj.
有灯照明的

hedge n.
树篱

mellow adj.
较柔和的

meadow n.
草甸

humanity n.
全人类

veil n.
帷幕

ribbon n.
带状物；光带

twinkle v.
闪烁

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In fact, insects _____ keep their backs to whatever shines brightest, normally the sky, to stay level.

02 A lamp, a _____ sky, hijacks this built-in habit, leaving insects eddying in its halo or inverted above it.

03 Females _____ about half as often under red or blue light, scarcely at all under white, and never under amber.

04 Street lamps rigged up in seven of 14 Swiss meadows cut _____ pollinator visits by 62%.

文章主线

- 迷向 昆虫本能地背朝最亮处以保持平稳；2024年研究发现，灯冒充天空，扰乱这种习性，使昆虫绕灯打转或翻身。
- 求偶 萤火虫靠闪光求偶；实验中人造光使雄虫闪光约减半，雌虫在红蓝光下回应减半，白光下极少、琥珀光下全无。
- 幼虫 英国55处路边调查中，有灯树篱的蛾类毛虫比暗处少47%；白色LED的影响重于钠灯。
- 传粉 瑞士草甸架灯实验使夜间传粉访问减少62%、植物结实减少；白天传粉未能补足夜间损失。
- 夜空 全球逾八成人口受光污染影响，逾三分之一看不见银河；2011—2022年公众观星记录显示，夜空年均增亮约9.6%。

参考答案　1 instinctively 2 counterfeit 3 reciprocated 4 nocturnal

冰海求生：坚忍号传奇

Survival in the Ice: The Legend of Endurance

冰海中的精确计算，怎样换来生路？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 5 December 1914 Endurance departed South Georgia, a remote whaling outpost, for Antarctica's Weddell Sea. In January 1915 ice trapped the vessel. On 27 October, her hull buckling, the men abandoned ship. She foundered on 21 November; her captain, Frank Worsley, logged the spot.

The stranded men endured over five gruelling months on the ice. Their floe split on 9 April 1916; on 15 April their lifeboats reached Elephant Island, a bleak, uninhabited rock far from shipping lanes.

On 24 April Shackleton and five others embarked in the James Caird, a fragile lifeboat, on a perilous voyage of some 1,300 kilometres. As ferocious gales battered the boat, Worsley snatched only four sights of the sun, deducing the rest by dead reckoning from course, speed, and time elapsed. On 10 May they landed on South Georgia's deserted south coast.

Shackleton, Worsley, and Tom Crean traversed its mountainous interior in about 36 exhausting hours, reaching the Stromness whaling station on 20 May. Ice repelled three rescue attempts before the Yelcho, a Chilean tug, evacuated the 22 marooned men on 30 August. All 28 of Endurance's party survived their ordeal.

On 5 March 2022 searchers located the wreck 3,008 metres down, upright and intact. She lay about four miles south of the spot Worsley recorded in November 1915, months before the voyage: a slight discrepancy for a calculation on shifting ice.

Legend rightly salutes Shackleton for his charisma. Yet both ends of the story hinge on Worsley's meticulous arithmetic. Leadership held 28 men together. Precision put six ashore within reach of help and, a century on, brought their ship back into view.

中文译文

1914年12月5日，坚忍号驶离偏远的捕鲸据点南乔治亚岛，前往南极威德尔海。1915年1月，海冰困住了船。10月27日，船体受压变形，众人弃船；11月21日，坚忍号沉没，船长弗兰克·沃斯利记下了沉没地点。

自弃船起，被困的人们在冰上熬过五个多月的艰苦日子。1916年4月9日，他们脚下的浮冰裂开。4月15日，救生艇抵达象岛，那是一座荒凉、无人居住、远离航线的岩岛。

4月24日，沙克尔顿和另外五人登上单薄的救生艇詹姆斯·凯尔德号，开始约1300公里的危险航程。猛烈大风不断冲击小艇，沃斯利只抓住四次机会观测太阳定位，其余时间靠航向、航速和经过的时间作航位推算。5月10日，他们登上南乔治亚岛空无一人的南岸。

沙克尔顿、沃斯利和汤姆·克林用了约36小时，艰苦横越岛上多山的腹地，于5月20日抵达斯特罗姆内斯捕鲸站。浮冰逼退了三次救援尝试，直到8月30日，智利拖船耶乔号才将象岛上受困的22人接走。坚忍号这一队的28人全都挺过磨难、生还归来。

2022年3月5日，搜寻者在3008米深的海底发现沉船，它保持正立，完好无损。沉船位于沃斯利在1915年11月记录的位置以南约四英里；那是小艇求援航程开始前几个月记下的位置。在漂移冰面上完成的计算只有这点偏差，已属难得。

后世因沙克尔顿的个人感召力称颂他，确有道理。但故事的前后两端，都取决于沃斯利一丝不苟的计算。领导力让28人团结在一起，精确计算则让六人登岸，进入可以求援的范围；一个世纪后，又让他们的船重新进入世人的视野。

28人指坚忍号这一队；1915年记录的沉船位置早于1916年的求援航程。

核心词汇

先记这22项。连着下方的原文短语一起回想。

hull n.
船体
her hull buckling

buckle v.
受压弯曲；变形
her hull buckling

founder v.
沉没
foundered on 21 November

stranded adj.
受困的；滞留的
The stranded men

gruelling adj.
艰苦难熬的
over five gruelling months

floe n.
浮冰块
Their floe split

bleak adj.
荒凉的；凄冷的
a bleak, uninhabited rock

embark v.
登船；启程
embarked in the James Caird

perilous adj.
危险重重的
a perilous voyage

ferocious adj.
猛烈的；凶猛的
As ferocious gales

deduce v.
推算；推断
deducing the rest

dead reckoning 名词短语
航位推算，按航向等估计位置
by dead reckoning

traverse v.
横越；穿过
traversed its mountainous interior

repel v.
逼退；阻退
Ice repelled three rescue attempts

evacuate v.
撤出；疏散
evacuated the 22 marooned men

marooned adj.
困在荒岛上的
evacuated the 22 marooned men

ordeal n.
磨难；严酷考验
survived their ordeal

intact adj.
完好无损的
upright and intact

discrepancy n.
偏差；差异
a slight discrepancy

charisma n.
个人魅力；感召力
for his charisma

hinge on 动词短语
取决于
hinge on Worsley's

meticulous adj.
一丝不苟的；细致的
Worsley's meticulous arithmetic

文中词汇与搭配

补充词汇共45项。保留每个表达在本文中的意思。

depart v.
驶离；离开

remote adj.
偏远的

whaling n.
捕鲸

outpost n.
偏远据点；前哨站

for prep.
前往；开往

trap v.
困住

vessel n.
船舶

abandon v.
弃船；放弃

log v.
记入航海日志

endure v.
忍受；熬过

lifeboat n.
救生艇

uninhabited adj.
无人居住的

shipping lane 名词短语
航线

in prep.
表示所乘的船

fragile adj.
单薄易损的；脆弱的

on prep.
表示所踏上的航程

voyage n.
航程；航海

gale n.
大风；强风

batter v.
反复猛击

snatch v.
抓住机会；匆忙取得

sight of the sun 名词短语
观测太阳以确定位置

course n.
航向

文中词汇与搭配 续

补充词汇共45项。保留每个表达在本文中的意思。

time elapsed 名词短语
经过的时间

land on 动词短语
在……登陆；登上

deserted adj.
空无一人的

mountainous adj.
多山的

interior n.
腹地；内部

exhausting adj.
令人筋疲力尽的

rescue attempt 名词短语
救援尝试

tug n.
拖船

locate v.
找到；确定位置

wreck n.
沉船残骸

upright adj.
正立的；直立的

calculation n.
计算

shifting adj.
漂移的；不断移动的

rightly adv.
理所应当地；有道理地

salute v.
称颂；致敬

arithmetic n.
算术；计算

leadership n.
领导力

hold v.
使保持；使团结

together adv.
在一起；团结一致

precision n.
精准；精确性

ashore adv.
上岸；在岸上

bring v.
使进入视野；带来

back adv.
重新；返回

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 On 27 October, her hull _____, the men abandoned ship.

02 deducing the rest by _____ from course, speed, and time elapsed.

03 She lay about four miles south of the spot Worsley recorded in November 1915, months before the voyage: a slight _____ for a calculation on shifting ice.

04 Yet both ends of the story hinge on Worsley's _____ arithmetic.

文章主线

- 受困 1915年，坚忍号被南极海冰困住，10月弃船、11月沉没；船长沃斯利记下位置，众人自弃船起在浮冰上熬过五个多月。
- 求援 浮冰破裂后，众人乘救生艇抵象岛；岛远离航线，沙克尔顿等六人再乘小艇航行约1300公里，到南乔治亚岛求援。
- 导航 沃斯利仅四次观测太阳定位，其余靠航向、速度和时间推算；登岸后，三人再翻山约36小时找到捕鲸站。
- 获救 三次救援被冰逼退，智利拖船于1916年8月接回象岛22人，坚忍号28人全部生还。
- 寻回 2022年沉船在旧记录点以南约四英里处找到；沙克尔顿凝聚众人，沃斯利的计算则既助求援，也助寻船。

参考答案　1 buckling 2 dead reckoning 3 discrepancy 4 meticulous

一枚邮票改变通信

The Stamp That Changed the Post

一枚邮票怎样简化寄信，又怎样防止重复使用？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Before 1840, British postage was levied according to the number of sheets in a letter and the mileage of its carriage. The sum was usually payable on delivery, so the person receiving the correspondence, not the sender, settled it. Prices thus fluctuated from one postal item to the next.

In 1837 Rowland Hill published a booklet urging reform. Parliament enacted the campaigner's proposal in 1839, and on 10 January 1840 the Uniform Penny Post began. A letter of up to half an ounce now cost one penny to any destination in Britain, and the fee was prepaid by whoever sent it. One flat-rate charge replaced variation with simplicity.

The Penny Black, a one-penny denomination, was released on 1 May 1840, its validity starting on 6 May. Engraved on it was a youthful likeness of Queen Victoria, the reigning monarch. An aggregate of 68,808,000 were printed. The sheets had no holes for separation, so each stamp was cut out individually with scissors.

Mailings doubled, from about 80 million items in 1839 to over 169 million in 1840. The design had a drawback. Against black, the red cancelling mark was barely noticeable, lacking contrast, and the ink could be erased, leaving stamps vulnerable to reuse. In February 1841 the Penny Red, cancelled in black ink, arrived to deter tampering and fraud.

As the initiator of postage stamps, Britain set a precedent and earned an exemption. Its stamps are nameless, with no country inscribed on them. By international agreement, the sovereign's head suffices for identification, an implicit designation that other countries spell out in print.

中文译文

1840年以前，英国邮费按信内纸张数量与运输里程收取，通常在投递时付款。因此，结清邮费的往往是收信人，而非寄信人，每件邮件的价格也不相同。

1837年，罗兰·希尔出版小册子，呼吁改革。议会于1839年通过了他的提议，统一便士邮资在1840年1月10日开始实行：半盎司（约14克）以内的信，寄往英国境内任何地方都只收一便士，由寄信人预付。一项统一费率让原来各不相同的收费变得简单。

面值一便士的黑便士邮票于1840年5月1日发行，5月6日起可用于邮寄。邮票上刻有当时在位君主维多利亚女王的年轻肖像，共印制68,808,000枚。整版邮票没有用来分开的齿孔，每一枚都要用剪刀单独剪下。

邮寄量翻了一番，从1839年的约8000万件增加至1840年的1.69亿多件。但设计也有缺点：红色盖销印在黑底上缺少反差，很不显眼，而且红色墨迹能被擦掉，邮票因此容易被重复使用。1841年2月，改用黑色墨水盖销的红便士邮票问世，以阻止篡改和欺诈。

英国作为邮票首创国，开创先例，也获得一项特例：邮票上可以不印国名。按照国际协议，只凭君主头像就足以识别国家；这是一种隐含的国家标识，其他国家则需要把国名直接印出来。

核心词汇

先记这22项。连着下方的原文短语一起回想。

postage n.
邮费
British postage was levied

levy v.
征收
levied according to the number

mileage n.
里程
the mileage of its carriage

payable adj.
应支付的
usually payable on delivery

correspondence n.
信件
receiving the correspondence

fluctuate v.
波动；各不相同
Prices thus fluctuated

enact v.
立法通过
Parliament enacted

proposal n.
提议；方案
the campaigner's proposal

prepay v.
预付
the fee was prepaid

flat-rate charge 名词短语
统一费率收费
One flat-rate charge

denomination n.
面值
a one-penny denomination

validity n.
效力；有效期
its validity starting on 6 May

aggregate n.
总计；总数
aggregate of 68,808,000

mailing n.
邮寄件数；邮件
Mailings doubled

drawback n.
缺点
The design had a drawback

cancelling mark 名词短语
邮票盖销印记
the red cancelling mark

contrast n.
反差
lacking contrast

vulnerable to 形容词短语
容易遭到；易受……影响
vulnerable to reuse

deter v.
阻止；遏制
to deter tampering and fraud

fraud n.
欺诈
to deter tampering and fraud

precedent n.
先例
Britain set a precedent

exemption n.
豁免；特例
earned an exemption

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

sheet n.
纸张

carriage n.
运输

delivery n.
投递

sender n.
寄信人

postal adj.
邮政的

item n.
一件邮件；物品

booklet n.
小册子

campaigner n.
改革倡导者

ounce n.
盎司

penny n.
便士

destination n.
目的地

variation n.
差异；变化

simplicity n.
简便

engrave v.
雕刻；刻印

youthful adj.
年轻的

likeness n.
肖像

reigning adj.
在位的

monarch n.
君主

separation n.
分离

individually adv.
单独地

scissors n.
剪刀

item n.
一件件邮件；物品

noticeable adj.
显眼的

erase v.
擦除

reuse n.
重复使用

tampering n.
篡改；擅自改动

initiator n.
首创者；发起者

nameless adj.
此处指未印国名的

inscribe v.
刻印

sovereign n.
君主

suffice for 动词短语
足以用于

identification n.
识别

implicit adj.
隐含的

designation n.
标识；名称

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 and the fee was _____ by whoever sent it.

02 An _____ of 68,808,000 were printed.

03 The design had a _____.

04 Britain set a precedent and earned an _____.

文章主线

- 旧制 英国邮费原按信纸数量和运输距离计算，通常由收信人付款，每封信的价格不一。
- 改革 1840年1月起，希尔推动的改革让半盎司（约14克）以内的信在英国境内统一收一便士，由寄件人预付。
- 邮票 同年5月发行印有维多利亚女王头像的黑便士；当年邮寄量由上年约8000万件增至1.69亿多件。
- 改进 黑便士上的红色邮戳不显眼，红戳又可擦掉，使邮票容易被重用；1841年改发红便士并盖黑戳防重用。
- 特例 英国因首创邮票获国际特例，可不印国名，以君主头像标明国家；其他国家则要把国名印出。

参考答案　1 prepaid 2 aggregate 3 drawback 4 exemption

林下之网

The Web Beneath the Woods

树木的地下联系，哪些已有证据?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

One of ecology's most romantic ideas grew out of plastic bags. In a Canadian forest, researchers enclosed paper birch and Douglas fir seedlings in them, then bathed each species in carbon dioxide laced with a different tracer. Carbon shuttled both ways, apparently via fungi linking the trees' roots, yet the firs ran a slight surplus, larger in deeper shade.

Most terrestrial plants harbour such fungi, whose filaments forage for water and nutrients including nitrogen, swapping them for plant sugars. One fungus can knit plants into a shared network. The journal Nature unveiled the study on 7 August 1997, its cover offering a catchy coinage: the "wood-wide web".

The metaphor proved irresistible. Between 2015 and 2021, a forester's bestseller, an acclaimed novel, and an ecologist's memoir told readers that trees converse, warn one another, and, as "mother trees", nurture their kin. Yet the 1997 trees were seedlings, not elderly mothers.

In February 2023, three researchers combed through the field evidence. They judged it too sparse, inconsistent, or ambiguous to demonstrate that shared networks are widespread. In 26 field studies, networks helped and hindered seedlings about equally often, with neutral effects predominating. No peer-reviewed field study showed mature trees favouring their own offspring. Unsupported assertions in later papers had doubled over 25 years, with citations skewed towards positive results.

None of this disproves the networks' existence. In January 2025, defenders faulted the critique's methods. The critics concede that carbon and nutrients can migrate between trees underground. It is evidence unearthed in real forests, not the allure of a family saga, that must settle how much the networks convey, and whether it matters.

中文译文

生态学中一个格外浪漫的观念，竟起源于塑料袋。在加拿大的一片森林里，研究人员把纸桦和花旗松的幼苗罩进袋中，分别让两种幼苗接触掺有不同示踪物的二氧化碳。碳在两种树之间双向转移，似乎经由连接根系的真菌传递。不过，花旗松最终获得了少量净盈余，在遮荫更深的地方，盈余也更大。

大多数陆生植物都与这类真菌共存。真菌的菌丝寻找水分和包括氮在内的养分，再拿这些资源换取植物的糖分。一个真菌就能将多株植物连成共享网络。《自然》于1997年8月7日发表这项研究，封面还用上了一个好记的新词：“林木万维网”。

这个比喻实在太有吸引力。2015年至2021年间，一位护林人的畅销书、一部广受好评的小说和一位生态学家的回忆录，都告诉读者：树木能够交谈、相互示警，还会像“母树”一样养育亲族。可1997年实验中的树只是幼苗，并不是年迈的母亲。

2023年2月，三位研究者系统梳理了野外证据。他们认为，这些证据或太少、或互不一致、或含糊不清，不足以证明共享网络普遍存在。在26项野外研究中，帮助与妨碍幼苗的结果出现得大致同样频繁，而最多见的是没有明显影响。没有一项经同行评审的野外研究表明，成年树木会偏爱自己的后代。后续论文中缺乏依据的断言在25年间翻了一倍，引用也偏向正面结果。

这些批评并不能否定网络的存在。2025年1月，网络说的拥护者质疑了那篇批评文章的方法；批评者也承认，碳和养分能够在树木之间经地下迁移。网络究竟传输了多少、这种传输在生态上有多大作用，必须由真实森林中的证据来判定，不能取决于一段“树木家族传奇”有多动人。

“无明显影响最多”与“帮助、妨碍的次数接近”并不矛盾；复核未证明网络不存在。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ecology n.
生态学
One of ecology's most romantic ideas

seedling n.
幼苗
paper birch and Douglas fir seedlings

tracer n.
示踪物
a different tracer

apparently adv.
似乎；看来
apparently via fungi

surplus n.
盈余
a slight surplus

terrestrial adj.
陆生的
Most terrestrial plants

filament n.
丝状体；此处指菌丝
whose filaments forage for water

nutrient n.
养分
water and nutrients including nitrogen

metaphor n.
比喻
The metaphor proved irresistible

nurture v.
养育；培育
nurture their kin

comb through 动词短语
仔细梳理；彻底搜查
combed through the field evidence

sparse adj.
稀少的
too sparse, inconsistent, or ambiguous

inconsistent adj.
互不一致的
inconsistent, or ambiguous

ambiguous adj.
含糊不清的
inconsistent, or ambiguous

demonstrate v.
证明
to demonstrate that shared networks

widespread adj.
普遍存在的
shared networks are widespread

neutral adj.
无明显正负影响的
neutral effects predominating

peer-reviewed adj.
经同行评审的
No peer-reviewed field study

assertion n.
断言
Unsupported assertions

citation n.
引文；引用
with citations skewed

concede v.
承认（先前不愿承认的事）
The critics concede

convey v.
传输；输送
how much the networks convey

文中词汇与搭配

补充词汇共50项。保留每个表达在本文中的意思。

romantic adj.
富有浪漫色彩的

grow out of 动词短语
源自；由……发展而来

enclose v.
把……围住；罩住

in prep.
在……里面

bathe v.
使浸在；使充分接触

be laced with 动词短语
掺有

shuttle v.
往返移动

fungus n.
真菌

link v.
连接

slight adj.
少量的；轻微的

shade n.
遮荫

harbour v.
容纳；携带

forage for 动词短语
搜寻（水或养分）

nitrogen n.
氮

swap v.
交换

knit v.
联结；编织

into prep.
进入；连成

shared network n.
共享网络

unveil v.
公布；发表

catchy adj.
好记而吸引人的

coinage n.
新造词

irresistible adj.
难以抗拒的

bestseller n.
畅销书

acclaimed adj.
广受赞誉的

memoir n.
回忆录

文中词汇与搭配 续

补充词汇共50项。保留每个表达在本文中的意思。

converse v.
交谈

warn one another 动词短语
互相警告

kin n.
亲族

elderly adj.
年迈的

hinder v.
妨碍

predominate v.
占多数

mature adj.
成熟的；成年的

favour v.
偏爱

offspring n.
后代

unsupported adj.
缺乏依据的

double v.
翻倍

skew v.
使偏向

towards prep.
朝；偏向

positive result n.
正面结果

disprove v.
证明……不成立

existence n.
存在

defender n.
拥护者

fault v.
指责；挑出问题

critique n.
批评性分析

critic n.
批评者

migrate v.
迁移

unearth v.
发掘

allure n.
吸引力

saga n.
传奇故事

matter v.
有重要意义

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Carbon shuttled both ways, apparently via fungi linking the trees' roots, yet the firs ran a slight _____, larger in deeper shade.

02 The _____ proved irresistible.

03 In 26 field studies, networks helped and hindered seedlings about equally often, with _____ effects predominating.

04 The critics _____ that carbon and nutrients can migrate between trees underground.

文章主线

- 发现 1997年，示踪实验发现纸桦和花旗松幼苗双向交换碳，似乎经连接根系的真菌传递。
- 走红 真菌以水和养分换植物糖分，也能连接多株植物；“林木万维网”走红后，被演绎成母树哺育亲族。
- 复核 2023年复核认为网络普遍存在的证据不足；26项野外研究以无明显影响为主，帮助和阻碍幼苗的次数相近。
- 偏差 复核未找到母树偏爱亲生后代的同行评审野外证据；论文偏引正面结果，无依据断言在25年间翻倍。
- 争议 2025年拥护者质疑复核方法；批评者也承认树间有地下物质转移，但网络输送多少、生态作用多大仍待实证。

参考答案　1 surplus 2 metaphor 3 neutral 4 concede

向鸸鹋宣战

The Great Emu War

机枪为何没能解决鸸鹋危机?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Around Campion, Western Australia, most farmers were First World War veterans, settlers placed by a government scheme on marginal parcels of farmland. They were inexperienced growers, and a quarter had already quit. In 1932, after drought, roughly 20,000 emus swarmed in from arid country in flocks and trampled the wheat crops.

The farmers' plight took a military turn. In desperation, a delegation of ex-soldiers aired their grievances and asked the Defence Minister for machine-guns. He agreed, sending not a battalion but a detachment: Major Meredith, an artillery officer, with two soldiers, two Lewis guns, and 10,000 rounds of ammunition. The offensive began on 2 November 1932.

At a dam, about a thousand birds approached an ambush, but a gun jammed after a dozen fell. The flocks broke formation into small, agile groups, frustrating each tactical plan, and a truck-mounted gun proved futile. Six days of wasteful firing spent 2,500 rounds on perhaps 50 to 200 birds, a ludicrous return. After parliamentary questions and press ridicule, an embarrassing withdrawal came on 8 November, though shooting restarted within days.

Meredith was recalled on 10 December. His memo's inventory counted 986 kills from 9,860 rounds, a ratio of ten bullets to one bird. He likened the resilient birds' toughness to the armour of tanks. Five weeks of warfare had barely dented the hardy flocks' endurance.

Later pleas for reinforcements were refused, and intervention gave way to a bounty sanctioned by the government. This payout per bird gave farmers compensation and an incentive for extermination. Farmers claimed over 57,000 payments in six months of 1934, more than fifty-seven times Meredith's 986 kills.

中文译文

在西澳大利亚州坎皮恩一带，多数农民是第一次世界大战的退伍军人；政府通过一项安置计划，让这些定居者在一块块贫瘠农地上定居。他们缺少种植经验，已有四分之一放弃了耕作。1932年旱灾过后，约两万只鸸鹋从干旱地区成群涌入，践踏小麦庄稼。

农民的困境由此走向了军事解决。在绝望之中，一支由退伍士兵组成的代表团公开诉说不满，向国防部长请求机枪。部长答应了；派来的却不是一个营，而是一支小分队：炮兵军官梅雷迪思少校、两名士兵、两挺刘易斯机枪，以及一万发弹药。行动于1932年11月2日开始。

在一座水坝旁，约一千只鸸鹋接近伏击地点；但只打倒十来只，一挺机枪便卡了壳。鸟群散成灵活的小群，使一个个战术计划落空，卡车上架设的机枪也未能奏效。六天中低效的射击消耗了2500发子弹，可能只打死50至200只鸟，成效实在令人啼笑皆非。议会提出质询，报界加以嘲讽，部队于11月8日尴尬撤退，不过几天后便又恢复了射击。

梅雷迪思于12月10日被召回。他的备忘录列出的清单显示，9860发子弹打死986只鸟，相当于每只鸟耗费十发子弹。他把这些鸟顽强的抗击打能力比作坦克装甲。五个星期的战事，几乎没能削弱这些顽强鸟群的耐力。

此后，请求增援的恳求都遭到拒绝，军事干预让位于政府批准的悬赏制度。按每只鸟发放的赏金，既给农民提供了补偿，也成为扑杀鸸鹋的激励。1934年的六个月内，农民申领了五万七千多笔赏金，数量是梅雷迪思所报986只击杀数的五十七倍多。

核心词汇

先记这22项。连着下方的原文短语一起回想。

veteran n.
退伍军人
First World War veterans

marginal adj.
贫瘠的；不适宜耕作的
on marginal parcels of farmland.

arid adj.
干旱的
from arid country in flocks

plight n.
困境；苦境
farmers' plight

delegation n.
代表团
a delegation of ex-soldiers

grievance n.
不满；委屈
their grievances

detachment n.
小分队；分遣队
not a battalion but a detachment

ammunition n.
弹药
10,000 rounds of ammunition.

ambush n.
伏击；埋伏
approached an ambush

agile adj.
灵活的；敏捷的
small, agile groups

frustrate v.
使受挫；使落空
frustrating each tactical plan

futile adj.
徒劳的；无效的
proved futile.

ludicrous adj.
荒唐可笑的
a ludicrous return.

withdrawal n.
撤退；撤离
an embarrassing withdrawal

ratio n.
比例；比率
a ratio of ten bullets to one bird.

resilient adj.
顽强的；承受力强的
the resilient birds' toughness

reinforcement n.
增援部队；援兵
pleas for reinforcements

intervention n.
干预；介入
intervention gave way to a bounty

give way to phr. v.
让位于；被……取代
intervention gave way to a bounty

bounty n.
赏金；悬赏
a bounty sanctioned by the government.

compensation n.
补偿；赔偿
gave farmers compensation

incentive n.
激励；诱因
an incentive for extermination.

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

settler n.
定居者；移民

parcel n.
一块（土地）

farmland n.
农地

inexperienced adj.
缺乏经验的

grower n.
种植者

swarm v.
蜂拥而至

in flocks phr.
成群地（尤指鸟）

trample v.
践踏；踩坏

crop n.
庄稼；农作物

in desperation phr.
在绝望之中

air v.
公开表达

machine-gun n.
机枪

battalion n.
营（军队编制）

artillery n.
炮兵；火炮

offensive n.
进攻；攻势

jam v.
卡住；卡壳

formation n.
队形

tactical adj.
战术的

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

wasteful adj.
浪费的

parliamentary adj.
议会的

ridicule n.
嘲笑；嘲讽

embarrassing adj.
令人尴尬的

memo n.
备忘录

inventory n.
清单

bullet n.
子弹

liken v.
把……比作

toughness n.
坚韧；耐打程度

to prep.
与liken连用，表示比作

armour n.
装甲

warfare n.
战事；战争

hardy adj.
顽强的；耐受力强的

endurance n.
耐力；持久力

plea n.
恳求

sanction v.
批准；准许

payout n.
支付的款项

extermination n.
彻底消灭；扑灭

payment n.
支付款项

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The farmers' _____ took a military turn.

02 After parliamentary questions and press ridicule, an embarrassing _____ came on 8 November, though shooting restarted within days.

03 His memo's inventory counted 986 kills from 9,860 rounds, a _____ of ten bullets to one bird.

04 Later pleas for _____ were refused, and intervention gave way to a bounty sanctioned by the government.

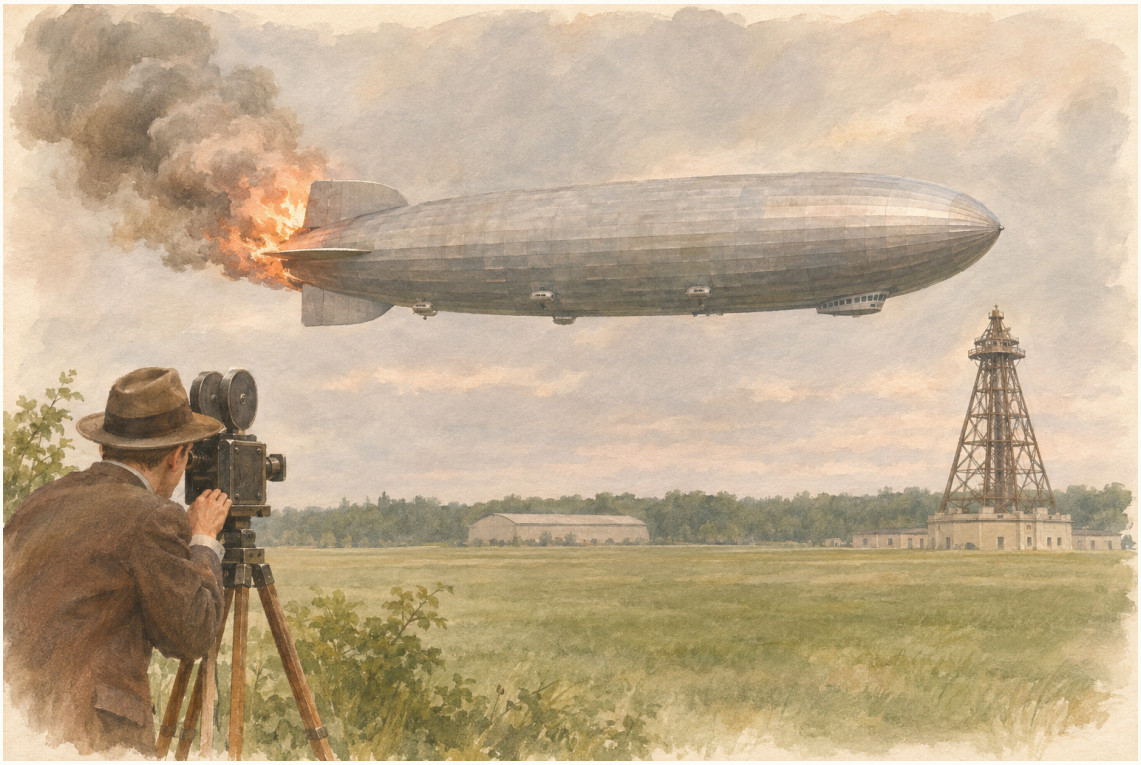
文章主线

- 困境 1932年旱灾后，约两万只鸸鹋涌入西澳农田、践踏庄稼，当地退伍军人农户向军方求助。
- 出兵 军方派一名少校和两名士兵，携两挺机枪、一万发子弹，试图控制鸟群。
- 受挫 鸟群分散成灵活小群，机枪卡壳，车载射击也无效，行动受到媒体和议会嘲讽。
- 撤军 部队一度撤离又复战，五周后被召回；报告称9860发子弹击杀986只，仍未解决鸟患。
- 转向 政府拒绝增援，改按只发赏金；1934年半年内农民申领逾5.7万笔，远超军方战果。

参考答案　1 plight 2 withdrawal 3 ratio 4 reinforcements

兴登堡号的 34 秒

The Last Flight of the Hindenburg



一场被拍下的大火，怎样终结了飞艇客运时代？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1936 the colossal Hindenburg made 17 Atlantic round trips, a marvel of luxury and glamour. Its amenities included a lounge and, that year, a grand piano. An American embargo on helium export had left the ship relying on hydrogen. Because that gas was combustible, smokers sat behind an airlock with one electric lighter.

The airship left Frankfurt on 3 May 1937. Hours behind its timetable, it met thunderstorms at Lakehurst and circled for over an hour before its descent towards the mooring mast. At about 7:25 pm on 6 May, flames appeared near the stern. The inferno incinerated the ship in about half a minute, leaving a crumpled frame.

Of the 97 aboard, 62 survived, a proportion close to two-thirds, while a minority of 35 died, as did one man on the ground. A 14-year-old cabin boy was drenched by a deluge from a burst water tank, and the soaking saved him. He dropped through a hatch and scrambled clear, escaping only narrowly.

Newsreel cameras and photographers caught the disaster. A radio correspondent's eyewitness commentary was not live; audiences heard it the next day. The cause has not been definitively established, but static electricity igniting leaking hydrogen is the leading explanation. A theory blaming the doped fabric has been rejected.

The 1933 crash of USS Akron killed 73 yet won less notoriety. The filmed fire left a fateful image that abruptly wrecked confidence. As public perception soured, Graf Zeppelin was withdrawn, later scrapped, and a glamorous epoch of aviation closed. The doomed airship's first journey to North America had begun on 6 May 1936, exactly a year before the fire.

中文译文

1936年，庞大的兴登堡号完成17次跨大西洋往返，堪称奢华而迷人的奇观。艇内设有休息厅，那一年还配过一架三角钢琴。由于美国禁止出口氦气，飞艇只能依靠氢气升空。氢气易燃，吸烟者只能待在气闸隔开的区域，共用一个电点烟器。

1937年5月3日，飞艇离开法兰克福。它抵达莱克赫斯特时已晚点数小时，又遇上雷暴，盘旋一个多小时后才开始向系留塔下降。5月6日晚约7点25分，尾部附近出现火焰。大火在约半分钟内焚毁飞艇，只留下一副扭曲的骨架。

艇上97人中，62人生还，接近三分之二；35人遇难，地面另有一名男子死亡。一名14岁的艇上服务生被破裂水箱涌出的水浇透，这一身湿水救了他。他从舱口落下，连滚带爬逃到安全处，险些未能脱身。

新闻纪录片摄像机和摄影师拍下了灾难。电台记者的现场目击解说并非直播，听众到第二天才听到。起火原因至今没有定论，主要解释是静电点燃泄漏的氢气。把起火归咎于蒙皮织物涂层的说法已被否定。

1933年阿克伦号失事造成73人死亡，却没有那么广为人知。被拍下的大火留下了一幅影响深远的画面，骤然摧毁公众信心。随着公众看法恶化，格拉夫·齐柏林号退出运营，后来被拆解，航空史上一个充满魅力的时代就此落幕。兴登堡号首次飞往北美的航程始于1936年5月6日，恰好是失火前整整一年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

colossal adj.
庞大的
the colossal Hindenburg

amenity n.
便利舒适设施
Its amenities included a lounge

embargo n.
禁运
An American embargo on helium export

combustible adj.
易燃的
that gas was combustible

descent n.
下降
its descent towards the mooring mast

mooring mast 名词短语
飞艇系留塔
towards the mooring mast

stern n.
飞艇尾部
near the stern

inferno n.
猛烈的大火
The inferno incinerated the ship

incinerate v.
焚毁
The inferno incinerated the ship

crumple v.
压皱；扭曲变形
leaving a crumpled frame

proportion n.
比例
a proportion close to two-thirds

drench v.
浇透；使湿透
was drenched by a deluge

deluge n.
大量涌出的水
a deluge from a burst water tank

hatch n.
舱口
He dropped through a hatch

correspondent n.
新闻记者
A radio correspondent's

eyewitness adj.
亲眼目击的
eyewitness commentary was not live

commentary n.
现场解说
eyewitness commentary was not live

definitively adv.
确切地；有定论地
has not been definitively established

static electricity 名词短语
静电
static electricity igniting

leading explanation 名词短语
主要解释
is the leading explanation

perception n.
看法；观感
As public perception soured

epoch n.
时代
a glamorous epoch of aviation

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

round trip 名词短语
往返航程

marvel n.
令人惊叹的事物

luxury n.
奢华

glamour n.
魅力

lounge n.
休息厅

helium n.
氦气

export n.
出口

hydrogen n.
氢气

smoker n.
吸烟者

sit behind 动词短语
坐在……隔开的后方

electric lighter 名词短语
电点烟器

airship n.
飞艇

timetable n.
时刻表

thunderstorm n.
雷暴

circle v.
盘旋

before prep.
在……之前

flame n.
火焰

frame n.
骨架；框架

two-thirds 数词
三分之二

a minority of 名词短语
少于半数的

cabin boy 名词短语
艇上少年服务生

burst v.
爆裂

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

soaking n.
全身湿透

scramble v.
急忙攀爬；连滚带爬

narrowly adv.
勉强；险些未能

photographer n.
摄影师

audience n.
听众

ignite v.
点燃

leak v.
泄漏

theory n.
解释；假说

doped adj.
涂有涂料的

fabric n.
蒙皮织物

reject v.
否定；不接受

notoriety n.
因坏事而广为人知

fateful adj.
影响深远而不幸的

image n.
画面

abruptly adv.
突然地

confidence n.
信心

sour v.
变坏；恶化

withdraw v.
撤出运营

scrap v.
报废拆解

glamorous adj.
富有魅力的

aviation n.
航空

doomed adj.
注定毁灭的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 An American _____ on helium export had left the ship relying on hydrogen.

02 Because that gas was _____,

03 The cause has not been _____ established,

04 As public _____ soured,

文章主线

- 隐患 兴登堡号是豪华跨洋飞艇；美国禁运氦气，使它只能靠易燃氢气浮空，留下安全隐患。
- 失事 1937年5月6日，它在美国准备停靠系留塔时尾部起火，约半分钟就烧成扭曲骨架。
- 伤亡 艇上97人中62人生还、35人遇难；地面另有1人死亡，遇难者合计36人。
- 原因 火因尚无定论，主要解释是静电点燃泄漏氢气；归咎外层布料涂层的说法被否定。
- 落幕 1933年另一飞艇事故死73人，却不及兴登堡号出名；这次影像和次日播出的录音重创信心，飞艇客运时代落幕。

参考答案　1 embargo 2 combustible 3 definitively 4 perception

比信先到家的鸟

The Bird That Beat the Letter Home

能辨方向的鸟，怎样知道自己在哪里、家又在哪里？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 3 June 1952 Rosario Mazzeo released a Manx shearwater in Boston. It came by rail and airliner from its burrow on Skokholm, an isle off Wales, and Mazzeo posted a letter at once. After a roughly 3,200-mile odyssey across the unbroken Atlantic, the feathered traveller was back on its doorstep twelve and a half days later. The letter arrived afterwards.

The first hints of a compass were celestial. In the 1950s Kramer kept starlings cooped in cages and moved the apparent sun with mirrors. The duped birds swivelled to match, allowing for the sun's daily arc. In the 1960s Emlen curtained off stars in a planetarium's vault. His buntings steered by stars around the pole-star, the axis of the nightly whirl.

In 1972 the Wiltschkos showed that robins read Earth's magnetic field by inclination, not polarity, unlike a sailor's needle. In 1978 Schulten proposed a light-triggered reaction to decode the field. Proposed in 2000, cryptochrome, a protein in the retina, is now the foremost suspect. In 2021 robin cryptochrome proved more finely tuned in the laboratory than that of chickens or pigeons. Proof in a living bird is awaited.

In 2014 came an eerie, man-made twist. At Oldenburg University, robins in campus huts lost that sense. In huts clad in grounded aluminium, they reclaimed their bearings. Removing the grounding, or piping radio noise in, bewildered them again in double-blind tests. The interference, a murmur from everyday appliances, sat far below WHO safety limits.

How a wanderer knows where it is remains unanswered. The compass code is partly cracked, but the atlas that steered the shearwater has yet to be read.

中文译文

1952年6月3日，罗萨里奥·马泽奥在波士顿放飞一只大西洋鹱。它原先栖居在威尔士外海斯科克霍姆岛的巢穴中，经火车和客机运到这里；马泽奥放飞后立即寄出一封信。跨越约3200英里连绵无际的大西洋后，这位长着羽毛的旅客于十二天半后回到巢门口。信却更晚才到。

最早的“罗盘”线索来自天体。20世纪50年代，克雷默把椋鸟关在笼中，用镜子改变鸟看到的太阳位置。受骗的鸟随之转向，同时还能把太阳每日运行的轨迹考虑进去。20世纪60年代，埃姆伦在天文馆穹顶上遮挡部分星星；他的鸲鸟依靠北极星周围的星星辨向，那里是夜空看似每夜旋转的中心。

1972年，维尔奇科夫妇证明，知更鸟依靠地磁倾角，而不是磁场极性来感知地磁，这与水手的罗盘指针不同。1978年，舒尔滕提出，它们可能利用光触发的反应解读磁场。2000年，有人提出视网膜中的隐花色素蛋白可能参与其中，如今它已成为头号候选者。2021年，实验室研究显示，知更鸟隐花色素的磁感反应，比鸡或鸽子的更精细；但仍需在活鸟体内证明这一机制。

2014年，事情出现了一个奇异的人为转折。奥尔登堡大学校园小屋中的知更鸟失去了这种定向感；在包覆接地铝板的小屋内，它们却恢复了方向感。双盲实验中，无论断开接地，还是引入无线电噪声，都让鸟再次迷向。干扰来自日常电器产生的微弱电磁噪声，强度远低于世卫组织的安全限值。

鸟儿究竟怎样知道自己身在何处，仍无答案。罗盘的密码已被破解一部分，但指引那只大西洋鹱回家的“地图”，还没有人读懂。

地磁倾角是地磁方向与水平面的夹角。文中的世卫安全限值针对人体暴露，不能据此判断鸟类定向是否受干扰。

核心词汇

先记这22项。连着下方的原文短语一起回想。

odyssey n.
漫长艰辛的旅程
a roughly 3,200-mile odyssey

celestial adj.
天体的；天空的
hints of a compass were celestial

apparent sun 名词短语
鸟所看到的太阳位置
moved the apparent sun with mirrors

allow for 动词短语
考虑到；作出相应调整
allowing for the sun's daily arc

axis n.
轴心
the axis of the nightly whirl

magnetic field 名词短语
磁场
Earth's magnetic field

inclination n.
地磁倾角；地磁方向与水平面的夹角
by inclination, not polarity

polarity n.
磁场的南北极性
by inclination, not polarity

light-triggered adj.
由光触发的
a light-triggered reaction

decode v.
解读；破解
to decode the field

retina n.
视网膜
a protein in the retina

foremost adj.
最主要的；首要的
the foremost suspect

suspect n.
待验证的主要候选对象
the foremost suspect

await v.
等待；有待取得
Proof in a living bird is awaited

reclaim v.
找回；恢复
they reclaimed their bearings

bearings n.
方向感；方位
they reclaimed their bearings

bewilder v.
使迷向；使不知所措
bewildered them again

interference n.
电磁干扰
The interference, a murmur

appliance n.
家用电器
everyday appliances

unanswered adj.
尚无答案的
remains unanswered

crack v.
破解
The compass code is partly cracked

atlas n.
地图集；此处比喻导航地图
the atlas that steered the shearwater

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

release v.
放飞；释放

rail n.
铁路；火车运输

airliner n.
民航客机

burrow n.
地洞；洞穴式巢穴

isle n.
小岛

post a letter 动词短语
寄出一封信

at once 介词短语
立即

unbroken adj.
连续无间的；连绵的

Atlantic 专名
大西洋

feathered adj.
长着羽毛的

traveller n.
旅行者

doorstep n.
门口；此处比喻巢口

afterwards adv.
后来；随后

hint n.
线索；迹象

coop v.
关在狭小空间里

cage n.
笼子

dupe v.
欺骗

swivel v.
转动方向

arc n.
弧线；运行轨迹

curtain off 动词短语
用帘子遮挡

vault n.
穹顶

steer v.
辨向；调整方向

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

pole-star n.
北极星

nightly adj.
每夜的

whirl n.
旋转

robin n.
知更鸟

sailor n.
水手

needle n.
罗盘指针

finely adv.
精细地

tune v.
调节；使敏锐

pigeon n.
鸽子

eerie adj.
奇异得令人不安的

man-made adj.
人为的

twist n.
意外转折

hut n.
小屋

sense n.
感知能力

clad in 分词短语
包覆着

aluminium n.
铝

pipe v.
输送；引入

in adv.
进来；与pipe连用

murmur n.
微弱声响；此处比喻微弱噪声

safety limit 名词短语
安全限值

wanderer n.
漫游者

have yet to 动词短语
尚未

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The first hints of a compass were _____.

02 robins read Earth's magnetic field by _____, not polarity

03 Proof in a living bird is _____.

04 The compass code is partly _____

文章主线

- 归巢 大西洋鹱被运到波士顿放飞，12天半跨越约3200英里回巢，比放飞者随即寄出的信还早。
- 罗盘 用镜子改变鸟看到的太阳位置、在天文馆遮星的实验，分别证实不同鸟类能借太阳和星空辨向。
- 机制 知更鸟按地磁倾角辨向；眼内隐花色素蛋白在实验室显示磁感潜力，可能通过光触发的反应感磁。
- 干扰 双盲实验中，低于世卫限值的电器电磁干扰也会使知更鸟迷向；接地铝板恢复辨向，断开接地又迷失。
- 未解 隐花色素感磁尚待活鸟验证；能辨方向不等于知道自己在哪，鸟如何确定回巢路线仍未解。

参考答案　1 celestial 2 inclination 3 awaited 4 cracked

大恶臭逼出了下水道

The Stink That Built London's Sewers

下水道本为消除臭味，为什么还能减少疾病?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In the fierce heat of summer 1858, the curtains of the new Houses of Parliament hung heavy with chloride of lime, a bleach to make the air tolerable. The Thames festered, and MPs contemplated moving out.

The river had become London's lavatory. Antiquated sewers and outlets discharged the untreated sewage of some three million city dwellers into its currents, where the excrement stewed and decayed. The vile fumes grew so overpowering that the episode earned a name, the Great Stink.

Disgust found a voice on 15 July, when the Chancellor of the Exchequer, Benjamin Disraeli, denounced the river as a hellish pool of unbearable horrors. A bill funding new sewers was fast-tracked into law in about 18 days.

The engineer Joseph Bazalgette masterminded the response. His 82 miles of stout intercepting sewers caught the flow short of the river and diverted it east, downstream of the city. With some 1,100 miles of street sewers as tributaries, the infrastructure took 318 million bricks. Having calculated an adequate calibre for each pipe, he doubled it. The backbone lies buried underneath the newly built Victoria, Albert, and Chelsea Embankments. Substantially complete by 1875, much of the network is in use today.

Yet that sanitation rested on a fallacy. The predominant belief, miasma, blamed disease on foul air and vapours, so the sewers were intended principally to suppress the smell. They worked for an unintended reason, guarding the purity of the water Londoners drank.

Cholera resurfaced in 1866 and killed about 5,600 people in the East End, the one locality not yet connected.

A remedy aimed at the nose protected the gut.

中文译文

1858年夏天酷热难耐。新议会大厦的窗帘浸了含氯漂白粉，沉甸甸地垂着，只为让空气勉强可以忍受。泰晤士河腐败发臭，议员们考虑搬离。

这条河已成伦敦的厕所。老旧下水道和排污口，把约300万市民未经处理的污水排进河中，粪便在暑热中沤积、腐烂。恶臭强烈得令人无法忍受，这场事件因此有了一个名字：“大恶臭”。

7月15日，财政大臣本杰明·迪斯雷利替这份厌恶发了声，公开斥责河流是一潭地狱般、恐怖得让人无法忍受的污水。为新下水道拨款的法案加速审议，约18天便成为法律。

工程师约瑟夫·巴泽尔杰特主持设计这套应对工程。82英里坚固的截流下水道，在污水流入河中之前将其截住，改道向东，送往城市下游。约1100英里的街道下水道汇入这套系统，整项基础设施用了3.18亿块砖。他算出各条管道所需的内径后，又将直径加倍。系统主干埋在当时新建的维多利亚、阿尔伯特和切尔西河堤之下。管网到1875年基本建成，大部分至今仍在使用。

然而，这项公共卫生工程建立在一个错误观念之上。当时占主导的“瘴气说”把疾病归因于污浊空气与蒸气，因此修下水道主要是为了压住臭味。它们却因另一个并非设计初衷的原因而奏效：保护了伦敦人的饮用水水质。

1866年霍乱再次出现，在伦敦东区夺走约5600条生命；那里正是唯一尚未接入管网的地区。

原本冲着鼻子去的补救措施，保护的却是肠胃。

核心词汇

先记这22项。连着下方的原文短语一起回想。

tolerable adj.
可忍受的
to make the air tolerable

antiquated adj.
陈旧过时的
Antiquated sewers and outlets

discharge v.
排放
discharged the untreated sewage

untreated adj.
未经处理的
the untreated sewage

sewage n.
污水
the untreated sewage

denounce v.
公开斥责
denounced the river

bill n.
法案
A bill funding new sewers

fast-track v.
加速办理；加快审议
fast-tracked into law

mastermind v.
主持策划
masterminded the response

intercepting sewer 名词短语
截流下水道
stout intercepting sewers

divert v.
使改道
diverted it east

infrastructure n.
基础设施
the infrastructure took

adequate adj.
足够的
an adequate calibre

calibre n.
管道内径
an adequate calibre for each pipe

sanitation n.
公共卫生设施
that sanitation rested on a fallacy

fallacy n.
错误观念；谬误
rested on a fallacy

predominant adj.
占主导的
The predominant belief

suppress v.
压制；消除
to suppress the smell

unintended adj.
非本意的；未预期的
an unintended reason

purity n.
洁净
the purity of the water

resurface v.
再次出现
Cholera resurfaced in 1866

aim at 动词短语
针对；旨在改善
A remedy aimed at the nose

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

fierce heat 名词短语
酷热

Houses of Parliament 专名
英国议会大厦

lime n.
石灰；文中为含氯漂白剂成分

bleach n.
漂白剂

fester v.
腐败发臭

contemplate v.
考虑

lavatory n.
厕所

outlet n.
排污口

dweller n.
居民

current n.
水流

excrement n.
粪便

stew v.
在闷热中沤积

decay v.
腐败；腐烂

vile adj.
恶心的；恶劣的

fume n.
难闻的气体

overpowering adj.
强烈得难以忍受的

episode n.
事件；一段经历

stink n.
恶臭

disgust n.
厌恶

Chancellor of the Exchequer 专名
英国财政大臣

as prep.
是；引出斥责中的定性

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

hellish adj.
地狱般的

unbearable adj.
难以忍受的

stout adj.
坚固的

short of 介词短语
在到达……之前

downstream adv.
向下游

tributary n.
支流；汇入主干的支管

brick n.
砖块

double v.
使加倍

backbone n.
主干

bury v.
埋在地下

underneath prep.
在……下面

newly adv.
新近地

substantially adv.
基本上；大体上

in use 介词短语
在使用中

rest on 动词短语
建立在……之上

belief n.
观念；看法

foul air 名词短语
污浊空气

vapour n.
蒸气

principally adv.
主要地

locality n.
地区

gut n.
肠胃

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The engineer Joseph Bazalgette _____ the response.

02 Having calculated an adequate calibre for each pipe, he _____ it.

03 Yet that sanitation rested on a _____.

04 A remedy _____ the nose protected the gut.

文章主线

- 危机 1858年酷暑，伦敦未经处理的污水在泰晤士河发臭，恶臭逼得议员考虑搬离议会。
- 行动 财政大臣公开抨击后，为新下水道拨款的法案约18天获通过，工程师巴泽尔杰特主持建设。
- 工程 82英里截流管拦下原本入河的污水，改送城市下游；按需要算出的管径再加倍。
- 误判 修下水道本为消除人们误认为会致病的臭气；实际作用是减少饮用水遭污水污染。
- 对照 1866年，唯一尚未接网的东区有约5600人死于霍乱；管网1875年基本完工，多数设施至今仍用。

参考答案　1 masterminded 2 doubled 3 fallacy 4 aimed at

原子弹之父的审判

The Trial of the Father of the Atomic Bomb

造出原子弹的科学家，为何被撤销涉密资格？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

J. Robert Oppenheimer, born in New York on 22 April 1904, was an accomplished physicist and prominent lecturer at Berkeley and Caltech. From April 1943 he oversaw a secretive weapons installation at Los Alamos, whose bombs were the collaborative creation of a thousands-strong workforce.

In the pre-dawn darkness of 16 July 1945, the first nuclear test illuminated the desolate New Mexico landscape with a blinding glare, its explosive force roughly 20 kilotons. On 6 and 9 August 1945 atomic bombings devastated Hiroshima and Nagasaki, where year-end fatalities numbered well over 100,000 and about 70,000 respectively.

Oppenheimer left Los Alamos in October 1945 and from 1947 presided over Princeton's Institute for Advanced Study. In October 1949 the advisory committee he chaired, expressing misgivings, unanimously counselled against a crash programme for the hydrogen bomb, a vastly more destructive weapon. In December 1953 a presidential directive severed his access to confidential documentation.

From 12 April to 6 May 1954 he was cross-examined at a security hearing, an administrative proceeding, not a criminal prosecution. On 29 June 1954 the Atomic Energy Commission voted 4 to 1 against letting him regain his clearance, two days before his consultancy contract was due to expire. The ruling did not find him disloyal, but nonetheless questioned his character and associates.

On 2 December 1963 President Johnson bestowed on him the Fermi Award, which Kennedy had authorised before his assassination. Oppenheimer died on 18 February 1967. In December 2022 the energy secretary officially vacated the 1954 determination, calling the process flawed and contrary to the commission's rules. The stigma of the revocation had persisted for fifty-five years beyond his lifetime.

中文译文

J.罗伯特·奥本海默于1904年4月22日生于纽约。他是卓有成就的物理学家，也是伯克利和加州理工学院的知名教师。从1943年4月起，他主管洛斯阿拉莫斯的一处秘密武器设施，那里的原子弹是数千名工作人员协作的成果。

1945年7月16日黎明前，首次核试验的刺眼强光照亮了新墨西哥州荒凉的地貌，爆炸威力约为2万吨TNT当量。1945年8月6日和9日，原子弹分别重创广岛、长崎；截至当年年底，两地死亡人数分别远超10万和约7万。

奥本海默于1945年10月离开洛斯阿拉莫斯，从1947年起执掌位于普林斯顿的高等研究院。1949年10月，他主持的顾问委员会表达疑虑，一致建议反对一项加速研制氢弹的紧急计划；这种武器的破坏力要大得多。1953年12月，一道总统指令切断了他接触机密文件的权限。

1954年4月12日至5月6日，他在安全听证会上接受盘问。这是行政程序，而非刑事起诉。6月29日，原子能委员会以4比1投票反对恢复他的涉密资格，距离他的顾问合同到期仅剩两天。裁决没有认定他不忠，却仍然质疑他的品格和交往对象。

1963年12月2日，约翰逊总统向他颁授费米奖；肯尼迪遇刺前已经批准授奖。奥本海默于1967年2月18日去世。2022年12月，能源部长正式撤销1954年的裁定，称当年的程序存在缺陷，违反了委员会的规定。撤销涉密资格所带来的污名，在他去世后又延续了55年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

accomplished adj.
卓有成就的
an accomplished physicist

prominent adj.
知名的；杰出的
prominent lecturer

oversee v.
主管；监督
he oversaw

collaborative adj.
共同协作的
the collaborative creation

workforce n.
全体工作人员；劳动力
a thousands-strong workforce

illuminate v.
照亮
the first nuclear test illuminated

devastate v.
重创；摧毁
devastated Hiroshima and Nagasaki

fatality n.
死亡人数；死者
year-end fatalities

preside over 动词短语
主持；掌管
presided over Princeton's Institute

misgiving n.
疑虑；担忧
expressing misgivings

unanimously adv.
一致地
unanimously counselled against

counsel against 动词短语
建议不要；劝阻
counselled against a crash programme

directive n.
指令
a presidential directive

confidential adj.
机密的
confidential documentation

hearing n.
听证会
a security hearing

administrative adj.
行政的
an administrative proceeding

prosecution n.
刑事起诉
not a criminal prosecution

clearance n.
接触机密的许可
regain his clearance

expire v.
到期
was due to expire

vacate v.
撤销裁定
vacated the 1954 determination

flawed adj.
有缺陷的
calling the process flawed

stigma n.
污名
The stigma of the revocation

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

lecturer n.
授课教师

secretive adj.
秘密的；隐秘的

installation n.
大型专用设施

bomb n.
炸弹

creation n.
创造成果

thousands-strong adj.
有数千人的

pre-dawn adj.
黎明前的

desolate adj.
荒凉的

landscape n.
地貌；景观

blinding glare 名词短语
刺眼的强光

explosive adj.
爆炸的

bombing n.
轰炸

year-end adj.
截至年底的；年末的

respectively adv.
分别

Oppenheimer 专有名词
奥本海默

advisory adj.
顾问的；咨询的

crash programme 名词短语
紧急加速计划

hydrogen n.
氢

vastly adv.
大大地

destructive adj.
破坏性强的

presidential adj.
总统的

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

sever v.
切断

access to 名词短语
接触或使用……的权限

documentation n.
文件；书面资料

cross-examine v.
交叉盘问；详细询问

proceeding n.
程序；审理过程

atomic energy 名词短语
原子能

regain v.
重新获得

consultancy n.
顾问工作

disloyal adj.
不忠的

nonetheless adv.
尽管如此；仍然

associate n.
交往的人；同事

bestow on 动词短语
授予

authorise v.
批准；授权

assassination n.
暗杀；遇刺

secretary n.
此处指部长

officially adv.
正式地

determination n.
裁定；正式决定

contrary to 介词短语
违反；与……相反

revocation n.
撤销许可等

persist v.
持续存在

lifetime n.
一生

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the first nuclear test _____ the desolate New Mexico landscape

02 expressing _____,

03 an _____ proceeding, not a criminal prosecution.

04 calling the process _____ and contrary to the commission's rules.

文章主线

- 造弹 物理学家奥本海默领导数千人造出原子弹，1945年完成首次核试验，随后原子弹用于轰炸广岛、长崎。
- 分歧 战后，他主持的顾问委员会一致反对紧急推进氢弹计划；氢弹是一种比原子弹破坏力更强的核武器。
- 听证 1954年他接受安全听证，这是行政审查，并非刑事起诉；裁定也未认定他不忠。
- 失权 原子能委员会以4比1拒绝恢复他接触机密资料的资格，质疑其品格和交往，使其背负污名。
- 撤销 1963年虽获费米奖，1954年裁定直到2022年才由能源部长以程序违规为由撤销，距他去世已55年。

参考答案　1 illuminated 2 misgivings 3 administrative 4 flawed

一颗刺果钩出的发明

The Burrs That Became Velcro

衣服上的刺果，怎样变成了日常搭扣？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1941 George de Mestral, a Swiss engineer, returned from hunting in the Alps with his hound, an Irish Pointer. Prickly burdock seed-heads had hitched themselves to his clothes and lay tangled in the dog's fur. Under a microscope he saw thousands of tiny barbs, each a hook grasping the small loops in cloth.

Few took his idea seriously at first. In Lyon, then a weaving centre, a cotton version proved too frail to survive abrasion. Working with a textiles firm, he turned to nylon, whose hooks were stiff enough to hold yet easy to pull apart. The trial-and-error at the looms and mills lasted about ten years.

The name Velcro merged two French words, velours, meaning velvet, and crochet, meaning hook. A US patent duly followed in 1955. In 1959 a New York fashion show sent apparel stitched with the fasteners down the runway. The public was less charmed than expected, mostly because of the aesthetic.

The underdog proved its utility in outer space. NASA used it in the Apollo programme to keep gear anchored in weightlessness, so that nothing would hover in mid-air. Apollo 11 was equipped with a sizeable amount, about 3,300 square inches. NASA did not invent the nylon clasp.

When the patent ran out in 1978, cut-price knock-offs from other countries overran the market, and a niche product went mainstream. In 2017, to stop its brand-name melting into a generic noun, the company asked people to call such mimics hook and loop fasteners. In effect, the firm was asking the public to stop using its own name for rivals' goods.

中文译文

1941年，瑞士工程师乔治·德·梅斯特拉尔带着自己的猎犬——一只爱尔兰指示犬——从阿尔卑斯山狩猎归来。带刺的牛蒡种子头钩在衣服上，也缠进狗毛里。他用显微镜观察，发现上面有成千上万根微小倒刺，每根都像钩子，扣住织物上的小毛圈。

起初，几乎没人认真对待他的想法。他在当时的纺织中心里昂试制棉制搭扣，但它太脆弱，经不住磨损。与一家纺织公司合作时，他改用尼龙：这种材料的钩子足够挺硬，既能扣牢，又容易拉开。织机旁、工厂里的反复试验持续了约十年。

Velcro这个名字合并了两个法语词：velours意为天鹅绒，crochet意为钩子。1955年，美国专利随之获批。1959年，纽约一场时装秀让缝有这类搭扣的服装登上T台。然而公众的反应不如预期热烈，主要原因是外观。

这件起初不被看好的发明，在外太空证明了用途。美国航天局在阿波罗计划中用它把装备固定住，避免装备在失重环境里漂浮。阿波罗11号配备的搭扣面积相当可观，约3300平方英寸。但这种尼龙扣件并非美国航天局发明。

1978年专利到期后，其他国家的廉价仿品涌入市场，小众产品也走向大众。2017年，为防品牌名逐渐变成整类产品的通用名词，公司要求人们把这类仿品称为“钩毛搭扣”。实际上，它是在请公众不要再用自家的名字称呼竞争对手的商品。

核心词汇

先记这22项。连着下方的原文短语一起回想。

barb n.
倒刺
thousands of tiny barbs

grasp v.
抓住；扣住
a hook grasping the small loops

loop n.
毛圈；环
the small loops in cloth

frail adj.
脆弱的
a cotton version proved too frail

abrasion n.
摩擦造成的磨损
to survive abrasion

nylon n.
尼龙
he turned to nylon

stiff adj.
挺硬的；不易弯曲的
hooks were stiff enough to hold

pull apart 动词短语
拉开；扯开
easy to pull apart

trial and error n.
反复试验；试错
The trial-and-error

merge v.
合并
merged two French words

apparel n.
服装
apparel stitched with the fasteners

fastener n.
扣件；搭扣
apparel stitched with the fasteners

aesthetic n.
外观；审美风格
mostly because of the aesthetic

utility n.
实用性；用途
proved its utility in outer space

anchor v.
固定住
keep gear anchored in weightlessness

hover v.
悬浮；漂浮
hover in mid-air

equip with 动词短语
配备
was equipped with a sizeable amount

run out 动词短语
期限届满
the patent ran out in 1978

niche adj.
小众的
a niche product went mainstream

mainstream adj.
主流的；大众化的
a niche product went mainstream

generic adj.
通用的；非特定品牌的
a generic noun

in effect 副词短语
实际上
In effect, the firm

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

Swiss adj.
瑞士的

hound n.
猎犬

prickly adj.
多刺的

seed-head n.
含种子的花序；种子头

hitch v.
钩住；挂住

tangle v.
缠结

fur n.
动物皮毛

hook n.
钩子

take v.
与seriously连用，认真对待

seriously adv.
认真地

weaving n.
织造；纺织

textile n.
纺织品

loom n.
织机

mill n.
工厂；纺织厂

velvet n.
天鹅绒

duly adv.
如期；按预期

stitch v.
缝上

runway n.
时装表演的T台

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

charm v.
使着迷；吸引

mostly adv.
主要地；大多

underdog n.
不被看好的一方

outer space 名词短语
外太空

Apollo programme 专有名词
阿波罗计划

gear n.
装备

in weightlessness 介词短语
在失重环境中

mid-air n.
半空中

sizeable adj.
相当大的；可观的

clasp n.
扣件

cut-price adj.
廉价的；降价的

knock-off n.
仿制品

overrun v.
大量涌入；占满

brand name n.
品牌名

melt into 动词短语
逐渐融为；变成

noun n.
名词

mimic n.
仿制品

rival n.
竞争对手

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Under a microscope he saw thousands of tiny _____,

02 he turned to _____, whose hooks were stiff enough to hold yet easy to pull apart.

03 The underdog proved its _____ in outer space.

04 a niche product went _____.

文章主线

- 启发 瑞士工程师梅斯特拉尔发现刺果用微钩扣住衣物毛圈，便仿照它设计可反复扣合、拉开的搭扣。
- 试制 棉制搭扣不耐磨，他改用尼龙，反复试验约十年，1955年获得美国专利。
- 应用 公众主要因外观而反应冷淡；美国航天局在阿波罗计划中用搭扣固定失重中的装备，并非搭扣的发明者。
- 普及 1978年专利到期后，廉价仿品涌入市场，原本小众的搭扣逐渐成为日常用品。
- 正名 Velcro是品牌名；2017年品牌方要求将仿品称为钩毛搭扣，避免自家商标泛指整类产品。

参考答案　1 barbs 2 nylon 3 utility 4 mainstream

破解霍乱之谜的医生

The Doctor Who Solved the Cholera Mystery

破解霍乱之谜，关键是地图还是饮水？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In the first ten days of September 1854, cholera ravaged an overcrowded corner of London's Soho; upwards of 500 people perished. Most doctors attributed epidemics to miasma, foul air arising from filth. John Snow, a physician who had been convinced since 1849 that cholera's poison was swallowed, not inhaled, suspected the Broad Street pump.

Finding little visible impurity, he hesitated, then consulted the death register and questioned bereaved families. Most victims lived near the pump and drank from it. At his urging, parish officials removed the handle, but daily fatal attacks had already plunged from a peak of 143 to 12. He later mapped the deaths.

In 1855 Edmund Parkes, a professor of clinical medicine, interpreted the cluster on that map as a poison drifting through the air. Yet a map could chart where people died, not what they drank.

The crucial evidence, gathered in 1854, lay in exceptions. Amid the deaths in the surrounding streets, a workhouse with its own well lost a mere five of 535 inmates. More than 70 nearby brewery workers, with a beer allowance and their own well, escaped severe cholera.

Conversely, a Hampstead widow who had not set foot in Soho for months had the pump's water carted to her daily. She died, as did a visiting niece who drank it. Hampstead then had no cholera.

In 1855 a government inquiry declined to condemn the pump, contending its water had at most absorbed airborne poison. Meanwhile, the sceptical curate Henry Whitehead set out to refute Snow but found that those who drank from the pump were far more likely to succumb to the disease than others, vindicating him.

中文译文

1854年9月的头十天，霍乱在伦敦苏豪区一片人口过密的街区肆虐，500多人丧生。多数医生把流行病归因于瘴气，也就是污秽中产生的恶臭空气。医生约翰·斯诺自1849年起便确信，霍乱的致病物是被吞入人体，而不是随空气吸入；他因此怀疑宽街的公共水泵。

泵水中几乎没有肉眼可见的杂质，这让他一度犹豫。随后，他查阅死亡登记册，走访失去亲人的家庭。多数死者住在水泵附近，也喝过那里的水。在他的敦促下，教区官员拆下了泵柄；但在此之前，每天新发的致死病例数，就已经从高峰的143例骤降到12例。他后来才把死亡地点标在地图上。

1855年，临床医学教授埃德蒙·帕克斯把地图上的死亡聚集现象，解释成毒物在空气中飘散的结果。然而，地图只能记录人们死在哪里，并不能说明他们喝过什么。

真正关键的证据早在1854年就已收集到，存在于那些例外之中。周边街巷死亡接连发生，一所自备水井的济贫院，535名被收容者中却只有5人死亡。附近啤酒厂的70多名工人，既享有啤酒配给，也有自备水井，没有患上重症霍乱。

与之相反，汉普斯特德的一位寡妇，已经好几个月没有踏入苏豪区，却每天让人把宽街水泵的水用车运到家中。她去世了；来探望她、也喝了这些水的侄甥女，同样去世了。而汉普斯特德当时并没有流行霍乱。

1855年，政府的一次调查拒绝认定水泵应当负责，认为泵水充其量只是吸收了空气中的毒物。与此同时，原本持怀疑态度的助理牧师亨利·怀特黑德，着手寻找证据来驳倒斯诺，却发现喝过泵水的人，死于这种疾病的可能性远高于其他人，反倒印证了斯诺的判断。

niece可指侄女或外甥女，原文未进一步区分，译文以“侄甥女”统称。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ravage v.
肆虐；严重毁坏
cholera ravaged an overcrowded corner

attribute v.
归因于
doctors attributed epidemics

foul adj.
恶臭的；污浊的
foul air arising from filth.

physician n.
医生
John Snow, a physician

convince v.
使确信；此处指深信不疑
convinced since 1849

consult v.
查阅
then consulted the death register

bereaved adj.
失去亲人的
questioned bereaved families.

fatal adj.
致死的
daily fatal attacks

plunge v.
骤降
had already plunged

interpret v.
解释；解读
interpreted the cluster

cluster n.
聚集；密集的一群
the cluster on that map

crucial adj.
至关重要的
The crucial evidence

exception n.
例外
lay in exceptions.

conversely adv.
相反地；反过来
Conversely, a Hampstead widow

inquiry n.
调查；官方调查
a government inquiry

decline to phr. v.
拒绝做……
declined to condemn the pump

contend v.
主张；坚持认为
contending its water had

airborne adj.
经空气传播的
airborne poison.

sceptical adj.
持怀疑态度的
the sceptical curate Henry Whitehead

refute v.
驳倒；反驳
set out to refute Snow

succumb to phr. v.
死于（疾病）；屈服于
succumb to the disease

vindicate v.
证明……正确；印证
vindicating him.

文中词汇与搭配

补充词汇共35项。保留每个表达在本文中的意思。

overcrowded adj.
人口过密的；过度拥挤的

upwards of phr.
超过

perish v.
丧生

epidemic n.
流行病；疫情

to prep.
与attribute连用，表示归因于

arise v.
产生；源于

filth n.
污秽

swallow v.
吞入

inhale v.
吸入

hesitate v.
犹豫

register n.
登记册

victim n.
受害者；此处指死者

urging n.
敦促；催促

parish n.
教区

peak n.
峰值；最高点

clinical medicine n.
临床医学

poison n.
毒物；毒素

drift v.
飘散

文中词汇与搭配 续

补充词汇共35项。保留每个表达在本文中的意思。

chart v.
绘制；记录分布

lie in phr. v.
在于；存在于

amid prep.
在……之中

surrounding adj.
周围的

workhouse n.
济贫院

a mere phr.
仅仅；区区

inmate n.
被收容者

allowance n.
配给；补助

widow n.
寡妇

set foot in phr.
踏入

cart v.
用车运送

niece n.
侄女；外甥女

then adv.
当时

condemn v.
谴责；认定应负责任

absorb v.
吸收

meanwhile adv.
与此同时

set out to phr. v.
着手去；打算去

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 At his urging, parish officials removed the handle, but daily _____ attacks had already plunged from a peak of 143 to 12.

02 In 1855 Edmund Parkes, a professor of clinical medicine, interpreted the _____ on that map as a poison drifting through the air.

03 The _____ evidence, gathered in 1854, lay in exceptions.

04 _____, a Hampstead widow who had not set foot in Soho for months had the pump's water carted to her daily.

文章主线

- 旧说 1854年伦敦苏豪区霍乱十日夺命逾500人；多数医生归因臭气，斯诺则怀疑饮水传播。
- 追查 他查死亡记录、访问家属，发现许多死者喝过宽街水泵的水，促使官员拆下泵柄。
- 对照 自备井的济贫院仅少数死亡，啤酒厂无人患重症；远处无疫情地区，却有两人喝泵水后死亡。
- 辨别 每日致死病例数在拆泵柄前已大幅减少；地图只标死亡地点，关键是比较人们是否喝过泵水。
- 验证 官方调查仍坚持空气传播；原想驳倒斯诺的牧师怀特黑德却发现喝泵水者更易死亡，支持了斯诺。

参考答案　1 fatal 2 cluster 3 crucial 4 Conversely

拥挤的苍穹

The Crowded Heavens

两颗卫星相撞，为什么多年后仍要避让？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 10 February 2009, nearly 790 kilometres above Siberia, two satellites slammed together at about 42,000 kilometres an hour. Iridium 33 belonged to an American satellite-phone constellation; Kosmos 2251, a Russian military satellite, had been derelict since about 1995. Both shattered in the first accidental collision between two intact satellites.

Neither craft swerved. The Russian hulk, lacking propulsion, could not. A public collision-warning service had forecast a narrow miss of about 584 metres, yet the predicted encounter was not among the week's closest; such warnings were frequent and imprecise. One owner's discarded relic demolished another's working asset.

The wreckage, however, endangered everyone else. The impact flung out more than 2,000 catalogued fragments and a multitude of others too small to pinpoint. At that height air drag is feeble, so many would linger for decades. In January 2012, the International Space Station fired its thrusters to dodge an Iridium 33 remnant.

In June 1978, two NASA scientists envisaged a vicious spiral, later named the Kessler syndrome: each collision would scatter fragments, raising the odds of subsequent crashes. The 2009 crash illustrated that premise without triggering a runaway chain reaction. By the end of 2024, surveillance networks tracked about 40,000 objects, only some 11,000 of them functional.

Such a cascade would unfold over decades, though, and is not inevitable; the risk can be mitigated. Operators now routinely manoeuvre satellites out of harm's way. In September 2022, America's Federal Communications Commission, which licenses satellites, mandated disposal of low-orbit craft within five years of retirement instead of 25. Years after the crash, spacecraft whose owners had no part in it kept swerving around its splinters.

中文译文

2009年2月10日，西伯利亚上空近790公里处，两颗卫星以约每小时4.2万公里的相对速度猛撞在一起。铱星33号属于美国的一个卫星电话星座；俄罗斯军用卫星宇宙2251号则约从1995年起便已失效。两星双双解体，这是两颗完整卫星之间首次发生意外相撞。

两颗卫星都没有转向避让。俄方那颗失效卫星缺乏推进能力，无法躲开。一个公开的碰撞预警服务曾预测，两星会相距约584米擦过，但这次预计交会还不在当周距离最近的几次之列；这样的预警既频繁，又不精确。一方丢弃的旧物，摧毁了另一方仍在工作的资产。

碎片却危及所有其他航天器。撞击抛出了两千多块已编目碎片，还有大量小得无法精确定位的碎片。那个高度空气阻力很弱，许多碎片会在轨道上滞留数十年。2012年1月，国际空间站启动推进器，避开了一块铱星33号的残片。

1978年6月，两位美国航天局科学家设想了一种后来被称为“凯斯勒综合征”的恶性循环：每次碰撞都会散出碎片，提高后续相撞的概率。2009年的事故展示了这一机制的基础，却没有引发失控的连锁反应。截至2024年底，监测网络跟踪着约4万个在轨物体，其中只有约1.1万个正常运行。

不过，这样的连锁过程会历时数十年，并非不可避免，风险也能减轻。卫星运营方如今会经常调整卫星轨道，避开危险。2022年9月，负责颁发卫星许可的美国联邦通信委员会要求，其监管的低轨航天器在退役后五年内完成处置，而不再是25年。事故多年后，那些所有者与此事毫无关系的航天器，仍要一再转向躲避当年的碎片。

核心词汇

先记这22项。连着下方的原文短语一起回想。

derelict adj.
已废弃失效的
had been derelict since about 1995

collision n.
碰撞
the first accidental collision

propulsion n.
推进力；推进能力
lacking propulsion

narrow miss 名词短语
险些相撞；近距离擦过
a narrow miss of about 584 metres

imprecise adj.
不精确的
frequent and imprecise

discard v.
弃置
One owner's discarded relic

asset n.
资产
another's working asset

wreckage n.
残骸；碎片
The wreckage, however

catalogue v.
编目；登记
more than 2,000 catalogued fragments

pinpoint v.
精确定位
too small to pinpoint

air drag 名词短语
空气阻力
air drag is feeble

linger v.
滞留；迟迟不消失
many would linger for decades

dodge v.
躲避
to dodge an Iridium 33 remnant

envisage v.
设想
scientists envisaged a vicious spiral

subsequent adj.
后续的
the odds of subsequent crashes

premise n.
前提；基本设想
illustrated that premise

trigger v.
引发
without triggering

chain reaction 名词短语
连锁反应
a runaway chain reaction

surveillance n.
监视；监测
surveillance networks tracked

functional adj.
正常运行的
11,000 of them functional

inevitable adj.
不可避免的
is not inevitable

disposal n.
处置
mandated disposal of low-orbit craft

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

slam v.
猛撞

belong to 动词短语
属于

constellation n.
卫星星座；卫星群

shatter v.
粉碎；解体

accidental adj.
意外的

swerve v.
突然转向；转向避让

hulk n.
废弃的大型躯壳

encounter n.
交会；相遇

frequent adj.
频繁的

relic n.
遗留旧物

demolish v.
摧毁

endanger v.
危及

impact n.
撞击

fling out 动词短语
抛出；甩出

a multitude of 名词短语
大量的

feeble adj.
微弱的

space station 名词短语
空间站

fire v.
启动；点火

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

thruster n.
推进器

remnant n.
残片；遗留部分

vicious spiral 名词短语
恶性循环

syndrome n.
综合征

scatter v.
散布；使散开

raise the odds of 动词短语
提高……发生的概率

illustrate v.
例证；说明

runaway adj.
失控的

unfold v.
逐步展开

mitigate v.
减轻；缓解

routinely adv.
例行地；经常地

manoeuvre v.
操纵；调整运行轨迹

out of harm's way 介词短语
避开危险；在安全处

mandate v.
强制要求

retirement n.
退役

instead of 介词短语
而不是；代替

have no part in 动词短语
与……无关；未参与

splinter n.
碎片

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the first accidental _____ between two intact satellites.

02 The Russian hulk, lacking _____, could not.

03 a multitude of others too small to _____.

04 and is not _____; the risk can be mitigated.

文章主线

- 相撞 2009年，两颗完整卫星首次意外相撞：失效的宇宙2251号撞上仍工作的铱星33号。
- 预警 公开预警频繁且不准，预计两星会隔约584米擦过，并非当周最近交会；失效卫星也无法避让。
- 现状 碰撞留下两千多块已编目碎片；截至2024年底，全球跟踪约4万个在轨物体，只有约1.1万个正常运行。
- 风险 高空阻力小，碎片可滞留数十年，又增加碰撞风险；此次未触发失控连锁反应，恶性循环并非必然。
- 应对 卫星运营方加强避让；美国联邦通信委员会把其监管低轨卫星的退役后处置期限，从25年缩至5年。

参考答案　1 collision 2 propulsion 3 pinpoint 4 inevitable

被降级的冥王星

The Planet That Was Demoted

冥王星本身没变，为什么会失去行星身份？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 18 February 1930, comparing photographic plates at the Lowell Observatory, the astronomer Clyde Tombaugh perceived a dim dot that shifted between them. An eleven-year-old English girl put forward the name, after the Roman deity of the underworld. For 76 years Pluto held the ninth slot on the planetary roster.

In January 2005, astronomers found a far-off body, later named Eris, at first thought larger than Pluto. NASA briefly called it the tenth planet. The contender posed an awkward dilemma. The term "planet" had hitherto been used loosely, and with more icy worlds likely to be found, a precise definition became a necessity.

On 24 August 2006 the International Astronomical Union, convened in Prague, voted that a planet must revolve around the Sun, be nearly spherical, and have cleared the neighbourhood around its orbit. Pluto failed that last criterion and was reclassified as a dwarf planet, leaving eight. Pluto was physically unaltered; only the boundary had moved.

The vote was valid, yet its legitimacy drew criticism: only 424 of some 10,000 members took part. Hundreds of dissenting astronomers signed a petition. The American Dialect Society made "plutoed", playful slang for "to devalue", its word of 2006, and New Mexico and Illinois lawmakers passed resolutions affirming that Pluto was a planet.

Amid the controversy, an irony was already in transit. New Horizons was launched on 19 January 2006, seven months before the vote, carrying a pinch of Tombaugh's ashes. On 14 July 2015 the unmanned craft flew past Pluto, transmitting images of an enormous heart-shaped plain, fittingly named Tombaugh Regio. The discoverer's ashes swept past the heart that bears his name.

中文译文

1930年2月18日，天文学家克莱德·汤博在洛厄尔天文台比较照相底片时，发现一个暗淡的小点在不同底片之间移动。一名11岁的英国女孩建议，以罗马神话中的冥界之神为它命名。此后76年，冥王星一直占据行星名单上的第九个位置。

2005年1月，天文学家发现一个遥远天体，后来命名为厄里斯，起初认为它比冥王星更大。NASA曾短暂地称它为第十大行星。这个新候选者带来一道棘手难题：“行星”此前一直是个使用宽泛的称呼，而随着更多冰冷天体可能被发现，给它下一个精确定义成了必要之举。

2006年8月24日，国际天文学联合会在布拉格开会，投票决定：行星必须绕太阳运行，形状近似球体，并已在轨道邻域占据引力主导地位。冥王星不符合最后一条标准，被重新归为矮行星，太阳系行星由此只剩八颗。冥王星本身丝毫未变，变化的只是分类界线。

表决有效，但其正当性遭到批评：约一万名会员中，只有424人参与。数百名持异议的天文学家联名请愿。美国方言学会把戏谑地表示“贬低”的俚语“plutoed”选为2006年度词汇；新墨西哥州和伊利诺伊州的立法者则通过决议，确认冥王星是行星。

争论之中，一件颇具反讽意味的事早已在路上。新视野号于2006年1月19日发射，比那场表决早七个月，携带着一小撮汤博的骨灰。2015年7月14日，无人探测器飞掠冥王星，传回巨大心形区域的图像；这个区域恰以汤博的名字命名为“汤博区”。发现者的骨灰，掠过了那颗以他命名的“心”。

“清理轨道邻域”指在该区域占据引力主导地位，并非清空所有天体。英文plain是对心形地貌的概括；译文用“心形区域”，因为整个汤博区并不全是平原。

核心词汇

先记这22项。连着下方的原文短语一起回想。

perceive v.
察觉；注意到
perceived a dim dot

put forward 动词短语
提出；提议
put forward the name

contender n.
竞争者；候选者
The contender posed an awkward dilemma

dilemma n.
两难困境；棘手难题
an awkward dilemma

precise adj.
精确的
a precise definition

definition n.
定义
a precise definition

necessity n.
必要之事
became a necessity

convene v.
开会；召开会议
convened in Prague

revolve around 动词短语
绕……运行
revolve around the Sun

spherical adj.
球形的
be nearly spherical

criterion n.
标准
that last criterion

reclassify as 动词短语
重新归类为
reclassified as a dwarf planet

unaltered adj.
未改变的
Pluto was physically unaltered

boundary n.
界线；此处指分类边界
only the boundary had moved

valid adj.
有效的
The vote was valid

legitimacy n.
正当性
its legitimacy drew criticism

dissent v.（分词作定语）
持异议
dissenting astronomers

petition n.
请愿书
signed a petition

affirm v.
确认；明确主张
affirming that Pluto was a planet

controversy n.
争议
Amid the controversy

irony n.
具有反讽意味的事
an irony was already in transit

transmit v.
传送；传回
transmitting images

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

photographic adj.
照相的

astronomer n.
天文学家

dim adj.
暗淡的

dot n.
小点

deity n.
神灵

underworld n.
冥界

slot n.
位置；名额

planetary adj.
行星的

roster n.
名单

far-off adj.
遥远的

awkward adj.
棘手的；尴尬的

hitherto adv.
此前；迄今为止

loosely adv.
宽泛地；不严格地

icy adj.
冰冷的；覆冰的

neighbourhood n.
周边区域；此处指轨道邻域

dwarf adj.
矮的；用于矮行星名称

physically adv.
在实体上；在物理上

criticism n.
批评

playful adj.
戏谑的

slang n.
俚语

devalue v.
贬低；使贬值

lawmaker n.
立法者

resolution n.
决议

in transit 介词短语
在途中

a pinch of 数量短语
一小撮

unmanned adj.
无人的；无人驾驶的

enormous adj.
巨大的

heart-shaped adj.
心形的

fittingly adv.
恰如其分地

discoverer n.
发现者

sweep past 动词短语
飞掠而过

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The contender posed an awkward _____.

02 Pluto failed that last _____ and was reclassified as a dwarf planet, leaving eight.

03 Pluto was physically _____; only the boundary had moved.

04 The vote was valid, yet its _____ drew criticism: only 424 of some 10,000 members took part.

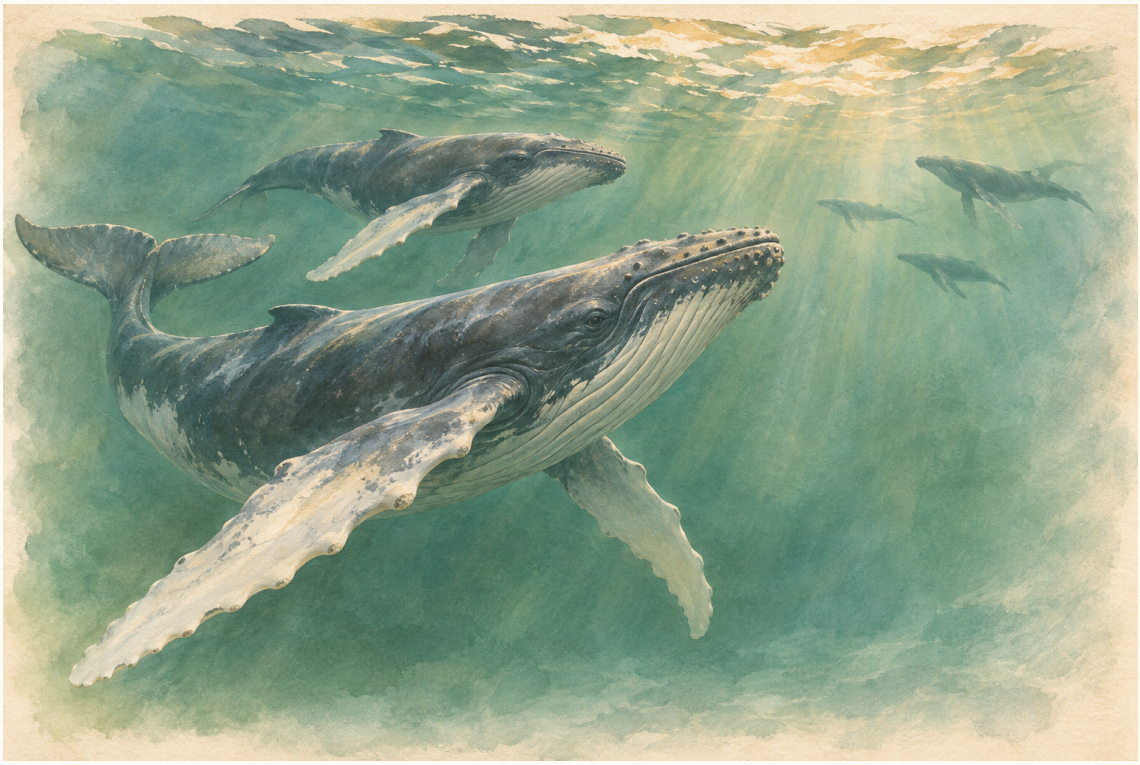
文章主线

- 发现 1930年汤博发现冥王星，此后76年，它一直被列为太阳系的第九颗行星。
- 重审 2005年发现的厄里斯起初被认为比冥王星更大，预期还有更多同类天体，行星需要明确标准。
- 改类 2006年新规要求行星绕日、近球形并靠引力主导轨道邻域；冥王星未过第三条，改列矮行星，行星剩八颗。
- 争议 国际天文学联合会的表决有效，但约一万会员仅424人参与，正当性受质疑；冥王星本身未变。
- 探访 无人探测器新视野号携少量汤博骨灰，于2015年飞掠冥王星，传回以这位发现者命名的心形区域图像。

参考答案　1 dilemma 2 criterion 3 unaltered 4 legitimacy

鲸歌易帜

A Change of Anthem in the Deep



新歌如何跨过海洋，传遍鲸群？

英文原文

Each winter, male humpback whales, and only males, fill their breeding grounds with a haunting melody. Sound units constitute phrases, and phrases build themes in an elaborate hierarchy. Each rendition lasts minutes, and the song can recur for hours, probably for courtship. All the males in a population sing an identical version, which evolves subtly each year.

That is the rule. Off eastern Australia, it was broken. From 1995 to 1998, researchers used hydrophones, or underwater microphones, to record humpbacks on their seasonal migration along the coastline. In the mid-1990s, merely two of 82 singers emitted a distinctive new song, utterly unlike the reigning chorus.

Within about two years, virtually every singer had converted, and by 1998 the old song had been displaced wholesale. A study published in Nature in November 2000 found that the new song resembled that of the Indian Ocean population off western Australia, apparently ferried east by a handful of visiting performers.

Nor was it a freak. In April 2011, a study in Current Biology analysed over a decade of recordings from six South Pacific populations. Song types primarily arose off eastern Australia, then rippled eastward through successive populations to French Polynesia, generally crossing the region within about two years.

Such rapid change, too fast for genetic evolution, points to imitation rather than inheritance, a persuasive case of cultural transmission. Why males sing, and why they overhaul their songs, remain controversial. One idea is that novelty itself spurs change. Some scholars are reluctant to define this as culture in the human sense. Whatever the label, a few outsiders off eastern Australia reshaped an entire population's music.

中文译文

每到冬季，雄性座头鲸——而且只有雄性——会让繁殖地回荡着令人难忘的旋律。声音单元组成乐句，乐句再构成主题，形成复杂精细的层级结构。每次歌唱持续数分钟，又能连续重复数小时，很可能是为了求偶。同一种群中的所有雄性都唱着相同版本，这个版本每年都会发生细微变化。

这是通常的规律，但在澳大利亚东部海域，规律被打破了。1995至1998年，研究者用水听器，也就是水下麦克风，记录沿岸季节性迁徙的座头鲸。20世纪90年代中期录到的82头歌者中，只有两头唱出了一首全新的歌，与当时占主流的合唱完全不同。

大约两年内，几乎每头歌者都换了歌；到1998年，旧歌已被彻底取代。2000年11月发表于《自然》的一项研究发现，新歌与澳大利亚西部近海的印度洋种群所唱的歌相似，似乎是少数来访歌者把它向东带来的。

这也并非孤例。2011年4月，《当代生物学》发表的一项研究分析了南太平洋六个种群十余年的录音。歌型大多起源于澳大利亚东部海域，随后依次经由一个个种群向东传播，直到法属波利尼西亚，通常约两年就能横跨这片区域。

这种变化快得难以用基因演化解释，指向的是模仿而非遗传，为文化传递提供了一个有说服力的例证。但雄性为何歌唱、为何全面更换歌曲，仍有争议。一种观点认为，新奇本身就会推动变化。有些学者不愿把它定义为人类意义上的文化。无论采用什么名称，澳大利亚东部海域的少数外来者，确实重塑了整个种群的音乐。

核心词汇

先记这22项。连着下方的原文短语一起回想。

constitute v.
构成
Sound units constitute phrases,

elaborate adj.
复杂精细的
themes in an elaborate hierarchy.

hierarchy n.
层级结构
themes in an elaborate hierarchy.

recur v.
反复出现
and the song can recur for hours,

courtship n.
求偶
for hours, probably for courtship.

identical adj.
完全相同的
population sing an identical version,

migration n.
迁徙
on their seasonal migration along the

distinctive adj.
与众不同的
emitted a distinctive new song,

virtually adv.
几乎
two years, virtually every singer had

displace v.
取代
old song had been displaced wholesale.

resemble v.
与……相似
the new song resembled that of the

a handful of n.phr.
少数；一小批
by a handful of visiting performers.

primarily adv.
主要地
Song types primarily arose off eastern

successive adj.
相继的
through successive populations to

evolution n.
演化
too fast for genetic evolution,

imitation n.
模仿
to imitation rather than inheritance,

inheritance n.
遗传
to imitation rather than inheritance,

transmission n.
传递；传播
case of cultural transmission.

overhaul v.
全面改造
why they overhaul their songs, remain

controversial adj.
存在争议的
their songs, remain controversial.

novelty n.
新奇
is that novelty itself spurs change.

reluctant adj.
不情愿的
Some scholars are reluctant to define

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

male adj.
雄性的

humpback whale n.phr.
座头鲸

breeding ground n.phr.
繁殖地

haunting adj.
萦绕心头的；难忘的

melody n.
旋律

sound unit n.phr.
声音单元

theme n.
主题；主题段落

rendition n.
一次演唱；演绎

population n.
生物种群

evolve v.
演变

subtly adv.
细微地；不易察觉地

off prep.
在……近海

researcher n.
研究者

microphone n.
麦克风

seasonal adj.
季节性的

coastline n.
海岸线

merely adv.
仅仅

emit v.
发出

utterly unlike adj.phr.
完全不同于

reigning adj.
当前占主导的

chorus n.
合唱

convert v.
转变；此处指改唱

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

wholesale adv.
全面地；全盘地

ferry v.
带来；运送

performer n.
表演者；此处指歌者

freak n.
罕见的例外

analyse v.
分析

recording n.
录音

Pacific proper n.
太平洋

ripple v.
如涟漪般扩散

eastward adv.
向东

Polynesia proper n.
波利尼西亚

generally adv.
通常

region n.
区域

rapid change n.phr.
快速变化

persuasive adj.
有说服力的

cultural adj.
文化的

spur v.
推动；促进

scholar n.
学者

define as phr.v.
与as连用：把……界定为

as prep.
标示被归入的类别

outsider n.
外来者

reshape v.
重塑

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Sound units _____ phrases,

02 _____ every singer had converted,

03 _____ visiting performers.

04 Some scholars are _____ to define this as culture in the human sense.

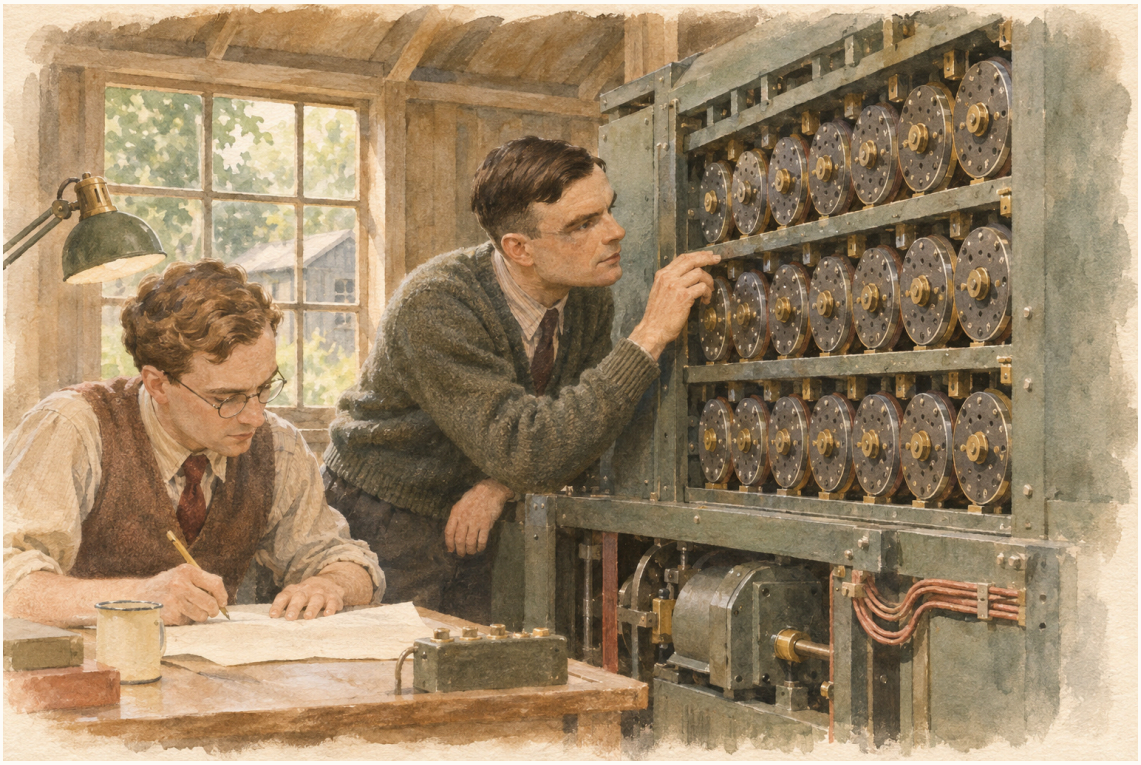
文章主线

- 常态 同一种群的雄性座头鲸唱相同版本的歌，通常每年只作细微变化，歌唱可能用于求偶。
- 换歌 澳洲东部录到的82头鲸中，最初仅两头唱全新歌；约两年后，几乎整个种群都改唱新歌。
- 来源 新歌与澳洲西部种群的歌相似，研究推测是少量来访鲸把它带到了东部。
- 传播 六个南太平洋种群的十余年录音显示，歌型主要从澳洲东部逐群向东扩散，通常约两年传到法属波利尼西亚。
- 解释 变化太快，支持模仿而非遗传的解释；为何换歌、是否等同人类文化，仍有争议。

参考答案　1 constitute 2 virtually 3 a handful of 4 reluctant

为机器点燃思想的人

The Man Who Gave Machines a Mind



图灵的贡献，为何没能保护他?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Born in June 1912, Alan Turing had a formidable intellect by his early twenties. In a paper read in November 1936, he outlined a "universal machine": a device able to execute any computation, given the right instructions. That abstract construct, rooted in mathematical logic, became the theoretical blueprint of the modern computer.

On 4 September 1939 he joined the classified codebreaking effort at Bletchley Park, leading work on naval Enigma, Germany's encryption machine. Polish mathematicians had deciphered Enigma by late 1932 and shared their insights with Britain and France in July 1939. His team's machines exploited them and, operational from March 1940, automated the laborious search for each day's rotor settings.

In October 1950, writing in the philosophy journal Mind, he posed a provocative question: "Can machines think?" He proposed a practical substitute, the imitation game, now widely called the Turing test. An interrogator exchanges typed messages with two concealed players, one human and one machine. If the interrogator cannot consistently distinguish them, the machine passes.

His exceptional wartime contribution did not protect him. On 31 March 1952 he was convicted of "gross indecency", the charge then used against gay men. He forfeited his security clearance and, to avert imprisonment, submitted to hormone injections. He died of cyanide poisoning on 7 June 1954. A coroner's inquest ruled it suicide, though some doubt it.

Recognition lagged. On 10 September 2009 Britain's prime minister publicly apologised, and on 24 December 2013 Turing was posthumously granted a royal pardon. On 23 June 2021, his birthday, a fifty-pound note bearing his portrait entered circulation. The country that prosecuted him put his face on its money.

中文译文

艾伦·图灵生于1912年6月，二十出头便展现出非凡才智。1936年11月宣读的一篇论文中，他提出“通用机器”的轮廓：只要给出正确指令，这种装置就能执行任何计算。这一植根于数理逻辑的抽象构想，后来成为现代计算机的理论蓝图。

1939年9月4日，图灵加入布莱切利园的机密破译工作，领导针对德国海军恩尼格玛加密机的研究。波兰数学家早在1932年底就已破译恩尼格玛，并于1939年7月向英国和法国分享了思路。图灵团队的机器借鉴这些成果，从1940年3月投入运行，将每天搜寻转子设置的繁重工作自动化。

1950年10月，图灵在哲学期刊《心灵》上提出一个发人深思的问题：“机器能思考吗？”他建议用一种实际可行的替代办法来讨论这个问题，即“模仿游戏”，如今通常称为图灵测试。提问者通过打字与两名隐藏的参与者交流，一方是人，另一方是机器。如果提问者无法稳定地区分二者，机器就算通过测试。

卓越的战时贡献并未保护图灵。1952年3月31日，他因“严重猥亵”罪被定罪，这是当时用来惩治男同性恋者的罪名。他失去接触机密资料的资格，并为避免入狱而接受激素注射。1954年6月7日，他死于氰化物中毒。验尸官的死因调查裁定为自杀，但有人对此存疑。

认可来得很迟。2009年9月10日，英国首相公开道歉；2013年12月24日，图灵获得身后皇家赦免。2021年6月23日，他生日当天，印有他肖像的50英镑纸币开始流通。曾经起诉他的国家，将他的面容印在了自己的货币上。

图灵测试是行为区分标准，不能直接等同于机器具有意识的证明。

核心词汇

先记这22项。连着下方的原文短语一起回想。

intellect n.
才智；智力
a formidable intellect

execute v.
执行（指令或任务）
able to execute any computation

computation n.
计算
able to execute any computation

abstract adj.
抽象的
That abstract construct

construct n.
理论构想；概念模型
That abstract construct

blueprint n.
蓝图；设计基础
the theoretical blueprint

classified adj.
机密的；保密的
the classified codebreaking effort

encryption n.
加密
Germany's encryption machine

decipher v.
破译；解读
had deciphered Enigma

insight n.
深刻见解；关键思路
shared their insights

operational adj.
投入运行的
operational from March 1940

automate v.
使自动化
automated the laborious search

laborious adj.
费力耗时的
automated the laborious search

provocative adj.
发人深思的；引起争议的
a provocative question

substitute n.
替代办法；替代物
a practical substitute

consistently adv.
稳定地；始终如一地
cannot consistently distinguish them

distinguish v.
区分；辨别
cannot consistently distinguish them

be convicted of 动词短语
被判犯有……罪
he was convicted of

forfeit v.
丧失；被剥夺
He forfeited his security clearance

avert v.
避免；防止
to avert imprisonment

posthumously adv.
在死后；身后地
posthumously granted

pardon n.
赦免
a royal pardon

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

formidable adj.
非凡的；强大的

outline v.
概述；勾勒

device n.
装置；设备

be rooted in 动词短语
根源于；植根于

mathematical adj.
数学的

logic n.
逻辑

theoretical adj.
理论的

codebreaking effort 名词短语
密码破译工作

naval adj.
海军的

exploit v.
利用；运用

rotor n.
转子

philosophy n.
哲学

pose v.
提出（问题）

propose v.
提议；建议

practical adj.
切实可行的

widely adv.
广泛地；普遍地

interrogator n.
提问者；询问者

concealed adj.
隐藏的

exceptional adj.
杰出的；非凡的

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

wartime adj.
战时的

contribution n.
贡献

security clearance 名词短语
涉密资格；接触机密的许可

imprisonment n.
监禁；入狱

submit to 动词短语
接受；屈从于

hormone n.
激素

injection n.
注射

cyanide poisoning 名词短语
氰化物中毒

coroner n.
验尸官

suicide n.
自杀

recognition n.
认可；承认

lag v.
滞后；迟迟未到

publicly adv.
公开地

apologise v.
道歉

be aux.
构成过去时被动语态

grant v.
授予；给予

portrait n.
肖像

circulation n.
（货币等的）流通

prosecute v.
起诉；提起公诉

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That abstract construct, rooted in mathematical logic, became the theoretical _____ of the modern computer.

02 His team's machines exploited them and, operational from March 1940, automated the _____ search for each day's rotor settings.

03 If the interrogator cannot consistently _____ them, the machine passes.

04 Turing was _____ granted a royal pardon.

文章主线

- 理论 1936年，图灵提出按指令执行各种计算的通用机器模型，为现代计算机奠定理论基础。
- 破译 图灵团队借鉴波兰的破译成果，1940年用机器自动搜索德国海军密码机每天的设置，加速破译。
- 测试 1950年，他提出模仿游戏：通过打字交谈，提问者若不能稳定分清人和机器，机器就通过测试。
- 迫害 1952年图灵因同性恋行为获罪，被取消涉密资格，为免入狱接受激素注射；1954年死于氰化物中毒。
- 追认 其死被判为自杀，但有人质疑。英国2009年道歉、2013年给予身后赦免；2021年肖像登上50镑钞票。

参考答案　1 blueprint 2 laborious 3 distinguish 4 posthumously

没有到来的零日

The Day Zero That Never Came

开普敦怎样避开了水龙头大范围关闭的零日？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In January 2018, Cape Town issued a stark alert: Day Zero could arrive in April. The term denoted a grim threshold, reached when water in the city's reservoirs, as a percentage of capacity, fell to 13.5 per cent. Most taps would then be shut, and residents would queue at about 200 designated collection points.

The prospect was genuine. Three consecutive years of meagre rainfall, from 2015 to 2017, had diminished the reservoirs serving a metropolitan area of about four million people. By early 2018 the dams were roughly a fifth full. The countdown turned a chronic shortage into an imminent deadline.

The city paired mandatory restrictions with unprecedented transparency. From 1 February 2018, each person was allotted 50 litres a day. Officials disclosed dam levels and household consumption; the published tallies made thrift visible and let neighbours benchmark their use. Daily use was more than halved, from about 1.2 billion litres in 2015 to about 500 million by early 2018.

Other forces intervened. In February 2018, upstream farmers released water from their irrigation dams; with farms drawing less, Day Zero was postponed to July. In March 2018, officials judged Day Zero unlikely that year. Winter rains subsequently replenished the reservoirs, which were about half full by July. The 50-litre limit was eased in October 2018.

The burden, critics argued, fell unevenly. Townships, poor areas segregated by race under apartheid, had long made do with scant water, yet steeper tariffs squeezed them hardest. Others blamed alarmist warnings for tourist cancellations. Households, farmers, and rain kept the collection points idle, but the sacrifice weighed heaviest on those with the least to give.

中文译文

2018年1月，开普敦发出严峻警报：“零日”可能在4月到来。这个词指一道令人忧惧的门槛：城市各水库蓄水量降至总容量的13.5%。届时，大多数水龙头将关闭，居民必须到约200个指定取水点排队领水。

这并非虚构的危险。2015至2017年连续三年降雨稀少，为约400万人口的都会区供水的水库蓄水持续下降。到2018年初，蓄水量只剩约五分之一。倒计时把长期缺水变成了一个迫在眉睫的最后期限。

市政府把强制限水与前所未有的信息公开结合起来。从2018年2月1日起，每人每天限用50升水。官员公布水库水位和家庭用水量，让节水成效看得见，也让邻居能相互比较用水情况。全市日用水量已从2015年的约12亿升降至2018年初的约5亿升，减少了一半以上。

其他因素也发挥了作用。2018年2月，上游农民从自有灌溉水库放水，再加上农场减少取水，“零日”被推迟至7月。到了3月，官员判断当年不大可能出现“零日”。随后到来的冬雨补充了水库，到7月时蓄水已恢复至约半满。2018年10月，每人每日50升的限制得到放宽。

批评者指出，负担并不均等。种族隔离时期按种族划出的贫困聚居区，长期以来就只能用少量水勉强生活，却最难承受提高后的水价。也有人指责，危言耸听的警报导致游客取消行程。居民、农民和雨水共同让取水点没有派上用场，但付出最沉重代价的，却是那些最没有余力付出的人。

核心词汇

先记这22项。连着下方的原文短语一起回想。

issue v.
发出；发布
Cape Town issued a stark alert

stark adj.
严峻鲜明的
a stark alert

denote v.
指；表示
The term denoted a grim threshold

threshold n.
门槛；临界点
a grim threshold

reservoir n.
水库
the city's reservoirs

capacity n.
容量
as a percentage of capacity

designate v.
指定
200 designated collection points

consecutive adj.
连续的
Three consecutive years

meagre adj.
少得可怜的；不足的
meagre rainfall

chronic adj.
长期持续的
a chronic shortage

imminent adj.
迫在眉睫的
an imminent deadline

mandatory adj.
强制性的
mandatory restrictions

unprecedented adj.
前所未有的
unprecedented transparency

transparency n.
公开透明
unprecedented transparency

allot v.
分配；配给
each person was allotted 50 litres

consumption n.
消耗量；用水量
household consumption

benchmark v.
与基准或他人比较
benchmark their use

postpone v.
推迟
Day Zero was postponed to July

replenish v.
补充；重新补给
replenished the reservoirs

burden n.
负担
The burden, critics argued

unevenly adv.
不均等地
fell unevenly

tariff n.
水价；收费标准
steeper tariffs squeezed them hardest

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

alert n.
警报

grim adj.
严酷的；令人忧惧的

percentage n.
百分比

resident n.
居民

prospect n.
可能出现的局面

genuine adj.
真实的；确实存在的

diminish v.
减少；使减少

metropolitan adj.
大都市的

dam n.
水坝；此处指水库

roughly adv.
大约

countdown n.
倒计时

shortage n.
短缺

restriction n.
限制措施

litre n.
升

official n.
官员

disclose v.
公布；披露

household n.
家庭

tally n.
统计数目

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

thrift n.
节约；节俭

visible adj.
看得见的

halve v.
减半

intervene v.
介入；发挥作用

upstream adj.
上游的

irrigation n.
灌溉

unlikely adj.
不大可能的

subsequently adv.
随后

ease v.
放宽；减轻

township n.
隔离制下的贫困聚居区

segregate v.
按种族隔离

apartheid n.
种族隔离制度

scant adj.
很少的；不足的

steep adj.
价格更高的

squeeze v.
使承受经济压力

alarmist adj.
危言耸听的

cancellation n.
取消；退订

idle adj.
闲置的

sacrifice n.
牺牲；付出的代价

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The term denoted a grim _____,

02 The city paired _____ restrictions with unprecedented transparency.

03 each person was _____ 50 litres a day.

04 Winter rains subsequently _____ the reservoirs, which were about half full by July.

文章主线

- 危机 连续三年少雨使开普敦水库告急，2018年初当局警告，四月可能迎来零日。
- 门槛 零日指蓄水降至容量的13.5%，届时大部分水龙头关闭，居民须到指定地点领水。
- 节水 日用水量由2015年约12亿升降至2018年初约5亿升；2月起限每人每日50升，并公开水位及用量。
- 转机 居民节水、农场放水及减少取水，使零日一再推迟；随后冬雨补充水库，避免了当年大范围停水。
- 负担 批评者指出，贫困社区原本用水就少，却最难承受上涨水价，节水代价并不均等。

参考答案　1 threshold 2 mandatory 3 allotted 4 replenished

改变人类命运的霉菌

The Mould That Changed the World

从一团霉菌到救命药，中间还差什么？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In September 1928 Alexander Fleming, a bacteriologist at St Mary's Hospital, London, returned from holiday to find that a stray mould had contaminated a dish of staphylococcus bacteria. Nothing grew around it. Popular mythology blames an open window; the spores probably originated in a laboratory downstairs, where a colleague cultivated moulds.

Fleming called the mould's juice penicillin and published in June 1929. Yet the crude juice was unstable and stubbornly hard to purify, and he and the chemists he enlisted lacked the expertise to refine it. He used it chiefly to isolate other microbes, and penicillin lay dormant, a curiosity rather than a cure.

A decade later, in Oxford, Howard Florey, an Australian pathologist, revived the work with Ernst Chain, a German-born chemist and Jewish refugee. Their colleague Norman Heatley improvised an ingenious apparatus to extract the compound. On 25 May 1940 eight mice received a lethal dose of bacteria; only the four also injected with penicillin lived. The Lancet, a prestigious journal, printed the outcome that August.

In February 1941 Albert Alexander, an Oxford constable gravely ill with infection, rallied on penicillin. When the scarce supply ran out, he died in March: the obstacle was quantity. From July 1941 a government laboratory in Peoria, Illinois, boosted yields about tenfold with corn-steep liquor, a starch by-product; later a mouldy market melon furnished the laboratory with a more prolific strain. By D-Day in June 1944 industrial fermentation met Allied needs.

In October 1945 Fleming, Florey, and Chain shared the Nobel prize; Heatley was overlooked. Fortune put the mould in Fleming's dish; tenacious teams and years of toil transformed it into a remedy.

中文译文

1928年9月，伦敦圣玛丽医院的细菌学家亚历山大·弗莱明度假回来，发现一团偶然出现的霉菌污染了一只培养葡萄球菌的培养皿，霉菌周围没有细菌生长。流行传说认为孢子从敞开的窗户飘入；它们其实很可能来自楼下的实验室，那里有位同事在培养霉菌。

弗莱明将霉菌产生的液汁称为青霉素，并于1929年6月发表研究。然而，粗制液汁不稳定，极难提纯；他和请来的化学家又缺乏精炼它的专业知识。他主要用它分离其他微生物，青霉素因此沉寂下来，仍是一件引人好奇的科学奇物，而非治病良药。

约十年后，在牛津，澳大利亚病理学家霍华德·弗洛里与德国出生的化学家、犹太难民恩斯特·钱恩重启研究。同事诺曼·希特利就地取材，制作了一套巧妙装置来提取这种化合物。1940年5月25日，八只小鼠接受致死剂量的细菌，只有同时注射青霉素的四只存活。权威期刊《柳叶刀》于同年8月发表了结果。

1941年2月，牛津警员艾伯特·亚历山大因感染病情危重，用上青霉素后有所好转。但稀缺的药物耗尽后，他于3月去世：难关在于产量。从1941年7月起，伊利诺伊州皮奥里亚的一家政府实验室利用淀粉生产副产物玉米浆，使产量达到原先的约十倍。后来，市场上的一只发霉甜瓜又提供了更高产的菌株。到1944年6月诺曼底登陆时，工业发酵已能满足盟军需求。

1945年10月，弗莱明、弗洛里和钱恩共同获得诺贝尔奖，希特利却被忽略了。机缘将霉菌送入弗莱明的培养皿，而多支团队锲而不舍、辛劳多年，才把它变成救人的良药。

核心词汇

先记这22项。连着下方的原文短语一起回想。

contaminate v.
污染；使混入杂质
a stray mould had contaminated

spore n.
孢子
the spores probably originated

penicillin n.
青霉素
the mould's juice penicillin

crude adj.
粗制的；未经精炼的
the crude juice

unstable adj.
不稳定的
the crude juice was unstable

purify v.
提纯；纯化
stubbornly hard to purify

expertise n.
专业知识；专长
lacked the expertise to refine it

isolate v.
分离；隔离
to isolate other microbes

revive v.
重启；使恢复活力
revived the work

improvise v.
就地取材制作；临时拼装
improvised an ingenious apparatus

ingenious adj.
巧妙的；独创的
an ingenious apparatus

apparatus n.
装置；设备
an ingenious apparatus

extract v.
提取
to extract the compound

lethal adj.
致死的
a lethal dose of bacteria

scarce adj.
稀缺的；不足的
the scarce supply

obstacle n.
障碍；难关
the obstacle was quantity

quantity n.
数量；产量
the obstacle was quantity

tenfold adv.
增至十倍
boosted yields about tenfold

prolific adj.
高产的；多产的
a more prolific strain

fermentation n.
发酵
industrial fermentation

tenacious adj.
锲而不舍的；顽强的
tenacious teams

remedy n.
良药；治疗手段
transformed it into a remedy

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

stray adj.
偶然出现的；零散的

mould n.
霉菌

bacterium n.
细菌（复数）

popular mythology 名词短语
流行传说

originate in 动词短语
起源于；来自

cultivate v.
培养；培育

stubbornly adv.
顽固地；格外难以

refine v.
精炼；提纯

chiefly adv.
主要地

microbe n.
微生物

lie dormant 动词短语
沉寂；暂未得到发展

curiosity n.
引人好奇的事物

cure n.
疗法；有效的治疗手段

chemist n.
化学家

refugee n.
难民

compound n.
化合物

dose n.
剂量

inject with 动词短语
注射（药物）

prestigious adj.
享有盛誉的

outcome n.
结果

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

constable n.
警员

gravely ill 形容词短语
病情危重的

infection n.
感染

rally v.
病情好转；恢复

run out 动词短语
用尽；耗尽

boost v.
提高；增加

corn-steep liquor 名词短语
玉米浆

starch n.
淀粉

mouldy adj.
发霉的

melon n.
甜瓜

furnish v.
提供；供应

with prep.
表示提供的东西

industrial adj.
工业的

Allied adj.
盟军的

overlook v.
忽视；未给予应有认可

fortune n.
机缘；运气

toil n.
辛劳；苦干

transform v.
使转化；改变

into prep.
变成；成为

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Yet the crude juice was unstable and stubbornly hard to _____, and he and the chemists he enlisted lacked the expertise to refine it.

02 Their colleague Norman Heatley _____ an ingenious apparatus to extract the compound.

03 When the scarce supply ran out, he died in March: the obstacle was _____.

04 By D-Day in June 1944 industrial _____ met Allied needs.

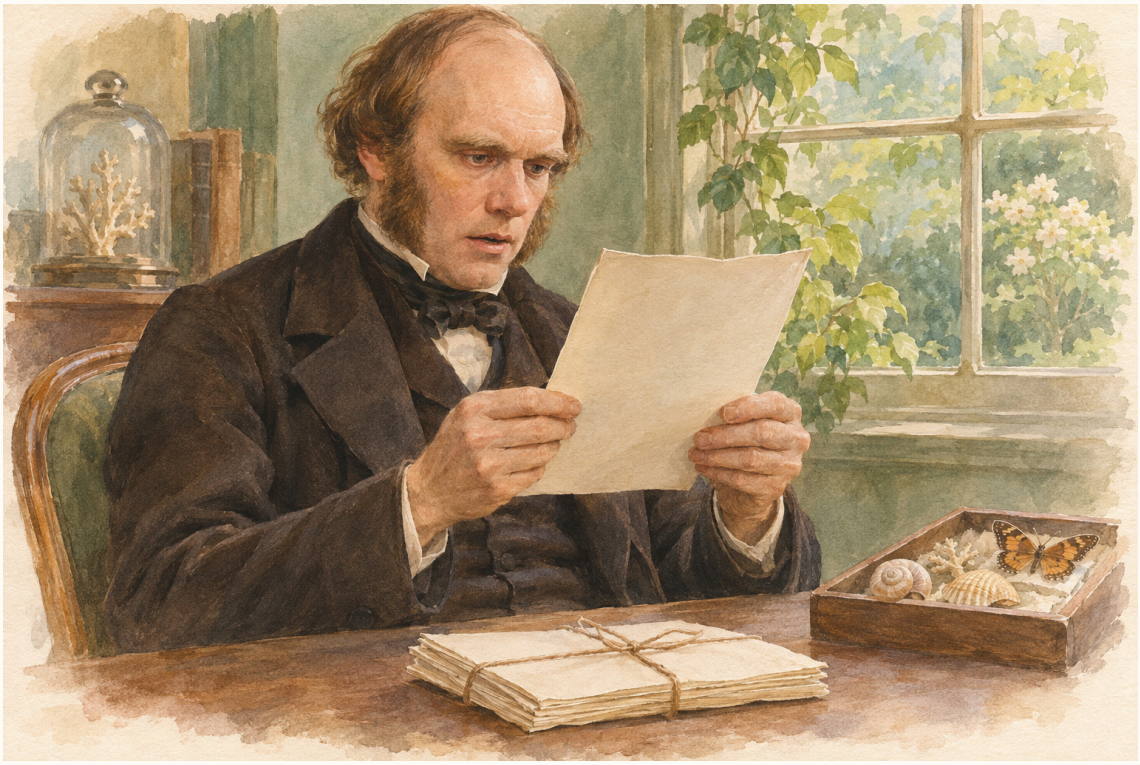
文章主线

- 偶然 1928年，弗莱明发现霉菌抑制细菌，把霉菌产物称作青霉素；孢子很可能来自楼下实验室，开窗说只是传闻。
- 难关 粗制青霉素不稳定、难提纯，最初多用于分离微生物；约十年后，弗洛里、钱恩和希特利等接力研究。
- 验证 8只感染小鼠中，只有用药的4只存活；1941年一名警员用药好转，却因药物耗尽死亡：瓶颈是产量。
- 量产 玉米浆使产量提高约十倍，发霉甜瓜又提供高产菌株；到1944年，工业发酵已能满足盟军用药。
- 归功 1945年弗莱明、弗洛里和钱恩获诺奖，希特利未入列；偶然发现经过多支团队多年攻关，才变成良药。

参考答案　1 purify 2 improvised 3 quantity 4 fermentation

一封信惊醒了达尔文

The Letter That Startled Darwin



华莱士的来信为何让达尔文不安?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Charles Darwin disembarked from HMS Beagle in October 1836. In September 1838 he began formulating the principle of natural selection: individuals better adapted to their surroundings survive and pass on their hereditary traits. By 1844 he had composed a 230-page manuscript, yet kept it unpublished, cautiously amassing evidence.

In February 1858, two decades after Darwin's first insight, Alfred Russel Wallace, a naturalist afflicted by fever in the Malay Archipelago, independently conceived the same idea. His concise essay reached Darwin on 18 June 1858. Its argument paralleled Darwin's own almost precisely; he told Charles Lyell he had never seen a more remarkable coincidence. His claim to priority seemed threatened.

Lyell and the botanist Joseph Hooker arranged for both papers to be read simultaneously at the Linnean Society, a scientific institution in London, on 1 July 1858. Neither author attended the proceedings. Wallace was abroad, and Darwin was grieving: scarlet fever had killed his infant son on 28 June. The compromise, struck without Wallace, may have kept rivalry over precedence at bay.

The papers' publication on 20 August 1858 met a muted reception. In May 1859 the Society's president remarked in his yearly address that the year had been barren of revolutionary discoveries. The sensation came on 24 November 1859 with Darwin's treatise, On the Origin of Species: booksellers had subscribed for every copy of its first impression, 1,250 in all.

Wallace had readily signified his acceptance in October 1858, assuring Hooker in a cordial letter of his satisfaction with the arrangement. In 1889 he published a lengthy exposition of the idea and entitled it Darwinism. Wallace never named the theory after himself.

中文译文

1836年10月，查尔斯·达尔文从小猎犬号下船。1838年9月，他开始形成自然选择原理：更能适应环境的个体存活下来，把遗传性状传给后代。到1844年，他已写出230页手稿，却没有发表，仍谨慎地积累证据。

1858年2月，也就是达尔文最初产生这个想法约二十年后，博物学家阿尔弗雷德·拉塞尔·华莱士在马来群岛身患热病时，独立想到了同一原理。他的简短论文于1858年6月18日送达达尔文手中，论证与达尔文自己的几乎完全一致。达尔文告诉查尔斯·莱尔，他从未见过如此惊人的巧合。他对首创地位的主张，似乎受到了威胁。

莱尔和植物学家约瑟夫·胡克安排两人的论文于1858年7月1日在伦敦的科学机构林奈学会上同场宣读。两位作者都没有到场：华莱士身在海外，达尔文则正沉浸在悲痛中——6月28日，猩红热夺走了他幼子的生命。这个在华莱士未参与商定的情况下达成的折中办法，或许避免了一场首创权之争。

论文于1858年8月20日发表，反响平淡。1859年5月，学会会长甚至在年度致辞中说，这一年没有革命性的发现。真正的轰动直到1859年11月24日达尔文的专著《物种起源》出版时才到来：书商已把首印的1250册全部订走。

华莱士已于1858年10月欣然表示接受这一安排，并在写给胡克的亲切信件中说明自己对此感到满意。1889年，他又出版了一部详细阐述这一思想的著作，题名《达尔文主义》。他从未用自己的名字为这套理论命名。

核心词汇

先记这22项。连着下方的原文短语一起回想。

formulate v.
形成并表述
he began formulating the principle of

principle n.
原理；原则
he began formulating the principle of

adapted to adj.phr.
适应……的
individuals better adapted to their

hereditary adj.
遗传的
and pass on their hereditary traits.

trait n.
性状；特征
and pass on their hereditary traits.

manuscript n.
手稿
he had composed a 230-page manuscript,

amass v.
积累；大量收集
it unpublished, cautiously amassing

afflicted by adj.phr.
遭受……折磨的
a naturalist afflicted by fever in the

independently adv.
独立地
independently conceived the same idea.

conceive v.
构想出
independently conceived the same idea.

concise adj.
简明的
His concise essay reached Darwin on 18

coincidence n.
巧合
seen a more remarkable coincidence.

priority n.
首创地位；优先权
claim to priority seemed threatened.

simultaneously adv.
同时地
to be read simultaneously at the

compromise n.
折中方案
compromise, struck without Wallace,

rivalry n.
竞争
may have kept rivalry over precedence

precedence n.
领先地位；先后次序
may have kept rivalry over precedence

muted adj.
平淡的；不热烈的
20 August 1858 met a muted reception.

reception n.
反响；接受情况
20 August 1858 met a muted reception.

cordial adj.
亲切友好的
a cordial letter of his satisfaction

exposition n.
详细阐述
a lengthy exposition of the idea and

entitle v.
为……题名
exposition of the idea and entitled it

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

disembark v.
下船；登岸

selection n.
选择；筛选

surroundings n.
周围环境

pass on phr.v.
传递；传给后代

compose v.
撰写

unpublished adj.
未发表的

cautiously adv.
谨慎地

naturalist n.
博物学家

parallel v.
与……十分相似

precisely adv.
精确地；完全地

remarkable adj.
非同寻常的

threaten v.
威胁

botanist n.
植物学家

arrange for phr.v.
安排……发生

institution n.
机构

proceedings n.
会议；会议活动

grieve v.
悲痛；哀悼

scarlet adj.
猩红色的；构成猩红热一词

infant adj.
婴幼儿的

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

publication n.
发表；出版

remark v.
评论；说

yearly adj.
年度的

barren of adj.phr.
缺乏；毫无

revolutionary adj.
革命性的

sensation n.
轰动

treatise n.
专著

On the Origin of Species proper n.
《物种起源》

subscribe for phr.v.
预订；订购

impression n.
印次；一次印刷

readily adv.
欣然地；毫不迟疑地

signify v.
表示；表明

acceptance n.
接受

assure v.
向……保证

satisfaction n.
满意

arrangement n.
安排

lengthy adj.
篇幅很长的

name after phr.v.
与after连用：以……命名

after prep.
标示命名所依据的人

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 cautiously _____ evidence.

02 His claim to _____ seemed threatened.

03 met a _____ reception.

04 assuring Hooker in a _____ letter

文章主线

- 酝酿 达尔文1838年构想自然选择：更适应环境者存活并传下遗传性状；1844年已写成230页手稿，仍搜证未发表。
- 来信 华莱士独立构想同一原理，1858年来稿几乎与达尔文一致，令达尔文担心自己的首创地位。
- 折中 莱尔、胡克安排两人成果于1858年7月同场宣读；两位作者都未出席，华莱士也未参与商定办法。
- 反响 论文发表初期反响冷淡；1859年《物种起源》引起轰动，首印1250册已被书商全部订购。
- 认可 华莱士1858年10月事后致信认可安排；1889年又以《达尔文主义》阐述自然选择，未用自己名字命名理论。

参考答案　1 amassing 2 priority 3 muted 4 cordial

入侵澳洲的毒蟾

The Toad That Invaded Australia

引进的灭虫帮手，为什么成了生态难题？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Greyback cane beetles, whose larvae devour roots, were plaguing lucrative sugar plantations in Queensland, north-eastern Australia. In June 1935 Queensland imported 102 cane toads, bulky amphibians native to the Americas, from Hawaii. That August about 2,400 of their young were released. Biological control, setting one species against a pest, had an ambitious new recruit.

The hasty plan met opposition. In November 1935 an insect expert, Walter Froggatt, persuaded officials to halt releases; after Queensland's lobbying, they resumed in September 1936. The scheme flopped: the toads barely dented beetle numbers, perhaps because the insects stayed high up on cane stalks.

The toads thrived instead. With few foes and fertile females spawning thousands of eggs, they multiplied unchecked. Their invasion frontier, creeping about ten kilometres a year at first, accelerated to about 50 a year in the tropical north. In February 2009 the first known toad breached Western Australia's border.

Toxin glands in their skin make them a deadly meal. Numbers of goannas (large lizards) and northern quolls (small marsupial hunters) have slumped in places; some local quoll populations were wiped out. The toads have wreaked havoc, but the damage is uneven; some species have prospered.

Biological control can triumph. From 1926 the cactus moth, Cactoblastis, cleared most prickly pear, an invasive cactus, from Queensland and New South Wales within about a decade. Tested first, its larvae starved rather than eat fifteen other plants; the toads faced no such test. Under an October 1984 law, recruits now face stringent scrutiny. The blunder was born of haste and sealed by lobbying that overrode a warning: an untested ally, once loose, may never be recalled.

中文译文

在澳大利亚东北部昆士兰州，灰背蔗甲虫的幼虫啃食根系，困扰着利润丰厚的甘蔗种植业。1935年6月，当地从夏威夷引进102只甘蔗蟾蜍。这些体型笨重的两栖动物原产美洲。同年8月，约2400只繁育出的幼蟾被放到野外。用一种生物控制另一种害虫的生物防治，就这样迎来了一位雄心勃勃的“新兵”。

这项草率的计划遭到反对。1935年11月，昆虫专家沃尔特·弗罗格特说服官员暂停释放，但昆士兰州游说后，释放又在1936年9月恢复。计划最终失败：蟾蜍几乎没有减少甲虫数量，可能因为这些昆虫待在高高的甘蔗茎秆上。

蟾蜍自身反而繁盛起来。它们天敌很少，雌蟾能产下数千枚卵，于是数量不受控制地增加。入侵前沿最初每年推进约10公里，在热带北部加速到每年约50公里。2009年2月，已知的第一只蟾蜍越过了西澳大利亚州边界。

蟾蜍皮肤里有毒腺，捕食者吃下它们可能送命。一些地区的巨蜥和北袋鼬数量大幅下降，某些地方的袋鼬种群甚至消失了。巨蜥是大型蜥蜴，北袋鼬则是小型有袋类捕食者。蟾蜍造成了严重生态破坏，但影响并不均匀，一些物种反而兴盛起来。

生物防治也有成功的例子。自1926年起，仙人掌螟在约十年间清除了昆士兰州和新南威尔士州大部分入侵的仙人掌属植物。它的幼虫事先经过检验，宁可饿死也不吃另外15种植物；蟾蜍却没有接受过这样的测试。根据1984年10月通过的法律，如今引入生物防治物种需要严格审查。这次失误始于仓促，又因游说压过警告而成为定局：一个未经检验的“盟友”，一旦放出，就可能再也召不回来。

原文只说甲虫在高处“可能”解释防治失败；生态损害也并非对所有物种相同。

核心词汇

先记这22项。连着下方的原文短语一起回想。

larva n.
幼虫
whose larvae devour roots

devour v.
大量啃食
larvae devour roots

lucrative adj.
利润丰厚的
lucrative sugar plantations

amphibian n.
两栖动物
bulky amphibians

native to 形容词短语
原产于
native to the Americas

species n.
物种
setting one species against a pest

pest n.
害虫；有害生物
one species against a pest

hasty adj.
草率仓促的
The hasty plan met opposition

opposition n.
反对
The hasty plan met opposition

persuade v.
说服
persuaded officials to halt releases

lobbying n.
游说
after Queensland's lobbying

resume v.
恢复；重新开始
they resumed in September 1936

thrive v.
繁盛
The toads thrived instead

unchecked adj.
未受遏制的
they multiplied unchecked

invasion n.
入侵
Their invasion frontier

accelerate v.
加速
accelerated to about 50 a year

toxin n.
毒素
Toxin glands in their skin

wipe out 动词短语
彻底消灭
quoll populations were wiped out

invasive adj.
入侵性的
an invasive cactus

stringent adj.
严格的
stringent scrutiny

scrutiny n.
仔细审查
stringent scrutiny

override v.
压过；推翻
lobbying that overrode a warning

文中词汇与搭配

补充词汇共40项。保留每个表达在本文中的意思。

beetle n.
甲虫

plague v.
困扰

plantation n.
种植园

import v.
引进

bulky adj.
体积庞大笨重的

release v.
释放

biological adj.
生物的

ambitious adj.
雄心勃勃的

recruit n.
新成员；新兵

halt v.
停止

scheme n.
计划

flop v.
彻底失败

dent v.
削减；产生影响

stalk n.
茎秆

foe n.
敌手；此处指天敌

fertile adj.
能繁殖的

spawn v.
产卵

multiply v.
大量繁殖

frontier n.
前沿

tropical adj.
热带的

文中词汇与搭配 续

补充词汇共40项。保留每个表达在本文中的意思。

breach v.
突破；越过

gland n.
腺体

lizard n.
蜥蜴

marsupial adj.
有袋类的

slump v.
锐减

wreak v.
造成（破坏）

havoc n.
严重破坏

uneven adj.
不均等的

prosper v.
兴盛

triumph v.
大获成功

moth n.
蛾

prickly adj.
多刺的

cactus n.
仙人掌

starve v.
饿死

blunder n.
严重错误

haste n.
仓促

untested adj.
未经检验的

ally n.
盟友

loose adj.
被放出的；不受约束的

recall v.
召回

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Greyback cane beetles, whose larvae devour roots, were plaguing _____ sugar plantations in Queensland, north-eastern Australia.

02 The hasty plan met _____.

03 With few foes and fertile females spawning thousands of eggs, they multiplied _____.

04 Under an October 1984 law, recruits now face stringent _____.

文章主线

- 引进 1935年，澳大利亚昆士兰引进102只甘蔗蟾蜍，放出约2,400只后代，想控制啃食甘蔗根系的甲虫。
- 警告 昆虫专家曾促成暂停释放，地方游说却使1936年恢复；蟾蜍释放前缺少食性检验，警告最终被压过。
- 失控 甲虫几乎没减少，或因它们待在高处；蟾蜍却天敌少、产卵多，在热带北部每年扩散约50公里。
- 伤害 蟾蜍的皮肤毒素使巨蜥、北袋鼬在部分地区锐减，某些袋鼬种群消失；也有物种反而兴盛。
- 对照 仙人掌螟经检验不吃另15种植物，后来成功控制入侵仙人掌；1984年立法要求严格审查生物防治引种。

参考答案　1 lucrative 2 opposition 3 unchecked 4 scrutiny

从一道划痕到天花绝迹

From a Scratch to a World Without Smallpox

詹纳不是第一人，贡献究竟在哪里?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Smallpox, a contagious affliction, killed about three in ten of its sufferers and an estimated 300 million in the 20th century. An early safeguard, deliberately implanting matter from its sores in the healthy, was practised in China by the 16th century and reached England in 1721. The procedure itself could cause grave sickness or death, if less frequently than natural smallpox.

Rustic lore held that milkmaids with cowpox, a mild ailment of livestock, were immune to smallpox. In 1774 Benjamin Jesty, a Dorset farmer, employed cowpox to protect his wife and two sons. On 14 May 1796 Edward Jenner, a provincial practitioner, inserted fluid from a milkmaid's cowpox sore into a scratch on eight-year-old James Phipps.

On 1 July 1796 Jenner exposed the boy to smallpox; no symptoms ensued. In 1797 the Royal Society declined to publish his paper, judging his evidence unconvincing. In June 1798 he privately funded his book, the Inquiry, with supplementary cases. The word vaccine derives from the Latin vacca, cow.

Jenner was not the idea's originator. His contribution was a test, with results accessible in print, so other doctors could replicate and disseminate the technique. By today's standards, experimentation on a child without informed consent would be unethical. Jenner died on 26 January 1823, with smallpox far from eliminated.

In 1967 the WHO launched an intensified, concerted campaign to eradicate smallpox. The last naturally occurring case was diagnosed in Somalia on 26 October 1977. On 8 May 1980 the World Health Assembly formally declared the world free of smallpox, the only human disease ever wiped out. That pronouncement came almost 184 years after Phipps was vaccinated.

中文译文

天花是一种传染病，约有三成患者因此死亡；仅在20世纪，估计就夺走了3亿人的生命。一种早期预防办法，是把取自天花疮疹的物质接种给健康人。这种做法到16世纪已在中国使用，1721年传到英国。接种本身也可能造成重病或死亡，只是比自然感染天花时少见。

乡间相传，挤奶女工得过牛痘后，就会对天花免疫；牛痘是牲畜的一种轻微疾病。1774年，多塞特郡农夫本杰明·杰斯蒂就曾用牛痘来保护妻子和两个儿子。1796年5月14日，乡村医生爱德华·詹纳把取自一名挤奶女工牛痘疮的液体，接种到8岁男孩詹姆斯·菲普斯身上的一道划痕中。

1796年7月1日，詹纳让男孩接触天花，男孩没有出现症状。1797年，皇家学会认为证据缺乏说服力，拒绝发表他的论文。1798年6月，他自费出版《探究》，补充了更多病例。英语中表示疫苗的vaccine一词，来自拉丁语vacca，意思是牛。

詹纳不是这个想法的首创者。他的贡献在于开展试验，并把结果公开出版，让其他医生能够重复验证、传播这项技术。按今天的标准，在没有知情同意的情况下拿孩子做实验，是不合伦理的。詹纳于1823年1月26日去世，那时天花还远未被消除。

1967年，世界卫生组织发起更强有力的协同行动，决心根除天花。最后一例自然发生的天花病例，于1977年10月26日在索马里确诊。1980年5月8日，世界卫生大会正式宣布全球摆脱天花；它也是迄今唯一被彻底根除的人类疾病。此时，距菲普斯接种已过去近184年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

contagious adj.
有传染性的
Smallpox, a contagious affliction,

affliction n.
疾病；苦难
Smallpox, a contagious affliction,

safeguard n.
防护措施
safeguard, deliberately implanting

implant v.
植入；接种
safeguard, deliberately implanting

procedure n.
做法；操作程序
The procedure itself could cause grave

grave adj.
严重的
itself could cause grave sickness or

immune to adj.phr.
对……免疫的
of livestock, were immune to smallpox.

practitioner n.
执业医师
Jenner, a provincial practitioner,

ensue v.
随后发生
no symptoms ensued.

supplementary adj.
补充的
with supplementary cases.

vaccine n.
疫苗
The word vaccine derives from the

originator n.
首创者
Jenner was not the idea's originator.

accessible adj.
可以获得或查阅的
was a test, with results accessible in

replicate v.
重复验证；复制
so other doctors could replicate and

disseminate v.
传播；散布
could replicate and disseminate the

informed adj.
知情的
on a child without informed consent

consent n.
同意
on a child without informed consent

unethical adj.
不合伦理的
would be unethical.

concerted adj.
协同一致的
an intensified, concerted campaign to

eradicate v.
根除
campaign to eradicate smallpox.

diagnose v.
确诊
was diagnosed in Somalia on 26 October

pronouncement n.
正式宣告
That pronouncement came almost 184

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

smallpox n.
天花

sufferer n.
患者

deliberately adv.
有意地

matter n.
物质；此处指痘疮取材

sore n.
疮；疮疹

practise v.
施行；采用

sickness n.
疾病

frequently adv.
经常；频繁地

rustic adj.
乡间的

lore n.
民间传说；传统知识

mild adj.
轻微的

ailment n.
病症；小病

livestock n.
牲畜

employ v.
使用；采用

provincial adj.
地方的；外省的

insert v.
放入；植入

fluid n.
液体

scratch n.
划痕

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

expose to phr.v.
与to连用：使接触

to prep.
标示所接触的对象

symptom n.
症状

publish v.
发表；出版

unconvincing adj.
缺乏说服力的

privately adv.
由私人出资地；自费地

fund v.
出资；资助

derive from phr.v.
源于

Latin n.
拉丁语

technique n.
技术；方法

experimentation n.
试验；实验

eliminate v.
消除

intensified adj.
加强了的

naturally adv.
自然地

occur v.
发生

formally adv.
正式地

wipe out phr.v.
彻底消灭

vaccinate v.
接种疫苗

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Smallpox, a _____ affliction,

02 no symptoms _____.

03 so other doctors could _____ and disseminate the technique.

04 an intensified, concerted campaign to _____ smallpox.

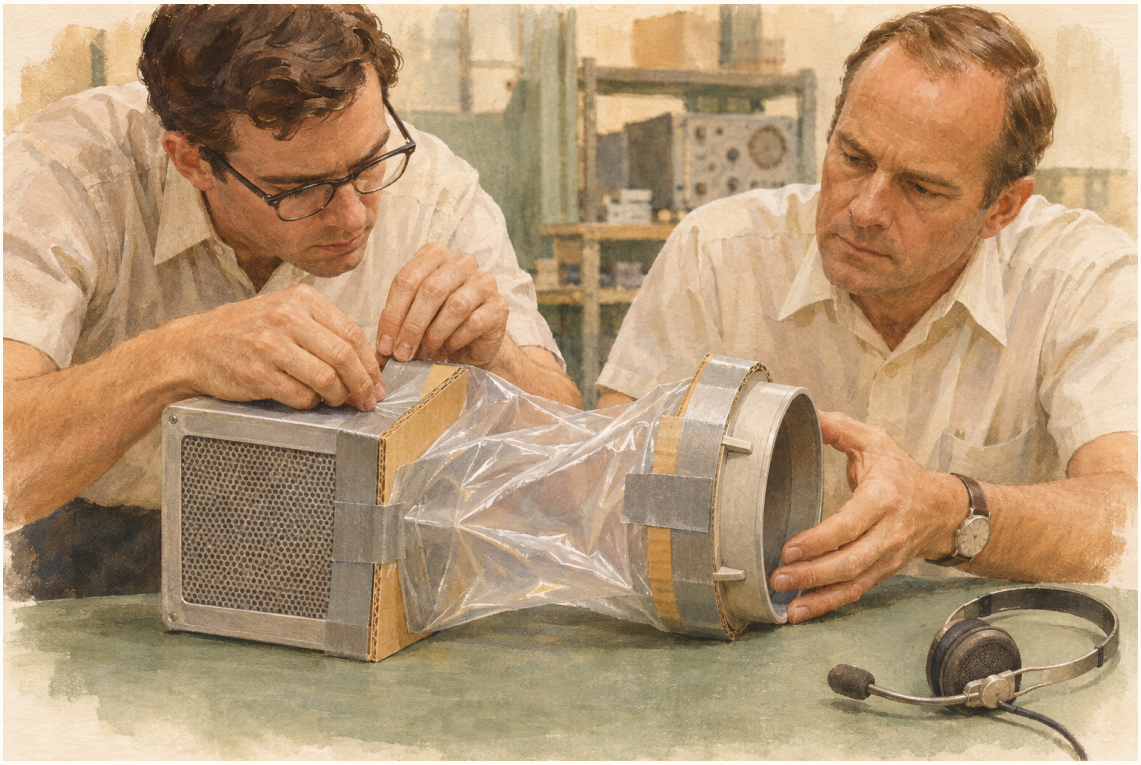
文章主线

- 背景 天花约致三成患者死亡；早期人痘术把天花疮中物质接种给健康人作预防，风险虽低于自然染病，仍可能致重病或死亡。
- 先例 乡间传牛痘可防天花；农夫杰斯蒂1774年已用牛痘保护家人，詹纳并非首创。
- 试验 詹纳1796年给男孩种牛痘，再让他接触天花而未发病；无知情同意的儿童试验按今天标准不合伦理。
- 推广 皇家学会嫌证据不足拒稿后，詹纳自费公布试验和补充病例，使其他医生能够重复验证、传播技术。
- 根除 世卫1967年强化行动，1977年出现最后自然病例；1980年世界卫生大会宣布根除天花，迄今唯一被根除的人类疾病。

参考答案　1 contagious 2 ensued 3 replicate 4 eradicate

阿波罗13号：成功的失败

Apollo 13: The Successful Failure



方形滤芯怎样接进圆形插槽，救回三名宇航员？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On April 11th 1970 NASA launched Apollo 13, meant for the third crewed lunar landing. On April 13th, some 200,000 miles out, an oxygen tank exploded in the service module, home to most of the ship's oxygen and power. The blast crippled the spacecraft; the landing was aborted.

The astronauts took refuge in the lunar module, Aquarius, built to support two men for about two days. Using its engine, the crew steered the craft onto a trajectory around the Moon, whose gravity swung it home. They rationed water, conserved power, and shivered in a bitterly cold cabin.

A quieter hazard emerged: with a third man aboard, carbon dioxide was accumulating. The command module, the crew's capsule, used square filter cartridges; Aquarius's sockets were round. From plastic bags, cardboard, and tape on board, ground engineers devised an adapter, then talked the crew through assembling it. On April 17th the capsule splashed down in the Pacific.

Yet the fault had been built in on Earth. A NASA review board traced the explosion to the tank's thermostat switches, designed for the ship's 28-volt supply. In 1965 the spacecraft's builder rewrote the specification to let the heaters run on 65-volt launch-site power; the switches were never upgraded, a mismatch neither that contractor nor NASA detected. Weeks before launch, to drain the stubborn tank, a heater ran about eight hours. The switches probably malfunctioned; unchecked heat scorched the insulation on the tank's wiring; in flight, that insulation ignited.

NASA called Apollo 13 a "successful failure". The paradox holds: a flight undone by a five-year-old engineering oversight was salvaged within four days by ingenuity on the ground.

中文译文

1970年4月11日，NASA发射阿波罗13号，原计划完成第三次载人登月。4月13日，飞船已飞出约20万英里，服务舱内一个氧气罐爆炸。飞船大部分氧气和电力都由服务舱提供；爆炸使航天器严重受损，登月任务被取消。

宇航员转入“水瓶座”号登月舱避难，而它原本只能维持两个人生存约两天。机组用登月舱发动机调整航向，使飞船沿绕月轨迹飞行，再借月球引力转向地球。他们限量用水、节约电力，在严寒的舱内瑟瑟发抖。

另一种不易察觉的危险出现了：舱内多了第三个人，二氧化碳不断积聚。机组乘坐的指令舱使用方形滤芯，水瓶座号的插槽却是圆的。地面工程师利用飞船上现有的塑料袋、硬纸板和胶带设计转接装置，再通过通话一步步指导机组组装。4月17日，指令舱在太平洋溅落。

然而，隐患早在地球上就已埋下。NASA审查委员会将爆炸原因追溯到罐内的温控开关：它们按飞船28伏电源设计。1965年，航天器承建商修改技术规格，允许加热器使用发射场65伏电源，开关却始终没有升级。承包商和NASA都未发现这种不匹配。发射前几周，为排空那个难以排空的储罐，一台加热器连续运行约八小时。温控开关很可能失灵，失去控制的高温烧焦罐内电线的绝缘层；飞行途中，绝缘材料起火。

NASA把阿波罗13号称为“成功的失败”。这看似矛盾的评价确实成立：一次飞行毁于五年前的工程疏忽，却在四天内靠地面人员的巧思得以挽救。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cripple v.
使严重受损
The blast crippled the spacecraft

abort v.
中止；取消
the landing was aborted

refuge n.
避难；庇护
took refuge in the lunar module

trajectory n.
轨迹；飞行路线
a trajectory around the Moon

ration v.
限量供应；限量使用
They rationed water

hazard n.
危险；隐患
A quieter hazard emerged

emerge v.
出现
A quieter hazard emerged

accumulate v.
积聚
carbon dioxide was accumulating

devise v.
设计出；想出
ground engineers devised an adapter

adapter n.
转接装置
devised an adapter

assemble v.
组装
through assembling it

fault n.
缺陷；故障
the fault had been built in on Earth

thermostat n.
温控器
the tank's thermostat switches

specification n.
技术规格
rewrote the specification

mismatch n.
不匹配
a mismatch

drain v.
排空
to drain the stubborn tank

malfunction v.
失灵；发生故障
The switches probably malfunctioned

insulation n.
绝缘材料
the insulation on the tank's wiring

paradox n.
看似矛盾却成立的事
The paradox holds

oversight n.
疏忽
a five-year-old engineering oversight

salvage v.
挽救
was salvaged within four days

ingenuity n.
巧思；足智多谋
by ingenuity on the ground

文中词汇与搭配

补充词汇共52项。保留每个表达在本文中的意思。

launch v.
发射

crewed adj.
载人的

lunar landing 名词短语
登月

oxygen tank 名词短语
氧气罐

explode v.
爆炸

service module 名词短语
服务舱

blast n.
爆炸；爆炸冲击

spacecraft n.
航天器

astronaut n.
宇航员

take v.
采取；与refuge连用指避难

in prep.
在……中；在此引出避难处

lunar module 名词短语
登月舱

engine n.
发动机

steer v.
操纵；引导航向

craft n.
航天器

onto prep.
进入；在此指进入某条轨迹

swing v.
使转向；甩回

conserve power 动词短语
节约电力

shiver v.
因寒冷而发抖

bitterly adv.
极其；非常

cabin n.
舱室

carbon dioxide 名词短语
二氧化碳

capsule n.
太空舱；返回舱

filter n.
过滤器

cartridge n.
可替换的滤芯

socket n.
插槽；接口

文中词汇与搭配 续

补充词汇共52项。保留每个表达在本文中的意思。

cardboard n.
硬纸板

tape n.
胶带

talk v.
口头指导；与through连用

through prep.
贯穿整个过程；与talk连用

splash down 动词短语
在水上溅落

build in 动词短语
预先嵌入；埋下

review board 名词短语
审查委员会

trace v.
追溯；查明来源

explosion n.
爆炸

to prep.
用于引出追溯到的原因

switch n.
开关

volt n.
伏特；电压单位

builder n.
承建商

rewrite v.
改写；修改

heater n.
加热器

run on 动词短语
靠某种能源运行

upgrade v.
升级

contractor n.
承包商

detect v.
发现；查出

stubborn adj.
难处理的；此处指难排空的

heater n.
加热器

unchecked adj.
未受控制的

scorch v.
烧焦

wiring n.
电线；线路

ignite v.
起火；燃烧

be undone by 动词短语
被……毁掉；因……而失败

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The blast crippled the spacecraft; the landing was _____.

02 They _____ water, conserved power, and shivered in a bitterly cold cabin.

03 A quieter _____ emerged: with a third man aboard, carbon dioxide was accumulating.

04 a flight undone by a five-year-old engineering _____

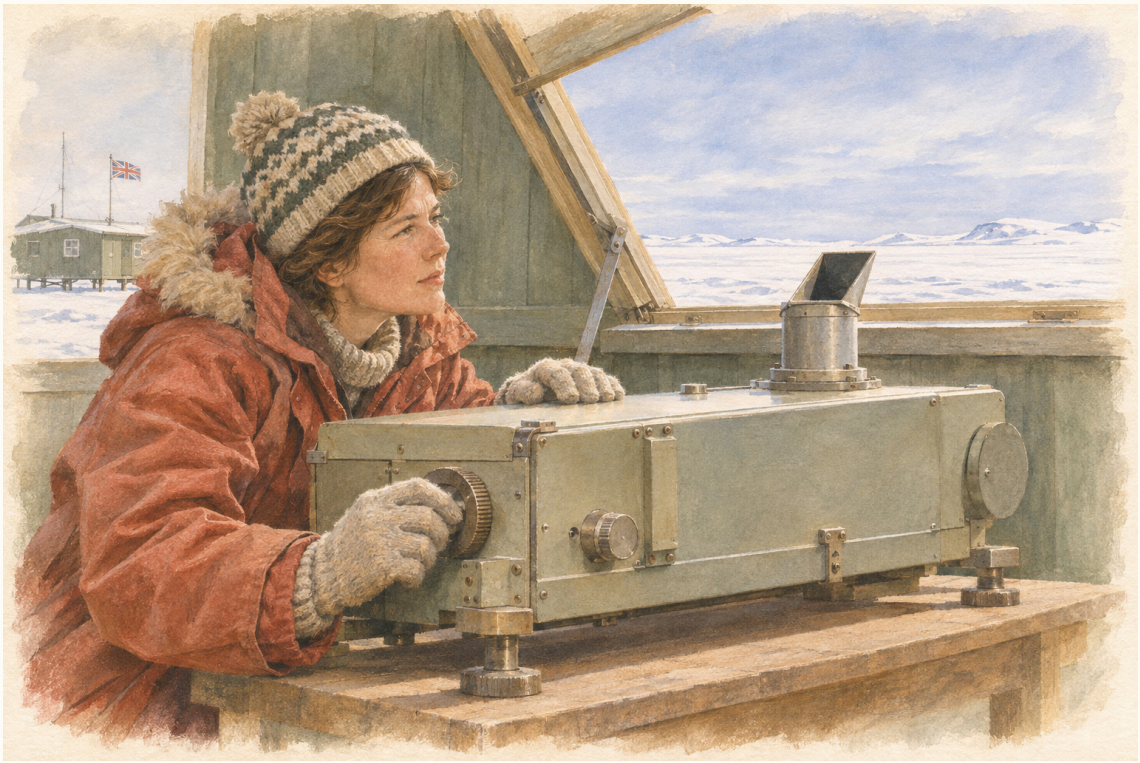
文章主线

- 事故 1970年，阿波罗13号氧气罐爆炸，登月取消，三名宇航员转入原本只供两人约两天的登月舱。
- 险情 机组缺水缺电、忍受严寒，二氧化碳不断积聚，清除它的方形滤芯却装不进圆形插槽。
- 解围 地面工程师用船上的塑料袋、纸板和胶带设计转接器，指导机组组装，让滤芯接上圆形插槽。
- 祸根 发射前，加热器用65伏地面电源，温控开关仍按28伏设计；开关疑失灵，约八小时加热损伤线路绝缘层，飞行中起火。
- 返航 机组用登月舱发动机调整绕月路线，借月球引力返航；爆炸四天后，三人安全溅落太平洋。

参考答案　1 aborted 2 rationed 3 hazard 4 oversight

拯救天空

Saving the Sky



臭氧层的低值，为什么被软件怀疑?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

A layer of ozone shields Earth from much of the sun's ultraviolet radiation. In June 1974 Mario Molina and Sherwood Rowland, chemists in California, warned in Nature of a menace in fridges and spray cans. Chlorofluorocarbons (CFCs) are so inert that they drift intact into the stratosphere, high in the atmosphere, where they can liberate chlorine, a catalyst that destroys ozone molecule after molecule. The pair foresaw gradual erosion.

The alarm, when it sounded, was abrupt. In May 1985 Joseph Farman and colleagues at the British Antarctic Survey reported that ozone over Halley Bay, their Antarctic station, had plummeted each southern spring since the mid-1970s. Their humble Dobson spectrophotometer gauged ozone from the ground. The staff were persistent, reading it in a painstaking routine of manual observation.

Yet NASA's Nimbus-7 satellite, a sophisticated eye in orbit, had recorded the same lows. Its software flagged them as dubious, since ozone that low had never been reliably measured. NASA says it was still verifying them when Farman published. The sensor had worked; expectation had not.

Remarkably, the response was swift. In September 1987 diplomats signed the Montreal Protocol, a binding accord to curb CFCs. Amendments tightened it: in June 1990 affluent countries pledged to phase out CFCs by 2000. In September 2009 the Protocol and its parent Vienna Convention became the UN's first treaties to win universal ratification.

In January 2023 a UN-backed assessment forecast recovery to 1980 levels around 2066 over Antarctica, 2045 over the Arctic, and 2040 elsewhere. Healing will take decades. The sky's rescue began with diligent observers who trusted that a baffling number was no glitch, and published it.

中文译文

臭氧层为地球挡住了太阳的大量紫外线。1974年6月，在加州工作的化学家马里奥·莫利纳和舍伍德·罗兰在《自然》发出警告：冰箱和喷雾罐里藏着一种威胁。氯氟烃，也就是CFCs，性质很不活泼，能完整地飘入高空的平流层，在那里释放氯。氯是一种催化剂，会接连破坏臭氧分子。两人预料到的是逐步发生的损耗。

真正的警报却来得很突然。1985年5月，约瑟夫·法曼和英国南极调查局的同事报告：自20世纪70年代中期起，他们南极哈雷湾考察站上空的臭氧，每逢南半球春季都会骤降。他们使用一台不起眼的多布森分光光度计，从地面测量臭氧。工作人员持之以恒，遵循细致的固定流程，靠人工观测读数。

然而，美国国家航空航天局的雨云7号卫星，这只精密的太空眼睛，也已测到同样的低值。由于此前从未可靠地测到这么低的臭氧含量，软件把这些读数标成可疑。NASA表示，法曼发表论文时，他们仍在核实数据。传感器没有出错，出错的是人的预期。

各国的反应迅速得令人瞩目。1987年9月，外交官签署了具有约束力的《蒙特利尔议定书》，限制氯氟烃的使用。之后的修正案进一步收紧管控：1990年6月，富裕国家承诺在2000年前逐步淘汰氯氟烃。2009年9月，议定书及其所依据的《维也纳公约》，成为联合国首批获得所有国家批准的条约。

2023年1月，一份由联合国支持的评估报告预测，臭氧层恢复到1980年水平的时间，南极上空约为2066年，北极约为2045年，其他地区约为2040年。愈合仍需数十年。拯救天空的起点，是一群勤勉的观测者：他们相信，那组令人费解的数字不是故障，于是将其发表。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ozone n.
臭氧
layer of ozone shields Earth from much

menace n.
威胁
of a menace in fridges and spray cans.

inert adj.
化学性质不活泼的
inert that they drift intact into the

intact adj.
完好无损的
inert that they drift intact into the

liberate v.
释放
where they can liberate chlorine,

catalyst n.
催化剂
catalyst that destroys ozone molecule

erosion n.
侵蚀；逐步损耗
The pair foresaw gradual erosion.

abrupt adj.
突然的
alarm, when it sounded, was abrupt.

plummet v.
骤降
plummeted each southern spring since

persistent adj.
坚持不懈的
The staff were persistent, reading it

painstaking adj.
细致认真的
it in a painstaking routine of manual

sophisticated adj.
精密复杂的
a sophisticated eye in orbit,

dubious adj.
可疑的
Its software flagged them as dubious,

verify v.
核实
was still verifying them when Farman

binding adj.
有约束力的
a binding accord to curb CFCs.

accord n.
协定
a binding accord to curb CFCs.

curb v.
限制；遏制
a binding accord to curb CFCs.

pledge v.
承诺
June 1990 affluent countries pledged

phase out phr.v.
逐步淘汰
countries pledged to phase out CFCs by

ratification n.
正式批准
to win universal ratification.

assessment n.
评估
2023 a UN-backed assessment forecast

baffling adj.
令人费解的
a baffling number was no glitch, and

文中词汇与搭配

补充词汇共40项。保留每个表达在本文中的意思。

layer n.
层

shield from phr.v.
与from连用：保护；遮挡

from prep.
标示所防护的危害

ultraviolet adj.
紫外线的

radiation n.
辐射

spray n.
喷雾

atmosphere n.
大气层

chlorine n.
氯

molecule n.
分子

gradual adj.
逐渐的

alarm n.
警报

Antarctic adj.
南极的

humble adj.
不起眼的；简朴的

gauge v.
测量

routine n.
固定流程；例行程序

manual adj.
人工的；手工的

observation n.
观测

satellite n.
卫星

in orbit prep.phr.
在轨道上运行

flag as phr.v.
与as连用：标记为

文中词汇与搭配 续

补充词汇共40项。保留每个表达在本文中的意思。

as prep.
标示被认定的类别

sensor n.
传感器

expectation n.
预期

remarkably adv.
引人注目地

swift adj.
迅速的

diplomat n.
外交官

protocol n.
议定书

amendment n.
修正案

tighten v.
收紧；加强

affluent adj.
富裕的

treaty n.
条约

universal adj.
全体的；覆盖所有国家的

forecast v.
预测

recovery n.
恢复

Antarctica proper n.
南极洲

Arctic proper n.
北极地区

healing n.
愈合；修复

rescue n.
拯救

diligent adj.
勤勉的

glitch n.
小故障

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a _____ that destroys ozone molecule after molecule.

02 Its software flagged them as _____,

03 a _____ accord to curb CFCs.

04 affluent countries pledged to _____ CFCs by 2000.

文章主线

- 预警 臭氧挡紫外线。1974年，化学家警告：冰箱和喷罐的氯氟烃可进入平流层，释放氯持续破坏臭氧。
- 发现 1985年南极地面观测报告：哈雷湾臭氧自70年代中期起，每逢南半球春季便骤降。
- 误判 卫星同样测到低值，软件因罕见而标为可疑；NASA称当时尚在核实，是预期出了错，并非仪器失灵。
- 合作 各国1987年签署《蒙特利尔议定书》，1990年富国承诺2000年前淘汰氯氟烃，议定书2009年获所有国家批准。
- 恢复 2023年评估预测恢复至1980年水平：其他地区约2040年、北极约2045年、南极约2066年。

参考答案　1 catalyst 2 dubious 3 binding 4 phase out

写给外星人的一张唱片

The Record Humanity Threw into the Stars

如果有人找到这张唱片，会认识地球吗?

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RECALL 回想巩固

英文原文

In late summer 1977, NASA fastened a disc wrought from gilded copper, 30 centimetres wide, to each Voyager spaceship before takeoff. Nobody could foresee who, if anyone, would play one.

Advisers under Carl Sagan of Cornell distilled our earthly abode into this compendium. Voices greet the listener in 55 languages, followed by the rumble of thunder, the whistle of wind, whales, footsteps, laughter, and a rhythmic heartbeat. About 90 minutes of eclectic music cover a spectrum of cultures and eras, notably Bach and Chuck Berry's Johnny B. Goode.

An array of 115 images, encoded in analogue form, sits beside printed words from two statesmen, Jimmy Carter and Kurt Waldheim. The sleeve gives guidance on playing the disc, plus a diagram fixing the Sun with pulsars as beacons.

On 14 February 1990, at Sagan's request, Voyager 1 pivoted its camera for a glance backwards across about six billion kilometres. Sixty frames formed a panorama of the solar system, a mosaic in which Earth was a glimmer about an eighth of a pixel across. They reached Earth's receivers between March and 1 May, each signal taking five and a half hours.

In a 1994 book, Sagan offered that pale dot as a perspective on our small and ephemeral place amid cosmic vastness.

Voyager 1 crossed into interstellar space in August 2012, the first human-made object there, and Voyager 2 followed in 2018. Both work on with a longevity unmatched at NASA, but depletion of their power should silence them within a few years.

The Voyagers' discs will glide on outwards through the void as messengers with a map leading back to the Sun.

中文译文

1977年夏末，美国航天局在两艘旅行者号探测器起飞前，各固定上一张直径30厘米的镀金铜唱片。谁也不知道，究竟会不会有谁、又会是谁来播放它们。

康奈尔大学卡尔·萨根带领的顾问团队，把地球家园浓缩成这份汇编。人声用55种语言向听者问候，接着是雷声、风声、鲸声、脚步声、笑声和有节奏的心跳。约90分钟的音乐包容了不同文化和时代，其中包括巴赫的作品，以及查克·贝里的《Johnny B. Goode》。

唱片中还有以模拟方式编码的115幅图像，并附有两位政治家吉米·卡特和库尔特·瓦尔德海姆的书面致辞。封套上给出了播放方法，以及一幅用脉冲星作信标、标定太阳位置的示意图。

1990年2月14日，应萨根的请求，旅行者1号转动相机，从约60亿公里外回望。60帧画面组成太阳系全景；地球在其中只是一点微光，宽度约为八分之一个像素。这些画面在3月至5月1日之间传回地球，信号单程需要五个半小时。

在1994年的一本书中，萨根借那颗暗淡的小点，帮助人们重新审视自己在浩瀚宇宙中渺小、短暂的位置。

旅行者1号于2012年8月进入星际空间，成为首个到达那里的人工物体；旅行者2号于2018年跟进。两者仍在工作，寿命在美国航天局的任务中无可匹敌；但随着电力耗尽，预计几年内就会沉寂。

旅行者号携带的唱片仍将继续向外穿越虚空，作为来自人类的信使，带着一幅指回太阳的地图。

文中的信号需五个半小时，是从当时旅行者1号的位置到地球的单程时间。

核心词汇

先记这22项。连着下方的原文短语一起回想。

fasten v.
固定；系牢
NASA fastened a disc

wrought adj.
制成的；加工成形的
a disc wrought from gilded copper

gilded adj.
镀金的
gilded copper

foresee v.
预见
Nobody could foresee

distil v.
提炼；浓缩
distilled our earthly abode

abode n.
住所；家园
our earthly abode

compendium n.
汇编
into this compendium

eclectic adj.
兼收并蓄的
About 90 minutes of eclectic music

encode v.
编码
encoded in analogue form

guidance n.
指引
guidance on playing the disc

beacon n.
信标
with pulsars as beacons

pivot v.
转动
Voyager 1 pivoted its camera

panorama n.
全景
a panorama of the solar system

mosaic n.
拼接图
a mosaic in which Earth was a glimmer

pixel n.
像素
an eighth of a pixel across

signal n.
信号
signal taking five and a half hours

perspective n.
看待问题的视角
as a perspective

ephemeral adj.
短暂的
our small and ephemeral place

interstellar adj.
恒星之间的
crossed into interstellar space

longevity n.
长寿；长久持续
a longevity unmatched at NASA

unmatched adj.
无可匹敌的
a longevity unmatched at NASA

depletion n.
耗竭
depletion of their power

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

disc n.
唱片；圆盘

copper n.
铜

centimetre n.
厘米

spaceship n.
宇宙飞船；航天器

takeoff n.
起飞

adviser n.
顾问

earthly adj.
地球上的

greet v.
向……问候

listener n.
听者

rumble n.
隆隆声

thunder n.
雷声

whistle n.
呼啸声

laughter n.
笑声

rhythmic adj.
有节奏的

heartbeat n.
心跳

a spectrum of 名词短语
一系列；多种多样的

era n.
时代

notably adv.
尤其

an array of 名词短语
一系列

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

statesman n.
政治家

sleeve n.
唱片封套

diagram n.
示意图

at prep.
应（某人的请求）

request n.
请求

glance n.
一瞥

backwards adv.
向后

frame n.
画面；帧

glimmer n.
微光

across adv.
从一侧到另一侧；宽

receiver n.
接收设备

pale adj.
暗淡的

cosmic adj.
宇宙的

vastness n.
浩瀚；辽阔

human-made adj.
人造的

glide on 动词短语
继续滑行

outwards adv.
向外

void n.
虚空

messenger n.
信使

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Nobody could _____ who, if anyone, would play one.

02 Advisers under Carl Sagan of Cornell distilled our earthly abode into this _____.

03 An array of 115 images, _____ in analogue form, sits beside printed words from two statesmen, Jimmy Carter and Kurt Waldheim.

04 In a 1994 book, Sagan offered that pale dot as a perspective on our small and _____ place amid cosmic vastness.

文章主线

- 出发 1977年，两艘旅行者号探测器各携一张镀金铜唱片，萨根团队以此介绍地球；会不会有人播放，仍是未知。
- 内容 唱片有55种语言的问候、自然和生活声响、约90分钟音乐与115幅图像，还附时任美国总统与联合国秘书长的文字致辞。
- 指引 封套附播放指引，并把脉冲星当作定位信标，标出太阳的位置，告诉潜在听者唱片来自哪里。
- 回望 1990年，旅行者1号在约60亿公里外回拍，地球仅约八分之一像素；萨根借此审视人类在宇宙中的渺小短暂。
- 远行 旅行者1号2012年成为首个进入星际空间的人造物，2号2018年跟进；电力终会耗尽，唱片仍将远行。

参考答案　1 foresee 2 compendium 3 encoded 4 ephemeral